

NMR spectra analysis pipeline using R

This is an initial R pipeline for importing and processing NMR spectral data from Bruker topspin software. It was run in Ubuntu 16.04 “Xenial”.

R packages

The key R packages incorporated in this pipeline are BATMAN, hyperspec, and specmine

Other online resources for spectral nmr analysis

There are also useful online tools for spectral analyses such as nmrML (nmrml.org) which has useful function for converting TopSpin peaks into format that could be used by other r packages, and NMRProcFlow, which in itself you can also import TopSpin data and convert into format that could be used by itself or other r packages for further pre-processing and stastical analyses.

Setting R environment and loading key libraries

```
setwd("/home/bajuna/rworksgenomics/anisha")
library(specmine)
library(hyperSpec)
library(batman)
```

Reading Bruker topspin Files

Read Bruker topspin file from directory using specmine

```
new_spec_read_bruker <- read_Bruker_files("~/rworksgenomics/anisha/examdata", zipped = F)
```

```
## Reading sample exam_DNMR_ipr2sic in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_ipr2sic
## Reading sample exam_DNMR_ipr2sic in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_ipr2sic
## Reading sample exam_DNMR_ipr2sic in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_ipr2sic
## Reading sample exam_DNMR_ipr2sic in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_ipr2sic
## Reading sample exam_DNMR_ipr2sic in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_ipr2sic
## Reading sample exam_DNMR_ipr2sic in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_ipr2sic
## Reading sample exam_DNMR_ipr2sic in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_ipr2sic
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
```

```
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam_DNMR_Me2NCOMe in /home/bajuna/rworksgenomics/anisha/examdata/exam_DNMR_Me2NCOMe
## Reading sample exam1d_13C in /home/bajuna/rworksgenomics/anisha/examdata/exam1d_13C/1/pdata/1
## Reading sample exam1d_13C in /home/bajuna/rworksgenomics/anisha/examdata/exam1d_13C/2/pdata/1
## Bruker file does not exist in datapath, or other problems with bruker files...
## Reading sample exam1d_13C in /home/bajuna/rworksgenomics/anisha/examdata/exam1d_13C/3/pdata/1
## Bruker file does not exist in datapath, or other problems with bruker files...
## Reading sample exam1d_13C in /home/bajuna/rworksgenomics/anisha/examdata/exam1d_13C/4/pdata/1
## Bruker file does not exist in datapath, or other problems with bruker files...
## Reading sample exam1d_1H in /home/bajuna/rworksgenomics/anisha/examdata/exam1d_1H/1/pdata/1
## Reading sample exam1d_1H in /home/bajuna/rworksgenomics/anisha/examdata/exam1d_1H/2/pdata/1
## Creating dataset (this may take a while)
## Done.
```

Check if the dataset is valid spec data

```
check_dataset(new_spec_read_bruker)
```

```
## Valid dataset
```

Summary of the imported data

```
sum_dataset(new_spec_read_bruker)
```

```
## Dataset summary:
## Valid dataset
## Description:
## Type of data: nmr-spectra
## Number of samples: 4
## Number of data points 50597
## Number of missing values in data: 126448
## Mean of data values: 116562.5
## Median of data values: 252.0625
## Standard deviation: 1792881
## Range of values: -3632858 125379201
## Quantiles:
##          0%          25%          50%          75%          100%
## -3.632858e+06 -4.749992e+03 2.520625e+02 2.124409e+04 1.253792e+08
```

Retrieve sample names

```
get_sample_names(new_spec_read_bruker)
```

```
## [1] "exam_DNMR_ipr2sic" "exam_DNMR_Me2NCOMe" "exam1d_13C"
## [4] "exam1d_1H"
```

Check the dataset whether is from the spectral data

```
is_spectra(new_spec_read_bruker)
```

```
## [1] TRUE
```

Missing values (NAs) imputation

Remove NAs using specmine

```
removed_na_new_spec_read_bruker <- impute_nas_mean(new_spec_read_bruker)
```

Check again summary of the data

```
sum_dataset(removed_na_new_spec_read Bruker)
```

```
## Dataset summary:
## Valid dataset
## Description:
## Type of data:  nmr-spectra
## Number of samples:  4
## Number of data points 50597
## Number of missing values in data:  0
## Mean of data values:  122763.1
## Median of data values:  118.625
## Standard deviation:  1879662
## Range of values:  -3632858 125379201
## Quantiles:
##           0%           25%           50%           75%           100%
## -3632858.500  -28529.750    118.625    51301.250 125379200.875
```

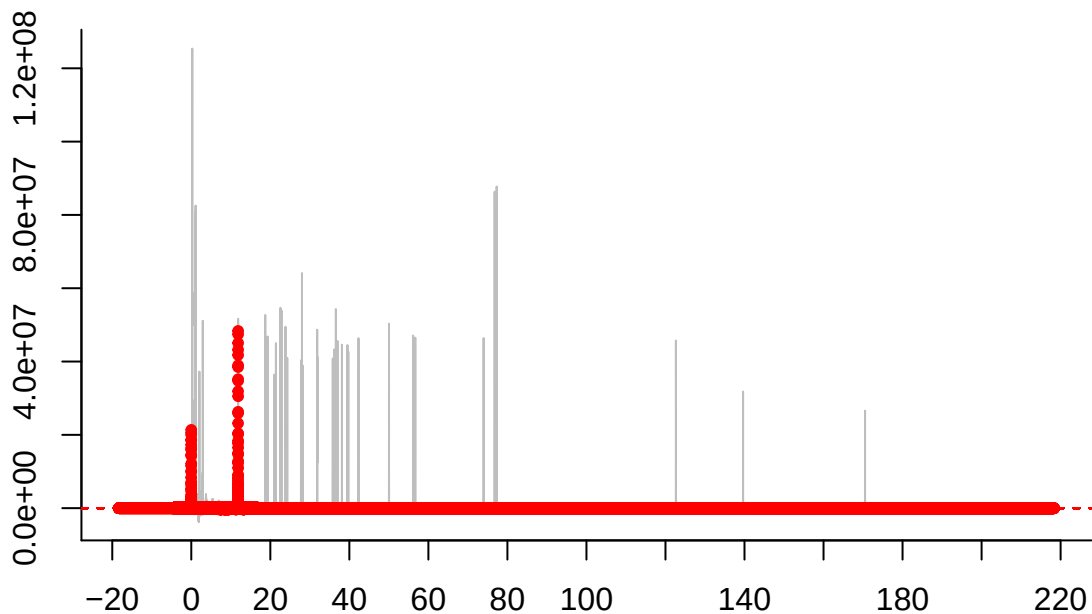
Remove NAs using hyperSpec.

Convert spec data first from specmine into hyperspec

```
hyperspec_read Bruker <- convert_to_hyperspec(new_spec_read Bruker)
```

Then remove NAs This produce a single plot

```
hyperspec_read Bruker_no_NA <- spc.NA.approx(hyperspec_read Bruker, debuglevel = 1)
```



PRE-PROCESSING

Background Correction

Compute background correction after removing missing values (NAs)

```
corrected_spec Bruker <- background_correction(removed_na_new_spec_read Bruker)
```

Baseline Correction

A. With specmine using imputed data

```
base_corrected_spec Bruker <- baseline_correction(removed_na_new_spec_read Bruker, method = "modpolyfit")
```

B. With hyperSpec using imputed data

```
baselines_hyperSpec <- spc.fit.poly.below(hyperspec_read Bruker_no_NA)
```

Normalise the spectra data

A. Using specmine with non NAs dataset : Two methods used

```
normalized_spectra_specmine <- normalize(removed_na_new_spec_read Bruker, method = "median")  
normalized_spectra_specmine1 <- normalize(removed_na_new_spec_read Bruker, method = "sum", constant = 1)
```

B. Using hyperspec with non NAs dataset

```
normalized_spectra_hyperspec <- normalize01(hyperspec_read Bruker_no_NA)
```

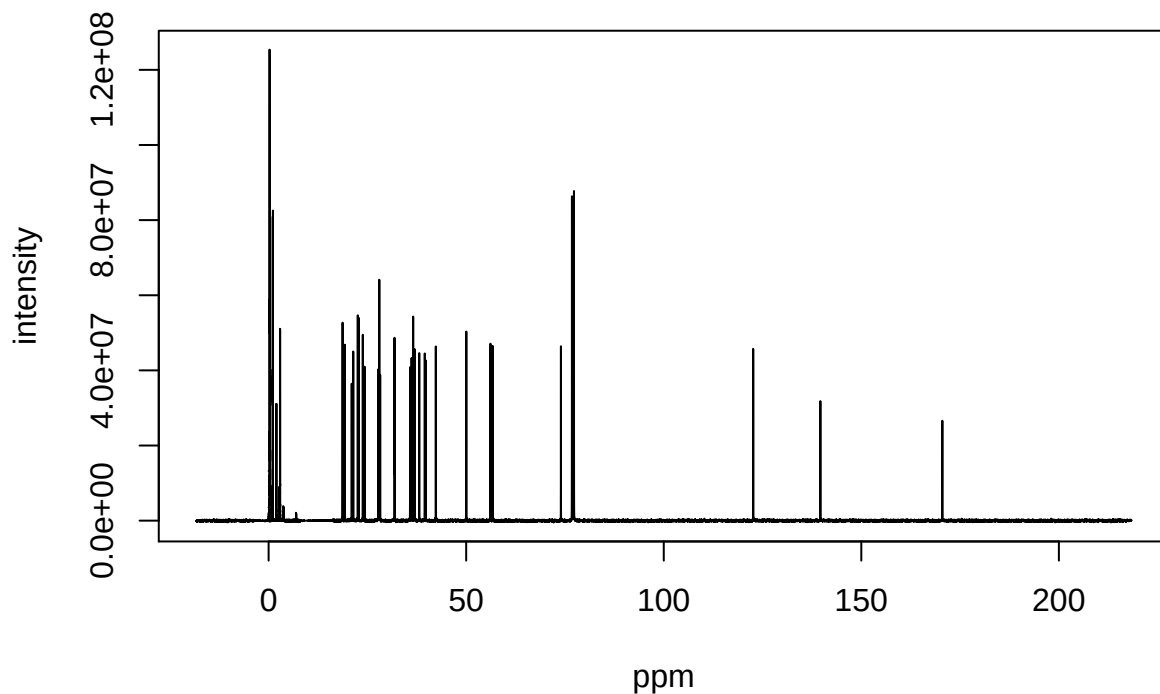
PLOTTING SPECTRA DATA

With specmine

Using dataset containing NAs

```
plot_spectra_simple(new_spec_read Bruker, samples = c("exam_DNMR_ipr2sic", "exam_DNMR_Me2NCOMe", "exam1"))
```

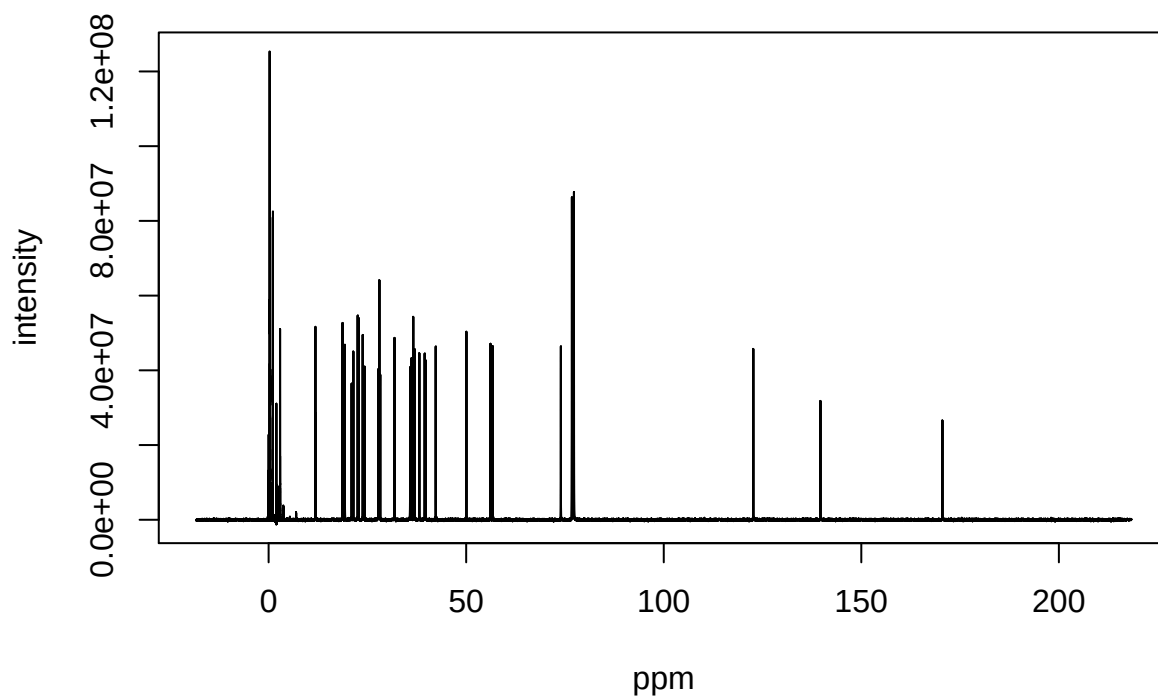
Unnormalized spectra with data containing NAs



Using dataset that doesn't contain NAs

```
plot_spectra_simple(removed_na_new_spec_read Bruker, samples = c("exam_DNMR_ipr2sic", "exam_DNMR_Me2NCOMe"))
```

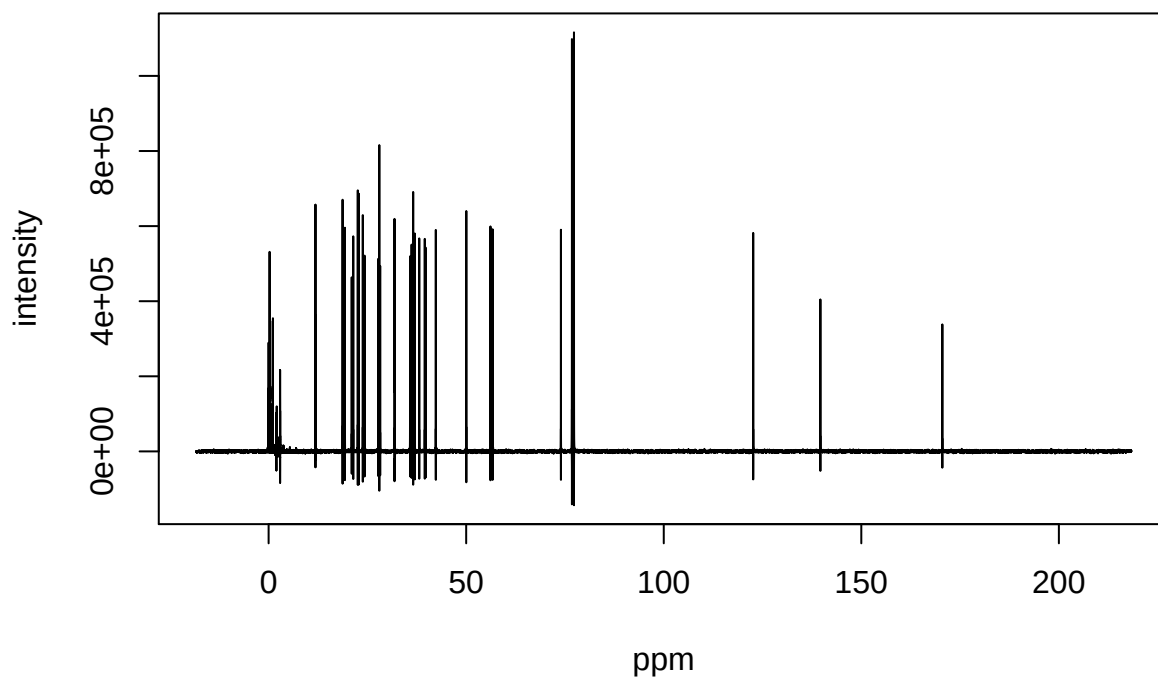
Unnormalized spectra with data containing no NAs



Plot normalised data

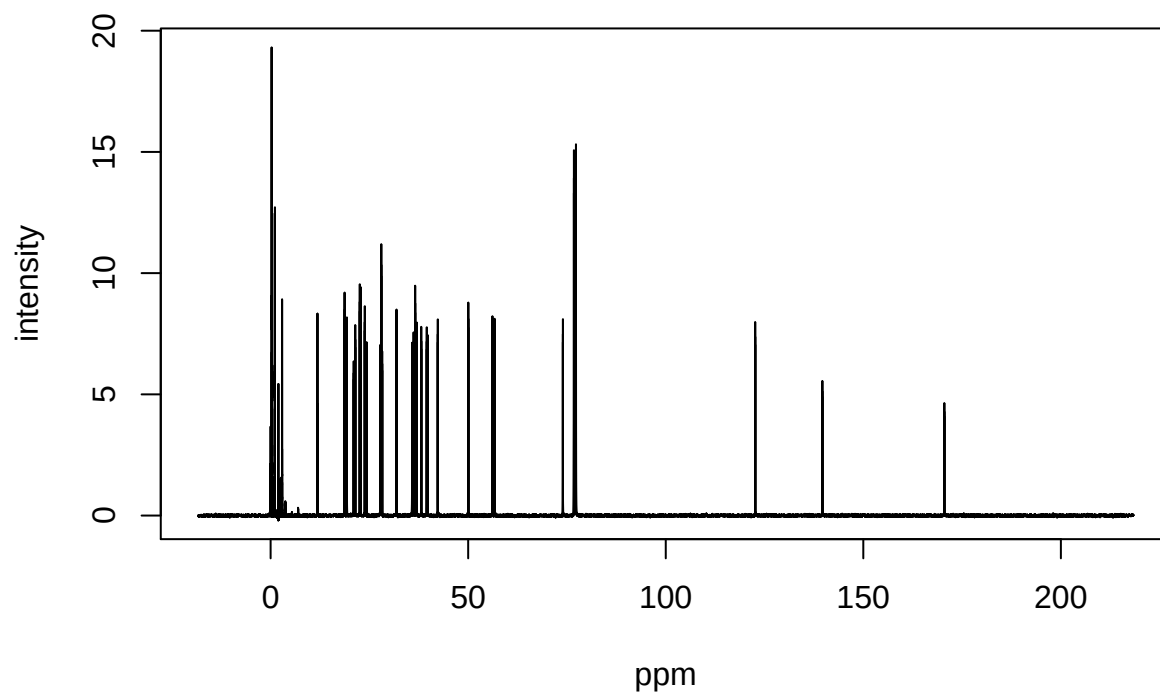
```
plot_spectra_simple(normalized_spectra_specmine, samples = c("exam_DNMR_ipr2sic", "exam_DNMR_Me2NCOMe",
```

Normalized spectra – median



```
plot_spectra_simple(normalized_spectra_specmine1, samples = c("exam_DNMR_ipr2sic", "exam_DNMR_Me2NCOMe"
```

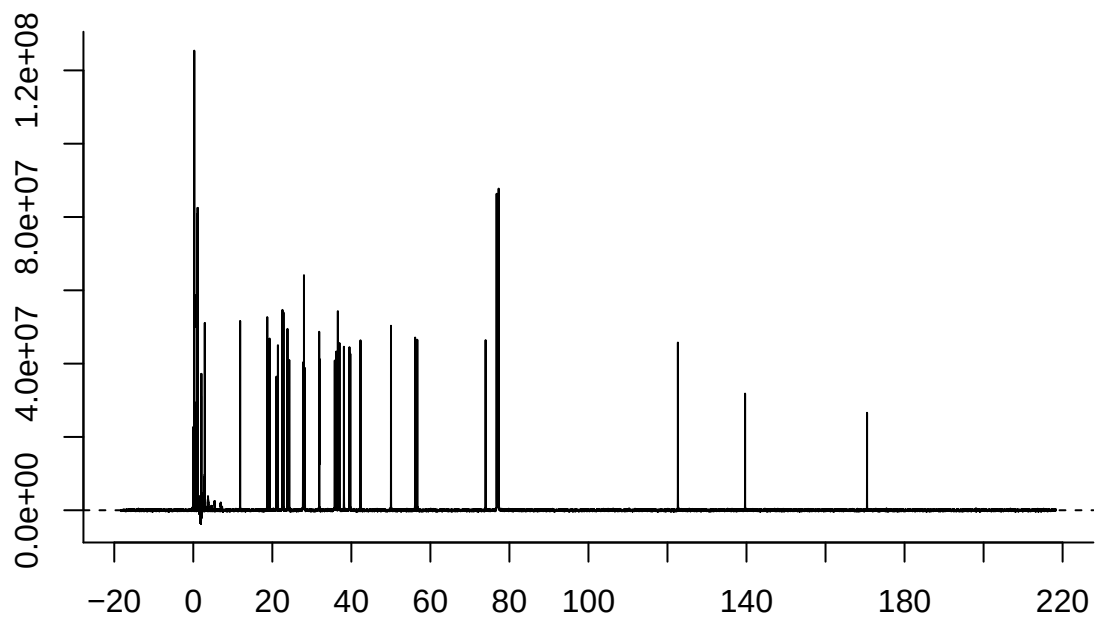
Normalized spectra – sum



With hyperspec

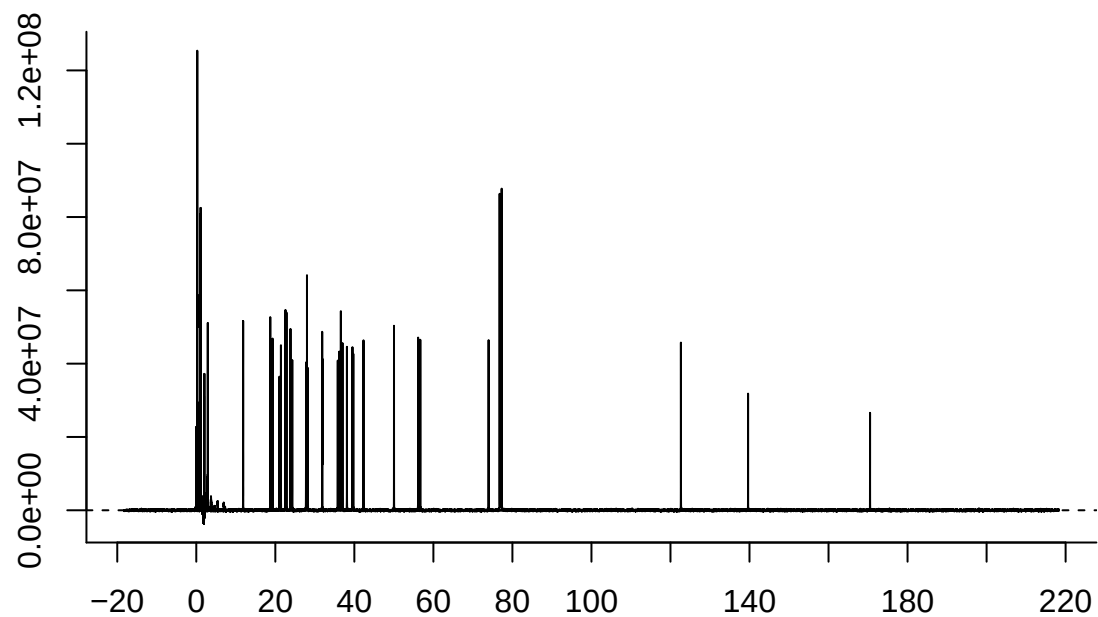
Containing NAs

```
plot(hyperspec_read_bruker)
```

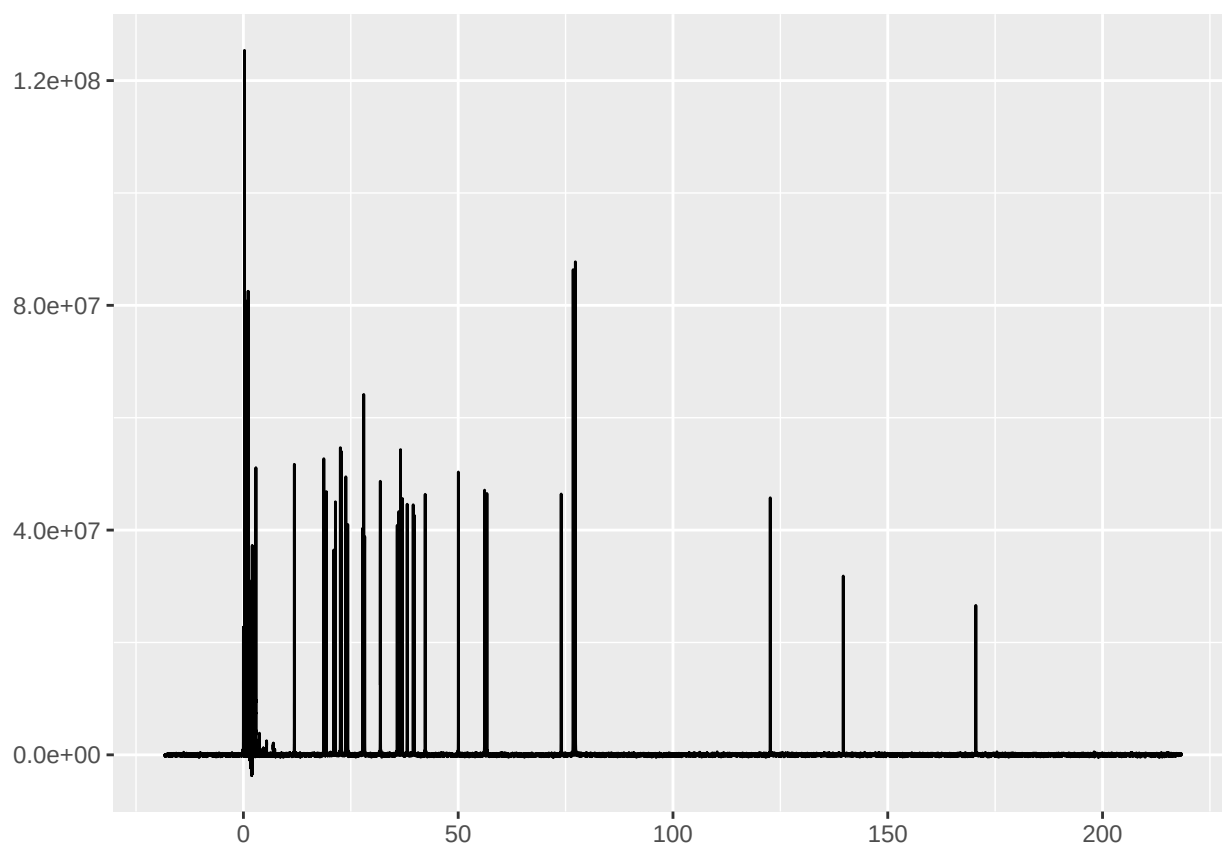


Containing no NAs

```
plot(hyperspec_read_bruker_no_NA)
```

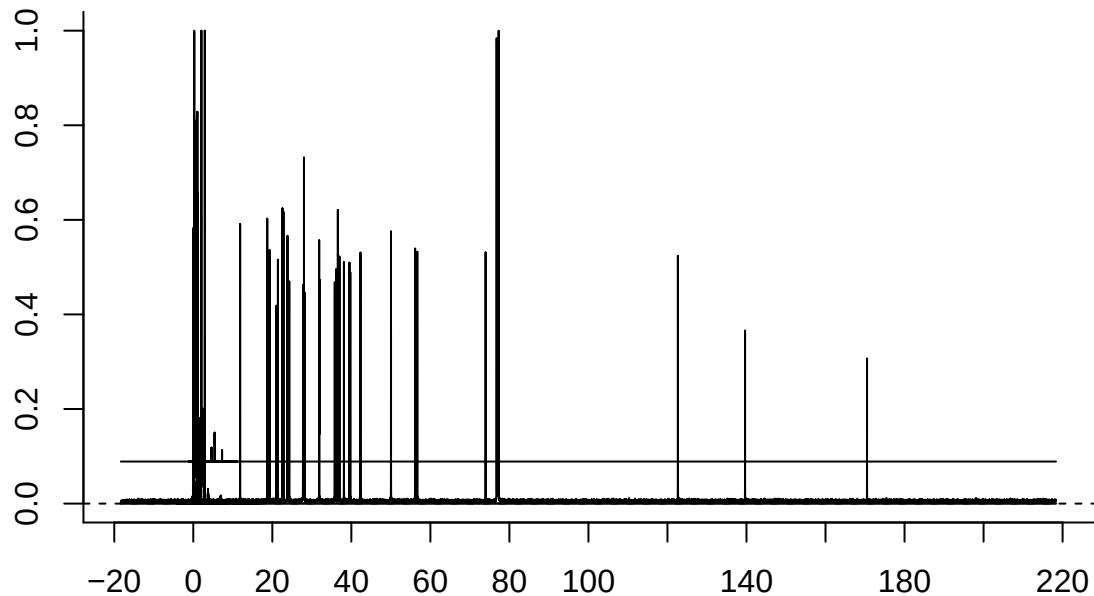


```
qplotspc(hyperspec_read_bruker_no_NA)
```



Plotting using normalized data with no NAs

```
plot(normalized_spectra_hyperspec)
```



EXPORT SPECTRA DATA INTO DATAFRAME

Alternative - 1

Get the data frame from the dataset

```
bruker_specm_df <- get_data_as_df(new_spec_read_bruker)
```

Write to the hard drive as csv file

```
write.csv(bruker_specm_df, "mysamples.csv")
```

The data frame could be generated using the imputed data

```
bruker_specm_df1 <- get_data_as_df(removed_na_new_spec_read_bruker)
```

Write to the new dataframe to the hard drive as csv file

```
write.csv(bruker_specm_df1, "mysamples1.csv")
```

Alternative - 2

```
bruker_hyperspec_df <- as.data.frame(hyperspec_read_bruker_no_NA, row.names = NULL)
```

And then write to csv file

```
write.csv(bruker_hyperspec_df, "mysample2.csv")
```

ALIGNING SPECTRA

Detecting lower peaks with minimum intensity baseline and align them.

```
align_peaks <- detect_nmr_peaks_from_dataset(normalized_spectra_specmine, baseline_tresh = 50000, ap.me
```

```
## [1] "j =7 peak =91 savedpeak = 859"    "j =7 peak =117 savedpeak = 885"
## [3] "j =7 peak =158 savedpeak = 926"   "j =7 peak =192 savedpeak = 960"
## [5] "j =7 peak =224 savedpeak = 992"   "j =7 peak =250 savedpeak = 1018"
```



```

## [7] "j =7 peak =261 savedpeak = 1029" "j =7 peak =297 savedpeak = 1065"
## [9] "j =7 peak =347 savedpeak = 1115" "j =7 peak =365 savedpeak = 1133"
## [11] "j =7 peak =388 savedpeak = 1156" "j =7 peak =417 savedpeak = 1185"
## [1] "j =8 peak =96 savedpeak = 992" "j =8 peak =122 savedpeak = 1018"
## [3] "j =8 peak =133 savedpeak = 1029" "j =8 peak =169 savedpeak = 1065"
## [5] "j =8 peak =219 savedpeak = 1115" "j =8 peak =237 savedpeak = 1133"
## [7] "j =8 peak =260 savedpeak = 1156" "j =8 peak =289 savedpeak = 1185"
## [9] "j =8 peak =310 savedpeak = 1206" "j =8 peak =334 savedpeak = 1230"
## [11] "j =8 peak =380 savedpeak = 1276"
## [1] "j =9 peak =91 savedpeak = 1115" "j =9 peak =109 savedpeak = 1133"
## [3] "j =9 peak =132 savedpeak = 1156" "j =9 peak =161 savedpeak = 1185"
## [5] "j =9 peak =182 savedpeak = 1206" "j =9 peak =206 savedpeak = 1230"
## [7] "j =9 peak =252 savedpeak = 1276" "j =9 peak =312 savedpeak = 1336"
## [9] "j =9 peak =353 savedpeak = 1377" "j =9 peak =392 savedpeak = 1416"
## [1] "j =10 peak =78 savedpeak = 1230" "j =10 peak =124 savedpeak = 1276"
## [3] "j =10 peak =184 savedpeak = 1336" "j =10 peak =225 savedpeak = 1377"
## [5] "j =10 peak =264 savedpeak = 1416" "j =10 peak =348 savedpeak = 1500"
## [7] "j =10 peak =379 savedpeak = 1531" "j =10 peak =407 savedpeak = 1559"
## [9] "j =10 peak =434 savedpeak = 1586"
## [1] "j =11 peak =97 savedpeak = 1377" "j =11 peak =136 savedpeak = 1416"
## [3] "j =11 peak =220 savedpeak = 1500" "j =11 peak =256 savedpeak = 1536"
## [5] "j =11 peak =279 savedpeak = 1559" "j =11 peak =306 savedpeak = 1586"
## [7] "j =11 peak =320 savedpeak = 1600" "j =11 peak =345 savedpeak = 1625"
## [9] "j =11 peak =383 savedpeak = 1663" "j =11 peak =425 savedpeak = 1705"
## [1] "j =12 peak =92 savedpeak = 1500" "j =12 peak =128 savedpeak = 1536"
## [3] "j =12 peak =151 savedpeak = 1559" "j =12 peak =178 savedpeak = 1586"
## [5] "j =12 peak =192 savedpeak = 1600" "j =12 peak =217 savedpeak = 1625"
## [7] "j =12 peak =255 savedpeak = 1663" "j =12 peak =297 savedpeak = 1705"
## [9] "j =12 peak =318 savedpeak = 1726" "j =12 peak =338 savedpeak = 1746"
## [11] "j =12 peak =365 savedpeak = 1773" "j =12 peak =400 savedpeak = 1808"
## [13] "j =12 peak =433 savedpeak = 1841"
## [1] "j =28 peak =76 savedpeak = 3532" "j =28 peak =112 savedpeak = 3568"
## [3] "j =28 peak =129 savedpeak = 3585" "j =28 peak =217 savedpeak = 3673"
## [5] "j =28 peak =239 savedpeak = 3695" "j =28 peak =259 savedpeak = 3715"
## [7] "j =28 peak =275 savedpeak = 3731" "j =28 peak =302 savedpeak = 3758"
## [9] "j =28 peak =349 savedpeak = 3805" "j =28 peak =385 savedpeak = 3841"
## [11] "j =28 peak =414 savedpeak = 3870" "j =28 peak =429 savedpeak = 3885"
## [1] "j =29 peak =89 savedpeak = 3673" "j =29 peak =111 savedpeak = 3695"
## [3] "j =29 peak =131 savedpeak = 3715" "j =29 peak =147 savedpeak = 3731"
## [5] "j =29 peak =174 savedpeak = 3758" "j =29 peak =221 savedpeak = 3805"
## [7] "j =29 peak =257 savedpeak = 3841" "j =29 peak =286 savedpeak = 3870"
## [9] "j =29 peak =301 savedpeak = 3885" "j =29 peak =329 savedpeak = 3913"
## [11] "j =29 peak =341 savedpeak = 3925" "j =29 peak =370 savedpeak = 3954"
## [13] "j =29 peak =388 savedpeak = 3972" "j =29 peak =411 savedpeak = 3995"
## [15] "j =29 peak =436 savedpeak = 4020"
## [1] "j =30 peak =93 savedpeak = 3805" "j =30 peak =129 savedpeak = 3841"
## [3] "j =30 peak =158 savedpeak = 3870" "j =30 peak =173 savedpeak = 3885"
## [5] "j =30 peak =201 savedpeak = 3913" "j =30 peak =213 savedpeak = 3925"
## [7] "j =30 peak =242 savedpeak = 3954" "j =30 peak =260 savedpeak = 3972"
## [9] "j =30 peak =283 savedpeak = 3995" "j =30 peak =308 savedpeak = 4020"
## [11] "j =30 peak =336 savedpeak = 4048" "j =30 peak =362 savedpeak = 4074"
## [13] "j =30 peak =372 savedpeak = 4084" "j =30 peak =394 savedpeak = 4106"
## [15] "j =30 peak =435 savedpeak = 4147"
## [1] "j =31 peak =85 savedpeak = 3925" "j =31 peak =114 savedpeak = 3954"

```

```

## [3] "j =31 peak =132 savedpeak = 3972" "j =31 peak =155 savedpeak = 3995"
## [5] "j =31 peak =180 savedpeak = 4020" "j =31 peak =208 savedpeak = 4048"
## [7] "j =31 peak =234 savedpeak = 4074" "j =31 peak =244 savedpeak = 4084"
## [9] "j =31 peak =266 savedpeak = 4106" "j =31 peak =307 savedpeak = 4147"
## [11] "j =31 peak =336 savedpeak = 4176" "j =31 peak =369 savedpeak = 4209"
## [13] "j =31 peak =400 savedpeak = 4240" "j =31 peak =410 savedpeak = 4250"
## [1] "j =32 peak =80 savedpeak = 4048" "j =32 peak =106 savedpeak = 4074"
## [3] "j =32 peak =116 savedpeak = 4084" "j =32 peak =138 savedpeak = 4106"
## [5] "j =32 peak =162 savedpeak = 4130" "j =32 peak =179 savedpeak = 4147"
## [7] "j =32 peak =208 savedpeak = 4176" "j =32 peak =241 savedpeak = 4209"
## [9] "j =32 peak =272 savedpeak = 4240" "j =32 peak =282 savedpeak = 4250"
## [11] "j =32 peak =310 savedpeak = 4278" "j =32 peak =332 savedpeak = 4300"
## [13] "j =32 peak =382 savedpeak = 4350" "j =32 peak =406 savedpeak = 4374"
## [1] "j =33 peak =80 savedpeak = 4176" "j =33 peak =113 savedpeak = 4209"
## [3] "j =33 peak =144 savedpeak = 4240" "j =33 peak =154 savedpeak = 4250"
## [5] "j =33 peak =182 savedpeak = 4278" "j =33 peak =204 savedpeak = 4300"
## [7] "j =33 peak =254 savedpeak = 4350" "j =33 peak =278 savedpeak = 4374"
## [9] "j =33 peak =329 savedpeak = 4425" "j =33 peak =368 savedpeak = 4464"
## [11] "j =33 peak =391 savedpeak = 4487" "j =33 peak =406 savedpeak = 4502"
## [1] "j =34 peak =126 savedpeak = 4350" "j =34 peak =150 savedpeak = 4374"
## [3] "j =34 peak =201 savedpeak = 4425" "j =34 peak =240 savedpeak = 4464"
## [5] "j =34 peak =263 savedpeak = 4487" "j =34 peak =278 savedpeak = 4502"
## [7] "j =34 peak =335 savedpeak = 4559" "j =34 peak =352 savedpeak = 4576"
## [9] "j =34 peak =407 savedpeak = 4631" "j =34 peak =431 savedpeak = 4655"
## [1] "j =35 peak =112 savedpeak = 4464" "j =35 peak =135 savedpeak = 4487"
## [3] "j =35 peak =150 savedpeak = 4502" "j =35 peak =207 savedpeak = 4559"
## [5] "j =35 peak =224 savedpeak = 4576" "j =35 peak =279 savedpeak = 4631"
## [7] "j =35 peak =303 savedpeak = 4655" "j =35 peak =323 savedpeak = 4675"
## [9] "j =35 peak =368 savedpeak = 4720" "j =35 peak =385 savedpeak = 4737"
## [11] "j =35 peak =412 savedpeak = 4764" "j =35 peak =433 savedpeak = 4785"
## [1] "j =36 peak =79 savedpeak = 4559" "j =36 peak =96 savedpeak = 4576"
## [3] "j =36 peak =151 savedpeak = 4631" "j =36 peak =175 savedpeak = 4655"
## [5] "j =36 peak =195 savedpeak = 4675" "j =36 peak =240 savedpeak = 4720"
## [7] "j =36 peak =257 savedpeak = 4737" "j =36 peak =284 savedpeak = 4764"
## [9] "j =36 peak =305 savedpeak = 4785" "j =36 peak =329 savedpeak = 4809"
## [11] "j =36 peak =346 savedpeak = 4826" "j =36 peak =377 savedpeak = 4857"
## [13] "j =36 peak =391 savedpeak = 4871"
## [1] "j =37 peak =112 savedpeak = 4720" "j =37 peak =129 savedpeak = 4737"
## [3] "j =37 peak =156 savedpeak = 4764" "j =37 peak =177 savedpeak = 4785"
## [5] "j =37 peak =201 savedpeak = 4809" "j =37 peak =218 savedpeak = 4826"
## [7] "j =37 peak =249 savedpeak = 4857" "j =37 peak =263 savedpeak = 4871"
## [9] "j =37 peak =347 savedpeak = 4955" "j =37 peak =364 savedpeak = 4972"
## [11] "j =37 peak =389 savedpeak = 4997" "j =37 peak =403 savedpeak = 5011"
## [13] "j =37 peak =425 savedpeak = 5033"
## [1] "j =38 peak =90 savedpeak = 4826" "j =38 peak =121 savedpeak = 4857"
## [3] "j =38 peak =135 savedpeak = 4871" "j =38 peak =219 savedpeak = 4955"
## [5] "j =38 peak =236 savedpeak = 4972" "j =38 peak =261 savedpeak = 4997"
## [7] "j =38 peak =275 savedpeak = 5011" "j =38 peak =297 savedpeak = 5033"
## [9] "j =38 peak =328 savedpeak = 5064" "j =38 peak =373 savedpeak = 5109"
## [11] "j =38 peak =412 savedpeak = 5148" "j =38 peak =434 savedpeak = 5170"
## [1] "j =39 peak =91 savedpeak = 4955" "j =39 peak =108 savedpeak = 4972"
## [3] "j =39 peak =133 savedpeak = 4997" "j =39 peak =147 savedpeak = 5011"
## [5] "j =39 peak =169 savedpeak = 5033" "j =39 peak =200 savedpeak = 5064"
## [7] "j =39 peak =245 savedpeak = 5109" "j =39 peak =284 savedpeak = 5148"

```

```

## [9] "j =39 peak =306 savedpeak = 5170" "j =39 peak =351 savedpeak = 5215"
## [11] "j =39 peak =387 savedpeak = 5251" "j =39 peak =406 savedpeak = 5270"
## [13] "j =39 peak =420 savedpeak = 5284"
## [1] "j =40 peak =117 savedpeak = 5109" "j =40 peak =156 savedpeak = 5148"
## [3] "j =40 peak =178 savedpeak = 5170" "j =40 peak =223 savedpeak = 5215"
## [5] "j =40 peak =259 savedpeak = 5251" "j =40 peak =278 savedpeak = 5270"
## [7] "j =40 peak =292 savedpeak = 5284" "j =40 peak =329 savedpeak = 5321"
## [9] "j =40 peak =341 savedpeak = 5333" "j =40 peak =357 savedpeak = 5349"
## [11] "j =40 peak =386 savedpeak = 5378" "j =40 peak =406 savedpeak = 5398"
## [13] "j =40 peak =425 savedpeak = 5417"
## [1] "j =41 peak =131 savedpeak = 5251" "j =41 peak =150 savedpeak = 5270"
## [3] "j =41 peak =164 savedpeak = 5284" "j =41 peak =201 savedpeak = 5321"
## [5] "j =41 peak =213 savedpeak = 5333" "j =41 peak =229 savedpeak = 5349"
## [7] "j =41 peak =258 savedpeak = 5378" "j =41 peak =278 savedpeak = 5398"
## [9] "j =41 peak =337 savedpeak = 5457" "j =41 peak =373 savedpeak = 5493"
## [11] "j =41 peak =397 savedpeak = 5517" "j =41 peak =410 savedpeak = 5530"
## [13] "j =41 peak =428 savedpeak = 5548"
## [1] "j =42 peak =101 savedpeak = 5349" "j =42 peak =130 savedpeak = 5378"
## [3] "j =42 peak =150 savedpeak = 5398" "j =42 peak =209 savedpeak = 5457"
## [5] "j =42 peak =245 savedpeak = 5493" "j =42 peak =269 savedpeak = 5517"
## [7] "j =42 peak =282 savedpeak = 5530" "j =42 peak =300 savedpeak = 5548"
## [9] "j =42 peak =327 savedpeak = 5575" "j =42 peak =351 savedpeak = 5599"
## [11] "j =42 peak =363 savedpeak = 5611" "j =42 peak =382 savedpeak = 5630"
## [13] "j =42 peak =414 savedpeak = 5662" "j =42 peak =435 savedpeak = 5683"
## [1] "j =43 peak =81 savedpeak = 5457" "j =43 peak =117 savedpeak = 5493"
## [3] "j =43 peak =141 savedpeak = 5517" "j =43 peak =154 savedpeak = 5530"
## [5] "j =43 peak =172 savedpeak = 5548" "j =43 peak =199 savedpeak = 5575"
## [7] "j =43 peak =223 savedpeak = 5599" "j =43 peak =235 savedpeak = 5611"
## [9] "j =43 peak =254 savedpeak = 5630" "j =43 peak =286 savedpeak = 5662"
## [11] "j =43 peak =307 savedpeak = 5683" "j =43 peak =336 savedpeak = 5712"
## [13] "j =43 peak =367 savedpeak = 5743" "j =43 peak =437 savedpeak = 5813"
## [1] "j =44 peak =95 savedpeak = 5599" "j =44 peak =102 savedpeak = 5606"
## [3] "j =44 peak =126 savedpeak = 5630" "j =44 peak =158 savedpeak = 5662"
## [5] "j =44 peak =239 savedpeak = 5743" "j =44 peak =323 savedpeak = 5827"
## [7] "j =44 peak =413 savedpeak = 5917"
## [1] "j =45 peak =317 savedpeak = 5949" "j =45 peak =359 savedpeak = 5991"
## [3] "j =45 peak =387 savedpeak = 6019"
## [1] "j =46 peak =393 savedpeak = 6153" "j =46 peak =425 savedpeak = 6185"
## [1] "j =47 peak =297 savedpeak = 6185"
## [1] "j =48 peak =169 savedpeak = 6185" "j =48 peak =376 savedpeak = 6392"
## [1] "j =49 peak =248 savedpeak = 6392"
## [1] "j =50 peak =120 savedpeak = 6392"
## [1] "j =51 peak =111 savedpeak = 6511" "j =51 peak =229 savedpeak = 6629"
## [1] "j =52 peak =101 savedpeak = 6629" "j =52 peak =428 savedpeak = 6956"
## [1] "j =53 peak =300 savedpeak = 6956" "j =53 peak =328 savedpeak = 6984"
## [3] "j =53 peak =386 savedpeak = 7042" "j =53 peak =410 savedpeak = 7066"
## [1] "j =54 peak =426 savedpeak = 7210"
## [1] "j =55 peak =298 savedpeak = 7210" "j =55 peak =325 savedpeak = 7237"
## [1] "j =56 peak =170 savedpeak = 7210" "j =56 peak =197 savedpeak = 7237"
## [1] "j =58 peak =94 savedpeak = 7390" "j =58 peak =164 savedpeak = 7460"
## [3] "j =58 peak =192 savedpeak = 7488"
## [1] "j =60 peak =81 savedpeak = 7633" "j =60 peak =155 savedpeak = 7707"
## [3] "j =60 peak =265 savedpeak = 7817" "j =60 peak =359 savedpeak = 7911"
## [5] "j =60 peak =410 savedpeak = 7962" "j =60 peak =426 savedpeak = 7978"

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## [1] "j =61 peak =137 savedpeak = 7817" "j =61 peak =217 savedpeak = 7897"
## [3] "j =61 peak =231 savedpeak = 7911" "j =61 peak =282 savedpeak = 7962"
## [5] "j =61 peak =298 savedpeak = 7978" "j =61 peak =348 savedpeak = 8028"
## [7] "j =61 peak =375 savedpeak = 8055" "j =61 peak =402 savedpeak = 8082"
## [1] "j =62 peak =423 savedpeak = 8231"
## [1] "j =63 peak =295 savedpeak = 8231"
## [1] "j =64 peak =167 savedpeak = 8231"
## [1] "j =66 peak =160 savedpeak = 8480" "j =66 peak =217 savedpeak = 8537"
## [3] "j =66 peak =239 savedpeak = 8559" "j =66 peak =270 savedpeak = 8590"
## [5] "j =66 peak =295 savedpeak = 8615" "j =66 peak =325 savedpeak = 8645"
## [7] "j =66 peak =349 savedpeak = 8669" "j =66 peak =373 savedpeak = 8693"
## [9] "j =66 peak =422 savedpeak = 8742"
## [1] "j =67 peak =294 savedpeak = 8742" "j =67 peak =383 savedpeak = 8831"
## [3] "j =67 peak =407 savedpeak = 8855" "j =67 peak =433 savedpeak = 8881"
## [1] "j =68 peak =166 savedpeak = 8742" "j =68 peak =255 savedpeak = 8831"
## [3] "j =68 peak =279 savedpeak = 8855" "j =68 peak =305 savedpeak = 8881"
## [5] "j =68 peak =329 savedpeak = 8905" "j =68 peak =355 savedpeak = 8931"
## [7] "j =68 peak =379 savedpeak = 8955"
## [1] "j =69 peak =127 savedpeak = 8831" "j =69 peak =151 savedpeak = 8855"
## [3] "j =69 peak =177 savedpeak = 8881" "j =69 peak =201 savedpeak = 8905"
## [5] "j =69 peak =227 savedpeak = 8931" "j =69 peak =251 savedpeak = 8955"
## [1] "j =70 peak =99 savedpeak = 8931" "j =70 peak =123 savedpeak = 8955"
## [3] "j =70 peak =337 savedpeak = 9169"
## [1] "j =71 peak =108 savedpeak = 9068" "j =71 peak =139 savedpeak = 9099"
## [3] "j =71 peak =182 savedpeak = 9142" "j =71 peak =209 savedpeak = 9169"
## [5] "j =71 peak =233 savedpeak = 9193" "j =71 peak =290 savedpeak = 9250"
## [7] "j =71 peak =324 savedpeak = 9284" "j =71 peak =379 savedpeak = 9339"
## [9] "j =71 peak =394 savedpeak = 9354" "j =71 peak =422 savedpeak = 9382"
## [11] "j =71 peak =437 savedpeak = 9397"
## [1] "j =72 peak =81 savedpeak = 9169" "j =72 peak =105 savedpeak = 9193"
## [3] "j =72 peak =162 savedpeak = 9250" "j =72 peak =196 savedpeak = 9284"
## [5] "j =72 peak =251 savedpeak = 9339" "j =72 peak =309 savedpeak = 9397"
## [7] "j =72 peak =333 savedpeak = 9421" "j =72 peak =401 savedpeak = 9489"
## [9] "j =72 peak =427 savedpeak = 9515"
## [1] "j =73 peak =123 savedpeak = 9339" "j =73 peak =330 savedpeak = 9546"
## [3] "j =73 peak =356 savedpeak = 9572" "j =73 peak =383 savedpeak = 9599"
## [5] "j =73 peak =411 savedpeak = 9627"
## [1] "j =75 peak =347 savedpeak = 9819" "j =75 peak =374 savedpeak = 9846"
## [3] "j =75 peak =401 savedpeak = 9873" "j =75 peak =429 savedpeak = 9901"
## [1] "j =76 peak =219 savedpeak = 9819" "j =76 peak =246 savedpeak = 9846"
## [3] "j =76 peak =273 savedpeak = 9873" "j =76 peak =301 savedpeak = 9901"
## [5] "j =76 peak =328 savedpeak = 9928"
## [1] "j =77 peak =91 savedpeak = 9819" "j =77 peak =118 savedpeak = 9846"
## [3] "j =77 peak =145 savedpeak = 9873" "j =77 peak =173 savedpeak = 9901"
## [5] "j =77 peak =200 savedpeak = 9928"
## [1] "j =78 peak =98 savedpeak = 9954"
## [1] "j =79 peak =105 savedpeak = 10089" "j =79 peak =134 savedpeak = 10118"
## [3] "j =79 peak =160 savedpeak = 10144" "j =79 peak =188 savedpeak = 10172"
## [5] "j =79 peak =222 savedpeak = 10206" "j =79 peak =241 savedpeak = 10225"
## [7] "j =79 peak =275 savedpeak = 10259" "j =79 peak =370 savedpeak = 10354"
## [1] "j =80 peak =95 savedpeak = 10207" "j =80 peak =116 savedpeak = 10228"
## [3] "j =80 peak =147 savedpeak = 10259" "j =80 peak =193 savedpeak = 10305"
## [5] "j =80 peak =221 savedpeak = 10333" "j =80 peak =241 savedpeak = 10353"
## [7] "j =80 peak =416 savedpeak = 10528"

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## [1] "j =81 peak =93 savedpeak = 10333"
## [2] "j =81 peak =113 savedpeak = 10353"
## [3] "j =81 peak =137 savedpeak = 10377"
## [4] "j =81 peak =208 savedpeak = 10448"
## [5] "j =81 peak =242 savedpeak = 10482"
## [6] "j =81 peak =262 savedpeak = 10502"
## [7] "j =81 peak =288 savedpeak = 10528"
## [8] "j =81 peak =319 savedpeak = 10559"
## [9] "j =81 peak =331 savedpeak = 10571"
## [10] "j =81 peak =355 savedpeak = 10595"
## [11] "j =81 peak =386 savedpeak = 10626"
## [12] "j =81 peak =412 savedpeak = 10652"
## [1] "j =82 peak =80 savedpeak = 10448"
## [2] "j =82 peak =114 savedpeak = 10482"
## [3] "j =82 peak =131 savedpeak = 10499"
## [4] "j =82 peak =160 savedpeak = 10528"
## [5] "j =82 peak =191 savedpeak = 10559"
## [6] "j =82 peak =203 savedpeak = 10571"
## [7] "j =82 peak =227 savedpeak = 10595"
## [8] "j =82 peak =261 savedpeak = 10629"
## [9] "j =82 peak =284 savedpeak = 10652"
## [10] "j =82 peak =330 savedpeak = 10698"
## [11] "j =82 peak =364 savedpeak = 10732"
## [12] "j =82 peak =383 savedpeak = 10751"
## [13] "j =82 peak =402 savedpeak = 10770"
## [14] "j =82 peak =426 savedpeak = 10794"
## [1] "j =83 peak =96 savedpeak = 10592"
## [2] "j =83 peak =133 savedpeak = 10629"
## [3] "j =83 peak =156 savedpeak = 10652"
## [4] "j =83 peak =180 savedpeak = 10676"
## [5] "j =83 peak =202 savedpeak = 10698"
## [6] "j =83 peak =236 savedpeak = 10732"
## [7] "j =83 peak =255 savedpeak = 10751"
## [8] "j =83 peak =273 savedpeak = 10769"
## [9] "j =83 peak =301 savedpeak = 10797"
## [10] "j =83 peak =327 savedpeak = 10823"
## [11] "j =83 peak =346 savedpeak = 10842"
## [12] "j =83 peak =377 savedpeak = 10873"
## [13] "j =83 peak =397 savedpeak = 10893"
## [14] "j =83 peak =423 savedpeak = 10919"
## [1] "j =84 peak =108 savedpeak = 10732"
## [2] "j =84 peak =127 savedpeak = 10751"
## [3] "j =84 peak =146 savedpeak = 10770"
## [4] "j =84 peak =173 savedpeak = 10797"
## [5] "j =84 peak =199 savedpeak = 10823"
## [6] "j =84 peak =218 savedpeak = 10842"
## [7] "j =84 peak =249 savedpeak = 10873"
## [8] "j =84 peak =266 savedpeak = 10890"
## [9] "j =84 peak =295 savedpeak = 10919"
## [10] "j =84 peak =326 savedpeak = 10950"
## [11] "j =84 peak =374 savedpeak = 10998"
## [12] "j =84 peak =386 savedpeak = 11010"
## [13] "j =84 peak =427 savedpeak = 11051"
## [1] "j =85 peak =90 savedpeak = 10842"

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## [2] "j =85 peak =121 savedpeak = 10873"
## [3] "j =85 peak =141 savedpeak = 10893"
## [4] "j =85 peak =167 savedpeak = 10919"
## [5] "j =85 peak =191 savedpeak = 10943"
## [6] "j =85 peak =246 savedpeak = 10998"
## [7] "j =85 peak =258 savedpeak = 11010"
## [8] "j =85 peak =299 savedpeak = 11051"
## [9] "j =85 peak =318 savedpeak = 11070"
## [10] "j =85 peak =342 savedpeak = 11094"
## [11] "j =85 peak =361 savedpeak = 11113"
## [12] "j =85 peak =391 savedpeak = 11143"
## [13] "j =85 peak =422 savedpeak = 11174"
## [14] "j =85 peak =433 savedpeak = 11185"
## [1] "j =86 peak =118 savedpeak = 10998"
## [2] "j =86 peak =130 savedpeak = 11010"
## [3] "j =86 peak =171 savedpeak = 11051"
## [4] "j =86 peak =190 savedpeak = 11070"
## [5] "j =86 peak =214 savedpeak = 11094"
## [6] "j =86 peak =233 savedpeak = 11113"
## [7] "j =86 peak =263 savedpeak = 11143"
## [8] "j =86 peak =295 savedpeak = 11175"
## [9] "j =86 peak =305 savedpeak = 11185"
## [10] "j =86 peak =332 savedpeak = 11212"
## [11] "j =86 peak =382 savedpeak = 11262"
## [12] "j =86 peak =406 savedpeak = 11286"
## [13] "j =86 peak =435 savedpeak = 11315"
## [1] "j =87 peak =86 savedpeak = 11094"
## [2] "j =87 peak =105 savedpeak = 11113"
## [3] "j =87 peak =135 savedpeak = 11143"
## [4] "j =87 peak =166 savedpeak = 11174"
## [5] "j =87 peak =177 savedpeak = 11185"
## [6] "j =87 peak =201 savedpeak = 11209"
## [7] "j =87 peak =257 savedpeak = 11265"
## [8] "j =87 peak =281 savedpeak = 11289"
## [9] "j =87 peak =307 savedpeak = 11315"
## [10] "j =87 peak =344 savedpeak = 11352"
## [11] "j =87 peak =360 savedpeak = 11368"
## [12] "j =87 peak =377 savedpeak = 11385"
## [13] "j =87 peak =435 savedpeak = 11443"
## [1] "j =88 peak =76 savedpeak = 11212"
## [2] "j =88 peak =126 savedpeak = 11262"
## [3] "j =88 peak =153 savedpeak = 11289"
## [4] "j =88 peak =182 savedpeak = 11318"
## [5] "j =88 peak =211 savedpeak = 11347"
## [6] "j =88 peak =232 savedpeak = 11368"
## [7] "j =88 peak =246 savedpeak = 11382"
## [8] "j =88 peak =307 savedpeak = 11443"
## [9] "j =88 peak =333 savedpeak = 11469"
## [10] "j =88 peak =360 savedpeak = 11496"
## [11] "j =88 peak =381 savedpeak = 11517"
## [12] "j =88 peak =421 savedpeak = 11557"
## [1] "j =89 peak =83 savedpeak = 11347"
## [2] "j =89 peak =104 savedpeak = 11368"
## [3] "j =89 peak =118 savedpeak = 11382"

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## [4] "j =89 peak =179 savedpeak = 11443"
## [5] "j =89 peak =205 savedpeak = 11469"
## [6] "j =89 peak =232 savedpeak = 11496"
## [7] "j =89 peak =253 savedpeak = 11517"
## [8] "j =89 peak =296 savedpeak = 11560"
## [9] "j =89 peak =323 savedpeak = 11587"
## [10] "j =89 peak =351 savedpeak = 11615"
## [11] "j =89 peak =368 savedpeak = 11632"
## [12] "j =89 peak =397 savedpeak = 11661"
## [1] "j =90 peak =77 savedpeak = 11469"
## [2] "j =90 peak =101 savedpeak = 11493"
## [3] "j =90 peak =125 savedpeak = 11517"
## [4] "j =90 peak =168 savedpeak = 11560"
## [5] "j =90 peak =195 savedpeak = 11587"
## [6] "j =90 peak =223 savedpeak = 11615"
## [7] "j =90 peak =240 savedpeak = 11632"
## [8] "j =90 peak =266 savedpeak = 11658"
## [9] "j =90 peak =320 savedpeak = 11712"
## [10] "j =90 peak =351 savedpeak = 11743"
## [11] "j =90 peak =360 savedpeak = 11752"
## [12] "j =90 peak =386 savedpeak = 11778"
## [13] "j =90 peak =437 savedpeak = 11829"
## [1] "j =91 peak =95 savedpeak = 11615"
## [2] "j =91 peak =112 savedpeak = 11632"
## [3] "j =91 peak =138 savedpeak = 11658"
## [4] "j =91 peak =192 savedpeak = 11712"
## [5] "j =91 peak =223 savedpeak = 11743"
## [6] "j =91 peak =232 savedpeak = 11752"
## [7] "j =91 peak =258 savedpeak = 11778"
## [8] "j =91 peak =306 savedpeak = 11826"
## [9] "j =91 peak =345 savedpeak = 11865"
## [10] "j =91 peak =364 savedpeak = 11884"
## [11] "j =91 peak =383 savedpeak = 11903"
## [12] "j =91 peak =410 savedpeak = 11930"
## [1] "j =92 peak =95 savedpeak = 11743"
## [2] "j =92 peak =100 savedpeak = 11748"
## [3] "j =92 peak =130 savedpeak = 11778"
## [4] "j =92 peak =178 savedpeak = 11826"
## [5] "j =92 peak =217 savedpeak = 11865"
## [6] "j =92 peak =231 savedpeak = 11879"
## [7] "j =92 peak =255 savedpeak = 11903"
## [8] "j =92 peak =282 savedpeak = 11930"
## [9] "j =92 peak =311 savedpeak = 11959"
## [10] "j =92 peak =345 savedpeak = 11993"
## [11] "j =92 peak =356 savedpeak = 12004"
## [12] "j =92 peak =383 savedpeak = 12031"
## [13] "j =92 peak =407 savedpeak = 12055"
## [1] "j =93 peak =89 savedpeak = 11865"
## [2] "j =93 peak =103 savedpeak = 11879"
## [3] "j =93 peak =127 savedpeak = 11903"
## [4] "j =93 peak =151 savedpeak = 11927"
## [5] "j =93 peak =183 savedpeak = 11959"
## [6] "j =93 peak =217 savedpeak = 11993"
## [7] "j =93 peak =228 savedpeak = 12004"

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## [8] "j =93 peak =255 savedpeak = 12031"
## [9] "j =93 peak =276 savedpeak = 12052"
## [10] "j =93 peak =315 savedpeak = 12091"
## [11] "j =93 peak =351 savedpeak = 12127"
## [12] "j =93 peak =377 savedpeak = 12153"
## [13] "j =93 peak =403 savedpeak = 12179"
## [14] "j =93 peak =432 savedpeak = 12208"
## [1] "j =94 peak =89 savedpeak = 11993"
## [2] "j =94 peak =100 savedpeak = 12004"
## [3] "j =94 peak =127 savedpeak = 12031"
## [4] "j =94 peak =151 savedpeak = 12055"
## [5] "j =94 peak =223 savedpeak = 12127"
## [6] "j =94 peak =250 savedpeak = 12154"
## [7] "j =94 peak =275 savedpeak = 12179"
## [8] "j =94 peak =304 savedpeak = 12208"
## [9] "j =94 peak =331 savedpeak = 12235"
## [10] "j =94 peak =350 savedpeak = 12254"
## [11] "j =94 peak =376 savedpeak = 12280"
## [12] "j =94 peak =398 savedpeak = 12302"
## [13] "j =94 peak =417 savedpeak = 12321"
## [1] "j =95 peak =95 savedpeak = 12127"
## [2] "j =95 peak =121 savedpeak = 12153"
## [3] "j =95 peak =147 savedpeak = 12179"
## [4] "j =95 peak =176 savedpeak = 12208"
## [5] "j =95 peak =203 savedpeak = 12235"
## [6] "j =95 peak =222 savedpeak = 12254"
## [7] "j =95 peak =248 savedpeak = 12280"
## [8] "j =95 peak =270 savedpeak = 12302"
## [9] "j =95 peak =285 savedpeak = 12317"
## [10] "j =95 peak =321 savedpeak = 12353"
## [11] "j =95 peak =337 savedpeak = 12369"
## [12] "j =95 peak =357 savedpeak = 12389"
## [13] "j =95 peak =390 savedpeak = 12422"
## [14] "j =95 peak =421 savedpeak = 12453"
## [1] "j =96 peak =94 savedpeak = 12254"
## [2] "j =96 peak =120 savedpeak = 12280"
## [3] "j =96 peak =142 savedpeak = 12302"
## [4] "j =96 peak =161 savedpeak = 12321"
## [5] "j =96 peak =193 savedpeak = 12353"
## [6] "j =96 peak =205 savedpeak = 12365"
## [7] "j =96 peak =229 savedpeak = 12389"
## [8] "j =96 peak =262 savedpeak = 12422"
## [9] "j =96 peak =293 savedpeak = 12453"
## [10] "j =96 peak =310 savedpeak = 12470"
## [11] "j =96 peak =341 savedpeak = 12501"
## [12] "j =96 peak =370 savedpeak = 12530"
## [13] "j =96 peak =387 savedpeak = 12547"
## [1] "j =97 peak =81 savedpeak = 12369"
## [2] "j =97 peak =101 savedpeak = 12389"
## [3] "j =97 peak =131 savedpeak = 12419"
## [4] "j =97 peak =165 savedpeak = 12453"
## [5] "j =97 peak =182 savedpeak = 12470"
## [6] "j =97 peak =213 savedpeak = 12501"
## [7] "j =97 peak =242 savedpeak = 12530"

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## [8] "j =97 peak =256 savedpeak = 12544"
## [9] "j =97 peak =317 savedpeak = 12605"
## [10] "j =97 peak =365 savedpeak = 12653"
## [11] "j =97 peak =386 savedpeak = 12674"
## [12] "j =97 peak =432 savedpeak = 12720"
## [1] "j =98 peak =85 savedpeak = 12501" "j =98 peak =114 savedpeak = 12530"
## [3] "j =98 peak =131 savedpeak = 12547" "j =98 peak =190 savedpeak = 12606"
## [5] "j =98 peak =237 savedpeak = 12653" "j =98 peak =258 savedpeak = 12674"
## [7] "j =98 peak =306 savedpeak = 12722" "j =98 peak =364 savedpeak = 12780"
## [9] "j =98 peak =405 savedpeak = 12821"
## [1] "j =99 peak =236 savedpeak = 12780" "j =99 peak =327 savedpeak = 12871"
## [3] "j =99 peak =382 savedpeak = 12926" "j =99 peak =406 savedpeak = 12950"
## [1] "j =100 peak =425 savedpeak = 13097"
## [1] "j =101 peak =297 savedpeak = 13097"
## [2] "j =101 peak =390 savedpeak = 13190"
## [1] "j =102 peak =166 savedpeak = 13094"
## [2] "j =102 peak =262 savedpeak = 13190"
## [1] "j =103 peak =139 savedpeak = 13195"
## [1] "j =104 peak =222 savedpeak = 13406"
## [1] "j =105 peak =90 savedpeak = 13402"
## [2] "j =105 peak =126 savedpeak = 13438"
## [3] "j =105 peak =155 savedpeak = 13467"
## [4] "j =105 peak =195 savedpeak = 13507"
## [5] "j =105 peak =226 savedpeak = 13538"
## [6] "j =105 peak =250 savedpeak = 13562"
## [7] "j =105 peak =272 savedpeak = 13584"
## [8] "j =105 peak =301 savedpeak = 13613"
## [9] "j =105 peak =325 savedpeak = 13637"
## [10] "j =105 peak =350 savedpeak = 13662"
## [11] "j =105 peak =373 savedpeak = 13685"
## [12] "j =105 peak =418 savedpeak = 13730"
## [13] "j =105 peak =437 savedpeak = 13749"
## [1] "j =106 peak =98 savedpeak = 13538"
## [2] "j =106 peak =122 savedpeak = 13562"
## [3] "j =106 peak =141 savedpeak = 13581"
## [4] "j =106 peak =173 savedpeak = 13613"
## [5] "j =106 peak =197 savedpeak = 13637"
## [6] "j =106 peak =221 savedpeak = 13661"
## [7] "j =106 peak =245 savedpeak = 13685"
## [8] "j =106 peak =255 savedpeak = 13695"
## [9] "j =106 peak =290 savedpeak = 13730"
## [10] "j =106 peak =310 savedpeak = 13750"
## [11] "j =106 peak =338 savedpeak = 13778"
## [12] "j =106 peak =351 savedpeak = 13791"
## [13] "j =106 peak =367 savedpeak = 13807"
## [14] "j =106 peak =407 savedpeak = 13847"
## [15] "j =106 peak =425 savedpeak = 13865"
## [1] "j =107 peak =93 savedpeak = 13661"
## [2] "j =107 peak =117 savedpeak = 13685"
## [3] "j =107 peak =127 savedpeak = 13695"
## [4] "j =107 peak =162 savedpeak = 13730"
## [5] "j =107 peak =181 savedpeak = 13749"
## [6] "j =107 peak =210 savedpeak = 13778"
## [7] "j =107 peak =223 savedpeak = 13791"

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## [8] "j =107 peak =244 savedpeak = 13812"
## [9] "j =107 peak =276 savedpeak = 13844"
## [10] "j =107 peak =297 savedpeak = 13865"
## [11] "j =107 peak =340 savedpeak = 13908"
## [12] "j =107 peak =367 savedpeak = 13935"
## [13] "j =107 peak =398 savedpeak = 13966"
## [1] "j =108 peak =82 savedpeak = 13778"
## [2] "j =108 peak =95 savedpeak = 13791"
## [3] "j =108 peak =111 savedpeak = 13807"
## [4] "j =108 peak =148 savedpeak = 13844"
## [5] "j =108 peak =169 savedpeak = 13865"
## [6] "j =108 peak =212 savedpeak = 13908"
## [7] "j =108 peak =236 savedpeak = 13932"
## [8] "j =108 peak =270 savedpeak = 13966"
## [9] "j =108 peak =308 savedpeak = 14004"
## [10] "j =108 peak =334 savedpeak = 14030"
## [11] "j =108 peak =361 savedpeak = 14057"
## [12] "j =108 peak =395 savedpeak = 14091"
## [13] "j =108 peak =435 savedpeak = 14131"
## [1] "j =109 peak =86 savedpeak = 13910"
## [2] "j =109 peak =108 savedpeak = 13932"
## [3] "j =109 peak =142 savedpeak = 13966"
## [4] "j =109 peak =180 savedpeak = 14004"
## [5] "j =109 peak =206 savedpeak = 14030"
## [6] "j =109 peak =233 savedpeak = 14057"
## [7] "j =109 peak =271 savedpeak = 14095"
## [8] "j =109 peak =307 savedpeak = 14131"
## [9] "j =109 peak =326 savedpeak = 14150"
## [10] "j =109 peak =343 savedpeak = 14167"
## [11] "j =109 peak =363 savedpeak = 14187"
## [12] "j =109 peak =401 savedpeak = 14225"
## [13] "j =109 peak =420 savedpeak = 14244"
## [1] "j =110 peak =78 savedpeak = 14030"
## [2] "j =110 peak =105 savedpeak = 14057"
## [3] "j =110 peak =143 savedpeak = 14095"
## [4] "j =110 peak =179 savedpeak = 14131"
## [5] "j =110 peak =198 savedpeak = 14150"
## [6] "j =110 peak =215 savedpeak = 14167"
## [7] "j =110 peak =235 savedpeak = 14187"
## [8] "j =110 peak =273 savedpeak = 14225"
## [9] "j =110 peak =292 savedpeak = 14244"
## [10] "j =110 peak =312 savedpeak = 14264"
## [11] "j =110 peak =338 savedpeak = 14290"
## [12] "j =110 peak =350 savedpeak = 14302"
## [13] "j =110 peak =388 savedpeak = 14340"
## [14] "j =110 peak =407 savedpeak = 14359"
## [15] "j =110 peak =432 savedpeak = 14384"
## [1] "j =111 peak =87 savedpeak = 14167"
## [2] "j =111 peak =107 savedpeak = 14187"
## [3] "j =111 peak =145 savedpeak = 14225"
## [4] "j =111 peak =164 savedpeak = 14244"
## [5] "j =111 peak =184 savedpeak = 14264"
## [6] "j =111 peak =210 savedpeak = 14290"
## [7] "j =111 peak =222 savedpeak = 14302"

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## [8] "j =111 peak =260 savedpeak = 14340"
## [9] "j =111 peak =279 savedpeak = 14359"
## [10] "j =111 peak =304 savedpeak = 14384"
## [11] "j =111 peak =323 savedpeak = 14403"
## [12] "j =111 peak =345 savedpeak = 14425"
## [13] "j =111 peak =357 savedpeak = 14437"
## [14] "j =111 peak =380 savedpeak = 14460"
## [15] "j =111 peak =393 savedpeak = 14473"
## [16] "j =111 peak =431 savedpeak = 14511"
## [1] "j =112 peak =82 savedpeak = 14290"
## [2] "j =112 peak =94 savedpeak = 14302"
## [3] "j =112 peak =132 savedpeak = 14340"
## [4] "j =112 peak =151 savedpeak = 14359"
## [5] "j =112 peak =176 savedpeak = 14384"
## [6] "j =112 peak =195 savedpeak = 14403"
## [7] "j =112 peak =217 savedpeak = 14425"
## [8] "j =112 peak =229 savedpeak = 14437"
## [9] "j =112 peak =252 savedpeak = 14460"
## [10] "j =112 peak =265 savedpeak = 14473"
## [11] "j =112 peak =303 savedpeak = 14511"
## [12] "j =112 peak =331 savedpeak = 14539"
## [13] "j =112 peak =360 savedpeak = 14568"
## [14] "j =112 peak =375 savedpeak = 14583"
## [15] "j =112 peak =394 savedpeak = 14602"
## [16] "j =112 peak =421 savedpeak = 14629"
## [1] "j =113 peak =89 savedpeak = 14425"
## [2] "j =113 peak =101 savedpeak = 14437"
## [3] "j =113 peak =124 savedpeak = 14460"
## [4] "j =113 peak =137 savedpeak = 14473"
## [5] "j =113 peak =175 savedpeak = 14511"
## [6] "j =113 peak =203 savedpeak = 14539"
## [7] "j =113 peak =232 savedpeak = 14568"
## [8] "j =113 peak =247 savedpeak = 14583"
## [9] "j =113 peak =266 savedpeak = 14602"
## [10] "j =113 peak =293 savedpeak = 14629"
## [11] "j =113 peak =312 savedpeak = 14648"
## [12] "j =113 peak =331 savedpeak = 14667"
## [13] "j =113 peak =357 savedpeak = 14693"
## [14] "j =113 peak =369 savedpeak = 14705"
## [15] "j =113 peak =396 savedpeak = 14732"
## [16] "j =113 peak =417 savedpeak = 14753"
## [1] "j =114 peak =104 savedpeak = 14568"
## [2] "j =114 peak =119 savedpeak = 14583"
## [3] "j =114 peak =138 savedpeak = 14602"
## [4] "j =114 peak =165 savedpeak = 14629"
## [5] "j =114 peak =184 savedpeak = 14648"
## [6] "j =114 peak =203 savedpeak = 14667"
## [7] "j =114 peak =229 savedpeak = 14693"
## [8] "j =114 peak =241 savedpeak = 14705"
## [9] "j =114 peak =268 savedpeak = 14732"
## [10] "j =114 peak =289 savedpeak = 14753"
## [11] "j =114 peak =318 savedpeak = 14782"
## [12] "j =114 peak =361 savedpeak = 14825"
## [13] "j =114 peak =373 savedpeak = 14837"

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## [14] "j =114 peak =412 savedpeak = 14876"
## [1] "j =115 peak =101 savedpeak = 14693"
## [2] "j =115 peak =113 savedpeak = 14705"
## [3] "j =115 peak =140 savedpeak = 14732"
## [4] "j =115 peak =161 savedpeak = 14753"
## [5] "j =115 peak =190 savedpeak = 14782"
## [6] "j =115 peak =233 savedpeak = 14825"
## [7] "j =115 peak =245 savedpeak = 14837"
## [8] "j =115 peak =284 savedpeak = 14876"
## [9] "j =115 peak =343 savedpeak = 14935"
## [10] "j =115 peak =384 savedpeak = 14976"
## [11] "j =115 peak =416 savedpeak = 15008"
## [1] "j =116 peak =105 savedpeak = 14825"
## [2] "j =116 peak =117 savedpeak = 14837"
## [3] "j =116 peak =156 savedpeak = 14876"
## [4] "j =116 peak =215 savedpeak = 14935"
## [5] "j =116 peak =256 savedpeak = 14976"
## [6] "j =116 peak =288 savedpeak = 15008"
## [7] "j =116 peak =321 savedpeak = 15041"
## [8] "j =116 peak =343 savedpeak = 15063"
## [9] "j =116 peak =389 savedpeak = 15109"
## [10] "j =116 peak =406 savedpeak = 15126"
## [11] "j =116 peak =425 savedpeak = 15145"
## [1] "j =117 peak =87 savedpeak = 14935"
## [2] "j =117 peak =128 savedpeak = 14976"
## [3] "j =117 peak =160 savedpeak = 15008"
## [4] "j =117 peak =193 savedpeak = 15041"
## [5] "j =117 peak =215 savedpeak = 15063"
## [6] "j =117 peak =261 savedpeak = 15109"
## [7] "j =117 peak =278 savedpeak = 15126"
## [8] "j =117 peak =297 savedpeak = 15145"
## [9] "j =117 peak =325 savedpeak = 15173"
## [10] "j =117 peak =337 savedpeak = 15185"
## [11] "j =117 peak =366 savedpeak = 15214"
## [12] "j =117 peak =395 savedpeak = 15243"
## [13] "j =117 peak =419 savedpeak = 15267"
## [14] "j =117 peak =429 savedpeak = 15277"
## [1] "j =119 peak =81 savedpeak = 15185"
## [2] "j =119 peak =110 savedpeak = 15214"
## [3] "j =119 peak =139 savedpeak = 15243"
## [4] "j =119 peak =163 savedpeak = 15267"
## [5] "j =119 peak =173 savedpeak = 15277"
## [6] "j =119 peak =199 savedpeak = 15303"
## [7] "j =119 peak =228 savedpeak = 15332"
## [8] "j =119 peak =276 savedpeak = 15380"
## [9] "j =119 peak =334 savedpeak = 15438"
## [10] "j =119 peak =393 savedpeak = 15497"
## [11] "j =119 peak =425 savedpeak = 15529"
## [1] "j =120 peak =100 savedpeak = 15332"
## [2] "j =120 peak =148 savedpeak = 15380"
## [3] "j =120 peak =164 savedpeak = 15396"
## [4] "j =120 peak =206 savedpeak = 15438"
## [5] "j =120 peak =265 savedpeak = 15497"
## [6] "j =120 peak =297 savedpeak = 15529"

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## [7] "j =120 peak =311 savedpeak = 15543"
## [8] "j =120 peak =330 savedpeak = 15562"
## [9] "j =120 peak =362 savedpeak = 15594"
## [10] "j =120 peak =385 savedpeak = 15617"
## [11] "j =120 peak =407 savedpeak = 15639"
## [12] "j =120 peak =429 savedpeak = 15661"
## [1] "j =121 peak =78 savedpeak = 15438"
## [2] "j =121 peak =137 savedpeak = 15497"
## [3] "j =121 peak =145 savedpeak = 15505"
## [4] "j =121 peak =169 savedpeak = 15529"
## [5] "j =121 peak =183 savedpeak = 15543"
## [6] "j =121 peak =202 savedpeak = 15562"
## [7] "j =121 peak =234 savedpeak = 15594"
## [8] "j =121 peak =257 savedpeak = 15617"
## [9] "j =121 peak =279 savedpeak = 15639"
## [10] "j =121 peak =301 savedpeak = 15661"
## [11] "j =121 peak =317 savedpeak = 15677"
## [12] "j =121 peak =327 savedpeak = 15687"
## [13] "j =121 peak =344 savedpeak = 15704"
## [14] "j =121 peak =361 savedpeak = 15721"
## [15] "j =121 peak =382 savedpeak = 15742"
## [16] "j =121 peak =416 savedpeak = 15776"
## [1] "j =122 peak =106 savedpeak = 15594"
## [2] "j =122 peak =129 savedpeak = 15617"
## [3] "j =122 peak =151 savedpeak = 15639"
## [4] "j =122 peak =173 savedpeak = 15661"
## [5] "j =122 peak =189 savedpeak = 15677"
## [6] "j =122 peak =199 savedpeak = 15687"
## [7] "j =122 peak =216 savedpeak = 15704"
## [8] "j =122 peak =233 savedpeak = 15721"
## [9] "j =122 peak =254 savedpeak = 15742"
## [10] "j =122 peak =288 savedpeak = 15776"
## [11] "j =122 peak =321 savedpeak = 15809"
## [12] "j =122 peak =360 savedpeak = 15848"
## [13] "j =122 peak =387 savedpeak = 15875"
## [14] "j =122 peak =411 savedpeak = 15899"
## [15] "j =122 peak =434 savedpeak = 15922"
## [1] "j =123 peak =88 savedpeak = 15704"
## [2] "j =123 peak =105 savedpeak = 15721"
## [3] "j =123 peak =126 savedpeak = 15742"
## [4] "j =123 peak =160 savedpeak = 15776"
## [5] "j =123 peak =193 savedpeak = 15809"
## [6] "j =123 peak =232 savedpeak = 15848"
## [7] "j =123 peak =259 savedpeak = 15875"
## [8] "j =123 peak =283 savedpeak = 15899"
## [9] "j =123 peak =306 savedpeak = 15922"
## [10] "j =123 peak =323 savedpeak = 15939"
## [11] "j =123 peak =343 savedpeak = 15959"
## [12] "j =123 peak =361 savedpeak = 15977"
## [13] "j =123 peak =383 savedpeak = 15999"
## [14] "j =123 peak =407 savedpeak = 16023"
## [15] "j =123 peak =426 savedpeak = 16042"
## [1] "j =124 peak =104 savedpeak = 15848"
## [2] "j =124 peak =131 savedpeak = 15875"

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## [3] "j =124 peak =155 savedpeak = 15899"
## [4] "j =124 peak =178 savedpeak = 15922"
## [5] "j =124 peak =195 savedpeak = 15939"
## [6] "j =124 peak =215 savedpeak = 15959"
## [7] "j =124 peak =233 savedpeak = 15977"
## [8] "j =124 peak =255 savedpeak = 15999"
## [9] "j =124 peak =279 savedpeak = 16023"
## [10] "j =124 peak =298 savedpeak = 16042"
## [11] "j =124 peak =322 savedpeak = 16066"
## [12] "j =124 peak =341 savedpeak = 16085"
## [13] "j =124 peak =375 savedpeak = 16119"
## [14] "j =124 peak =406 savedpeak = 16150"
## [15] "j =124 peak =433 savedpeak = 16177"
## [1] "j =125 peak =87 savedpeak = 15959"
## [2] "j =125 peak =105 savedpeak = 15977"
## [3] "j =125 peak =127 savedpeak = 15999"
## [4] "j =125 peak =151 savedpeak = 16023"
## [5] "j =125 peak =170 savedpeak = 16042"
## [6] "j =125 peak =194 savedpeak = 16066"
## [7] "j =125 peak =213 savedpeak = 16085"
## [8] "j =125 peak =247 savedpeak = 16119"
## [9] "j =125 peak =278 savedpeak = 16150"
## [10] "j =125 peak =305 savedpeak = 16177"
## [11] "j =125 peak =331 savedpeak = 16203"
## [12] "j =125 peak =351 savedpeak = 16223"
## [13] "j =125 peak =377 savedpeak = 16249"
## [14] "j =125 peak =413 savedpeak = 16285"
## [1] "j =126 peak =85 savedpeak = 16085"
## [2] "j =126 peak =119 savedpeak = 16119"
## [3] "j =126 peak =150 savedpeak = 16150"
## [4] "j =126 peak =177 savedpeak = 16177"
## [5] "j =126 peak =203 savedpeak = 16203"
## [6] "j =126 peak =223 savedpeak = 16223"
## [7] "j =126 peak =249 savedpeak = 16249"
## [8] "j =126 peak =285 savedpeak = 16285"
## [9] "j =126 peak =326 savedpeak = 16326"
## [10] "j =126 peak =352 savedpeak = 16352"
## [11] "j =126 peak =376 savedpeak = 16376"
## [12] "j =126 peak =402 savedpeak = 16402"
## [1] "j =127 peak =95 savedpeak = 16223"
## [2] "j =127 peak =121 savedpeak = 16249"
## [3] "j =127 peak =157 savedpeak = 16285"
## [4] "j =127 peak =198 savedpeak = 16326"
## [5] "j =127 peak =224 savedpeak = 16352"
## [6] "j =127 peak =248 savedpeak = 16376"
## [7] "j =127 peak =274 savedpeak = 16402"
## [8] "j =127 peak =351 savedpeak = 16479"
## [9] "j =127 peak =409 savedpeak = 16537"
## [10] "j =127 peak =421 savedpeak = 16549"
## [1] "j =128 peak =96 savedpeak = 16352"
## [2] "j =128 peak =120 savedpeak = 16376"
## [3] "j =128 peak =146 savedpeak = 16402"
## [4] "j =128 peak =223 savedpeak = 16479"
## [5] "j =128 peak =281 savedpeak = 16537"

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## [6] "j =128 peak =293 savedpeak = 16549"
## [7] "j =128 peak =315 savedpeak = 16571"
## [8] "j =128 peak =367 savedpeak = 16623"
## [9] "j =128 peak =427 savedpeak = 16683"
## [1] "j =129 peak =95 savedpeak = 16479"
## [2] "j =129 peak =119 savedpeak = 16503"
## [3] "j =129 peak =153 savedpeak = 16537"
## [4] "j =129 peak =165 savedpeak = 16549"
## [5] "j =129 peak =187 savedpeak = 16571"
## [6] "j =129 peak =211 savedpeak = 16595"
## [7] "j =129 peak =239 savedpeak = 16623"
## [8] "j =129 peak =264 savedpeak = 16648"
## [9] "j =129 peak =299 savedpeak = 16683"
## [10] "j =129 peak =321 savedpeak = 16705"
## [11] "j =129 peak =342 savedpeak = 16726"
## [12] "j =129 peak =386 savedpeak = 16770"
## [13] "j =129 peak =408 savedpeak = 16792"
## [14] "j =129 peak =426 savedpeak = 16810"
## [1] "j =130 peak =83 savedpeak = 16595"
## [2] "j =130 peak =111 savedpeak = 16623"
## [3] "j =130 peak =136 savedpeak = 16648"
## [4] "j =130 peak =171 savedpeak = 16683"
## [5] "j =130 peak =193 savedpeak = 16705"
## [6] "j =130 peak =214 savedpeak = 16726"
## [7] "j =130 peak =258 savedpeak = 16770"
## [8] "j =130 peak =280 savedpeak = 16792"
## [9] "j =130 peak =298 savedpeak = 16810"
## [10] "j =130 peak =321 savedpeak = 16833"
## [11] "j =130 peak =337 savedpeak = 16849"
## [12] "j =130 peak =363 savedpeak = 16875"
## [13] "j =130 peak =388 savedpeak = 16900"
## [14] "j =130 peak =409 savedpeak = 16921"
## [1] "j =131 peak =86 savedpeak = 16726"
## [2] "j =131 peak =130 savedpeak = 16770"
## [3] "j =131 peak =152 savedpeak = 16792"
## [4] "j =131 peak =170 savedpeak = 16810"
## [5] "j =131 peak =193 savedpeak = 16833"
## [6] "j =131 peak =209 savedpeak = 16849"
## [7] "j =131 peak =235 savedpeak = 16875"
## [8] "j =131 peak =260 savedpeak = 16900"
## [9] "j =131 peak =281 savedpeak = 16921"
## [10] "j =131 peak =336 savedpeak = 16976"
## [11] "j =131 peak =358 savedpeak = 16998"
## [12] "j =131 peak =379 savedpeak = 17019"
## [13] "j =131 peak =403 savedpeak = 17043"
## [14] "j =131 peak =416 savedpeak = 17056"
## [1] "j =132 peak =81 savedpeak = 16849"
## [2] "j =132 peak =107 savedpeak = 16875"
## [3] "j =132 peak =132 savedpeak = 16900"
## [4] "j =132 peak =153 savedpeak = 16921"
## [5] "j =132 peak =208 savedpeak = 16976"
## [6] "j =132 peak =230 savedpeak = 16998"
## [7] "j =132 peak =251 savedpeak = 17019"
## [8] "j =132 peak =275 savedpeak = 17043"

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## [9] "j =132 peak =288 savedpeak = 17056"
## [10] "j =132 peak =317 savedpeak = 17085"
## [11] "j =132 peak =353 savedpeak = 17121"
## [12] "j =132 peak =379 savedpeak = 17147"
## [13] "j =132 peak =400 savedpeak = 17168"
## [14] "j =132 peak =425 savedpeak = 17193"
## [1] "j =133 peak =80 savedpeak = 16976"
## [2] "j =133 peak =102 savedpeak = 16998"
## [3] "j =133 peak =123 savedpeak = 17019"
## [4] "j =133 peak =147 savedpeak = 17043"
## [5] "j =133 peak =160 savedpeak = 17056"
## [6] "j =133 peak =189 savedpeak = 17085"
## [7] "j =133 peak =225 savedpeak = 17121"
## [8] "j =133 peak =251 savedpeak = 17147"
## [9] "j =133 peak =272 savedpeak = 17168"
## [10] "j =133 peak =297 savedpeak = 17193"
## [11] "j =133 peak =327 savedpeak = 17223"
## [12] "j =133 peak =347 savedpeak = 17243"
## [13] "j =133 peak =359 savedpeak = 17255"
## [14] "j =133 peak =373 savedpeak = 17269"
## [15] "j =133 peak =395 savedpeak = 17291"
## [1] "j =134 peak =97 savedpeak = 17121"
## [2] "j =134 peak =123 savedpeak = 17147"
## [3] "j =134 peak =144 savedpeak = 17168"
## [4] "j =134 peak =169 savedpeak = 17193"
## [5] "j =134 peak =199 savedpeak = 17223"
## [6] "j =134 peak =219 savedpeak = 17243"
## [7] "j =134 peak =231 savedpeak = 17255"
## [8] "j =134 peak =245 savedpeak = 17269"
## [9] "j =134 peak =267 savedpeak = 17291"
## [10] "j =134 peak =312 savedpeak = 17336"
## [11] "j =134 peak =334 savedpeak = 17358"
## [12] "j =134 peak =358 savedpeak = 17382"
## [13] "j =134 peak =385 savedpeak = 17409"
## [14] "j =134 peak =403 savedpeak = 17427"
## [1] "j =135 peak =91 savedpeak = 17243"
## [2] "j =135 peak =103 savedpeak = 17255"
## [3] "j =135 peak =117 savedpeak = 17269"
## [4] "j =135 peak =139 savedpeak = 17291"
## [5] "j =135 peak =184 savedpeak = 17336"
## [6] "j =135 peak =206 savedpeak = 17358"
## [7] "j =135 peak =230 savedpeak = 17382"
## [8] "j =135 peak =257 savedpeak = 17409"
## [9] "j =135 peak =275 savedpeak = 17427"
## [10] "j =135 peak =314 savedpeak = 17466"
## [11] "j =135 peak =338 savedpeak = 17490"
## [12] "j =135 peak =379 savedpeak = 17531"
## [13] "j =135 peak =418 savedpeak = 17570"
## [1] "j =136 peak =78 savedpeak = 17358"
## [2] "j =136 peak =102 savedpeak = 17382"
## [3] "j =136 peak =129 savedpeak = 17409"
## [4] "j =136 peak =147 savedpeak = 17427"
## [5] "j =136 peak =186 savedpeak = 17466"
## [6] "j =136 peak =210 savedpeak = 17490"

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## [7] "j =136 peak =251 savedpeak = 17531"
## [8] "j =136 peak =290 savedpeak = 17570"
## [9] "j =136 peak =311 savedpeak = 17591"
## [10] "j =136 peak =333 savedpeak = 17613"
## [11] "j =136 peak =379 savedpeak = 17659"
## [12] "j =136 peak =388 savedpeak = 17668"
## [13] "j =136 peak =417 savedpeak = 17697"
## [1] "j =137 peak =82 savedpeak = 17490"
## [2] "j =137 peak =123 savedpeak = 17531"
## [3] "j =137 peak =162 savedpeak = 17570"
## [4] "j =137 peak =183 savedpeak = 17591"
## [5] "j =137 peak =205 savedpeak = 17613"
## [6] "j =137 peak =251 savedpeak = 17659"
## [7] "j =137 peak =260 savedpeak = 17668"
## [8] "j =137 peak =289 savedpeak = 17697"
## [9] "j =137 peak =312 savedpeak = 17720"
## [10] "j =137 peak =363 savedpeak = 17771"
## [11] "j =137 peak =380 savedpeak = 17788"
## [12] "j =137 peak =414 savedpeak = 17822"
## [13] "j =137 peak =426 savedpeak = 17834"
## [1] "j =138 peak =77 savedpeak = 17613"
## [2] "j =138 peak =123 savedpeak = 17659"
## [3] "j =138 peak =132 savedpeak = 17668"
## [4] "j =138 peak =161 savedpeak = 17697"
## [5] "j =138 peak =184 savedpeak = 17720"
## [6] "j =138 peak =235 savedpeak = 17771"
## [7] "j =138 peak =252 savedpeak = 17788"
## [8] "j =138 peak =286 savedpeak = 17822"
## [9] "j =138 peak =298 savedpeak = 17834"
## [10] "j =138 peak =314 savedpeak = 17850"
## [11] "j =138 peak =348 savedpeak = 17884"
## [12] "j =138 peak =372 savedpeak = 17908"
## [13] "j =138 peak =393 savedpeak = 17929"
## [14] "j =138 peak =412 savedpeak = 17948"
## [1] "j =177 peak =309 savedpeak = 22837"
## [2] "j =177 peak =432 savedpeak = 22960"
## [1] "j =178 peak =181 savedpeak = 22837"
## [2] "j =178 peak =304 savedpeak = 22960"
## [3] "j =178 peak =386 savedpeak = 23042"
## [1] "j =179 peak =176 savedpeak = 22960"
## [2] "j =179 peak =258 savedpeak = 23042"
## [1] "j =180 peak =130 savedpeak = 23042"
## [2] "j =180 peak =368 savedpeak = 23280"
## [3] "j =180 peak =422 savedpeak = 23334"
## [1] "j =181 peak =240 savedpeak = 23280"
## [2] "j =181 peak =294 savedpeak = 23334"
## [1] "j =182 peak =112 savedpeak = 23280"
## [2] "j =182 peak =166 savedpeak = 23334"
## [3] "j =182 peak =325 savedpeak = 23493"
## [4] "j =182 peak =360 savedpeak = 23528"
## [1] "j =183 peak =197 savedpeak = 23493"
## [2] "j =183 peak =232 savedpeak = 23528"
## [3] "j =183 peak =373 savedpeak = 23669"
## [4] "j =183 peak =435 savedpeak = 23731"

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## [1] "j =184 peak =104 savedpeak = 23528"
## [2] "j =184 peak =245 savedpeak = 23669"
## [3] "j =184 peak =307 savedpeak = 23731"
## [1] "j =185 peak =117 savedpeak = 23669"
## [2] "j =185 peak =179 savedpeak = 23731"
## [3] "j =185 peak =219 savedpeak = 23771"
## [1] "j =186 peak =91 savedpeak = 23771"
## [1] "j =187 peak =406 savedpeak = 24214"
## [1] "j =188 peak =278 savedpeak = 24214"
## [2] "j =188 peak =311 savedpeak = 24247"
## [3] "j =188 peak =342 savedpeak = 24278"
## [1] "j =189 peak =150 savedpeak = 24214"
## [2] "j =189 peak =183 savedpeak = 24247"
## [3] "j =189 peak =214 savedpeak = 24278"
## [1] "j =190 peak =86 savedpeak = 24278"
## [1] "j =192 peak =332 savedpeak = 24780"
## [1] "j =193 peak =204 savedpeak = 24780"
## [1] "j =194 peak =76 savedpeak = 24780"
## [1] "j =195 peak =82 savedpeak = 24914"
## [2] "j =195 peak =248 savedpeak = 25080"
## [3] "j =195 peak =380 savedpeak = 25212"
## [4] "j =195 peak =425 savedpeak = 25257"
## [1] "j =196 peak =366 savedpeak = 25326"
## [2] "j =196 peak =419 savedpeak = 25379"
## [1] "j =197 peak =238 savedpeak = 25326"
## [2] "j =197 peak =291 savedpeak = 25379"
## [3] "j =197 peak =346 savedpeak = 25434"
## [4] "j =197 peak =403 savedpeak = 25491"
## [1] "j =198 peak =110 savedpeak = 25326"
## [2] "j =198 peak =163 savedpeak = 25379"
## [3] "j =198 peak =218 savedpeak = 25434"
## [4] "j =198 peak =275 savedpeak = 25491"
## [5] "j =198 peak =431 savedpeak = 25647"
## [1] "j =199 peak =90 savedpeak = 25434"
## [2] "j =199 peak =147 savedpeak = 25491"
## [3] "j =199 peak =303 savedpeak = 25647"
## [1] "j =200 peak =175 savedpeak = 25647"
## [2] "j =200 peak =368 savedpeak = 25840"
## [3] "j =200 peak =398 savedpeak = 25870"
## [1] "j =201 peak =240 savedpeak = 25840"
## [2] "j =201 peak =270 savedpeak = 25870"
## [1] "j =202 peak =112 savedpeak = 25840"
## [2] "j =202 peak =142 savedpeak = 25870"
## [1] "j =203 peak =145 savedpeak = 26001"
## [2] "j =203 peak =235 savedpeak = 26091"
## [3] "j =203 peak =371 savedpeak = 26227"
## [4] "j =203 peak =412 savedpeak = 26268"
## [1] "j =204 peak =107 savedpeak = 26091"
## [2] "j =204 peak =243 savedpeak = 26227"
## [3] "j =204 peak =284 savedpeak = 26268"
## [1] "j =205 peak =115 savedpeak = 26227"
## [2] "j =205 peak =156 savedpeak = 26268"
## [1] "j =206 peak =80 savedpeak = 26320"
## [2] "j =206 peak =112 savedpeak = 26352"

```

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## [3] "j =206 peak =155 savedpeak = 26395"
## [4] "j =206 peak =190 savedpeak = 26430"
## [5] "j =206 peak =207 savedpeak = 26447"
## [6] "j =206 peak =233 savedpeak = 26473"
## [7] "j =206 peak =254 savedpeak = 26494"
## [8] "j =206 peak =291 savedpeak = 26531"
## [9] "j =206 peak =306 savedpeak = 26546"
## [10] "j =206 peak =325 savedpeak = 26565"
## [11] "j =206 peak =337 savedpeak = 26577"
## [12] "j =206 peak =359 savedpeak = 26599"
## [13] "j =206 peak =396 savedpeak = 26636"
## [14] "j =206 peak =421 savedpeak = 26661"
## [1] "j =211 peak =128 savedpeak = 27008"
## [2] "j =211 peak =416 savedpeak = 27296"
## [1] "j =212 peak =288 savedpeak = 27296"
## [1] "j =213 peak =160 savedpeak = 27296"
## [1] "j =216 peak =84 savedpeak = 27604"
## [2] "j =216 peak =96 savedpeak = 27616"
## [3] "j =216 peak =113 savedpeak = 27633"
## [4] "j =216 peak =153 savedpeak = 27673"
## [5] "j =216 peak =179 savedpeak = 27699"
## [6] "j =216 peak =238 savedpeak = 27758"
## [7] "j =216 peak =257 savedpeak = 27777"
## [8] "j =216 peak =270 savedpeak = 27790"
## [9] "j =216 peak =331 savedpeak = 27851"
## [10] "j =216 peak =347 savedpeak = 27867"
## [11] "j =216 peak =373 savedpeak = 27893"
## [12] "j =216 peak =412 savedpeak = 27932"
## [1] "j =217 peak =203 savedpeak = 27851"
## [2] "j =217 peak =244 savedpeak = 27892"
## [3] "j =217 peak =342 savedpeak = 27990"
## [4] "j =217 peak =362 savedpeak = 28010"
## [5] "j =217 peak =427 savedpeak = 28075"
## [1] "j =218 peak =116 savedpeak = 27892"
## [2] "j =218 peak =214 savedpeak = 27990"
## [3] "j =218 peak =234 savedpeak = 28010"
## [4] "j =218 peak =299 savedpeak = 28075"
## [5] "j =218 peak =366 savedpeak = 28142"
## [1] "j =219 peak =86 savedpeak = 27990"
## [2] "j =219 peak =106 savedpeak = 28010"
## [3] "j =219 peak =171 savedpeak = 28075"
## [4] "j =219 peak =238 savedpeak = 28142"
## [5] "j =219 peak =313 savedpeak = 28217"
## [6] "j =219 peak =377 savedpeak = 28281"
## [7] "j =219 peak =425 savedpeak = 28329"
## [1] "j =220 peak =110 savedpeak = 28142"
## [2] "j =220 peak =185 savedpeak = 28217"
## [3] "j =220 peak =249 savedpeak = 28281"
## [4] "j =220 peak =297 savedpeak = 28329"
## [1] "j =221 peak =121 savedpeak = 28281"
## [2] "j =221 peak =169 savedpeak = 28329"
## [3] "j =221 peak =358 savedpeak = 28518"
## [1] "j =237 peak =403 savedpeak = 30611"
## [1] "j =238 peak =275 savedpeak = 30611"

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## [2] "j =238 peak =430 savedpeak = 30766"
## [1] "j =239 peak =147 savedpeak = 30611"
## [2] "j =239 peak =302 savedpeak = 30766"
## [1] "j =240 peak =405 savedpeak = 30997"
## [1] "j =241 peak =277 savedpeak = 30997"
## [2] "j =241 peak =313 savedpeak = 31033"
## [3] "j =241 peak =348 savedpeak = 31068"
## [1] "j =242 peak =149 savedpeak = 30997"
## [2] "j =242 peak =185 savedpeak = 31033"
## [3] "j =242 peak =220 savedpeak = 31068"
## [1] "j =243 peak =92 savedpeak = 31068"
## [1] "j =244 peak =92 savedpeak = 31196"
## [2] "j =244 peak =118 savedpeak = 31222"
## [3] "j =244 peak =143 savedpeak = 31247"
## [4] "j =244 peak =154 savedpeak = 31258"
## [5] "j =244 peak =196 savedpeak = 31300"
## [6] "j =244 peak =253 savedpeak = 31357"
## [7] "j =244 peak =272 savedpeak = 31376"
## [8] "j =244 peak =298 savedpeak = 31402"
## [9] "j =244 peak =317 savedpeak = 31421"
## [10] "j =244 peak =359 savedpeak = 31463"
## [11] "j =244 peak =373 savedpeak = 31477"
## [12] "j =244 peak =403 savedpeak = 31507"
## [13] "j =244 peak =436 savedpeak = 31540"
## [1] "j =245 peak =125 savedpeak = 31357"
## [2] "j =245 peak =144 savedpeak = 31376"
## [3] "j =245 peak =170 savedpeak = 31402"
## [4] "j =245 peak =189 savedpeak = 31421"
## [5] "j =245 peak =231 savedpeak = 31463"
## [6] "j =245 peak =245 savedpeak = 31477"
## [7] "j =245 peak =275 savedpeak = 31507"
## [8] "j =245 peak =308 savedpeak = 31540"
## [9] "j =245 peak =327 savedpeak = 31559"
## [10] "j =245 peak =342 savedpeak = 31574"
## [11] "j =245 peak =367 savedpeak = 31599"
## [12] "j =245 peak =405 savedpeak = 31637"
## [13] "j =245 peak =424 savedpeak = 31656"
## [1] "j =246 peak =103 savedpeak = 31463"
## [2] "j =246 peak =117 savedpeak = 31477"
## [3] "j =246 peak =147 savedpeak = 31507"
## [4] "j =246 peak =180 savedpeak = 31540"
## [5] "j =246 peak =199 savedpeak = 31559"
## [6] "j =246 peak =214 savedpeak = 31574"
## [7] "j =246 peak =239 savedpeak = 31599"
## [8] "j =246 peak =277 savedpeak = 31637"
## [9] "j =246 peak =296 savedpeak = 31656"
## [10] "j =246 peak =335 savedpeak = 31695"
## [11] "j =246 peak =366 savedpeak = 31726"
## [12] "j =246 peak =388 savedpeak = 31748"
## [13] "j =246 peak =401 savedpeak = 31761"
## [14] "j =246 peak =436 savedpeak = 31796"
## [1] "j =247 peak =86 savedpeak = 31574"
## [2] "j =247 peak =111 savedpeak = 31599"
## [3] "j =247 peak =149 savedpeak = 31637"

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## [4] "j =247 peak =168 savedpeak = 31656"
## [5] "j =247 peak =207 savedpeak = 31695"
## [6] "j =247 peak =238 savedpeak = 31726"
## [7] "j =247 peak =260 savedpeak = 31748"
## [8] "j =247 peak =273 savedpeak = 31761"
## [9] "j =247 peak =308 savedpeak = 31796"
## [10] "j =247 peak =315 savedpeak = 31803"
## [11] "j =247 peak =348 savedpeak = 31836"
## [12] "j =247 peak =361 savedpeak = 31849"
## [13] "j =247 peak =386 savedpeak = 31874"
## [14] "j =247 peak =417 savedpeak = 31905"
## [1] "j =248 peak =79 savedpeak = 31695"
## [2] "j =248 peak =110 savedpeak = 31726"
## [3] "j =248 peak =132 savedpeak = 31748"
## [4] "j =248 peak =145 savedpeak = 31761"
## [5] "j =248 peak =180 savedpeak = 31796"
## [6] "j =248 peak =187 savedpeak = 31803"
## [7] "j =248 peak =220 savedpeak = 31836"
## [8] "j =248 peak =233 savedpeak = 31849"
## [9] "j =248 peak =258 savedpeak = 31874"
## [10] "j =248 peak =289 savedpeak = 31905"
## [11] "j =248 peak =342 savedpeak = 31958"
## [12] "j =248 peak =367 savedpeak = 31983"
## [13] "j =248 peak =394 savedpeak = 32010"
## [14] "j =248 peak =404 savedpeak = 32020"
## [15] "j =248 peak =427 savedpeak = 32043"
## [1] "j =249 peak =92 savedpeak = 31836"
## [2] "j =249 peak =105 savedpeak = 31849"
## [3] "j =249 peak =130 savedpeak = 31874"
## [4] "j =249 peak =161 savedpeak = 31905"
## [5] "j =249 peak =214 savedpeak = 31958"
## [6] "j =249 peak =239 savedpeak = 31983"
## [7] "j =249 peak =266 savedpeak = 32010"
## [8] "j =249 peak =276 savedpeak = 32020"
## [9] "j =249 peak =299 savedpeak = 32043"
## [10] "j =249 peak =329 savedpeak = 32073"
## [11] "j =249 peak =385 savedpeak = 32129"
## [12] "j =249 peak =404 savedpeak = 32148"
## [13] "j =249 peak =432 savedpeak = 32176"
## [1] "j =252 peak =103 savedpeak = 32231"
## [2] "j =252 peak =155 savedpeak = 32283"
## [3] "j =252 peak =167 savedpeak = 32295"
## [4] "j =252 peak =188 savedpeak = 32316"
## [5] "j =252 peak =216 savedpeak = 32344"
## [6] "j =252 peak =262 savedpeak = 32390"
## [7] "j =252 peak =275 savedpeak = 32403"
## [8] "j =252 peak =291 savedpeak = 32419"
## [9] "j =252 peak =317 savedpeak = 32445"
## [10] "j =252 peak =341 savedpeak = 32469"
## [11] "j =252 peak =365 savedpeak = 32493"
## [12] "j =252 peak =424 savedpeak = 32552"
## [1] "j =254 peak =85 savedpeak = 32469"
## [2] "j =254 peak =109 savedpeak = 32493"
## [3] "j =254 peak =168 savedpeak = 32552"

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## [4] "j =254 peak =186 savedpeak = 32570"
## [5] "j =254 peak =248 savedpeak = 32632"
## [6] "j =254 peak =272 savedpeak = 32656"
## [7] "j =254 peak =288 savedpeak = 32672"
## [8] "j =254 peak =321 savedpeak = 32705"
## [9] "j =254 peak =344 savedpeak = 32728"
## [10] "j =254 peak =368 savedpeak = 32752"
## [11] "j =254 peak =385 savedpeak = 32769"
## [12] "j =254 peak =403 savedpeak = 32787"
## [13] "j =254 peak =418 savedpeak = 32802"
## [1] "j =255 peak =120 savedpeak = 32632"
## [2] "j =255 peak =144 savedpeak = 32656"
## [3] "j =255 peak =160 savedpeak = 32672"
## [4] "j =255 peak =193 savedpeak = 32705"
## [5] "j =255 peak =216 savedpeak = 32728"
## [6] "j =255 peak =240 savedpeak = 32752"
## [7] "j =255 peak =257 savedpeak = 32769"
## [8] "j =255 peak =275 savedpeak = 32787"
## [9] "j =255 peak =290 savedpeak = 32802"
## [10] "j =255 peak =329 savedpeak = 32841"
## [11] "j =255 peak =375 savedpeak = 32887"
## [12] "j =255 peak =393 savedpeak = 32905"
## [13] "j =255 peak =433 savedpeak = 32945"
## [1] "j =259 peak =83 savedpeak = 33107"
## [2] "j =259 peak =114 savedpeak = 33138"
## [3] "j =259 peak =139 savedpeak = 33163"
## [4] "j =259 peak =189 savedpeak = 33213"
## [5] "j =259 peak =211 savedpeak = 33235"
## [6] "j =259 peak =229 savedpeak = 33253"
## [7] "j =259 peak =250 savedpeak = 33274"
## [8] "j =259 peak =283 savedpeak = 33307"
## [9] "j =259 peak =310 savedpeak = 33334"
## [10] "j =259 peak =339 savedpeak = 33363"
## [11] "j =259 peak =376 savedpeak = 33400"
## [12] "j =259 peak =403 savedpeak = 33427"
## [13] "j =259 peak =418 savedpeak = 33442"
## [1] "j =260 peak =83 savedpeak = 33235"
## [2] "j =260 peak =101 savedpeak = 33253"
## [3] "j =260 peak =122 savedpeak = 33274"
## [4] "j =260 peak =155 savedpeak = 33307"
## [5] "j =260 peak =182 savedpeak = 33334"
## [6] "j =260 peak =211 savedpeak = 33363"
## [7] "j =260 peak =248 savedpeak = 33400"
## [8] "j =260 peak =275 savedpeak = 33427"
## [9] "j =260 peak =290 savedpeak = 33442"
## [10] "j =260 peak =324 savedpeak = 33476"
## [11] "j =260 peak =340 savedpeak = 33492"
## [12] "j =260 peak =390 savedpeak = 33542"
## [13] "j =260 peak =431 savedpeak = 33583"
## [1] "j =269 peak =103 savedpeak = 34407"
## [2] "j =269 peak =117 savedpeak = 34421"
## [3] "j =269 peak =175 savedpeak = 34479"
## [4] "j =269 peak =206 savedpeak = 34510"
## [5] "j =269 peak =225 savedpeak = 34529"

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## [6] "j =269 peak =264 savedpeak = 34568"
## [7] "j =269 peak =291 savedpeak = 34595"
## [8] "j =269 peak =305 savedpeak = 34609"
## [9] "j =269 peak =334 savedpeak = 34638"
## [10] "j =269 peak =390 savedpeak = 34694"
## [11] "j =269 peak =406 savedpeak = 34710"
## [12] "j =269 peak =415 savedpeak = 34719"
## [13] "j =269 peak =431 savedpeak = 34735"
## [1] "j =270 peak =78 savedpeak = 34510"
## [2] "j =270 peak =97 savedpeak = 34529"
## [3] "j =270 peak =136 savedpeak = 34568"
## [4] "j =270 peak =163 savedpeak = 34595"
## [5] "j =270 peak =177 savedpeak = 34609"
## [6] "j =270 peak =206 savedpeak = 34638"
## [7] "j =270 peak =262 savedpeak = 34694"
## [8] "j =270 peak =278 savedpeak = 34710"
## [9] "j =270 peak =287 savedpeak = 34719"
## [10] "j =270 peak =303 savedpeak = 34735"
## [11] "j =270 peak =361 savedpeak = 34793"
## [12] "j =270 peak =379 savedpeak = 34811"
## [13] "j =270 peak =408 savedpeak = 34840"
## [14] "j =270 peak =425 savedpeak = 34857"
## [1] "j =271 peak =78 savedpeak = 34638"
## [2] "j =271 peak =150 savedpeak = 34710"
## [3] "j =271 peak =159 savedpeak = 34719"
## [4] "j =271 peak =175 savedpeak = 34735"
## [5] "j =271 peak =233 savedpeak = 34793"
## [6] "j =271 peak =251 savedpeak = 34811"
## [7] "j =271 peak =280 savedpeak = 34840"
## [8] "j =271 peak =297 savedpeak = 34857"
## [9] "j =271 peak =311 savedpeak = 34871"
## [10] "j =271 peak =327 savedpeak = 34887"
## [11] "j =271 peak =343 savedpeak = 34903"
## [12] "j =271 peak =384 savedpeak = 34944"
## [13] "j =271 peak =430 savedpeak = 34990"
## [1] "j =273 peak =87 savedpeak = 34903"
## [2] "j =273 peak =128 savedpeak = 34944"
## [3] "j =273 peak =174 savedpeak = 34990"
## [4] "j =273 peak =195 savedpeak = 35011"
## [5] "j =273 peak =218 savedpeak = 35034"
## [6] "j =273 peak =263 savedpeak = 35079"
## [7] "j =273 peak =297 savedpeak = 35113"
## [8] "j =273 peak =317 savedpeak = 35133"
## [9] "j =273 peak =351 savedpeak = 35167"
## [10] "j =273 peak =378 savedpeak = 35194"
## [11] "j =273 peak =435 savedpeak = 35251"
## [1] "j =274 peak =90 savedpeak = 35034"
## [2] "j =274 peak =135 savedpeak = 35079"
## [3] "j =274 peak =169 savedpeak = 35113"
## [4] "j =274 peak =189 savedpeak = 35133"
## [5] "j =274 peak =223 savedpeak = 35167"
## [6] "j =274 peak =250 savedpeak = 35194"
## [7] "j =274 peak =307 savedpeak = 35251"
## [8] "j =274 peak =326 savedpeak = 35270"

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## [9] "j =274 peak =365 savedpeak = 35309"
## [10] "j =274 peak =392 savedpeak = 35336"
## [11] "j =274 peak =412 savedpeak = 35356"
## [1] "j =278 peak =85 savedpeak = 35541"
## [2] "j =278 peak =97 savedpeak = 35553"
## [3] "j =278 peak =145 savedpeak = 35601"
## [4] "j =278 peak =182 savedpeak = 35638"
## [5] "j =278 peak =208 savedpeak = 35664"
## [6] "j =278 peak =245 savedpeak = 35701"
## [7] "j =278 peak =336 savedpeak = 35792"
## [8] "j =278 peak =361 savedpeak = 35817"
## [9] "j =278 peak =391 savedpeak = 35847"
## [10] "j =278 peak =411 savedpeak = 35867"
## [1] "j =279 peak =80 savedpeak = 35664"
## [2] "j =279 peak =117 savedpeak = 35701"
## [3] "j =279 peak =208 savedpeak = 35792"
## [4] "j =279 peak =233 savedpeak = 35817"
## [5] "j =279 peak =263 savedpeak = 35847"
## [6] "j =279 peak =283 savedpeak = 35867"
## [7] "j =279 peak =310 savedpeak = 35894"
## [8] "j =279 peak =375 savedpeak = 35959"
## [9] "j =279 peak =408 savedpeak = 35992"
## [10] "j =279 peak =426 savedpeak = 36010"
## [1] "j =280 peak =80 savedpeak = 35792"
## [2] "j =280 peak =105 savedpeak = 35817"
## [3] "j =280 peak =135 savedpeak = 35847"
## [4] "j =280 peak =155 savedpeak = 35867"
## [5] "j =280 peak =182 savedpeak = 35894"
## [6] "j =280 peak =247 savedpeak = 35959"
## [7] "j =280 peak =280 savedpeak = 35992"
## [8] "j =280 peak =298 savedpeak = 36010"
## [9] "j =280 peak =319 savedpeak = 36031"
## [10] "j =280 peak =339 savedpeak = 36051"
## [11] "j =280 peak =351 savedpeak = 36063"
## [12] "j =280 peak =390 savedpeak = 36102"
## [13] "j =280 peak =422 savedpeak = 36134"
## [1] "j =281 peak =76 savedpeak = 35916"
## [2] "j =281 peak =119 savedpeak = 35959"
## [3] "j =281 peak =152 savedpeak = 35992"
## [4] "j =281 peak =170 savedpeak = 36010"
## [5] "j =281 peak =191 savedpeak = 36031"
## [6] "j =281 peak =211 savedpeak = 36051"
## [7] "j =281 peak =223 savedpeak = 36063"
## [8] "j =281 peak =262 savedpeak = 36102"
## [9] "j =281 peak =294 savedpeak = 36134"
## [10] "j =281 peak =333 savedpeak = 36173"
## [11] "j =281 peak =374 savedpeak = 36214"
## [12] "j =281 peak =396 savedpeak = 36236"
## [1] "j =282 peak =83 savedpeak = 36051"
## [2] "j =282 peak =95 savedpeak = 36063"
## [3] "j =282 peak =134 savedpeak = 36102"
## [4] "j =282 peak =166 savedpeak = 36134"
## [5] "j =282 peak =205 savedpeak = 36173"
## [6] "j =282 peak =246 savedpeak = 36214"

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## [7] "j =282 peak =268 savedpeak = 36236"
## [8] "j =282 peak =315 savedpeak = 36283"
## [9] "j =282 peak =348 savedpeak = 36316"
## [10] "j =282 peak =364 savedpeak = 36332"
## [11] "j =282 peak =390 savedpeak = 36358"
## [12] "j =282 peak =406 savedpeak = 36374"
## [13] "j =282 peak =425 savedpeak = 36393"
## [1] "j =283 peak =77 savedpeak = 36173"
## [2] "j =283 peak =118 savedpeak = 36214"
## [3] "j =283 peak =140 savedpeak = 36236"
## [4] "j =283 peak =187 savedpeak = 36283"
## [5] "j =283 peak =220 savedpeak = 36316"
## [6] "j =283 peak =236 savedpeak = 36332"
## [7] "j =283 peak =262 savedpeak = 36358"
## [8] "j =283 peak =278 savedpeak = 36374"
## [9] "j =283 peak =297 savedpeak = 36393"
## [10] "j =283 peak =325 savedpeak = 36421"
## [11] "j =283 peak =348 savedpeak = 36444"
## [12] "j =283 peak =360 savedpeak = 36456"
## [13] "j =283 peak =382 savedpeak = 36478"
## [14] "j =283 peak =402 savedpeak = 36498"
## [15] "j =283 peak =415 savedpeak = 36511"
## [1] "j =289 peak =134 savedpeak = 36998"
## [2] "j =289 peak =275 savedpeak = 37139"
## [3] "j =289 peak =357 savedpeak = 37221"
## [1] "j =290 peak =147 savedpeak = 37139"
## [2] "j =290 peak =229 savedpeak = 37221"
## [3] "j =290 peak =357 savedpeak = 37349"
## [4] "j =290 peak =435 savedpeak = 37427"
## [1] "j =291 peak =101 savedpeak = 37221"
## [2] "j =291 peak =229 savedpeak = 37349"
## [3] "j =291 peak =307 savedpeak = 37427"
## [1] "j =292 peak =101 savedpeak = 37349"
## [2] "j =292 peak =179 savedpeak = 37427"
## [3] "j =292 peak =412 savedpeak = 37660"
## [1] "j =293 peak =107 savedpeak = 37483"
## [2] "j =293 peak =128 savedpeak = 37504"
## [3] "j =293 peak =168 savedpeak = 37544"
## [4] "j =293 peak =194 savedpeak = 37570"
## [5] "j =293 peak =240 savedpeak = 37616"
## [6] "j =293 peak =277 savedpeak = 37653"
## [7] "j =293 peak =307 savedpeak = 37683"
## [8] "j =293 peak =323 savedpeak = 37699"
## [9] "j =293 peak =369 savedpeak = 37745"
## [10] "j =293 peak =409 savedpeak = 37785"
## [11] "j =293 peak =423 savedpeak = 37799"
## [1] "j =294 peak =112 savedpeak = 37616"
## [2] "j =294 peak =155 savedpeak = 37659"
## [3] "j =294 peak =179 savedpeak = 37683"
## [4] "j =294 peak =195 savedpeak = 37699"
## [5] "j =294 peak =241 savedpeak = 37745"
## [6] "j =294 peak =281 savedpeak = 37785"
## [7] "j =294 peak =295 savedpeak = 37799"
## [8] "j =294 peak =320 savedpeak = 37824"

```

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## [9] "j =294 peak =344 savedpeak = 37848"
## [10] "j =294 peak =377 savedpeak = 37881"
## [11] "j =294 peak =406 savedpeak = 37910"
## [12] "j =294 peak =429 savedpeak = 37933"
## [1] "j =295 peak =113 savedpeak = 37745"
## [2] "j =295 peak =153 savedpeak = 37785"
## [3] "j =295 peak =167 savedpeak = 37799"
## [4] "j =295 peak =192 savedpeak = 37824"
## [5] "j =295 peak =216 savedpeak = 37848"
## [6] "j =295 peak =249 savedpeak = 37881"
## [7] "j =295 peak =278 savedpeak = 37910"
## [8] "j =295 peak =301 savedpeak = 37933"
## [9] "j =295 peak =316 savedpeak = 37948"
## [10] "j =295 peak =341 savedpeak = 37973"
## [11] "j =295 peak =361 savedpeak = 37993"
## [12] "j =295 peak =374 savedpeak = 38006"
## [13] "j =295 peak =393 savedpeak = 38025"
## [14] "j =295 peak =404 savedpeak = 38036"
## [15] "j =295 peak =422 savedpeak = 38054"
## [1] "j =296 peak =88 savedpeak = 37848"
## [2] "j =296 peak =121 savedpeak = 37881"
## [3] "j =296 peak =150 savedpeak = 37910"
## [4] "j =296 peak =173 savedpeak = 37933"
## [5] "j =296 peak =188 savedpeak = 37948"
## [6] "j =296 peak =213 savedpeak = 37973"
## [7] "j =296 peak =233 savedpeak = 37993"
## [8] "j =296 peak =246 savedpeak = 38006"
## [9] "j =296 peak =265 savedpeak = 38025"
## [10] "j =296 peak =276 savedpeak = 38036"
## [11] "j =296 peak =294 savedpeak = 38054"
## [12] "j =296 peak =329 savedpeak = 38089"
## [13] "j =296 peak =344 savedpeak = 38104"
## [14] "j =296 peak =385 savedpeak = 38145"
## [15] "j =296 peak =424 savedpeak = 38184"
## [1] "j =303 peak =98 savedpeak = 38754"
## [2] "j =303 peak =125 savedpeak = 38781"
## [3] "j =303 peak =164 savedpeak = 38820"
## [4] "j =303 peak =187 savedpeak = 38843"
## [5] "j =303 peak =228 savedpeak = 38884"
## [6] "j =303 peak =276 savedpeak = 38932"
## [7] "j =303 peak =297 savedpeak = 38953"
## [8] "j =303 peak =314 savedpeak = 38970"
## [9] "j =303 peak =359 savedpeak = 39015"
## [10] "j =303 peak =380 savedpeak = 39036"
## [11] "j =303 peak =393 savedpeak = 39049"
## [1] "j =304 peak =100 savedpeak = 38884"
## [2] "j =304 peak =148 savedpeak = 38932"
## [3] "j =304 peak =169 savedpeak = 38953"
## [4] "j =304 peak =186 savedpeak = 38970"
## [5] "j =304 peak =231 savedpeak = 39015"
## [6] "j =304 peak =252 savedpeak = 39036"
## [7] "j =304 peak =265 savedpeak = 39049"
## [8] "j =304 peak =310 savedpeak = 39094"
## [9] "j =304 peak =332 savedpeak = 39116"

```

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## [10] "j =304 peak =347 savedpeak = 39131"
## [11] "j =304 peak =369 savedpeak = 39153"
## [12] "j =304 peak =394 savedpeak = 39178"
## [13] "j =304 peak =419 savedpeak = 39203"
## [1] "j =305 peak =103 savedpeak = 39015"
## [2] "j =305 peak =124 savedpeak = 39036"
## [3] "j =305 peak =137 savedpeak = 39049"
## [4] "j =305 peak =182 savedpeak = 39094"
## [5] "j =305 peak =204 savedpeak = 39116"
## [6] "j =305 peak =219 savedpeak = 39131"
## [7] "j =305 peak =241 savedpeak = 39153"
## [8] "j =305 peak =266 savedpeak = 39178"
## [9] "j =305 peak =291 savedpeak = 39203"
## [10] "j =305 peak =315 savedpeak = 39227"
## [11] "j =305 peak =333 savedpeak = 39245"
## [12] "j =305 peak =380 savedpeak = 39292"
## [13] "j =305 peak =395 savedpeak = 39307"
## [14] "j =305 peak =431 savedpeak = 39343"
## [1] "j =306 peak =76 savedpeak = 39116"
## [2] "j =306 peak =91 savedpeak = 39131"
## [3] "j =306 peak =113 savedpeak = 39153"
## [4] "j =306 peak =138 savedpeak = 39178"
## [5] "j =306 peak =163 savedpeak = 39203"
## [6] "j =306 peak =187 savedpeak = 39227"
## [7] "j =306 peak =205 savedpeak = 39245"
## [8] "j =306 peak =252 savedpeak = 39292"
## [9] "j =306 peak =267 savedpeak = 39307"
## [10] "j =306 peak =303 savedpeak = 39343"
## [11] "j =306 peak =328 savedpeak = 39368"
## [12] "j =306 peak =364 savedpeak = 39404"
## [13] "j =306 peak =380 savedpeak = 39420"
## [14] "j =306 peak =404 savedpeak = 39444"
## [15] "j =306 peak =427 savedpeak = 39467"
## [1] "j =307 peak =77 savedpeak = 39245"
## [2] "j =307 peak =124 savedpeak = 39292"
## [3] "j =307 peak =139 savedpeak = 39307"
## [4] "j =307 peak =175 savedpeak = 39343"
## [5] "j =307 peak =200 savedpeak = 39368"
## [6] "j =307 peak =236 savedpeak = 39404"
## [7] "j =307 peak =252 savedpeak = 39420"
## [8] "j =307 peak =276 savedpeak = 39444"
## [9] "j =307 peak =299 savedpeak = 39467"
## [10] "j =307 peak =328 savedpeak = 39496"
## [11] "j =307 peak =347 savedpeak = 39515"
## [12] "j =307 peak =377 savedpeak = 39545"
## [13] "j =307 peak =397 savedpeak = 39565"
## [14] "j =307 peak =424 savedpeak = 39592"
## [1] "j =308 peak =107 savedpeak = 39403"
## [2] "j =308 peak =145 savedpeak = 39441"
## [3] "j =308 peak =198 savedpeak = 39494"
## [4] "j =308 peak =224 savedpeak = 39520"
## [5] "j =308 peak =269 savedpeak = 39565"
## [6] "j =308 peak =354 savedpeak = 39650"
## [7] "j =308 peak =406 savedpeak = 39702"

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## [1] "j =309 peak =141 savedpeak = 39565"
## [2] "j =309 peak =226 savedpeak = 39650"
## [3] "j =309 peak =278 savedpeak = 39702"
## [4] "j =309 peak =375 savedpeak = 39799"
## [5] "j =309 peak =399 savedpeak = 39823"
## [1] "j =310 peak =98 savedpeak = 39650"
## [2] "j =310 peak =150 savedpeak = 39702"
## [3] "j =310 peak =247 savedpeak = 39799"
## [4] "j =310 peak =271 savedpeak = 39823"
## [5] "j =310 peak =312 savedpeak = 39864"
## [6] "j =310 peak =341 savedpeak = 39893"
## [7] "j =310 peak =403 savedpeak = 39955"
## [8] "j =310 peak =427 savedpeak = 39979"
## [1] "j =311 peak =119 savedpeak = 39799"
## [2] "j =311 peak =143 savedpeak = 39823"
## [3] "j =311 peak =184 savedpeak = 39864"
## [4] "j =311 peak =213 savedpeak = 39893"
## [5] "j =311 peak =275 savedpeak = 39955"
## [6] "j =311 peak =299 savedpeak = 39979"
## [7] "j =311 peak =336 savedpeak = 40016"
## [8] "j =311 peak =373 savedpeak = 40053"
## [9] "j =311 peak =424 savedpeak = 40104"
## [1] "j =312 peak =80 savedpeak = 39888"
## [2] "j =312 peak =117 savedpeak = 39925"
## [3] "j =312 peak =140 savedpeak = 39948"
## [4] "j =312 peak =171 savedpeak = 39979"
## [5] "j =312 peak =210 savedpeak = 40018"
## [6] "j =312 peak =244 savedpeak = 40052"
## [7] "j =312 peak =260 savedpeak = 40068"
## [8] "j =312 peak =274 savedpeak = 40082"
## [9] "j =312 peak =297 savedpeak = 40105"
## [10] "j =312 peak =335 savedpeak = 40143"
## [11] "j =312 peak =370 savedpeak = 40178"
## [12] "j =312 peak =382 savedpeak = 40190"
## [13] "j =312 peak =410 savedpeak = 40218"
## [14] "j =312 peak =424 savedpeak = 40232"
## [15] "j =312 peak =433 savedpeak = 40241"
## [1] "j =313 peak =82 savedpeak = 40018"
## [2] "j =313 peak =116 savedpeak = 40052"
## [3] "j =313 peak =132 savedpeak = 40068"
## [4] "j =313 peak =146 savedpeak = 40082"
## [5] "j =313 peak =169 savedpeak = 40105"
## [6] "j =313 peak =207 savedpeak = 40143"
## [7] "j =313 peak =242 savedpeak = 40178"
## [8] "j =313 peak =254 savedpeak = 40190"
## [9] "j =313 peak =282 savedpeak = 40218"
## [10] "j =313 peak =296 savedpeak = 40232"
## [11] "j =313 peak =305 savedpeak = 40241"
## [12] "j =313 peak =330 savedpeak = 40266"
## [13] "j =313 peak =348 savedpeak = 40284"
## [14] "j =313 peak =365 savedpeak = 40301"
## [15] "j =313 peak =397 savedpeak = 40333"
## [16] "j =313 peak =410 savedpeak = 40346"
## [1] "j =315 peak =92 savedpeak = 40284"

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## [2] "j =315 peak =109 savedpeak = 40301"
## [3] "j =315 peak =141 savedpeak = 40333"
## [4] "j =315 peak =154 savedpeak = 40346"
## [5] "j =315 peak =186 savedpeak = 40378"
## [6] "j =315 peak =204 savedpeak = 40396"
## [7] "j =315 peak =228 savedpeak = 40420"
## [8] "j =315 peak =270 savedpeak = 40462"
## [9] "j =315 peak =310 savedpeak = 40502"
## [10] "j =315 peak =331 savedpeak = 40523"
## [11] "j =315 peak =347 savedpeak = 40539"
## [12] "j =315 peak =375 savedpeak = 40567"
## [13] "j =315 peak =413 savedpeak = 40605"
## [14] "j =315 peak =421 savedpeak = 40613"
## [1] "j =316 peak =76 savedpeak = 40396"
## [2] "j =316 peak =100 savedpeak = 40420"
## [3] "j =316 peak =142 savedpeak = 40462"
## [4] "j =316 peak =182 savedpeak = 40502"
## [5] "j =316 peak =203 savedpeak = 40523"
## [6] "j =316 peak =219 savedpeak = 40539"
## [7] "j =316 peak =247 savedpeak = 40567"
## [8] "j =316 peak =262 savedpeak = 40582"
## [9] "j =316 peak =285 savedpeak = 40605"
## [10] "j =316 peak =293 savedpeak = 40613"
## [11] "j =316 peak =322 savedpeak = 40642"
## [12] "j =316 peak =355 savedpeak = 40675"
## [13] "j =316 peak =417 savedpeak = 40737"
## [1] "j =317 peak =91 savedpeak = 40539"
## [2] "j =317 peak =119 savedpeak = 40567"
## [3] "j =317 peak =134 savedpeak = 40582"
## [4] "j =317 peak =157 savedpeak = 40605"
## [5] "j =317 peak =165 savedpeak = 40613"
## [6] "j =317 peak =194 savedpeak = 40642"
## [7] "j =317 peak =227 savedpeak = 40675"
## [8] "j =317 peak =289 savedpeak = 40737"
## [9] "j =317 peak =314 savedpeak = 40762"
## [10] "j =317 peak =357 savedpeak = 40805"
## [11] "j =317 peak =379 savedpeak = 40827"
## [12] "j =317 peak =402 savedpeak = 40850"
## [13] "j =317 peak =419 savedpeak = 40867"
## [14] "j =317 peak =437 savedpeak = 40885"
## [1] "j =318 peak =99 savedpeak = 40675"
## [2] "j =318 peak =161 savedpeak = 40737"
## [3] "j =318 peak =186 savedpeak = 40762"
## [4] "j =318 peak =229 savedpeak = 40805"
## [5] "j =318 peak =251 savedpeak = 40827"
## [6] "j =318 peak =274 savedpeak = 40850"
## [7] "j =318 peak =291 savedpeak = 40867"
## [8] "j =318 peak =309 savedpeak = 40885"
## [9] "j =318 peak =350 savedpeak = 40926"
## [10] "j =318 peak =387 savedpeak = 40963"
## [11] "j =318 peak =409 savedpeak = 40985"
## [12] "j =318 peak =436 savedpeak = 41012"
## [1] "j =319 peak =101 savedpeak = 40805"
## [2] "j =319 peak =123 savedpeak = 40827"

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## [3] "j =319 peak =146 savedpeak = 40850"
## [4] "j =319 peak =163 savedpeak = 40867"
## [5] "j =319 peak =181 savedpeak = 40885"
## [6] "j =319 peak =222 savedpeak = 40926"
## [7] "j =319 peak =259 savedpeak = 40963"
## [8] "j =319 peak =281 savedpeak = 40985"
## [9] "j =319 peak =308 savedpeak = 41012"
## [10] "j =319 peak =336 savedpeak = 41040"
## [11] "j =319 peak =370 savedpeak = 41074"
## [12] "j =319 peak =405 savedpeak = 41109"
## [13] "j =319 peak =415 savedpeak = 41119"
## [1] "j =320 peak =94 savedpeak = 40926"
## [2] "j =320 peak =131 savedpeak = 40963"
## [3] "j =320 peak =153 savedpeak = 40985"
## [4] "j =320 peak =180 savedpeak = 41012"
## [5] "j =320 peak =208 savedpeak = 41040"
## [6] "j =320 peak =242 savedpeak = 41074"
## [7] "j =320 peak =277 savedpeak = 41109"
## [8] "j =320 peak =287 savedpeak = 41119"
## [9] "j =320 peak =355 savedpeak = 41187"
## [10] "j =320 peak =382 savedpeak = 41214"
## [11] "j =320 peak =412 savedpeak = 41244"
## [1] "j =322 peak =99 savedpeak = 41187"
## [2] "j =322 peak =126 savedpeak = 41214"
## [3] "j =322 peak =156 savedpeak = 41244"
## [4] "j =322 peak =198 savedpeak = 41286"
## [5] "j =322 peak =215 savedpeak = 41303"
## [6] "j =322 peak =283 savedpeak = 41371"
## [7] "j =322 peak =316 savedpeak = 41404"
## [8] "j =322 peak =340 savedpeak = 41428"
## [9] "j =322 peak =368 savedpeak = 41456"
## [10] "j =322 peak =409 savedpeak = 41497"
## [1] "j =325 peak =84 savedpeak = 41556"
## [2] "j =325 peak =109 savedpeak = 41581"
## [3] "j =325 peak =141 savedpeak = 41613"
## [4] "j =325 peak =158 savedpeak = 41630"
## [5] "j =325 peak =182 savedpeak = 41654"
## [6] "j =325 peak =223 savedpeak = 41695"
## [7] "j =325 peak =241 savedpeak = 41713"
## [8] "j =325 peak =300 savedpeak = 41772"
## [9] "j =325 peak =323 savedpeak = 41795"
## [10] "j =325 peak =344 savedpeak = 41816"
## [11] "j =325 peak =351 savedpeak = 41823"
## [12] "j =325 peak =378 savedpeak = 41850"
## [13] "j =325 peak =416 savedpeak = 41888"
## [1] "j =326 peak =95 savedpeak = 41695"
## [2] "j =326 peak =113 savedpeak = 41713"
## [3] "j =326 peak =172 savedpeak = 41772"
## [4] "j =326 peak =195 savedpeak = 41795"
## [5] "j =326 peak =216 savedpeak = 41816"
## [6] "j =326 peak =223 savedpeak = 41823"
## [7] "j =326 peak =250 savedpeak = 41850"
## [8] "j =326 peak =288 savedpeak = 41888"
## [9] "j =326 peak =315 savedpeak = 41915"

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## [10] "j =326 peak =345 savedpeak = 41945"
## [11] "j =326 peak =390 savedpeak = 41990"
## [12] "j =326 peak =414 savedpeak = 42014"
## [13] "j =326 peak =429 savedpeak = 42029"
## [1] "j =327 peak =88 savedpeak = 41816"
## [2] "j =327 peak =95 savedpeak = 41823"
## [3] "j =327 peak =122 savedpeak = 41850"
## [4] "j =327 peak =160 savedpeak = 41888"
## [5] "j =327 peak =187 savedpeak = 41915"
## [6] "j =327 peak =217 savedpeak = 41945"
## [7] "j =327 peak =262 savedpeak = 41990"
## [8] "j =327 peak =286 savedpeak = 42014"
## [9] "j =327 peak =301 savedpeak = 42029"
## [10] "j =327 peak =324 savedpeak = 42052"
## [11] "j =327 peak =360 savedpeak = 42088"
## [12] "j =327 peak =380 savedpeak = 42108"
## [13] "j =327 peak =395 savedpeak = 42123"
## [1] "j =328 peak =89 savedpeak = 41945"
## [2] "j =328 peak =134 savedpeak = 41990"
## [3] "j =328 peak =158 savedpeak = 42014"
## [4] "j =328 peak =173 savedpeak = 42029"
## [5] "j =328 peak =196 savedpeak = 42052"
## [6] "j =328 peak =232 savedpeak = 42088"
## [7] "j =328 peak =252 savedpeak = 42108"
## [8] "j =328 peak =267 savedpeak = 42123"
## [9] "j =328 peak =311 savedpeak = 42167"
## [10] "j =328 peak =325 savedpeak = 42181"
## [11] "j =328 peak =350 savedpeak = 42206"
## [12] "j =328 peak =410 savedpeak = 42266"
## [1] "j =330 peak =94 savedpeak = 42206"
## [2] "j =330 peak =154 savedpeak = 42266"
## [3] "j =330 peak =183 savedpeak = 42295"
## [4] "j =330 peak =218 savedpeak = 42330"
## [5] "j =330 peak =279 savedpeak = 42391"
## [6] "j =330 peak =317 savedpeak = 42429"
## [7] "j =330 peak =334 savedpeak = 42446"
## [8] "j =330 peak =352 savedpeak = 42464"
## [9] "j =330 peak =368 savedpeak = 42480"
## [10] "j =330 peak =422 savedpeak = 42534"
## [1] "j =331 peak =90 savedpeak = 42330"
## [2] "j =331 peak =151 savedpeak = 42391"
## [3] "j =331 peak =189 savedpeak = 42429"
## [4] "j =331 peak =206 savedpeak = 42446"
## [5] "j =331 peak =224 savedpeak = 42464"
## [6] "j =331 peak =240 savedpeak = 42480"
## [7] "j =331 peak =294 savedpeak = 42534"
## [8] "j =331 peak =348 savedpeak = 42588"
## [9] "j =331 peak =365 savedpeak = 42605"
## [10] "j =331 peak =382 savedpeak = 42622"
## [11] "j =331 peak =408 savedpeak = 42648"
## [12] "j =331 peak =417 savedpeak = 42657"
## [13] "j =331 peak =437 savedpeak = 42677"
## [1] "j =332 peak =78 savedpeak = 42446"
## [2] "j =332 peak =96 savedpeak = 42464"

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## [3] "j =332 peak =112 savedpeak = 42480"
## [4] "j =332 peak =166 savedpeak = 42534"
## [5] "j =332 peak =220 savedpeak = 42588"
## [6] "j =332 peak =237 savedpeak = 42605"
## [7] "j =332 peak =254 savedpeak = 42622"
## [8] "j =332 peak =280 savedpeak = 42648"
## [9] "j =332 peak =289 savedpeak = 42657"
## [10] "j =332 peak =309 savedpeak = 42677"
## [11] "j =332 peak =334 savedpeak = 42702"
## [12] "j =332 peak =350 savedpeak = 42718"
## [13] "j =332 peak =362 savedpeak = 42730"
## [14] "j =332 peak =381 savedpeak = 42749"
## [15] "j =332 peak =402 savedpeak = 42770"
## [16] "j =332 peak =428 savedpeak = 42796"
## [1] "j =333 peak =92 savedpeak = 42588"
## [2] "j =333 peak =109 savedpeak = 42605"
## [3] "j =333 peak =126 savedpeak = 42622"
## [4] "j =333 peak =152 savedpeak = 42648"
## [5] "j =333 peak =161 savedpeak = 42657"
## [6] "j =333 peak =181 savedpeak = 42677"
## [7] "j =333 peak =206 savedpeak = 42702"
## [8] "j =333 peak =222 savedpeak = 42718"
## [9] "j =333 peak =234 savedpeak = 42730"
## [10] "j =333 peak =253 savedpeak = 42749"
## [11] "j =333 peak =274 savedpeak = 42770"
## [12] "j =333 peak =300 savedpeak = 42796"
## [13] "j =333 peak =319 savedpeak = 42815"
## [14] "j =333 peak =339 savedpeak = 42835"
## [15] "j =333 peak =354 savedpeak = 42850"
## [16] "j =333 peak =380 savedpeak = 42876"
## [17] "j =333 peak =395 savedpeak = 42891"
## [18] "j =333 peak =413 savedpeak = 42909"
## [19] "j =333 peak =433 savedpeak = 42929"
## [1] "j =334 peak =78 savedpeak = 42702"
## [2] "j =334 peak =94 savedpeak = 42718"
## [3] "j =334 peak =106 savedpeak = 42730"
## [4] "j =334 peak =125 savedpeak = 42749"
## [5] "j =334 peak =146 savedpeak = 42770"
## [6] "j =334 peak =172 savedpeak = 42796"
## [7] "j =334 peak =191 savedpeak = 42815"
## [8] "j =334 peak =211 savedpeak = 42835"
## [9] "j =334 peak =226 savedpeak = 42850"
## [10] "j =334 peak =252 savedpeak = 42876"
## [11] "j =334 peak =267 savedpeak = 42891"
## [12] "j =334 peak =285 savedpeak = 42909"
## [13] "j =334 peak =305 savedpeak = 42929"
## [14] "j =334 peak =354 savedpeak = 42978"
## [15] "j =334 peak =378 savedpeak = 43002"
## [16] "j =334 peak =422 savedpeak = 43046"
## [1] "j =335 peak =83 savedpeak = 42835"
## [2] "j =335 peak =98 savedpeak = 42850"
## [3] "j =335 peak =124 savedpeak = 42876"
## [4] "j =335 peak =139 savedpeak = 42891"
## [5] "j =335 peak =157 savedpeak = 42909"

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## [6] "j =335 peak =177 savedpeak = 42929"
## [7] "j =335 peak =226 savedpeak = 42978"
## [8] "j =335 peak =250 savedpeak = 43002"
## [9] "j =335 peak =294 savedpeak = 43046"
## [10] "j =335 peak =318 savedpeak = 43070"
## [11] "j =335 peak =346 savedpeak = 43098"
## [12] "j =335 peak =383 savedpeak = 43135"
## [13] "j =335 peak =421 savedpeak = 43173"
## [1] "j =337 peak =90 savedpeak = 43098"
## [2] "j =337 peak =127 savedpeak = 43135"
## [3] "j =337 peak =165 savedpeak = 43173"
## [4] "j =337 peak =198 savedpeak = 43206"
## [5] "j =337 peak =228 savedpeak = 43236"
## [6] "j =337 peak =258 savedpeak = 43266"
## [7] "j =337 peak =278 savedpeak = 43286"
## [8] "j =337 peak =316 savedpeak = 43324"
## [9] "j =337 peak =333 savedpeak = 43341"
## [10] "j =337 peak =367 savedpeak = 43375"
## [11] "j =337 peak =390 savedpeak = 43398"
## [12] "j =337 peak =411 savedpeak = 43419"
## [13] "j =337 peak =433 savedpeak = 43441"
## [1] "j =341 peak =117 savedpeak = 43637"
## [2] "j =341 peak =172 savedpeak = 43692"
## [3] "j =341 peak =209 savedpeak = 43729"
## [4] "j =341 peak =256 savedpeak = 43776"
## [5] "j =341 peak =312 savedpeak = 43832"
## [6] "j =341 peak =377 savedpeak = 43897"
## [7] "j =341 peak =401 savedpeak = 43921"
## [1] "j =342 peak =81 savedpeak = 43729"
## [2] "j =342 peak =128 savedpeak = 43776"
## [3] "j =342 peak =184 savedpeak = 43832"
## [4] "j =342 peak =249 savedpeak = 43897"
## [5] "j =342 peak =273 savedpeak = 43921"
## [6] "j =342 peak =328 savedpeak = 43976"
## [7] "j =342 peak =404 savedpeak = 44052"
## [1] "j =343 peak =121 savedpeak = 43897"
## [2] "j =343 peak =145 savedpeak = 43921"
## [3] "j =343 peak =200 savedpeak = 43976"
## [4] "j =343 peak =276 savedpeak = 44052"
## [5] "j =343 peak =338 savedpeak = 44114"
## [6] "j =343 peak =398 savedpeak = 44174"
## [1] "j =344 peak =148 savedpeak = 44052"
## [2] "j =344 peak =210 savedpeak = 44114"
## [3] "j =344 peak =270 savedpeak = 44174"
## [4] "j =344 peak =319 savedpeak = 44223"
## [5] "j =344 peak =343 savedpeak = 44247"
## [6] "j =344 peak =361 savedpeak = 44265"
## [7] "j =344 peak =380 savedpeak = 44284"
## [1] "j =345 peak =88 savedpeak = 44120"
## [2] "j =345 peak =109 savedpeak = 44141"
## [3] "j =345 peak =142 savedpeak = 44174"
## [4] "j =345 peak =159 savedpeak = 44191"
## [5] "j =345 peak =172 savedpeak = 44204"
## [6] "j =345 peak =189 savedpeak = 44221"

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## [7] "j =345 peak =215 savedpeak = 44247"
## [8] "j =345 peak =232 savedpeak = 44264"
## [9] "j =345 peak =252 savedpeak = 44284"
## [10] "j =345 peak =312 savedpeak = 44344"
## [11] "j =345 peak =359 savedpeak = 44391"
## [12] "j =345 peak =389 savedpeak = 44421"
## [13] "j =345 peak =418 savedpeak = 44450"
## [1] "j =346 peak =87 savedpeak = 44247"
## [2] "j =346 peak =104 savedpeak = 44264"
## [3] "j =346 peak =124 savedpeak = 44284"
## [4] "j =346 peak =184 savedpeak = 44344"
## [5] "j =346 peak =231 savedpeak = 44391"
## [6] "j =346 peak =261 savedpeak = 44421"
## [7] "j =346 peak =290 savedpeak = 44450"
## [8] "j =346 peak =317 savedpeak = 44477"
## [9] "j =346 peak =342 savedpeak = 44502"
## [10] "j =346 peak =355 savedpeak = 44515"
## [11] "j =346 peak =373 savedpeak = 44533"
## [12] "j =346 peak =397 savedpeak = 44557"
## [13] "j =346 peak =416 savedpeak = 44576"
## [14] "j =346 peak =432 savedpeak = 44592"
## [1] "j =347 peak =103 savedpeak = 44391"
## [2] "j =347 peak =133 savedpeak = 44421"
## [3] "j =347 peak =162 savedpeak = 44450"
## [4] "j =347 peak =189 savedpeak = 44477"
## [5] "j =347 peak =214 savedpeak = 44502"
## [6] "j =347 peak =227 savedpeak = 44515"
## [7] "j =347 peak =245 savedpeak = 44533"
## [8] "j =347 peak =269 savedpeak = 44557"
## [9] "j =347 peak =288 savedpeak = 44576"
## [10] "j =347 peak =304 savedpeak = 44592"
## [11] "j =347 peak =329 savedpeak = 44617"
## [12] "j =347 peak =348 savedpeak = 44636"
## [13] "j =347 peak =368 savedpeak = 44656"
## [14] "j =347 peak =401 savedpeak = 44689"
## [15] "j =347 peak =429 savedpeak = 44717"
## [1] "j =348 peak =86 savedpeak = 44502"
## [2] "j =348 peak =99 savedpeak = 44515"
## [3] "j =348 peak =117 savedpeak = 44533"
## [4] "j =348 peak =141 savedpeak = 44557"
## [5] "j =348 peak =160 savedpeak = 44576"
## [6] "j =348 peak =176 savedpeak = 44592"
## [7] "j =348 peak =201 savedpeak = 44617"
## [8] "j =348 peak =220 savedpeak = 44636"
## [9] "j =348 peak =240 savedpeak = 44656"
## [10] "j =348 peak =273 savedpeak = 44689"
## [11] "j =348 peak =301 savedpeak = 44717"
## [12] "j =348 peak =318 savedpeak = 44734"
## [13] "j =348 peak =338 savedpeak = 44754"
## [14] "j =348 peak =386 savedpeak = 44802"
## [15] "j =348 peak =417 savedpeak = 44833"
## [1] "j =349 peak =92 savedpeak = 44636"
## [2] "j =349 peak =112 savedpeak = 44656"
## [3] "j =349 peak =145 savedpeak = 44689"

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## [4] "j =349 peak =173 savedpeak = 44717"
## [5] "j =349 peak =190 savedpeak = 44734"
## [6] "j =349 peak =210 savedpeak = 44754"
## [7] "j =349 peak =258 savedpeak = 44802"
## [8] "j =349 peak =289 savedpeak = 44833"
## [9] "j =349 peak =337 savedpeak = 44881"
## [10] "j =349 peak =350 savedpeak = 44894"
## [11] "j =349 peak =370 savedpeak = 44914"
## [12] "j =349 peak =396 savedpeak = 44940"
## [13] "j =349 peak =408 savedpeak = 44952"
## [1] "j =356 peak =122 savedpeak = 45562"
## [2] "j =356 peak =142 savedpeak = 45582"
## [3] "j =356 peak =173 savedpeak = 45613"
## [4] "j =356 peak =211 savedpeak = 45651"
## [5] "j =356 peak =223 savedpeak = 45663"
## [6] "j =356 peak =261 savedpeak = 45701"
## [7] "j =356 peak =295 savedpeak = 45735"
## [8] "j =356 peak =330 savedpeak = 45770"
## [9] "j =356 peak =353 savedpeak = 45793"
## [10] "j =356 peak =382 savedpeak = 45822"
## [11] "j =356 peak =398 savedpeak = 45838"
## [1] "j =363 peak =83 savedpeak = 46419"
## [2] "j =363 peak =135 savedpeak = 46471"
## [3] "j =363 peak =162 savedpeak = 46498"
## [4] "j =363 peak =193 savedpeak = 46529"
## [5] "j =363 peak =206 savedpeak = 46542"
## [6] "j =363 peak =228 savedpeak = 46564"
## [7] "j =363 peak =242 savedpeak = 46578"
## [8] "j =363 peak =262 savedpeak = 46598"
## [9] "j =363 peak =296 savedpeak = 46632"
## [10] "j =363 peak =338 savedpeak = 46674"
## [11] "j =363 peak =365 savedpeak = 46701"
## [12] "j =363 peak =407 savedpeak = 46743"
## [13] "j =363 peak =430 savedpeak = 46766"
## [1] "j =369 peak =86 savedpeak = 47190"
## [2] "j =369 peak =153 savedpeak = 47257"
## [3] "j =369 peak =195 savedpeak = 47299"
## [4] "j =369 peak =239 savedpeak = 47343"
## [5] "j =369 peak =256 savedpeak = 47360"
## [6] "j =369 peak =292 savedpeak = 47396"
## [7] "j =369 peak =337 savedpeak = 47441"
## [8] "j =369 peak =364 savedpeak = 47468"
## [9] "j =369 peak =389 savedpeak = 47493"
## [10] "j =369 peak =430 savedpeak = 47534"
## [1] "j =370 peak =111 savedpeak = 47343"
## [2] "j =370 peak =128 savedpeak = 47360"
## [3] "j =370 peak =164 savedpeak = 47396"
## [4] "j =370 peak =209 savedpeak = 47441"
## [5] "j =370 peak =236 savedpeak = 47468"
## [6] "j =370 peak =261 savedpeak = 47493"
## [7] "j =370 peak =302 savedpeak = 47534"
## [8] "j =370 peak =322 savedpeak = 47554"
## [9] "j =370 peak =346 savedpeak = 47578"
## [10] "j =370 peak =358 savedpeak = 47590"

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## [11] "j =370 peak =380 savedpeak = 47612"
## [12] "j =370 peak =404 savedpeak = 47636"
## [13] "j =370 peak =424 savedpeak = 47656"
## [1] "j =371 peak =81 savedpeak = 47441"
## [2] "j =371 peak =108 savedpeak = 47468"
## [3] "j =371 peak =133 savedpeak = 47493"
## [4] "j =371 peak =174 savedpeak = 47534"
## [5] "j =371 peak =194 savedpeak = 47554"
## [6] "j =371 peak =218 savedpeak = 47578"
## [7] "j =371 peak =230 savedpeak = 47590"
## [8] "j =371 peak =252 savedpeak = 47612"
## [9] "j =371 peak =276 savedpeak = 47636"
## [10] "j =371 peak =296 savedpeak = 47656"
## [11] "j =371 peak =310 savedpeak = 47670"
## [12] "j =371 peak =344 savedpeak = 47704"
## [13] "j =371 peak =368 savedpeak = 47728"
## [14] "j =371 peak =402 savedpeak = 47762"
## [15] "j =371 peak =428 savedpeak = 47788"
## [1] "j =377 peak =112 savedpeak = 48240"
## [2] "j =377 peak =139 savedpeak = 48267"
## [3] "j =377 peak =164 savedpeak = 48292"
## [4] "j =377 peak =198 savedpeak = 48326"
## [5] "j =377 peak =225 savedpeak = 48353"
## [6] "j =377 peak =258 savedpeak = 48386"
## [7] "j =377 peak =302 savedpeak = 48430"
## [8] "j =377 peak =328 savedpeak = 48456"
## [9] "j =377 peak =362 savedpeak = 48490"
## [10] "j =377 peak =388 savedpeak = 48516"
## [11] "j =377 peak =406 savedpeak = 48534"
## [12] "j =377 peak =433 savedpeak = 48561"
## [1] "j =379 peak =106 savedpeak = 48490"
## [2] "j =379 peak =132 savedpeak = 48516"
## [3] "j =379 peak =150 savedpeak = 48534"
## [4] "j =379 peak =177 savedpeak = 48561"
## [5] "j =379 peak =195 savedpeak = 48579"
## [6] "j =379 peak =236 savedpeak = 48620"
## [7] "j =379 peak =259 savedpeak = 48643"
## [8] "j =379 peak =302 savedpeak = 48686"
## [9] "j =379 peak =315 savedpeak = 48699"
## [10] "j =379 peak =353 savedpeak = 48737"
## [11] "j =379 peak =394 savedpeak = 48778"
## [12] "j =379 peak =403 savedpeak = 48787"
## [13] "j =379 peak =435 savedpeak = 48819"
## [1] "j =380 peak =108 savedpeak = 48620"
## [2] "j =380 peak =117 savedpeak = 48629"
## [3] "j =380 peak =131 savedpeak = 48643"
## [4] "j =380 peak =174 savedpeak = 48686"
## [5] "j =380 peak =187 savedpeak = 48699"
## [6] "j =380 peak =225 savedpeak = 48737"
## [7] "j =380 peak =266 savedpeak = 48778"
## [8] "j =380 peak =275 savedpeak = 48787"
## [9] "j =380 peak =307 savedpeak = 48819"
## [10] "j =380 peak =345 savedpeak = 48857"
## [11] "j =380 peak =360 savedpeak = 48872"

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## [12] "j =380 peak =383 savedpeak = 48895"
## [13] "j =380 peak =428 savedpeak = 48940"
## [1] "j =381 peak =97 savedpeak = 48737"
## [2] "j =381 peak =138 savedpeak = 48778"
## [3] "j =381 peak =147 savedpeak = 48787"
## [4] "j =381 peak =179 savedpeak = 48819"
## [5] "j =381 peak =217 savedpeak = 48857"
## [6] "j =381 peak =232 savedpeak = 48872"
## [7] "j =381 peak =255 savedpeak = 48895"
## [8] "j =381 peak =300 savedpeak = 48940"
## [9] "j =381 peak =314 savedpeak = 48954"
## [10] "j =381 peak =354 savedpeak = 48994"
## [11] "j =381 peak =380 savedpeak = 49020"
## [12] "j =381 peak =410 savedpeak = 49050"
## [1] "j =382 peak =89 savedpeak = 48857"
## [2] "j =382 peak =104 savedpeak = 48872"
## [3] "j =382 peak =127 savedpeak = 48895"
## [4] "j =382 peak =172 savedpeak = 48940"
## [5] "j =382 peak =186 savedpeak = 48954"
## [6] "j =382 peak =226 savedpeak = 48994"
## [7] "j =382 peak =252 savedpeak = 49020"
## [8] "j =382 peak =282 savedpeak = 49050"
## [9] "j =382 peak =356 savedpeak = 49124"
## [10] "j =382 peak =382 savedpeak = 49150"
## [11] "j =382 peak =403 savedpeak = 49171"
## [12] "j =382 peak =419 savedpeak = 49187"
## [1] "j =390 peak =80 savedpeak = 49872"
## [2] "j =390 peak =114 savedpeak = 49906"
## [3] "j =390 peak =148 savedpeak = 49940"
## [4] "j =390 peak =164 savedpeak = 49956"
## [5] "j =390 peak =178 savedpeak = 49970"
## [6] "j =390 peak =213 savedpeak = 50005"
## [7] "j =390 peak =244 savedpeak = 50036"
## [8] "j =390 peak =275 savedpeak = 50067"
## [9] "j =390 peak =310 savedpeak = 50102"
## [10] "j =390 peak =330 savedpeak = 50122"
## [11] "j =390 peak =347 savedpeak = 50139"
## [12] "j =390 peak =368 savedpeak = 50160"
## [13] "j =390 peak =396 savedpeak = 50188"
## [14] "j =390 peak =413 savedpeak = 50205"
## [1] "j =391 peak =85 savedpeak = 50005"
## [2] "j =391 peak =116 savedpeak = 50036"
## [3] "j =391 peak =147 savedpeak = 50067"
## [4] "j =391 peak =182 savedpeak = 50102"
## [5] "j =391 peak =202 savedpeak = 50122"
## [6] "j =391 peak =219 savedpeak = 50139"
## [7] "j =391 peak =240 savedpeak = 50160"
## [8] "j =391 peak =268 savedpeak = 50188"
## [9] "j =391 peak =285 savedpeak = 50205"
## [10] "j =391 peak =319 savedpeak = 50239"
## [11] "j =391 peak =355 savedpeak = 50275"
## [12] "j =391 peak =373 savedpeak = 50293"
## [13] "j =391 peak =395 savedpeak = 50315"
## [1] "j =392 peak =91 savedpeak = 50139"

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## [2] "j =392 peak =112 savedpeak = 50160"
## [3] "j =392 peak =140 savedpeak = 50188"
## [4] "j =392 peak =157 savedpeak = 50205"
## [5] "j =392 peak =191 savedpeak = 50239"
## [6] "j =392 peak =227 savedpeak = 50275"
## [7] "j =392 peak =245 savedpeak = 50293"
## [8] "j =392 peak =267 savedpeak = 50315"
## [9] "j =392 peak =314 savedpeak = 50362"
## [10] "j =392 peak =327 savedpeak = 50375"
## [11] "j =392 peak =358 savedpeak = 50406"
## [12] "j =392 peak =378 savedpeak = 50426"
## [13] "j =392 peak =404 savedpeak = 50452"
## [14] "j =392 peak =413 savedpeak = 50461"
## [15] "j =392 peak =433 savedpeak = 50481"
## Spectrum exam_DNMR_ipr2sic has 30 peaks.
## [1] "j =1 peak =94 savedpeak = 94" "j =1 peak =117 savedpeak = 117"
## [3] "j =1 peak =135 savedpeak = 135" "j =1 peak =153 savedpeak = 153"
## [5] "j =1 peak =181 savedpeak = 181" "j =1 peak =204 savedpeak = 204"
## [7] "j =1 peak =219 savedpeak = 219" "j =1 peak =256 savedpeak = 256"
## [9] "j =1 peak =274 savedpeak = 274" "j =1 peak =294 savedpeak = 294"
## [11] "j =1 peak =335 savedpeak = 335" "j =1 peak =363 savedpeak = 363"
## [13] "j =1 peak =384 savedpeak = 384" "j =1 peak =407 savedpeak = 407"
## [15] "j =1 peak =430 savedpeak = 430"
## [1] "j =2 peak =76 savedpeak = 204" "j =2 peak =91 savedpeak = 219"
## [3] "j =2 peak =128 savedpeak = 256" "j =2 peak =166 savedpeak = 294"
## [5] "j =2 peak =207 savedpeak = 335" "j =2 peak =235 savedpeak = 363"
## [7] "j =2 peak =256 savedpeak = 384" "j =2 peak =279 savedpeak = 407"
## [9] "j =2 peak =302 savedpeak = 430" "j =2 peak =373 savedpeak = 501"
## [11] "j =2 peak =399 savedpeak = 527"
## [1] "j =3 peak =79 savedpeak = 335" "j =3 peak =107 savedpeak = 363"
## [3] "j =3 peak =128 savedpeak = 384" "j =3 peak =151 savedpeak = 407"
## [5] "j =3 peak =174 savedpeak = 430" "j =3 peak =245 savedpeak = 501"
## [7] "j =3 peak =271 savedpeak = 527" "j =3 peak =319 savedpeak = 575"
## [9] "j =3 peak =338 savedpeak = 594" "j =3 peak =379 savedpeak = 635"
## [11] "j =3 peak =411 savedpeak = 667" "j =3 peak =425 savedpeak = 681"
## [1] "j =4 peak =117 savedpeak = 501" "j =4 peak =143 savedpeak = 527"
## [3] "j =4 peak =191 savedpeak = 575" "j =4 peak =210 savedpeak = 594"
## [5] "j =4 peak =251 savedpeak = 635" "j =4 peak =283 savedpeak = 667"
## [7] "j =4 peak =297 savedpeak = 681" "j =4 peak =310 savedpeak = 694"
## [9] "j =4 peak =338 savedpeak = 722" "j =4 peak =365 savedpeak = 749"
## [11] "j =4 peak =397 savedpeak = 781" "j =4 peak =432 savedpeak = 816"
## [1] "j =5 peak =82 savedpeak = 594" "j =5 peak =123 savedpeak = 635"
## [3] "j =5 peak =155 savedpeak = 667" "j =5 peak =169 savedpeak = 681"
## [5] "j =5 peak =182 savedpeak = 694" "j =5 peak =210 savedpeak = 722"
## [7] "j =5 peak =237 savedpeak = 749" "j =5 peak =269 savedpeak = 781"
## [9] "j =5 peak =304 savedpeak = 816" "j =5 peak =316 savedpeak = 828"
## [11] "j =5 peak =337 savedpeak = 849" "j =5 peak =379 savedpeak = 891"
## [13] "j =5 peak =406 savedpeak = 918"
## [1] "j =6 peak =82 savedpeak = 722" "j =6 peak =109 savedpeak = 749"
## [3] "j =6 peak =141 savedpeak = 781" "j =6 peak =176 savedpeak = 816"
## [5] "j =6 peak =188 savedpeak = 828" "j =6 peak =209 savedpeak = 849"
## [7] "j =6 peak =251 savedpeak = 891" "j =6 peak =278 savedpeak = 918"
## [9] "j =6 peak =326 savedpeak = 966" "j =6 peak =360 savedpeak = 1000"
## [11] "j =6 peak =401 savedpeak = 1041" "j =6 peak =410 savedpeak = 1050"

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## [1] "j =13 peak =112 savedpeak = 1648" "j =13 peak =161 savedpeak = 1697"
## [3] "j =13 peak =180 savedpeak = 1716" "j =13 peak =200 savedpeak = 1736"
## [5] "j =13 peak =226 savedpeak = 1762" "j =13 peak =261 savedpeak = 1797"
## [7] "j =13 peak =279 savedpeak = 1815" "j =13 peak =314 savedpeak = 1850"
## [9] "j =13 peak =344 savedpeak = 1880" "j =13 peak =365 savedpeak = 1901"
## [11] "j =13 peak =391 savedpeak = 1927" "j =13 peak =404 savedpeak = 1940"
## [13] "j =13 peak =426 savedpeak = 1962"
## [1] "j =14 peak =98 savedpeak = 1762" "j =14 peak =120 savedpeak = 1784"
## [3] "j =14 peak =133 savedpeak = 1797" "j =14 peak =151 savedpeak = 1815"
## [5] "j =14 peak =186 savedpeak = 1850" "j =14 peak =216 savedpeak = 1880"
## [7] "j =14 peak =237 savedpeak = 1901" "j =14 peak =263 savedpeak = 1927"
## [9] "j =14 peak =276 savedpeak = 1940" "j =14 peak =298 savedpeak = 1962"
## [11] "j =14 peak =420 savedpeak = 2084"
## [1] "j =15 peak =88 savedpeak = 1880" "j =15 peak =109 savedpeak = 1901"
## [3] "j =15 peak =135 savedpeak = 1927" "j =15 peak =148 savedpeak = 1940"
## [5] "j =15 peak =170 savedpeak = 1962" "j =15 peak =292 savedpeak = 2084"
## [1] "j =16 peak =165 savedpeak = 2085" "j =16 peak =201 savedpeak = 2121"
## [1] "j =18 peak =82 savedpeak = 2258" "j =18 peak =183 savedpeak = 2359"
## [3] "j =18 peak =255 savedpeak = 2431" "j =18 peak =364 savedpeak = 2540"
## [5] "j =18 peak =436 savedpeak = 2612"
## [1] "j =20 peak =108 savedpeak = 2540" "j =20 peak =180 savedpeak = 2612"
## [3] "j =20 peak =267 savedpeak = 2699" "j =20 peak =310 savedpeak = 2742"
## [5] "j =20 peak =368 savedpeak = 2800"
## [1] "j =21 peak =139 savedpeak = 2699" "j =21 peak =182 savedpeak = 2742"
## [1] "j =27 peak =136 savedpeak = 3464" "j =27 peak =180 savedpeak = 3508"
## [3] "j =27 peak =223 savedpeak = 3551" "j =27 peak =273 savedpeak = 3601"
## [5] "j =27 peak =302 savedpeak = 3630"
## [1] "j =28 peak =95 savedpeak = 3551" "j =28 peak =145 savedpeak = 3601"
## [3] "j =28 peak =174 savedpeak = 3630"
## [1] "j =29 peak =292 savedpeak = 3876" "j =29 peak =328 savedpeak = 3912"
## [3] "j =29 peak =393 savedpeak = 3977"
## [1] "j =30 peak =164 savedpeak = 3876" "j =30 peak =200 savedpeak = 3912"
## [3] "j =30 peak =265 savedpeak = 3977" "j =30 peak =402 savedpeak = 4114"
## [1] "j =31 peak =137 savedpeak = 3977" "j =31 peak =274 savedpeak = 4114"
## [1] "j =33 peak =235 savedpeak = 4331" "j =33 peak =437 savedpeak = 4533"
## [1] "j =34 peak =107 savedpeak = 4331" "j =34 peak =309 savedpeak = 4533"
## [3] "j =34 peak =381 savedpeak = 4605"
## [1] "j =35 peak =181 savedpeak = 4533" "j =35 peak =253 savedpeak = 4605"
## [3] "j =35 peak =311 savedpeak = 4663" "j =35 peak =376 savedpeak = 4728"
## [1] "j =36 peak =125 savedpeak = 4605" "j =36 peak =183 savedpeak = 4663"
## [3] "j =36 peak =248 savedpeak = 4728" "j =36 peak =313 savedpeak = 4793"
## [5] "j =36 peak =400 savedpeak = 4880"
## [1] "j =37 peak =120 savedpeak = 4728" "j =37 peak =185 savedpeak = 4793"
## [3] "j =37 peak =272 savedpeak = 4880" "j =37 peak =358 savedpeak = 4966"
## [1] "j =38 peak =144 savedpeak = 4880" "j =38 peak =230 savedpeak = 4966"
## [3] "j =38 peak =411 savedpeak = 5147"
## [1] "j =39 peak =102 savedpeak = 4966" "j =39 peak =283 savedpeak = 5147"
## [1] "j =40 peak =104 savedpeak = 5096" "j =40 peak =155 savedpeak = 5147"
## [1] "j =42 peak =82 savedpeak = 5330" "j =42 peak =106 savedpeak = 5354"
## [3] "j =42 peak =211 savedpeak = 5459" "j =42 peak =278 savedpeak = 5526"
## [5] "j =42 peak =345 savedpeak = 5593" "j =42 peak =422 savedpeak = 5670"
## [1] "j =43 peak =83 savedpeak = 5459" "j =43 peak =150 savedpeak = 5526"
## [3] "j =43 peak =217 savedpeak = 5593" "j =43 peak =294 savedpeak = 5670"
## [5] "j =43 peak =336 savedpeak = 5712"

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## [1] "j =44 peak =89 savedpeak = 5593" "j =44 peak =166 savedpeak = 5670"
## [3] "j =44 peak =208 savedpeak = 5712" "j =44 peak =401 savedpeak = 5905"
## [1] "j =45 peak =80 savedpeak = 5712" "j =45 peak =273 savedpeak = 5905"
## [3] "j =45 peak =364 savedpeak = 5996" "j =45 peak =389 savedpeak = 6021"
## [5] "j =45 peak =403 savedpeak = 6035" "j =45 peak =435 savedpeak = 6067"
## [1] "j =46 peak =145 savedpeak = 5905" "j =46 peak =208 savedpeak = 5968"
## [3] "j =46 peak =236 savedpeak = 5996" "j =46 peak =261 savedpeak = 6021"
## [5] "j =46 peak =275 savedpeak = 6035" "j =46 peak =307 savedpeak = 6067"
## [7] "j =46 peak =380 savedpeak = 6140" "j =46 peak =436 savedpeak = 6196"
## [1] "j =47 peak =108 savedpeak = 5996" "j =47 peak =133 savedpeak = 6021"
## [3] "j =47 peak =179 savedpeak = 6067" "j =47 peak =252 savedpeak = 6140"
## [5] "j =47 peak =308 savedpeak = 6196" "j =47 peak =347 savedpeak = 6235"
## [7] "j =47 peak =365 savedpeak = 6253" "j =47 peak =386 savedpeak = 6274"
## [1] "j =48 peak =124 savedpeak = 6140" "j =48 peak =180 savedpeak = 6196"
## [3] "j =48 peak =219 savedpeak = 6235" "j =48 peak =237 savedpeak = 6253"
## [5] "j =48 peak =258 savedpeak = 6274" "j =48 peak =349 savedpeak = 6365"
## [7] "j =48 peak =416 savedpeak = 6432" "j =48 peak =437 savedpeak = 6453"
## [1] "j =49 peak =91 savedpeak = 6235" "j =49 peak =109 savedpeak = 6253"
## [3] "j =49 peak =130 savedpeak = 6274" "j =49 peak =221 savedpeak = 6365"
## [5] "j =49 peak =288 savedpeak = 6432" "j =49 peak =309 savedpeak = 6453"
## [7] "j =49 peak =400 savedpeak = 6544"
## [1] "j =50 peak =93 savedpeak = 6365" "j =50 peak =160 savedpeak = 6432"
## [3] "j =50 peak =181 savedpeak = 6453" "j =50 peak =272 savedpeak = 6544"
## [5] "j =50 peak =321 savedpeak = 6593" "j =50 peak =381 savedpeak = 6653"
## [7] "j =50 peak =419 savedpeak = 6691"
## [1] "j =51 peak =144 savedpeak = 6544" "j =51 peak =193 savedpeak = 6593"
## [3] "j =51 peak =253 savedpeak = 6653" "j =51 peak =291 savedpeak = 6691"
## [5] "j =51 peak =312 savedpeak = 6712"
## [1] "j =52 peak =125 savedpeak = 6653" "j =52 peak =163 savedpeak = 6691"
## [3] "j =52 peak =184 savedpeak = 6712" "j =52 peak =367 savedpeak = 6895"
## [5] "j =52 peak =437 savedpeak = 6965"
## [1] "j =53 peak =239 savedpeak = 6895" "j =53 peak =309 savedpeak = 6965"
## [3] "j =53 peak =348 savedpeak = 7004" "j =53 peak =369 savedpeak = 7025"
## [5] "j =53 peak =414 savedpeak = 7070"
## [1] "j =54 peak =111 savedpeak = 6895" "j =54 peak =181 savedpeak = 6965"
## [3] "j =54 peak =220 savedpeak = 7004" "j =54 peak =241 savedpeak = 7025"
## [5] "j =54 peak =286 savedpeak = 7070" "j =54 peak =332 savedpeak = 7116"
## [7] "j =54 peak =384 savedpeak = 7168" "j =54 peak =399 savedpeak = 7183"
## [9] "j =54 peak =409 savedpeak = 7193" "j =54 peak =423 savedpeak = 7207"
## [1] "j =58 peak =108 savedpeak = 7404" "j =58 peak =150 savedpeak = 7446"
## [3] "j =58 peak =174 savedpeak = 7470" "j =58 peak =244 savedpeak = 7540"
## [5] "j =58 peak =290 savedpeak = 7586" "j =58 peak =346 savedpeak = 7642"
## [7] "j =58 peak =430 savedpeak = 7726"
## [1] "j =59 peak =116 savedpeak = 7540" "j =59 peak =162 savedpeak = 7586"
## [3] "j =59 peak =225 savedpeak = 7649" "j =59 peak =302 savedpeak = 7726"
## [5] "j =59 peak =327 savedpeak = 7751" "j =59 peak =376 savedpeak = 7800"
## [7] "j =59 peak =435 savedpeak = 7859"
## [1] "j =60 peak =307 savedpeak = 7859" "j =60 peak =320 savedpeak = 7872"
## [3] "j =60 peak =409 savedpeak = 7961" "j =60 peak =423 savedpeak = 7975"
## [1] "j =78 peak =129 savedpeak = 9985" "j =78 peak =141 savedpeak = 9997"
## [3] "j =78 peak =228 savedpeak = 10084" "j =78 peak =265 savedpeak = 10121"
## [5] "j =78 peak =302 savedpeak = 10158" "j =78 peak =335 savedpeak = 10191"
## [7] "j =78 peak =366 savedpeak = 10222" "j =78 peak =401 savedpeak = 10257"
## [1] "j =79 peak =100 savedpeak = 10084"

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## [2] "j =79 peak =137 savedpeak = 10121"
## [3] "j =79 peak =171 savedpeak = 10155"
## [4] "j =79 peak =206 savedpeak = 10190"
## [5] "j =79 peak =241 savedpeak = 10225"
## [6] "j =79 peak =273 savedpeak = 10257"
## [7] "j =79 peak =318 savedpeak = 10302"
## [8] "j =79 peak =326 savedpeak = 10310"
## [9] "j =79 peak =350 savedpeak = 10334"
## [10] "j =79 peak =377 savedpeak = 10361"
## [11] "j =79 peak =401 savedpeak = 10385"
## [12] "j =79 peak =429 savedpeak = 10413"
## [1] "j =80 peak =78 savedpeak = 10190"
## [2] "j =80 peak =113 savedpeak = 10225"
## [3] "j =80 peak =145 savedpeak = 10257"
## [4] "j =80 peak =190 savedpeak = 10302"
## [5] "j =80 peak =201 savedpeak = 10313"
## [6] "j =80 peak =222 savedpeak = 10334"
## [7] "j =80 peak =249 savedpeak = 10361"
## [8] "j =80 peak =273 savedpeak = 10385"
## [9] "j =80 peak =296 savedpeak = 10408"
## [10] "j =80 peak =317 savedpeak = 10429"
## [11] "j =80 peak =331 savedpeak = 10443"
## [12] "j =80 peak =355 savedpeak = 10467"
## [13] "j =80 peak =390 savedpeak = 10502"
## [14] "j =80 peak =415 savedpeak = 10527"
## [1] "j =81 peak =94 savedpeak = 10334"
## [2] "j =81 peak =121 savedpeak = 10361"
## [3] "j =81 peak =145 savedpeak = 10385"
## [4] "j =81 peak =168 savedpeak = 10408"
## [5] "j =81 peak =189 savedpeak = 10429"
## [6] "j =81 peak =206 savedpeak = 10446"
## [7] "j =81 peak =227 savedpeak = 10467"
## [8] "j =81 peak =262 savedpeak = 10502"
## [9] "j =81 peak =287 savedpeak = 10527"
## [10] "j =81 peak =315 savedpeak = 10555"
## [11] "j =81 peak =346 savedpeak = 10586"
## [12] "j =81 peak =410 savedpeak = 10650"
## [13] "j =81 peak =431 savedpeak = 10671"
## [1] "j =83 peak =90 savedpeak = 10586" "j =83 peak =153 savedpeak = 10649"
## [3] "j =83 peak =178 savedpeak = 10674" "j =83 peak =213 savedpeak = 10709"
## [5] "j =83 peak =266 savedpeak = 10762" "j =83 peak =284 savedpeak = 10780"
## [7] "j =83 peak =357 savedpeak = 10853" "j =83 peak =396 savedpeak = 10892"
## [9] "j =83 peak =431 savedpeak = 10927"
## [1] "j =86 peak =103 savedpeak = 10983"
## [2] "j =86 peak =135 savedpeak = 11015"
## [3] "j =86 peak =163 savedpeak = 11043"
## [4] "j =86 peak =177 savedpeak = 11057"
## [5] "j =86 peak =201 savedpeak = 11081"
## [6] "j =86 peak =222 savedpeak = 11102"
## [7] "j =86 peak =233 savedpeak = 11113"
## [8] "j =86 peak =247 savedpeak = 11127"
## [9] "j =86 peak =268 savedpeak = 11148"
## [10] "j =86 peak =300 savedpeak = 11180"
## [11] "j =86 peak =342 savedpeak = 11222"

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## [12] "j =86 peak =377 savedpeak = 11257"
## [1] "j =87 peak =94 savedpeak = 11102"
## [2] "j =87 peak =105 savedpeak = 11113"
## [3] "j =87 peak =115 savedpeak = 11123"
## [4] "j =87 peak =140 savedpeak = 11148"
## [5] "j =87 peak =172 savedpeak = 11180"
## [6] "j =87 peak =217 savedpeak = 11225"
## [7] "j =87 peak =235 savedpeak = 11243"
## [8] "j =87 peak =249 savedpeak = 11257"
## [9] "j =87 peak =344 savedpeak = 11352"
## [10] "j =87 peak =361 savedpeak = 11369"
## [11] "j =87 peak =382 savedpeak = 11390"
## [12] "j =87 peak =403 savedpeak = 11411"
## [13] "j =87 peak =421 savedpeak = 11429"
## [1] "j =88 peak =86 savedpeak = 11222"
## [2] "j =88 peak =107 savedpeak = 11243"
## [3] "j =88 peak =121 savedpeak = 11257"
## [4] "j =88 peak =216 savedpeak = 11352"
## [5] "j =88 peak =233 savedpeak = 11369"
## [6] "j =88 peak =251 savedpeak = 11387"
## [7] "j =88 peak =275 savedpeak = 11411"
## [8] "j =88 peak =293 savedpeak = 11429"
## [9] "j =88 peak =314 savedpeak = 11450"
## [10] "j =88 peak =331 savedpeak = 11467"
## [11] "j =88 peak =387 savedpeak = 11523"
## [12] "j =88 peak =405 savedpeak = 11541"
## [13] "j =88 peak =423 savedpeak = 11559"
## [1] "j =89 peak =88 savedpeak = 11352"
## [2] "j =89 peak =105 savedpeak = 11369"
## [3] "j =89 peak =126 savedpeak = 11390"
## [4] "j =89 peak =147 savedpeak = 11411"
## [5] "j =89 peak =165 savedpeak = 11429"
## [6] "j =89 peak =186 savedpeak = 11450"
## [7] "j =89 peak =203 savedpeak = 11467"
## [8] "j =89 peak =256 savedpeak = 11520"
## [9] "j =89 peak =277 savedpeak = 11541"
## [10] "j =89 peak =291 savedpeak = 11555"
## [11] "j =89 peak =330 savedpeak = 11594"
## [12] "j =89 peak =368 savedpeak = 11632"
## [13] "j =89 peak =421 savedpeak = 11685"
## [1] "j =91 peak =112 savedpeak = 11632" "j =91 peak =165 savedpeak = 11685"
## [3] "j =91 peak =186 savedpeak = 11706" "j =91 peak =242 savedpeak = 11762"
## [5] "j =91 peak =281 savedpeak = 11801" "j =91 peak =305 savedpeak = 11825"
## [7] "j =91 peak =330 savedpeak = 11850" "j =91 peak =351 savedpeak = 11871"
## [9] "j =91 peak =386 savedpeak = 11906"
## [1] "j =92 peak =114 savedpeak = 11762"
## [2] "j =92 peak =153 savedpeak = 11801"
## [3] "j =92 peak =181 savedpeak = 11829"
## [4] "j =92 peak =202 savedpeak = 11850"
## [5] "j =92 peak =223 savedpeak = 11871"
## [6] "j =92 peak =258 savedpeak = 11906"
## [7] "j =92 peak =311 savedpeak = 11959"
## [8] "j =92 peak =346 savedpeak = 11994"
## [9] "j =92 peak =377 savedpeak = 12025"

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## [10] "j =92 peak =412 savedpeak = 12060"
## [1] "j =93 peak =77 savedpeak = 11853"
## [2] "j =93 peak =95 savedpeak = 11871"
## [3] "j =93 peak =130 savedpeak = 11906"
## [4] "j =93 peak =190 savedpeak = 11966"
## [5] "j =93 peak =218 savedpeak = 11994"
## [6] "j =93 peak =246 savedpeak = 12022"
## [7] "j =93 peak =281 savedpeak = 12057"
## [8] "j =93 peak =323 savedpeak = 12099"
## [9] "j =93 peak =334 savedpeak = 12110"
## [10] "j =93 peak =351 savedpeak = 12127"
## [11] "j =93 peak =372 savedpeak = 12148"
## [12] "j =93 peak =400 savedpeak = 12176"
## [13] "j =93 peak =428 savedpeak = 12204"
## [1] "j =94 peak =90 savedpeak = 11994"
## [2] "j =94 peak =121 savedpeak = 12025"
## [3] "j =94 peak =156 savedpeak = 12060"
## [4] "j =94 peak =191 savedpeak = 12095"
## [5] "j =94 peak =206 savedpeak = 12110"
## [6] "j =94 peak =223 savedpeak = 12127"
## [7] "j =94 peak =241 savedpeak = 12145"
## [8] "j =94 peak =276 savedpeak = 12180"
## [9] "j =94 peak =300 savedpeak = 12204"
## [10] "j =94 peak =313 savedpeak = 12217"
## [11] "j =94 peak =332 savedpeak = 12236"
## [12] "j =94 peak =353 savedpeak = 12257"
## [13] "j =94 peak =377 savedpeak = 12281"
## [14] "j =94 peak =406 savedpeak = 12310"
## [15] "j =94 peak =437 savedpeak = 12341"
## [1] "j =95 peak =78 savedpeak = 12110"
## [2] "j =95 peak =95 savedpeak = 12127"
## [3] "j =95 peak =116 savedpeak = 12148"
## [4] "j =95 peak =148 savedpeak = 12180"
## [5] "j =95 peak =172 savedpeak = 12204"
## [6] "j =95 peak =185 savedpeak = 12217"
## [7] "j =95 peak =200 savedpeak = 12232"
## [8] "j =95 peak =225 savedpeak = 12257"
## [9] "j =95 peak =246 savedpeak = 12278"
## [10] "j =95 peak =278 savedpeak = 12310"
## [11] "j =95 peak =309 savedpeak = 12341"
## [12] "j =95 peak =376 savedpeak = 12408"
## [13] "j =95 peak =397 savedpeak = 12429"
## [14] "j =95 peak =411 savedpeak = 12443"
## [15] "j =95 peak =432 savedpeak = 12464"
## [1] "j =96 peak =76 savedpeak = 12236"
## [2] "j =96 peak =97 savedpeak = 12257"
## [3] "j =96 peak =121 savedpeak = 12281"
## [4] "j =96 peak =150 savedpeak = 12310"
## [5] "j =96 peak =181 savedpeak = 12341"
## [6] "j =96 peak =248 savedpeak = 12408"
## [7] "j =96 peak =269 savedpeak = 12429"
## [8] "j =96 peak =286 savedpeak = 12446"
## [9] "j =96 peak =307 savedpeak = 12467"
## [10] "j =96 peak =325 savedpeak = 12485"

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## [11] "j =96 peak =360 savedpeak = 12520"
## [12] "j =96 peak =392 savedpeak = 12552"
## [13] "j =96 peak =423 savedpeak = 12583"
## [1] "j =97 peak =120 savedpeak = 12408"
## [2] "j =97 peak =141 savedpeak = 12429"
## [3] "j =97 peak =158 savedpeak = 12446"
## [4] "j =97 peak =197 savedpeak = 12485"
## [5] "j =97 peak =232 savedpeak = 12520"
## [6] "j =97 peak =267 savedpeak = 12555"
## [7] "j =97 peak =295 savedpeak = 12583"
## [8] "j =97 peak =341 savedpeak = 12629"
## [9] "j =97 peak =381 savedpeak = 12669"
## [10] "j =97 peak =408 savedpeak = 12696"
## [11] "j =97 peak =432 savedpeak = 12720"
## [1] "j =98 peak =101 savedpeak = 12517" "j =98 peak =136 savedpeak = 12552"
## [3] "j =98 peak =167 savedpeak = 12583" "j =98 peak =213 savedpeak = 12629"
## [5] "j =98 peak =280 savedpeak = 12696" "j =98 peak =308 savedpeak = 12724"
## [7] "j =98 peak =332 savedpeak = 12748" "j =98 peak =378 savedpeak = 12794"
## [9] "j =98 peak =428 savedpeak = 12844"
## [1] "j =99 peak =85 savedpeak = 12629"
## [2] "j =99 peak =152 savedpeak = 12696"
## [3] "j =99 peak =176 savedpeak = 12720"
## [4] "j =99 peak =204 savedpeak = 12748"
## [5] "j =99 peak =250 savedpeak = 12794"
## [6] "j =99 peak =300 savedpeak = 12844"
## [7] "j =99 peak =338 savedpeak = 12882"
## [8] "j =99 peak =362 savedpeak = 12906"
## [9] "j =99 peak =380 savedpeak = 12924"
## [10] "j =99 peak =401 savedpeak = 12945"
## [1] "j =100 peak =76 savedpeak = 12748"
## [2] "j =100 peak =122 savedpeak = 12794"
## [3] "j =100 peak =172 savedpeak = 12844"
## [4] "j =100 peak =210 savedpeak = 12882"
## [5] "j =100 peak =231 savedpeak = 12903"
## [6] "j =100 peak =252 savedpeak = 12924"
## [7] "j =100 peak =273 savedpeak = 12945"
## [8] "j =100 peak =329 savedpeak = 13001"
## [9] "j =100 peak =343 savedpeak = 13015"
## [10] "j =100 peak =367 savedpeak = 13039"
## [11] "j =100 peak =420 savedpeak = 13092"
## [1] "j =101 peak =82 savedpeak = 12882"
## [2] "j =101 peak =106 savedpeak = 12906"
## [3] "j =101 peak =127 savedpeak = 12927"
## [4] "j =101 peak =145 savedpeak = 12945"
## [5] "j =101 peak =201 savedpeak = 13001"
## [6] "j =101 peak =215 savedpeak = 13015"
## [7] "j =101 peak =236 savedpeak = 13036"
## [8] "j =101 peak =292 savedpeak = 13092"
## [9] "j =101 peak =331 savedpeak = 13131"
## [10] "j =101 peak =348 savedpeak = 13148"
## [11] "j =101 peak =366 savedpeak = 13166"
## [12] "j =101 peak =387 savedpeak = 13187"
## [13] "j =101 peak =408 savedpeak = 13208"
## [14] "j =101 peak =425 savedpeak = 13225"

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## [15] "j =101 peak =432 savedpeak = 13232"
## [1] "j =102 peak =87 savedpeak = 13015"
## [2] "j =102 peak =111 savedpeak = 13039"
## [3] "j =102 peak =161 savedpeak = 13089"
## [4] "j =102 peak =196 savedpeak = 13124"
## [5] "j =102 peak =220 savedpeak = 13148"
## [6] "j =102 peak =241 savedpeak = 13169"
## [7] "j =102 peak =259 savedpeak = 13187"
## [8] "j =102 peak =276 savedpeak = 13204"
## [9] "j =102 peak =297 savedpeak = 13225"
## [10] "j =102 peak =304 savedpeak = 13232"
## [11] "j =102 peak =326 savedpeak = 13254"
## [12] "j =102 peak =347 savedpeak = 13275"
## [13] "j =102 peak =385 savedpeak = 13313"
## [14] "j =102 peak =424 savedpeak = 13352"
## [1] "j =103 peak =92 savedpeak = 13148"
## [2] "j =103 peak =113 savedpeak = 13169"
## [3] "j =103 peak =131 savedpeak = 13187"
## [4] "j =103 peak =148 savedpeak = 13204"
## [5] "j =103 peak =166 savedpeak = 13222"
## [6] "j =103 peak =176 savedpeak = 13232"
## [7] "j =103 peak =198 savedpeak = 13254"
## [8] "j =103 peak =222 savedpeak = 13278"
## [9] "j =103 peak =257 savedpeak = 13313"
## [10] "j =103 peak =296 savedpeak = 13352"
## [11] "j =103 peak =317 savedpeak = 13373"
## [12] "j =103 peak =331 savedpeak = 13387"
## [13] "j =103 peak =352 savedpeak = 13408"
## [14] "j =103 peak =387 savedpeak = 13443"
## [15] "j =103 peak =398 savedpeak = 13454"
## [16] "j =103 peak =422 savedpeak = 13478"
## [1] "j =105 peak =96 savedpeak = 13408"
## [2] "j =105 peak =128 savedpeak = 13440"
## [3] "j =105 peak =142 savedpeak = 13454"
## [4] "j =105 peak =166 savedpeak = 13478"
## [5] "j =105 peak =187 savedpeak = 13499"
## [6] "j =105 peak =226 savedpeak = 13538"
## [7] "j =105 peak =243 savedpeak = 13555"
## [8] "j =105 peak =261 savedpeak = 13573"
## [9] "j =105 peak =285 savedpeak = 13597"
## [10] "j =105 peak =306 savedpeak = 13618"
## [11] "j =105 peak =328 savedpeak = 13640"
## [12] "j =105 peak =352 savedpeak = 13664"
## [13] "j =105 peak =370 savedpeak = 13682"
## [14] "j =105 peak =391 savedpeak = 13703"
## [15] "j =105 peak =408 savedpeak = 13720"
## [16] "j =105 peak =426 savedpeak = 13738"
## [1] "j =106 peak =101 savedpeak = 13541"
## [2] "j =106 peak =115 savedpeak = 13555"
## [3] "j =106 peak =133 savedpeak = 13573"
## [4] "j =106 peak =157 savedpeak = 13597"
## [5] "j =106 peak =178 savedpeak = 13618"
## [6] "j =106 peak =200 savedpeak = 13640"
## [7] "j =106 peak =221 savedpeak = 13661"

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## [8] "j =106 peak =242 savedpeak = 13682"
## [9] "j =106 peak =263 savedpeak = 13703"
## [10] "j =106 peak =280 savedpeak = 13720"
## [11] "j =106 peak =301 savedpeak = 13741"
## [12] "j =106 peak =357 savedpeak = 13797"
## [13] "j =106 peak =386 savedpeak = 13826"
## [14] "j =106 peak =410 savedpeak = 13850"
## [1] "j =107 peak =93 savedpeak = 13661"
## [2] "j =107 peak =117 savedpeak = 13685"
## [3] "j =107 peak =135 savedpeak = 13703"
## [4] "j =107 peak =152 savedpeak = 13720"
## [5] "j =107 peak =173 savedpeak = 13741"
## [6] "j =107 peak =226 savedpeak = 13794"
## [7] "j =107 peak =258 savedpeak = 13826"
## [8] "j =107 peak =282 savedpeak = 13850"
## [9] "j =107 peak =317 savedpeak = 13885"
## [10] "j =107 peak =335 savedpeak = 13903"
## [11] "j =107 peak =356 savedpeak = 13924"
## [12] "j =107 peak =391 savedpeak = 13959"
## [13] "j =107 peak =426 savedpeak = 13994"
## [1] "j =108 peak =101 savedpeak = 13797"
## [2] "j =108 peak =130 savedpeak = 13826"
## [3] "j =108 peak =151 savedpeak = 13847"
## [4] "j =108 peak =186 savedpeak = 13882"
## [5] "j =108 peak =207 savedpeak = 13903"
## [6] "j =108 peak =231 savedpeak = 13927"
## [7] "j =108 peak =266 savedpeak = 13962"
## [8] "j =108 peak =298 savedpeak = 13994"
## [9] "j =108 peak =316 savedpeak = 14012"
## [10] "j =108 peak =347 savedpeak = 14043"
## [11] "j =108 peak =382 savedpeak = 14078"
## [12] "j =108 peak =396 savedpeak = 14092"
## [1] "j =109 peak =82 savedpeak = 13906"
## [2] "j =109 peak =103 savedpeak = 13927"
## [3] "j =109 peak =138 savedpeak = 13962"
## [4] "j =109 peak =170 savedpeak = 13994"
## [5] "j =109 peak =191 savedpeak = 14015"
## [6] "j =109 peak =219 savedpeak = 14043"
## [7] "j =109 peak =254 savedpeak = 14078"
## [8] "j =109 peak =268 savedpeak = 14092"
## [9] "j =109 peak =321 savedpeak = 14145"
## [10] "j =109 peak =352 savedpeak = 14176"
## [11] "j =109 peak =388 savedpeak = 14212"
## [12] "j =109 peak =437 savedpeak = 14261"
## [1] "j =110 peak =91 savedpeak = 14043"
## [2] "j =110 peak =126 savedpeak = 14078"
## [3] "j =110 peak =140 savedpeak = 14092"
## [4] "j =110 peak =193 savedpeak = 14145"
## [5] "j =110 peak =224 savedpeak = 14176"
## [6] "j =110 peak =260 savedpeak = 14212"
## [7] "j =110 peak =309 savedpeak = 14261"
## [8] "j =110 peak =351 savedpeak = 14303"
## [9] "j =110 peak =382 savedpeak = 14334"
## [10] "j =110 peak =407 savedpeak = 14359"

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## [1] "j =111 peak =96 savedpeak = 14176"
## [2] "j =111 peak =132 savedpeak = 14212"
## [3] "j =111 peak =181 savedpeak = 14261"
## [4] "j =111 peak =223 savedpeak = 14303"
## [5] "j =111 peak =254 savedpeak = 14334"
## [6] "j =111 peak =279 savedpeak = 14359"
## [7] "j =111 peak =318 savedpeak = 14398"
## [8] "j =111 peak =339 savedpeak = 14419"
## [1] "j =112 peak =95 savedpeak = 14303"
## [2] "j =112 peak =126 savedpeak = 14334"
## [3] "j =112 peak =151 savedpeak = 14359"
## [4] "j =112 peak =190 savedpeak = 14398"
## [5] "j =112 peak =211 savedpeak = 14419"
## [6] "j =112 peak =358 savedpeak = 14566"
## [7] "j =112 peak =397 savedpeak = 14605"
## [1] "j =113 peak =83 savedpeak = 14419"
## [2] "j =113 peak =230 savedpeak = 14566"
## [3] "j =113 peak =269 savedpeak = 14605"
## [4] "j =113 peak =321 savedpeak = 14657"
## [5] "j =113 peak =377 savedpeak = 14713"
## [6] "j =113 peak =395 savedpeak = 14731"
## [1] "j =114 peak =102 savedpeak = 14566"
## [2] "j =114 peak =141 savedpeak = 14605"
## [3] "j =114 peak =193 savedpeak = 14657"
## [4] "j =114 peak =249 savedpeak = 14713"
## [5] "j =114 peak =267 savedpeak = 14731"
## [6] "j =114 peak =316 savedpeak = 14780"
## [7] "j =114 peak =365 savedpeak = 14829"
## [1] "j =115 peak =121 savedpeak = 14713"
## [2] "j =115 peak =139 savedpeak = 14731"
## [3] "j =115 peak =188 savedpeak = 14780"
## [4] "j =115 peak =237 savedpeak = 14829"
## [5] "j =115 peak =314 savedpeak = 14906"
## [6] "j =115 peak =409 savedpeak = 15001"
## [1] "j =116 peak =432 savedpeak = 15152"
## [1] "j =117 peak =331 savedpeak = 15179"
## [2] "j =117 peak =374 savedpeak = 15222"
## [3] "j =117 peak =411 savedpeak = 15259"
## [1] "j =118 peak =203 savedpeak = 15179"
## [2] "j =118 peak =246 savedpeak = 15222"
## [3] "j =118 peak =283 savedpeak = 15259"
## [4] "j =118 peak =311 savedpeak = 15287"
## [5] "j =118 peak =348 savedpeak = 15324"
## [1] "j =119 peak =155 savedpeak = 15259"
## [2] "j =119 peak =220 savedpeak = 15324"
## [3] "j =119 peak =321 savedpeak = 15425"
## [4] "j =119 peak =371 savedpeak = 15475"
## [1] "j =120 peak =92 savedpeak = 15324"
## [2] "j =120 peak =193 savedpeak = 15425"
## [3] "j =120 peak =243 savedpeak = 15475"
## [1] "j =121 peak =115 savedpeak = 15475"
## [2] "j =121 peak =195 savedpeak = 15555"
## [3] "j =121 peak =317 savedpeak = 15677"
## [4] "j =121 peak =368 savedpeak = 15728"

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## [5] "j =121 peak =411 savedpeak = 15771"
## [1] "j =127 peak =214 savedpeak = 16342"
## [2] "j =127 peak =257 savedpeak = 16385"
## [3] "j =127 peak =358 savedpeak = 16486"
## [4] "j =127 peak =416 savedpeak = 16544"
## [1] "j =128 peak =86 savedpeak = 16342"
## [2] "j =128 peak =129 savedpeak = 16385"
## [3] "j =128 peak =202 savedpeak = 16458"
## [4] "j =128 peak =230 savedpeak = 16486"
## [5] "j =128 peak =288 savedpeak = 16544"
## [6] "j =128 peak =382 savedpeak = 16638"
## [1] "j =129 peak =102 savedpeak = 16486"
## [2] "j =129 peak =160 savedpeak = 16544"
## [3] "j =129 peak =254 savedpeak = 16638"
## [4] "j =129 peak =334 savedpeak = 16718"
## [1] "j =134 peak =322 savedpeak = 17346"
## [2] "j =134 peak =423 savedpeak = 17447"
## [1] "j =135 peak =194 savedpeak = 17346"
## [2] "j =135 peak =295 savedpeak = 17447"
## [3] "j =135 peak =411 savedpeak = 17563"
## [1] "j =136 peak =167 savedpeak = 17447"
## [2] "j =136 peak =283 savedpeak = 17563"
## [1] "j =145 peak =106 savedpeak = 18538"
## [2] "j =145 peak =279 savedpeak = 18711"
## [3] "j =145 peak =395 savedpeak = 18827"
## [1] "j =146 peak =151 savedpeak = 18711"
## [2] "j =146 peak =267 savedpeak = 18827"
## [3] "j =146 peak =332 savedpeak = 18892"
## [4] "j =146 peak =411 savedpeak = 18971"
## [1] "j =147 peak =139 savedpeak = 18827"
## [2] "j =147 peak =204 savedpeak = 18892"
## [3] "j =147 peak =341 savedpeak = 19029"
## [4] "j =147 peak =427 savedpeak = 19115"
## [1] "j =148 peak =76 savedpeak = 18892"
## [2] "j =148 peak =213 savedpeak = 19029"
## [3] "j =148 peak =321 savedpeak = 19137"
## [4] "j =148 peak =379 savedpeak = 19195"
## [1] "j =149 peak =193 savedpeak = 19137"
## [2] "j =149 peak =251 savedpeak = 19195"
## [3] "j =149 peak =330 savedpeak = 19274"
## [4] "j =149 peak =359 savedpeak = 19303"
## [5] "j =149 peak =431 savedpeak = 19375"
## [1] "j =150 peak =123 savedpeak = 19195"
## [2] "j =150 peak =202 savedpeak = 19274"
## [3] "j =150 peak =231 savedpeak = 19303"
## [4] "j =150 peak =303 savedpeak = 19375"
## [5] "j =150 peak =397 savedpeak = 19469"
## [1] "j =151 peak =103 savedpeak = 19303"
## [2] "j =151 peak =175 savedpeak = 19375"
## [3] "j =151 peak =269 savedpeak = 19469"
## [4] "j =151 peak =356 savedpeak = 19556"
## [1] "j =152 peak =141 savedpeak = 19469"
## [2] "j =152 peak =228 savedpeak = 19556"
## [3] "j =152 peak =322 savedpeak = 19650"

```



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## [1] "j =153 peak =100 savedpeak = 19556"
## [2] "j =153 peak =194 savedpeak = 19650"
## [3] "j =153 peak =324 savedpeak = 19780"
## [4] "j =153 peak =403 savedpeak = 19859"
## [1] "j =154 peak =196 savedpeak = 19780"
## [2] "j =154 peak =275 savedpeak = 19859"
## [3] "j =154 peak =319 savedpeak = 19903"
## [1] "j =158 peak =182 savedpeak = 20278"
## [1] "j =159 peak =285 savedpeak = 20509"
## [1] "j =160 peak =157 savedpeak = 20509"
## [2] "j =160 peak =215 savedpeak = 20567"
## [3] "j =160 peak =352 savedpeak = 20704"
## [1] "j =161 peak =224 savedpeak = 20704"
## [2] "j =161 peak =419 savedpeak = 20899"
## [1] "j =162 peak =96 savedpeak = 20704"
## [2] "j =162 peak =140 savedpeak = 20748"
## [3] "j =162 peak =291 savedpeak = 20899"
## [4] "j =162 peak =320 savedpeak = 20928"
## [5] "j =162 peak =342 savedpeak = 20950"
## [1] "j =163 peak =163 savedpeak = 20899"
## [2] "j =163 peak =214 savedpeak = 20950"
## [3] "j =163 peak =358 savedpeak = 21094"
## [4] "j =163 peak =387 savedpeak = 21123"
## [5] "j =163 peak =431 savedpeak = 21167"
## [1] "j =164 peak =259 savedpeak = 21123"
## [2] "j =164 peak =382 savedpeak = 21246"
## [3] "j =164 peak =425 savedpeak = 21289"
## [1] "j =166 peak =126 savedpeak = 21246"
## [2] "j =166 peak =169 savedpeak = 21289"
## [3] "j =166 peak =249 savedpeak = 21369"
## [4] "j =166 peak =350 savedpeak = 21470"
## [5] "j =166 peak =393 savedpeak = 21513"
## [6] "j =166 peak =415 savedpeak = 21535"
## [1] "j =167 peak =121 savedpeak = 21369"
## [2] "j =167 peak =222 savedpeak = 21470"
## [3] "j =167 peak =265 savedpeak = 21513"
## [4] "j =167 peak =287 savedpeak = 21535"
## [1] "j =168 peak =94 savedpeak = 21470"
## [2] "j =168 peak =137 savedpeak = 21513"
## [3] "j =168 peak =159 savedpeak = 21535"
## [4] "j =168 peak =202 savedpeak = 21578"
## [5] "j =168 peak =325 savedpeak = 21701"
## [6] "j =168 peak =347 savedpeak = 21723"
## [7] "j =168 peak =383 savedpeak = 21759"
## [1] "j =169 peak =197 savedpeak = 21701"
## [2] "j =169 peak =334 savedpeak = 21838"
## [1] "j =170 peak =206 savedpeak = 21838"
## [2] "j =170 peak =358 savedpeak = 21990"
## [3] "j =170 peak =387 savedpeak = 22019"
## [4] "j =170 peak =437 savedpeak = 22069"
## [1] "j =173 peak =97 savedpeak = 22113"
## [2] "j =173 peak =167 savedpeak = 22183"
## [3] "j =173 peak =180 savedpeak = 22196"
## [4] "j =173 peak =263 savedpeak = 22279"

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## [5] "j =173 peak =341 savedpeak = 22357"
## [1] "j =174 peak =135 savedpeak = 22279"
## [2] "j =174 peak =213 savedpeak = 22357"
## [3] "j =174 peak =347 savedpeak = 22491"
## [4] "j =174 peak =388 savedpeak = 22532"
## [5] "j =174 peak =414 savedpeak = 22558"
## [6] "j =174 peak =437 savedpeak = 22581"
## [1] "j =175 peak =373 savedpeak = 22645"
## [2] "j =175 peak =389 savedpeak = 22661"
## [3] "j =175 peak =410 savedpeak = 22682"
## [4] "j =175 peak =430 savedpeak = 22702"
## [1] "j =176 peak =245 savedpeak = 22645"
## [2] "j =176 peak =261 savedpeak = 22661"
## [3] "j =176 peak =282 savedpeak = 22682"
## [4] "j =176 peak =302 savedpeak = 22702"
## [5] "j =176 peak =325 savedpeak = 22725"
## [6] "j =176 peak =343 savedpeak = 22743"
## [7] "j =176 peak =358 savedpeak = 22758"
## [8] "j =176 peak =378 savedpeak = 22778"
## [9] "j =176 peak =406 savedpeak = 22806"
## [10] "j =176 peak =426 savedpeak = 22826"
## [1] "j =207 peak =98 savedpeak = 26466"
## [2] "j =207 peak =117 savedpeak = 26485"
## [3] "j =207 peak =140 savedpeak = 26508"
## [4] "j =207 peak =172 savedpeak = 26540"
## [5] "j =207 peak =191 savedpeak = 26559"
## [6] "j =207 peak =207 savedpeak = 26575"
## [7] "j =207 peak =224 savedpeak = 26592"
## [8] "j =207 peak =259 savedpeak = 26627"
## [9] "j =207 peak =276 savedpeak = 26644"
## [10] "j =207 peak =305 savedpeak = 26673"
## [11] "j =207 peak =327 savedpeak = 26695"
## [12] "j =207 peak =345 savedpeak = 26713"
## [13] "j =207 peak =392 savedpeak = 26760"
## [14] "j =207 peak =416 savedpeak = 26784"
## [15] "j =207 peak =432 savedpeak = 26800"
## [1] "j =208 peak =79 savedpeak = 26575"
## [2] "j =208 peak =96 savedpeak = 26592"
## [3] "j =208 peak =131 savedpeak = 26627"
## [4] "j =208 peak =148 savedpeak = 26644"
## [5] "j =208 peak =177 savedpeak = 26673"
## [6] "j =208 peak =199 savedpeak = 26695"
## [7] "j =208 peak =217 savedpeak = 26713"
## [8] "j =208 peak =264 savedpeak = 26760"
## [9] "j =208 peak =288 savedpeak = 26784"
## [10] "j =208 peak =304 savedpeak = 26800"
## [11] "j =208 peak =320 savedpeak = 26816"
## [12] "j =208 peak =386 savedpeak = 26882"
## [13] "j =208 peak =429 savedpeak = 26925"
## [1] "j =209 peak =89 savedpeak = 26713"
## [2] "j =209 peak =136 savedpeak = 26760"
## [3] "j =209 peak =160 savedpeak = 26784"
## [4] "j =209 peak =176 savedpeak = 26800"
## [5] "j =209 peak =192 savedpeak = 26816"

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## [6] "j =209 peak =258 savedpeak = 26882"
## [7] "j =209 peak =301 savedpeak = 26925"
## [8] "j =209 peak =320 savedpeak = 26944"
## [9] "j =209 peak =351 savedpeak = 26975"
## [10] "j =209 peak =367 savedpeak = 26991"
## [11] "j =209 peak =396 savedpeak = 27020"
## [12] "j =209 peak =415 savedpeak = 27039"
## [1] "j =215 peak =87 savedpeak = 27479"
## [2] "j =215 peak =112 savedpeak = 27504"
## [3] "j =215 peak =140 savedpeak = 27532"
## [4] "j =215 peak =162 savedpeak = 27554"
## [5] "j =215 peak =188 savedpeak = 27580"
## [6] "j =215 peak =218 savedpeak = 27610"
## [7] "j =215 peak =251 savedpeak = 27643"
## [8] "j =215 peak =274 savedpeak = 27666"
## [9] "j =215 peak =305 savedpeak = 27697"
## [10] "j =215 peak =320 savedpeak = 27712"
## [11] "j =215 peak =374 savedpeak = 27766"
## [12] "j =215 peak =394 savedpeak = 27786"
## [1] "j =222 peak =81 savedpeak = 28369"
## [2] "j =222 peak =103 savedpeak = 28391"
## [3] "j =222 peak =135 savedpeak = 28423"
## [4] "j =222 peak =160 savedpeak = 28448"
## [5] "j =222 peak =185 savedpeak = 28473"
## [6] "j =222 peak =207 savedpeak = 28495"
## [7] "j =222 peak =246 savedpeak = 28534"
## [8] "j =222 peak =269 savedpeak = 28557"
## [9] "j =222 peak =306 savedpeak = 28594"
## [10] "j =222 peak =320 savedpeak = 28608"
## [11] "j =222 peak =338 savedpeak = 28626"
## [12] "j =222 peak =363 savedpeak = 28651"
## [13] "j =222 peak =385 savedpeak = 28673"
## [14] "j =222 peak =399 savedpeak = 28687"
## [15] "j =222 peak =424 savedpeak = 28712"
## [1] "j =223 peak =79 savedpeak = 28495"
## [2] "j =223 peak =118 savedpeak = 28534"
## [3] "j =223 peak =141 savedpeak = 28557"
## [4] "j =223 peak =178 savedpeak = 28594"
## [5] "j =223 peak =192 savedpeak = 28608"
## [6] "j =223 peak =210 savedpeak = 28626"
## [7] "j =223 peak =235 savedpeak = 28651"
## [8] "j =223 peak =257 savedpeak = 28673"
## [9] "j =223 peak =271 savedpeak = 28687"
## [10] "j =223 peak =296 savedpeak = 28712"
## [11] "j =223 peak =322 savedpeak = 28738"
## [12] "j =223 peak =349 savedpeak = 28765"
## [13] "j =223 peak =374 savedpeak = 28790"
## [14] "j =223 peak =395 savedpeak = 28811"
## [15] "j =223 peak =418 savedpeak = 28834"
## [1] "j =224 peak =82 savedpeak = 28626"
## [2] "j =224 peak =107 savedpeak = 28651"
## [3] "j =224 peak =129 savedpeak = 28673"
## [4] "j =224 peak =143 savedpeak = 28687"
## [5] "j =224 peak =168 savedpeak = 28712"

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## [6] "j =224 peak =194 savedpeak = 28738"
## [7] "j =224 peak =221 savedpeak = 28765"
## [8] "j =224 peak =246 savedpeak = 28790"
## [9] "j =224 peak =267 savedpeak = 28811"
## [10] "j =224 peak =290 savedpeak = 28834"
## [11] "j =224 peak =315 savedpeak = 28859"
## [12] "j =224 peak =352 savedpeak = 28896"
## [13] "j =224 peak =375 savedpeak = 28919"
## [14] "j =224 peak =433 savedpeak = 28977"
## [1] "j =225 peak =93 savedpeak = 28765"
## [2] "j =225 peak =118 savedpeak = 28790"
## [3] "j =225 peak =139 savedpeak = 28811"
## [4] "j =225 peak =162 savedpeak = 28834"
## [5] "j =225 peak =187 savedpeak = 28859"
## [6] "j =225 peak =224 savedpeak = 28896"
## [7] "j =225 peak =247 savedpeak = 28919"
## [8] "j =225 peak =305 savedpeak = 28977"
## [9] "j =225 peak =327 savedpeak = 28999"
## [10] "j =225 peak =360 savedpeak = 29032"
## [11] "j =225 peak =382 savedpeak = 29054"
## [12] "j =225 peak =403 savedpeak = 29075"
## [13] "j =225 peak =421 savedpeak = 29093"
## [1] "j =226 peak =96 savedpeak = 28896"
## [2] "j =226 peak =119 savedpeak = 28919"
## [3] "j =226 peak =177 savedpeak = 28977"
## [4] "j =226 peak =199 savedpeak = 28999"
## [5] "j =226 peak =232 savedpeak = 29032"
## [6] "j =226 peak =254 savedpeak = 29054"
## [7] "j =226 peak =275 savedpeak = 29075"
## [8] "j =226 peak =293 savedpeak = 29093"
## [9] "j =226 peak =317 savedpeak = 29117"
## [10] "j =226 peak =338 savedpeak = 29138"
## [11] "j =226 peak =362 savedpeak = 29162"
## [12] "j =226 peak =371 savedpeak = 29171"
## [13] "j =226 peak =384 savedpeak = 29184"
## [14] "j =226 peak =412 savedpeak = 29212"
## [1] "j =227 peak =104 savedpeak = 29032"
## [2] "j =227 peak =126 savedpeak = 29054"
## [3] "j =227 peak =147 savedpeak = 29075"
## [4] "j =227 peak =165 savedpeak = 29093"
## [5] "j =227 peak =189 savedpeak = 29117"
## [6] "j =227 peak =210 savedpeak = 29138"
## [7] "j =227 peak =234 savedpeak = 29162"
## [8] "j =227 peak =243 savedpeak = 29171"
## [9] "j =227 peak =256 savedpeak = 29184"
## [10] "j =227 peak =284 savedpeak = 29212"
## [11] "j =227 peak =313 savedpeak = 29241"
## [12] "j =227 peak =346 savedpeak = 29274"
## [13] "j =227 peak =372 savedpeak = 29300"
## [14] "j =227 peak =384 savedpeak = 29312"
## [15] "j =227 peak =420 savedpeak = 29348"
## [1] "j =228 peak =82 savedpeak = 29138"
## [2] "j =228 peak =106 savedpeak = 29162"
## [3] "j =228 peak =115 savedpeak = 29171"

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## [4] "j =228 peak =128 savedpeak = 29184"
## [5] "j =228 peak =156 savedpeak = 29212"
## [6] "j =228 peak =185 savedpeak = 29241"
## [7] "j =228 peak =218 savedpeak = 29274"
## [8] "j =228 peak =244 savedpeak = 29300"
## [9] "j =228 peak =256 savedpeak = 29312"
## [10] "j =228 peak =292 savedpeak = 29348"
## [11] "j =228 peak =310 savedpeak = 29366"
## [12] "j =228 peak =370 savedpeak = 29426"
## [13] "j =228 peak =385 savedpeak = 29441"
## [14] "j =228 peak =400 savedpeak = 29456"
## [1] "j =229 peak =90 savedpeak = 29274"
## [2] "j =229 peak =116 savedpeak = 29300"
## [3] "j =229 peak =128 savedpeak = 29312"
## [4] "j =229 peak =164 savedpeak = 29348"
## [5] "j =229 peak =182 savedpeak = 29366"
## [6] "j =229 peak =242 savedpeak = 29426"
## [7] "j =229 peak =257 savedpeak = 29441"
## [8] "j =229 peak =272 savedpeak = 29456"
## [9] "j =229 peak =314 savedpeak = 29498"
## [10] "j =229 peak =361 savedpeak = 29545"
## [11] "j =229 peak =385 savedpeak = 29569"
## [12] "j =229 peak =425 savedpeak = 29609"
## [1] "j =230 peak =114 savedpeak = 29426"
## [2] "j =230 peak =129 savedpeak = 29441"
## [3] "j =230 peak =144 savedpeak = 29456"
## [4] "j =230 peak =186 savedpeak = 29498"
## [5] "j =230 peak =233 savedpeak = 29545"
## [6] "j =230 peak =257 savedpeak = 29569"
## [7] "j =230 peak =297 savedpeak = 29609"
## [8] "j =230 peak =318 savedpeak = 29630"
## [9] "j =230 peak =329 savedpeak = 29641"
## [10] "j =230 peak =367 savedpeak = 29679"
## [11] "j =230 peak =409 savedpeak = 29721"
## [12] "j =230 peak =422 savedpeak = 29734"
## [1] "j =231 peak =105 savedpeak = 29545"
## [2] "j =231 peak =129 savedpeak = 29569"
## [3] "j =231 peak =169 savedpeak = 29609"
## [4] "j =231 peak =190 savedpeak = 29630"
## [5] "j =231 peak =201 savedpeak = 29641"
## [6] "j =231 peak =239 savedpeak = 29679"
## [7] "j =231 peak =281 savedpeak = 29721"
## [8] "j =231 peak =294 savedpeak = 29734"
## [9] "j =231 peak =317 savedpeak = 29757"
## [10] "j =231 peak =362 savedpeak = 29802"
## [11] "j =231 peak =398 savedpeak = 29838"
## [12] "j =231 peak =435 savedpeak = 29875"
## [1] "j =232 peak =111 savedpeak = 29679"
## [2] "j =232 peak =153 savedpeak = 29721"
## [3] "j =232 peak =166 savedpeak = 29734"
## [4] "j =232 peak =189 savedpeak = 29757"
## [5] "j =232 peak =234 savedpeak = 29802"
## [6] "j =232 peak =270 savedpeak = 29838"
## [7] "j =232 peak =307 savedpeak = 29875"

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## [8] "j =232 peak =362 savedpeak = 29930"
## [9] "j =232 peak =378 savedpeak = 29946"
## [10] "j =232 peak =394 savedpeak = 29962"
## [1] "j =233 peak =106 savedpeak = 29802"
## [2] "j =233 peak =142 savedpeak = 29838"
## [3] "j =233 peak =179 savedpeak = 29875"
## [4] "j =233 peak =234 savedpeak = 29930"
## [5] "j =233 peak =250 savedpeak = 29946"
## [6] "j =233 peak =266 savedpeak = 29962"
## [7] "j =233 peak =315 savedpeak = 30011"
## [8] "j =233 peak =332 savedpeak = 30028"
## [9] "j =233 peak =353 savedpeak = 30049"
## [10] "j =233 peak =380 savedpeak = 30076"
## [11] "j =233 peak =429 savedpeak = 30125"
## [1] "j =234 peak =106 savedpeak = 29930"
## [2] "j =234 peak =122 savedpeak = 29946"
## [3] "j =234 peak =138 savedpeak = 29962"
## [4] "j =234 peak =187 savedpeak = 30011"
## [5] "j =234 peak =204 savedpeak = 30028"
## [6] "j =234 peak =225 savedpeak = 30049"
## [7] "j =234 peak =252 savedpeak = 30076"
## [8] "j =234 peak =301 savedpeak = 30125"
## [9] "j =234 peak =332 savedpeak = 30156"
## [10] "j =234 peak =358 savedpeak = 30182"
## [11] "j =234 peak =381 savedpeak = 30205"
## [12] "j =234 peak =402 savedpeak = 30226"
## [13] "j =234 peak =416 savedpeak = 30240"
## [14] "j =234 peak =433 savedpeak = 30257"
## [1] "j =235 peak =76 savedpeak = 30028"
## [2] "j =235 peak =97 savedpeak = 30049"
## [3] "j =235 peak =124 savedpeak = 30076"
## [4] "j =235 peak =173 savedpeak = 30125"
## [5] "j =235 peak =204 savedpeak = 30156"
## [6] "j =235 peak =230 savedpeak = 30182"
## [7] "j =235 peak =253 savedpeak = 30205"
## [8] "j =235 peak =274 savedpeak = 30226"
## [9] "j =235 peak =288 savedpeak = 30240"
## [10] "j =235 peak =305 savedpeak = 30257"
## [11] "j =235 peak =324 savedpeak = 30276"
## [12] "j =235 peak =341 savedpeak = 30293"
## [13] "j =235 peak =403 savedpeak = 30355"
## [14] "j =235 peak =417 savedpeak = 30369"
## [1] "j =236 peak =76 savedpeak = 30156"
## [2] "j =236 peak =102 savedpeak = 30182"
## [3] "j =236 peak =125 savedpeak = 30205"
## [4] "j =236 peak =146 savedpeak = 30226"
## [5] "j =236 peak =177 savedpeak = 30257"
## [6] "j =236 peak =196 savedpeak = 30276"
## [7] "j =236 peak =213 savedpeak = 30293"
## [8] "j =236 peak =275 savedpeak = 30355"
## [9] "j =236 peak =289 savedpeak = 30369"
## [10] "j =236 peak =332 savedpeak = 30412"
## [11] "j =236 peak =380 savedpeak = 30460"
## [12] "j =236 peak =432 savedpeak = 30512"

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## [1] "j =250 peak =104 savedpeak = 31976"
## [2] "j =250 peak =121 savedpeak = 31993"
## [3] "j =250 peak =143 savedpeak = 32015"
## [4] "j =250 peak =163 savedpeak = 32035"
## [5] "j =250 peak =186 savedpeak = 32058"
## [6] "j =250 peak =218 savedpeak = 32090"
## [7] "j =250 peak =254 savedpeak = 32126"
## [8] "j =250 peak =264 savedpeak = 32136"
## [9] "j =250 peak =299 savedpeak = 32171"
## [10] "j =250 peak =317 savedpeak = 32189"
## [11] "j =250 peak =356 savedpeak = 32228"
## [12] "j =250 peak =388 savedpeak = 32260"
## [13] "j =250 peak =401 savedpeak = 32273"
## [14] "j =250 peak =418 savedpeak = 32290"
## [15] "j =250 peak =434 savedpeak = 32306"
## [1] "j =251 peak =90 savedpeak = 32090"
## [2] "j =251 peak =106 savedpeak = 32106"
## [3] "j =251 peak =126 savedpeak = 32126"
## [4] "j =251 peak =136 savedpeak = 32136"
## [5] "j =251 peak =171 savedpeak = 32171"
## [6] "j =251 peak =189 savedpeak = 32189"
## [7] "j =251 peak =228 savedpeak = 32228"
## [8] "j =251 peak =260 savedpeak = 32260"
## [9] "j =251 peak =273 savedpeak = 32273"
## [10] "j =251 peak =290 savedpeak = 32290"
## [11] "j =251 peak =306 savedpeak = 32306"
## [12] "j =251 peak =335 savedpeak = 32335"
## [13] "j =251 peak =382 savedpeak = 32382"
## [14] "j =251 peak =413 savedpeak = 32413"
## [15] "j =251 peak =428 savedpeak = 32428"
## [1] "j =253 peak =79 savedpeak = 32335"
## [2] "j =253 peak =126 savedpeak = 32382"
## [3] "j =253 peak =157 savedpeak = 32413"
## [4] "j =253 peak =172 savedpeak = 32428"
## [5] "j =253 peak =208 savedpeak = 32464"
## [6] "j =253 peak =220 savedpeak = 32476"
## [7] "j =253 peak =242 savedpeak = 32498"
## [8] "j =253 peak =288 savedpeak = 32544"
## [9] "j =253 peak =309 savedpeak = 32565"
## [10] "j =253 peak =341 savedpeak = 32597"
## [11] "j =253 peak =362 savedpeak = 32618"
## [12] "j =253 peak =387 savedpeak = 32643"
## [13] "j =253 peak =410 savedpeak = 32666"
## [14] "j =253 peak =429 savedpeak = 32685"
## [1] "j =256 peak =78 savedpeak = 32718"
## [2] "j =256 peak =107 savedpeak = 32747"
## [3] "j =256 peak =141 savedpeak = 32781"
## [4] "j =256 peak =151 savedpeak = 32791"
## [5] "j =256 peak =191 savedpeak = 32831"
## [6] "j =256 peak =220 savedpeak = 32860"
## [7] "j =256 peak =256 savedpeak = 32896"
## [8] "j =256 peak =301 savedpeak = 32941"
## [9] "j =256 peak =321 savedpeak = 32961"
## [10] "j =256 peak =351 savedpeak = 32991"

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## [11] "j =256 peak =371 savedpeak = 33011"
## [12] "j =256 peak =396 savedpeak = 33036"
## [1] "j =257 peak =92 savedpeak = 32860"
## [2] "j =257 peak =128 savedpeak = 32896"
## [3] "j =257 peak =173 savedpeak = 32941"
## [4] "j =257 peak =193 savedpeak = 32961"
## [5] "j =257 peak =223 savedpeak = 32991"
## [6] "j =257 peak =243 savedpeak = 33011"
## [7] "j =257 peak =268 savedpeak = 33036"
## [8] "j =257 peak =348 savedpeak = 33116"
## [9] "j =257 peak =388 savedpeak = 33156"
## [10] "j =257 peak =429 savedpeak = 33197"
## [1] "j =258 peak =95 savedpeak = 32991"
## [2] "j =258 peak =115 savedpeak = 33011"
## [3] "j =258 peak =140 savedpeak = 33036"
## [4] "j =258 peak =220 savedpeak = 33116"
## [5] "j =258 peak =260 savedpeak = 33156"
## [6] "j =258 peak =307 savedpeak = 33203"
## [7] "j =258 peak =331 savedpeak = 33227"
## [8] "j =258 peak =348 savedpeak = 33244"
## [9] "j =258 peak =383 savedpeak = 33279"
## [10] "j =258 peak =400 savedpeak = 33296"
## [11] "j =258 peak =416 savedpeak = 33312"
## [1] "j =262 peak =124 savedpeak = 33532"
## [2] "j =262 peak =167 savedpeak = 33575"
## [3] "j =262 peak =192 savedpeak = 33600"
## [4] "j =262 peak =203 savedpeak = 33611"
## [5] "j =262 peak =227 savedpeak = 33635"
## [6] "j =262 peak =263 savedpeak = 33671"
## [7] "j =262 peak =284 savedpeak = 33692"
## [8] "j =262 peak =324 savedpeak = 33732"
## [9] "j =262 peak =357 savedpeak = 33765"
## [10] "j =262 peak =391 savedpeak = 33799"
## [11] "j =262 peak =402 savedpeak = 33810"
## [12] "j =262 peak =435 savedpeak = 33843"
## [1] "j =263 peak =99 savedpeak = 33635"
## [2] "j =263 peak =135 savedpeak = 33671"
## [3] "j =263 peak =156 savedpeak = 33692"
## [4] "j =263 peak =196 savedpeak = 33732"
## [5] "j =263 peak =229 savedpeak = 33765"
## [6] "j =263 peak =263 savedpeak = 33799"
## [7] "j =263 peak =274 savedpeak = 33810"
## [8] "j =263 peak =307 savedpeak = 33843"
## [9] "j =263 peak =336 savedpeak = 33872"
## [10] "j =263 peak =388 savedpeak = 33924"
## [11] "j =263 peak =414 savedpeak = 33950"
## [12] "j =263 peak =428 savedpeak = 33964"
## [1] "j =264 peak =101 savedpeak = 33765"
## [2] "j =264 peak =135 savedpeak = 33799"
## [3] "j =264 peak =146 savedpeak = 33810"
## [4] "j =264 peak =179 savedpeak = 33843"
## [5] "j =264 peak =208 savedpeak = 33872"
## [6] "j =264 peak =260 savedpeak = 33924"
## [7] "j =264 peak =286 savedpeak = 33950"

```



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## [8] "j =264 peak =300 savedpeak = 33964"
## [9] "j =264 peak =329 savedpeak = 33993"
## [10] "j =264 peak =375 savedpeak = 34039"
## [11] "j =264 peak =398 savedpeak = 34062"
## [12] "j =264 peak =413 savedpeak = 34077"
## [13] "j =264 peak =437 savedpeak = 34101"
## [1] "j =265 peak =80 savedpeak = 33872"
## [2] "j =265 peak =132 savedpeak = 33924"
## [3] "j =265 peak =158 savedpeak = 33950"
## [4] "j =265 peak =172 savedpeak = 33964"
## [5] "j =265 peak =201 savedpeak = 33993"
## [6] "j =265 peak =247 savedpeak = 34039"
## [7] "j =265 peak =270 savedpeak = 34062"
## [8] "j =265 peak =285 savedpeak = 34077"
## [9] "j =265 peak =309 savedpeak = 34101"
## [10] "j =265 peak =332 savedpeak = 34124"
## [11] "j =265 peak =356 savedpeak = 34148"
## [12] "j =265 peak =376 savedpeak = 34168"
## [13] "j =265 peak =400 savedpeak = 34192"
## [1] "j =266 peak =119 savedpeak = 34039"
## [2] "j =266 peak =142 savedpeak = 34062"
## [3] "j =266 peak =157 savedpeak = 34077"
## [4] "j =266 peak =181 savedpeak = 34101"
## [5] "j =266 peak =204 savedpeak = 34124"
## [6] "j =266 peak =228 savedpeak = 34148"
## [7] "j =266 peak =248 savedpeak = 34168"
## [8] "j =266 peak =272 savedpeak = 34192"
## [9] "j =266 peak =315 savedpeak = 34235"
## [10] "j =266 peak =378 savedpeak = 34298"
## [11] "j =266 peak =412 savedpeak = 34332"
## [1] "j =267 peak =76 savedpeak = 34124"
## [2] "j =267 peak =100 savedpeak = 34148"
## [3] "j =267 peak =120 savedpeak = 34168"
## [4] "j =267 peak =144 savedpeak = 34192"
## [5] "j =267 peak =187 savedpeak = 34235"
## [6] "j =267 peak =250 savedpeak = 34298"
## [7] "j =267 peak =284 savedpeak = 34332"
## [8] "j =267 peak =310 savedpeak = 34358"
## [9] "j =267 peak =363 savedpeak = 34411"
## [10] "j =267 peak =383 savedpeak = 34431"
## [11] "j =267 peak =423 savedpeak = 34471"
## [1] "j =268 peak =122 savedpeak = 34298"
## [2] "j =268 peak =156 savedpeak = 34332"
## [3] "j =268 peak =182 savedpeak = 34358"
## [4] "j =268 peak =206 savedpeak = 34382"
## [5] "j =268 peak =235 savedpeak = 34411"
## [6] "j =268 peak =255 savedpeak = 34431"
## [7] "j =268 peak =295 savedpeak = 34471"
## [8] "j =268 peak =313 savedpeak = 34489"
## [9] "j =268 peak =341 savedpeak = 34517"
## [10] "j =268 peak =361 savedpeak = 34537"
## [11] "j =268 peak =378 savedpeak = 34554"
## [12] "j =268 peak =400 savedpeak = 34576"
## [13] "j =268 peak =428 savedpeak = 34604"

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```

## [1] "j =275 peak =100 savedpeak = 35172"
## [2] "j =275 peak =165 savedpeak = 35237"
## [3] "j =275 peak =186 savedpeak = 35258"
## [4] "j =275 peak =210 savedpeak = 35282"
## [5] "j =275 peak =232 savedpeak = 35304"
## [6] "j =275 peak =252 savedpeak = 35324"
## [7] "j =275 peak =276 savedpeak = 35348"
## [8] "j =275 peak =305 savedpeak = 35377"
## [9] "j =275 peak =325 savedpeak = 35397"
## [10] "j =275 peak =378 savedpeak = 35450"
## [11] "j =275 peak =406 savedpeak = 35478"
## [12] "j =275 peak =420 savedpeak = 35492"
## [1] "j =276 peak =82 savedpeak = 35282"
## [2] "j =276 peak =104 savedpeak = 35304"
## [3] "j =276 peak =124 savedpeak = 35324"
## [4] "j =276 peak =148 savedpeak = 35348"
## [5] "j =276 peak =177 savedpeak = 35377"
## [6] "j =276 peak =197 savedpeak = 35397"
## [7] "j =276 peak =250 savedpeak = 35450"
## [8] "j =276 peak =278 savedpeak = 35478"
## [9] "j =276 peak =292 savedpeak = 35492"
## [10] "j =276 peak =310 savedpeak = 35510"
## [11] "j =276 peak =346 savedpeak = 35546"
## [12] "j =276 peak =363 savedpeak = 35563"
## [13] "j =276 peak =388 savedpeak = 35588"
## [14] "j =276 peak =433 savedpeak = 35633"
## [1] "j =277 peak =122 savedpeak = 35450"
## [2] "j =277 peak =150 savedpeak = 35478"
## [3] "j =277 peak =164 savedpeak = 35492"
## [4] "j =277 peak =182 savedpeak = 35510"
## [5] "j =277 peak =218 savedpeak = 35546"
## [6] "j =277 peak =235 savedpeak = 35563"
## [7] "j =277 peak =260 savedpeak = 35588"
## [8] "j =277 peak =305 savedpeak = 35633"
## [9] "j =277 peak =322 savedpeak = 35650"
## [10] "j =277 peak =371 savedpeak = 35699"
## [11] "j =277 peak =432 savedpeak = 35760"
## [1] "j =284 peak =79 savedpeak = 36303"
## [2] "j =284 peak =102 savedpeak = 36326"
## [3] "j =284 peak =119 savedpeak = 36343"
## [4] "j =284 peak =138 savedpeak = 36362"
## [5] "j =284 peak =160 savedpeak = 36384"
## [6] "j =284 peak =179 savedpeak = 36403"
## [7] "j =284 peak =192 savedpeak = 36416"
## [8] "j =284 peak =210 savedpeak = 36434"
## [9] "j =284 peak =227 savedpeak = 36451"
## [10] "j =284 peak =260 savedpeak = 36484"
## [11] "j =284 peak =299 savedpeak = 36523"
## [12] "j =284 peak =330 savedpeak = 36554"
## [13] "j =284 peak =347 savedpeak = 36571"
## [14] "j =284 peak =405 savedpeak = 36629"
## [15] "j =284 peak =415 savedpeak = 36639"
## [16] "j =284 peak =435 savedpeak = 36659"
## [1] "j =285 peak =82 savedpeak = 36434"

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## [2] "j =285 peak =99 savedpeak = 36451"
## [3] "j =285 peak =132 savedpeak = 36484"
## [4] "j =285 peak =171 savedpeak = 36523"
## [5] "j =285 peak =202 savedpeak = 36554"
## [6] "j =285 peak =219 savedpeak = 36571"
## [7] "j =285 peak =277 savedpeak = 36629"
## [8] "j =285 peak =287 savedpeak = 36639"
## [9] "j =285 peak =307 savedpeak = 36659"
## [10] "j =285 peak =358 savedpeak = 36710"
## [11] "j =285 peak =387 savedpeak = 36739"
## [12] "j =285 peak =411 savedpeak = 36763"
## [1] "j =286 peak =91 savedpeak = 36571"
## [2] "j =286 peak =149 savedpeak = 36629"
## [3] "j =286 peak =159 savedpeak = 36639"
## [4] "j =286 peak =179 savedpeak = 36659"
## [5] "j =286 peak =230 savedpeak = 36710"
## [6] "j =286 peak =259 savedpeak = 36739"
## [7] "j =286 peak =283 savedpeak = 36763"
## [8] "j =286 peak =325 savedpeak = 36805"
## [9] "j =286 peak =354 savedpeak = 36834"
## [10] "j =286 peak =377 savedpeak = 36857"
## [11] "j =286 peak =415 savedpeak = 36895"
## [1] "j =287 peak =102 savedpeak = 36710"
## [2] "j =287 peak =131 savedpeak = 36739"
## [3] "j =287 peak =155 savedpeak = 36763"
## [4] "j =287 peak =197 savedpeak = 36805"
## [5] "j =287 peak =226 savedpeak = 36834"
## [6] "j =287 peak =249 savedpeak = 36857"
## [7] "j =287 peak =287 savedpeak = 36895"
## [8] "j =287 peak =325 savedpeak = 36933"
## [9] "j =287 peak =345 savedpeak = 36953"
## [10] "j =287 peak =377 savedpeak = 36985"
## [11] "j =287 peak =403 savedpeak = 37011"
## [12] "j =287 peak =429 savedpeak = 37037"
## [1] "j =288 peak =98 savedpeak = 36834"
## [2] "j =288 peak =121 savedpeak = 36857"
## [3] "j =288 peak =159 savedpeak = 36895"
## [4] "j =288 peak =197 savedpeak = 36933"
## [5] "j =288 peak =217 savedpeak = 36953"
## [6] "j =288 peak =249 savedpeak = 36985"
## [7] "j =288 peak =275 savedpeak = 37011"
## [8] "j =288 peak =301 savedpeak = 37037"
## [9] "j =288 peak =321 savedpeak = 37057"
## [10] "j =288 peak =348 savedpeak = 37084"
## [11] "j =288 peak =363 savedpeak = 37099"
## [12] "j =288 peak =386 savedpeak = 37122"
## [13] "j =288 peak =420 savedpeak = 37156"
## [1] "j =297 peak =95 savedpeak = 37983"
## [2] "j =297 peak =111 savedpeak = 37999"
## [3] "j =297 peak =129 savedpeak = 38017"
## [4] "j =297 peak =156 savedpeak = 38044"
## [5] "j =297 peak =185 savedpeak = 38073"
## [6] "j =297 peak =210 savedpeak = 38098"
## [7] "j =297 peak =253 savedpeak = 38141"

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## [8] "j =297 peak =287 savedpeak = 38175"
## [9] "j =297 peak =318 savedpeak = 38206"
## [10] "j =297 peak =337 savedpeak = 38225"
## [11] "j =297 peak =353 savedpeak = 38241"
## [12] "j =297 peak =391 savedpeak = 38279"
## [13] "j =297 peak =410 savedpeak = 38298"
## [1] "j =298 peak =82 savedpeak = 38098"
## [2] "j =298 peak =125 savedpeak = 38141"
## [3] "j =298 peak =159 savedpeak = 38175"
## [4] "j =298 peak =190 savedpeak = 38206"
## [5] "j =298 peak =209 savedpeak = 38225"
## [6] "j =298 peak =225 savedpeak = 38241"
## [7] "j =298 peak =263 savedpeak = 38279"
## [8] "j =298 peak =282 savedpeak = 38298"
## [9] "j =298 peak =311 savedpeak = 38327"
## [10] "j =298 peak =338 savedpeak = 38354"
## [11] "j =298 peak =363 savedpeak = 38379"
## [12] "j =298 peak =404 savedpeak = 38420"
## [13] "j =298 peak =429 savedpeak = 38445"
## [1] "j =299 peak =81 savedpeak = 38225"
## [2] "j =299 peak =97 savedpeak = 38241"
## [3] "j =299 peak =135 savedpeak = 38279"
## [4] "j =299 peak =154 savedpeak = 38298"
## [5] "j =299 peak =183 savedpeak = 38327"
## [6] "j =299 peak =210 savedpeak = 38354"
## [7] "j =299 peak =235 savedpeak = 38379"
## [8] "j =299 peak =276 savedpeak = 38420"
## [9] "j =299 peak =301 savedpeak = 38445"
## [10] "j =299 peak =340 savedpeak = 38484"
## [11] "j =299 peak =361 savedpeak = 38505"
## [12] "j =299 peak =384 savedpeak = 38528"
## [13] "j =299 peak =420 savedpeak = 38564"
## [14] "j =299 peak =435 savedpeak = 38579"
## [1] "j =300 peak =82 savedpeak = 38354"
## [2] "j =300 peak =107 savedpeak = 38379"
## [3] "j =300 peak =148 savedpeak = 38420"
## [4] "j =300 peak =173 savedpeak = 38445"
## [5] "j =300 peak =212 savedpeak = 38484"
## [6] "j =300 peak =233 savedpeak = 38505"
## [7] "j =300 peak =256 savedpeak = 38528"
## [8] "j =300 peak =292 savedpeak = 38564"
## [9] "j =300 peak =307 savedpeak = 38579"
## [10] "j =300 peak =320 savedpeak = 38592"
## [11] "j =300 peak =405 savedpeak = 38677"
## [12] "j =300 peak =429 savedpeak = 38701"
## [1] "j =301 peak =84 savedpeak = 38484"
## [2] "j =301 peak =105 savedpeak = 38505"
## [3] "j =301 peak =128 savedpeak = 38528"
## [4] "j =301 peak =164 savedpeak = 38564"
## [5] "j =301 peak =179 savedpeak = 38579"
## [6] "j =301 peak =192 savedpeak = 38592"
## [7] "j =301 peak =277 savedpeak = 38677"
## [8] "j =301 peak =301 savedpeak = 38701"
## [9] "j =301 peak =337 savedpeak = 38737"

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## [10] "j =301 peak =374 savedpeak = 38774"
## [11] "j =301 peak =400 savedpeak = 38800"
## [12] "j =301 peak =434 savedpeak = 38834"
## [1] "j =302 peak =107 savedpeak = 38635"
## [2] "j =302 peak =149 savedpeak = 38677"
## [3] "j =302 peak =173 savedpeak = 38701"
## [4] "j =302 peak =209 savedpeak = 38737"
## [5] "j =302 peak =246 savedpeak = 38774"
## [6] "j =302 peak =272 savedpeak = 38800"
## [7] "j =302 peak =306 savedpeak = 38834"
## [8] "j =302 peak =351 savedpeak = 38879"
## [9] "j =302 peak =367 savedpeak = 38895"
## [10] "j =302 peak =411 savedpeak = 38939"
## [11] "j =302 peak =432 savedpeak = 38960"
## [1] "j =314 peak =86 savedpeak = 40150"
## [2] "j =314 peak =121 savedpeak = 40185"
## [3] "j =314 peak =146 savedpeak = 40210"
## [4] "j =314 peak =156 savedpeak = 40220"
## [5] "j =314 peak =172 savedpeak = 40236"
## [6] "j =314 peak =199 savedpeak = 40263"
## [7] "j =314 peak =231 savedpeak = 40295"
## [8] "j =314 peak =259 savedpeak = 40323"
## [9] "j =314 peak =277 savedpeak = 40341"
## [10] "j =314 peak =298 savedpeak = 40362"
## [11] "j =314 peak =317 savedpeak = 40381"
## [12] "j =314 peak =349 savedpeak = 40413"
## [13] "j =314 peak =386 savedpeak = 40450"
## [14] "j =314 peak =412 savedpeak = 40476"
## [1] "j =321 peak =92 savedpeak = 41052"
## [2] "j =321 peak =120 savedpeak = 41080"
## [3] "j =321 peak =139 savedpeak = 41099"
## [4] "j =321 peak =156 savedpeak = 41116"
## [5] "j =321 peak =184 savedpeak = 41144"
## [6] "j =321 peak =230 savedpeak = 41190"
## [7] "j =321 peak =257 savedpeak = 41217"
## [8] "j =321 peak =293 savedpeak = 41253"
## [9] "j =321 peak =336 savedpeak = 41296"
## [10] "j =321 peak =369 savedpeak = 41329"
## [11] "j =321 peak =387 savedpeak = 41347"
## [12] "j =321 peak =405 savedpeak = 41365"
## [13] "j =321 peak =423 savedpeak = 41383"
## [14] "j =321 peak =434 savedpeak = 41394"
## [1] "j =323 peak =80 savedpeak = 41296"
## [2] "j =323 peak =113 savedpeak = 41329"
## [3] "j =323 peak =131 savedpeak = 41347"
## [4] "j =323 peak =149 savedpeak = 41365"
## [5] "j =323 peak =167 savedpeak = 41383"
## [6] "j =323 peak =178 savedpeak = 41394"
## [7] "j =323 peak =197 savedpeak = 41413"
## [8] "j =323 peak =222 savedpeak = 41438"
## [9] "j =323 peak =297 savedpeak = 41513"
## [10] "j =323 peak =316 savedpeak = 41532"
## [11] "j =323 peak =337 savedpeak = 41553"
## [12] "j =323 peak =353 savedpeak = 41569"

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## [13] "j =323 peak =380 savedpeak = 41596"
## [14] "j =323 peak =404 savedpeak = 41620"
## [1] "j =324 peak =94 savedpeak = 41438"
## [2] "j =324 peak =144 savedpeak = 41488"
## [3] "j =324 peak =169 savedpeak = 41513"
## [4] "j =324 peak =188 savedpeak = 41532"
## [5] "j =324 peak =209 savedpeak = 41553"
## [6] "j =324 peak =225 savedpeak = 41569"
## [7] "j =324 peak =252 savedpeak = 41596"
## [8] "j =324 peak =276 savedpeak = 41620"
## [9] "j =324 peak =317 savedpeak = 41661"
## [10] "j =324 peak =344 savedpeak = 41688"
## [11] "j =324 peak =364 savedpeak = 41708"
## [12] "j =324 peak =417 savedpeak = 41761"
## [1] "j =329 peak =100 savedpeak = 42084"
## [2] "j =329 peak =113 savedpeak = 42097"
## [3] "j =329 peak =135 savedpeak = 42119"
## [4] "j =329 peak =147 savedpeak = 42131"
## [5] "j =329 peak =194 savedpeak = 42178"
## [6] "j =329 peak =212 savedpeak = 42196"
## [7] "j =329 peak =240 savedpeak = 42224"
## [8] "j =329 peak =248 savedpeak = 42232"
## [9] "j =329 peak =295 savedpeak = 42279"
## [10] "j =329 peak =319 savedpeak = 42303"
## [11] "j =329 peak =333 savedpeak = 42317"
## [12] "j =329 peak =385 savedpeak = 42369"
## [13] "j =329 peak =411 savedpeak = 42395"
## [1] "j =336 peak =90 savedpeak = 42970"
## [2] "j =336 peak =106 savedpeak = 42986"
## [3] "j =336 peak =127 savedpeak = 43007"
## [4] "j =336 peak =175 savedpeak = 43055"
## [5] "j =336 peak =240 savedpeak = 43120"
## [6] "j =336 peak =270 savedpeak = 43150"
## [7] "j =336 peak =295 savedpeak = 43175"
## [8] "j =336 peak =353 savedpeak = 43233"
## [9] "j =336 peak =367 savedpeak = 43247"
## [10] "j =336 peak =396 savedpeak = 43276"
## [11] "j =336 peak =432 savedpeak = 43312"
## [1] "j =338 peak =97 savedpeak = 43233"
## [2] "j =338 peak =111 savedpeak = 43247"
## [3] "j =338 peak =140 savedpeak = 43276"
## [4] "j =338 peak =176 savedpeak = 43312"
## [5] "j =338 peak =202 savedpeak = 43338"
## [6] "j =338 peak =265 savedpeak = 43401"
## [7] "j =338 peak =301 savedpeak = 43437"
## [8] "j =338 peak =317 savedpeak = 43453"
## [9] "j =338 peak =344 savedpeak = 43480"
## [10] "j =338 peak =366 savedpeak = 43502"
## [11] "j =338 peak =398 savedpeak = 43534"
## [1] "j =339 peak =137 savedpeak = 43401"
## [2] "j =339 peak =173 savedpeak = 43437"
## [3] "j =339 peak =189 savedpeak = 43453"
## [4] "j =339 peak =216 savedpeak = 43480"
## [5] "j =339 peak =238 savedpeak = 43502"

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## [6] "j =339 peak =270 savedpeak = 43534"
## [7] "j =339 peak =328 savedpeak = 43592"
## [8] "j =339 peak =352 savedpeak = 43616"
## [9] "j =339 peak =379 savedpeak = 43643"
## [10] "j =339 peak =421 savedpeak = 43685"
## [1] "j =350 peak =79 savedpeak = 44751"
## [2] "j =350 peak =107 savedpeak = 44779"
## [3] "j =350 peak =142 savedpeak = 44814"
## [4] "j =350 peak =185 savedpeak = 44857"
## [5] "j =350 peak =213 savedpeak = 44885"
## [6] "j =350 peak =258 savedpeak = 44930"
## [7] "j =350 peak =274 savedpeak = 44946"
## [8] "j =350 peak =294 savedpeak = 44966"
## [9] "j =350 peak =346 savedpeak = 45018"
## [10] "j =350 peak =364 savedpeak = 45036"
## [11] "j =350 peak =375 savedpeak = 45047"
## [12] "j =350 peak =400 savedpeak = 45072"
## [13] "j =350 peak =429 savedpeak = 45101"
## [1] "j =351 peak =85 savedpeak = 44885"
## [2] "j =351 peak =130 savedpeak = 44930"
## [3] "j =351 peak =146 savedpeak = 44946"
## [4] "j =351 peak =166 savedpeak = 44966"
## [5] "j =351 peak =218 savedpeak = 45018"
## [6] "j =351 peak =236 savedpeak = 45036"
## [7] "j =351 peak =247 savedpeak = 45047"
## [8] "j =351 peak =272 savedpeak = 45072"
## [9] "j =351 peak =301 savedpeak = 45101"
## [10] "j =351 peak =340 savedpeak = 45140"
## [11] "j =351 peak =370 savedpeak = 45170"
## [12] "j =351 peak =405 savedpeak = 45205"
## [13] "j =351 peak =425 savedpeak = 45225"
## [1] "j =352 peak =90 savedpeak = 45018"
## [2] "j =352 peak =108 savedpeak = 45036"
## [3] "j =352 peak =119 savedpeak = 45047"
## [4] "j =352 peak =144 savedpeak = 45072"
## [5] "j =352 peak =173 savedpeak = 45101"
## [6] "j =352 peak =212 savedpeak = 45140"
## [7] "j =352 peak =242 savedpeak = 45170"
## [8] "j =352 peak =277 savedpeak = 45205"
## [9] "j =352 peak =297 savedpeak = 45225"
## [10] "j =352 peak =324 savedpeak = 45252"
## [11] "j =352 peak =368 savedpeak = 45296"
## [12] "j =352 peak =387 savedpeak = 45315"
## [1] "j =353 peak =84 savedpeak = 45140"
## [2] "j =353 peak =114 savedpeak = 45170"
## [3] "j =353 peak =149 savedpeak = 45205"
## [4] "j =353 peak =169 savedpeak = 45225"
## [5] "j =353 peak =196 savedpeak = 45252"
## [6] "j =353 peak =240 savedpeak = 45296"
## [7] "j =353 peak =259 savedpeak = 45315"
## [8] "j =353 peak =332 savedpeak = 45388"
## [9] "j =353 peak =367 savedpeak = 45423"
## [10] "j =353 peak =384 savedpeak = 45440"
## [11] "j =353 peak =412 savedpeak = 45468"

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## [12] "j =353 peak =429 savedpeak = 45485"
## [1] "j =355 peak =76 savedpeak = 45388"
## [2] "j =355 peak =111 savedpeak = 45423"
## [3] "j =355 peak =128 savedpeak = 45440"
## [4] "j =355 peak =156 savedpeak = 45468"
## [5] "j =355 peak =173 savedpeak = 45485"
## [6] "j =355 peak =187 savedpeak = 45499"
## [7] "j =355 peak =217 savedpeak = 45529"
## [8] "j =355 peak =259 savedpeak = 45571"
## [9] "j =355 peak =283 savedpeak = 45595"
## [10] "j =355 peak =345 savedpeak = 45657"
## [11] "j =355 peak =364 savedpeak = 45676"
## [12] "j =355 peak =402 savedpeak = 45714"
## [1] "j =357 peak =89 savedpeak = 45657"
## [2] "j =357 peak =108 savedpeak = 45676"
## [3] "j =357 peak =146 savedpeak = 45714"
## [4] "j =357 peak =190 savedpeak = 45758"
## [5] "j =357 peak =211 savedpeak = 45779"
## [6] "j =357 peak =231 savedpeak = 45799"
## [7] "j =357 peak =240 savedpeak = 45808"
## [8] "j =357 peak =259 savedpeak = 45827"
## [9] "j =357 peak =273 savedpeak = 45841"
## [10] "j =357 peak =292 savedpeak = 45860"
## [11] "j =357 peak =325 savedpeak = 45893"
## [12] "j =357 peak =342 savedpeak = 45910"
## [13] "j =357 peak =362 savedpeak = 45930"
## [14] "j =357 peak =381 savedpeak = 45949"
## [15] "j =357 peak =424 savedpeak = 45992"
## [1] "j =358 peak =83 savedpeak = 45779"
## [2] "j =358 peak =103 savedpeak = 45799"
## [3] "j =358 peak =112 savedpeak = 45808"
## [4] "j =358 peak =131 savedpeak = 45827"
## [5] "j =358 peak =145 savedpeak = 45841"
## [6] "j =358 peak =164 savedpeak = 45860"
## [7] "j =358 peak =197 savedpeak = 45893"
## [8] "j =358 peak =214 savedpeak = 45910"
## [9] "j =358 peak =234 savedpeak = 45930"
## [10] "j =358 peak =253 savedpeak = 45949"
## [11] "j =358 peak =296 savedpeak = 45992"
## [12] "j =358 peak =315 savedpeak = 46011"
## [13] "j =358 peak =329 savedpeak = 46025"
## [14] "j =358 peak =352 savedpeak = 46048"
## [15] "j =358 peak =374 savedpeak = 46070"
## [16] "j =358 peak =411 savedpeak = 46107"
## [1] "j =359 peak =86 savedpeak = 45910"
## [2] "j =359 peak =106 savedpeak = 45930"
## [3] "j =359 peak =125 savedpeak = 45949"
## [4] "j =359 peak =168 savedpeak = 45992"
## [5] "j =359 peak =187 savedpeak = 46011"
## [6] "j =359 peak =201 savedpeak = 46025"
## [7] "j =359 peak =224 savedpeak = 46048"
## [8] "j =359 peak =246 savedpeak = 46070"
## [9] "j =359 peak =283 savedpeak = 46107"
## [10] "j =359 peak =340 savedpeak = 46164"

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## [11] "j =359 peak =375 savedpeak = 46199"
## [12] "j =359 peak =392 savedpeak = 46216"
## [13] "j =359 peak =426 savedpeak = 46250"
## [1] "j =360 peak =96 savedpeak = 46048"
## [2] "j =360 peak =118 savedpeak = 46070"
## [3] "j =360 peak =155 savedpeak = 46107"
## [4] "j =360 peak =212 savedpeak = 46164"
## [5] "j =360 peak =247 savedpeak = 46199"
## [6] "j =360 peak =264 savedpeak = 46216"
## [7] "j =360 peak =298 savedpeak = 46250"
## [8] "j =360 peak =310 savedpeak = 46262"
## [9] "j =360 peak =351 savedpeak = 46303"
## [10] "j =360 peak =388 savedpeak = 46340"
## [11] "j =360 peak =420 savedpeak = 46372"
## [1] "j =361 peak =84 savedpeak = 46164"
## [2] "j =361 peak =119 savedpeak = 46199"
## [3] "j =361 peak =136 savedpeak = 46216"
## [4] "j =361 peak =170 savedpeak = 46250"
## [5] "j =361 peak =182 savedpeak = 46262"
## [6] "j =361 peak =223 savedpeak = 46303"
## [7] "j =361 peak =260 savedpeak = 46340"
## [8] "j =361 peak =292 savedpeak = 46372"
## [9] "j =361 peak =336 savedpeak = 46416"
## [10] "j =361 peak =348 savedpeak = 46428"
## [11] "j =361 peak =409 savedpeak = 46489"
## [12] "j =361 peak =437 savedpeak = 46517"
## [1] "j =362 peak =95 savedpeak = 46303"
## [2] "j =362 peak =132 savedpeak = 46340"
## [3] "j =362 peak =164 savedpeak = 46372"
## [4] "j =362 peak =208 savedpeak = 46416"
## [5] "j =362 peak =220 savedpeak = 46428"
## [6] "j =362 peak =281 savedpeak = 46489"
## [7] "j =362 peak =309 savedpeak = 46517"
## [8] "j =362 peak =325 savedpeak = 46533"
## [9] "j =362 peak =341 savedpeak = 46549"
## [10] "j =362 peak =362 savedpeak = 46570"
## [11] "j =362 peak =415 savedpeak = 46623"
## [1] "j =364 peak =85 savedpeak = 46549"
## [2] "j =364 peak =106 savedpeak = 46570"
## [3] "j =364 peak =127 savedpeak = 46591"
## [4] "j =364 peak =159 savedpeak = 46623"
## [5] "j =364 peak =188 savedpeak = 46652"
## [6] "j =364 peak =201 savedpeak = 46665"
## [7] "j =364 peak =223 savedpeak = 46687"
## [8] "j =364 peak =261 savedpeak = 46725"
## [9] "j =364 peak =294 savedpeak = 46758"
## [10] "j =364 peak =310 savedpeak = 46774"
## [11] "j =364 peak =386 savedpeak = 46850"
## [1] "j =365 peak =95 savedpeak = 46687"
## [2] "j =365 peak =133 savedpeak = 46725"
## [3] "j =365 peak =166 savedpeak = 46758"
## [4] "j =365 peak =182 savedpeak = 46774"
## [5] "j =365 peak =258 savedpeak = 46850"
## [6] "j =365 peak =328 savedpeak = 46920"

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## [7] "j =365 peak =351 savedpeak = 46943"
## [8] "j =365 peak =375 savedpeak = 46967"
## [9] "j =365 peak =390 savedpeak = 46982"
## [10] "j =365 peak =411 savedpeak = 47003"
## [11] "j =365 peak =434 savedpeak = 47026"
## [1] "j =366 peak =130 savedpeak = 46850"
## [2] "j =366 peak =200 savedpeak = 46920"
## [3] "j =366 peak =223 savedpeak = 46943"
## [4] "j =366 peak =247 savedpeak = 46967"
## [5] "j =366 peak =262 savedpeak = 46982"
## [6] "j =366 peak =283 savedpeak = 47003"
## [7] "j =366 peak =306 savedpeak = 47026"
## [8] "j =366 peak =363 savedpeak = 47083"
## [9] "j =366 peak =396 savedpeak = 47116"
## [10] "j =366 peak =414 savedpeak = 47134"
## [11] "j =366 peak =433 savedpeak = 47153"
## [1] "j =367 peak =95 savedpeak = 46943"
## [2] "j =367 peak =119 savedpeak = 46967"
## [3] "j =367 peak =134 savedpeak = 46982"
## [4] "j =367 peak =155 savedpeak = 47003"
## [5] "j =367 peak =178 savedpeak = 47026"
## [6] "j =367 peak =235 savedpeak = 47083"
## [7] "j =367 peak =268 savedpeak = 47116"
## [8] "j =367 peak =286 savedpeak = 47134"
## [9] "j =367 peak =305 savedpeak = 47153"
## [10] "j =367 peak =335 savedpeak = 47183"
## [11] "j =367 peak =358 savedpeak = 47206"
## [12] "j =367 peak =388 savedpeak = 47236"
## [13] "j =367 peak =434 savedpeak = 47282"
## [1] "j =368 peak =107 savedpeak = 47083"
## [2] "j =368 peak =140 savedpeak = 47116"
## [3] "j =368 peak =158 savedpeak = 47134"
## [4] "j =368 peak =177 savedpeak = 47153"
## [5] "j =368 peak =207 savedpeak = 47183"
## [6] "j =368 peak =230 savedpeak = 47206"
## [7] "j =368 peak =260 savedpeak = 47236"
## [8] "j =368 peak =306 savedpeak = 47282"
## [9] "j =368 peak =336 savedpeak = 47312"
## [10] "j =368 peak =371 savedpeak = 47347"
## [11] "j =368 peak =395 savedpeak = 47371"
## [12] "j =368 peak =404 savedpeak = 47380"
## [1] "j =372 peak =99 savedpeak = 47587"
## [2] "j =372 peak =141 savedpeak = 47629"
## [3] "j =372 peak =157 savedpeak = 47645"
## [4] "j =372 peak =178 savedpeak = 47666"
## [5] "j =372 peak =207 savedpeak = 47695"
## [6] "j =372 peak =243 savedpeak = 47731"
## [7] "j =372 peak =291 savedpeak = 47779"
## [8] "j =372 peak =317 savedpeak = 47805"
## [9] "j =372 peak =341 savedpeak = 47829"
## [10] "j =372 peak =362 savedpeak = 47850"
## [1] "j =373 peak =79 savedpeak = 47695"
## [2] "j =373 peak =115 savedpeak = 47731"
## [3] "j =373 peak =163 savedpeak = 47779"

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## [4] "j =373 peak =189 savedpeak = 47805"
## [5] "j =373 peak =213 savedpeak = 47829"
## [6] "j =373 peak =234 savedpeak = 47850"
## [7] "j =373 peak =313 savedpeak = 47929"
## [8] "j =373 peak =348 savedpeak = 47964"
## [9] "j =373 peak =375 savedpeak = 47991"
## [10] "j =373 peak =414 savedpeak = 48030"
## [1] "j =375 peak =92 savedpeak = 47964"
## [2] "j =375 peak =119 savedpeak = 47991"
## [3] "j =375 peak =158 savedpeak = 48030"
## [4] "j =375 peak =191 savedpeak = 48063"
## [5] "j =375 peak =204 savedpeak = 48076"
## [6] "j =375 peak =217 savedpeak = 48089"
## [7] "j =375 peak =241 savedpeak = 48113"
## [8] "j =375 peak =268 savedpeak = 48140"
## [9] "j =375 peak =287 savedpeak = 48159"
## [10] "j =375 peak =320 savedpeak = 48192"
## [11] "j =375 peak =360 savedpeak = 48232"
## [12] "j =375 peak =376 savedpeak = 48248"
## [13] "j =375 peak =425 savedpeak = 48297"
## [1] "j =376 peak =76 savedpeak = 48076"
## [2] "j =376 peak =89 savedpeak = 48089"
## [3] "j =376 peak =113 savedpeak = 48113"
## [4] "j =376 peak =140 savedpeak = 48140"
## [5] "j =376 peak =159 savedpeak = 48159"
## [6] "j =376 peak =192 savedpeak = 48192"
## [7] "j =376 peak =232 savedpeak = 48232"
## [8] "j =376 peak =248 savedpeak = 48248"
## [9] "j =376 peak =297 savedpeak = 48297"
## [10] "j =376 peak =332 savedpeak = 48332"
## [11] "j =376 peak =371 savedpeak = 48371"
## [12] "j =376 peak =399 savedpeak = 48399"
## [1] "j =378 peak =76 savedpeak = 48332"
## [2] "j =378 peak =115 savedpeak = 48371"
## [3] "j =378 peak =143 savedpeak = 48399"
## [4] "j =378 peak =195 savedpeak = 48451"
## [5] "j =378 peak =248 savedpeak = 48504"
## [6] "j =378 peak =282 savedpeak = 48538"
## [7] "j =378 peak =318 savedpeak = 48574"
## [8] "j =378 peak =348 savedpeak = 48604"
## [9] "j =378 peak =381 savedpeak = 48637"
## [10] "j =378 peak =402 savedpeak = 48658"
## [1] "j =383 peak =87 savedpeak = 48983"
## [2] "j =383 peak =103 savedpeak = 48999"
## [3] "j =383 peak =132 savedpeak = 49028"
## [4] "j =383 peak =192 savedpeak = 49088"
## [5] "j =383 peak =217 savedpeak = 49113"
## [6] "j =383 peak =244 savedpeak = 49140"
## [7] "j =383 peak =264 savedpeak = 49160"
## [8] "j =383 peak =282 savedpeak = 49178"
## [9] "j =383 peak =299 savedpeak = 49195"
## [10] "j =383 peak =337 savedpeak = 49233"
## [11] "j =383 peak =351 savedpeak = 49247"
## [12] "j =383 peak =382 savedpeak = 49278"

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## [13] "j =383 peak =397 savedpeak = 49293"
## [1] "j =385 peak =81 savedpeak = 49233"
## [2] "j =385 peak =95 savedpeak = 49247"
## [3] "j =385 peak =126 savedpeak = 49278"
## [4] "j =385 peak =141 savedpeak = 49293"
## [5] "j =385 peak =218 savedpeak = 49370"
## [6] "j =385 peak =281 savedpeak = 49433"
## [7] "j =385 peak =310 savedpeak = 49462"
## [8] "j =385 peak =319 savedpeak = 49471"
## [9] "j =385 peak =357 savedpeak = 49509"
## [10] "j =385 peak =396 savedpeak = 49548"
## [11] "j =385 peak =414 savedpeak = 49566"
## [1] "j =386 peak =90 savedpeak = 49370"
## [2] "j =386 peak =153 savedpeak = 49433"
## [3] "j =386 peak =182 savedpeak = 49462"
## [4] "j =386 peak =191 savedpeak = 49471"
## [5] "j =386 peak =229 savedpeak = 49509"
## [6] "j =386 peak =268 savedpeak = 49548"
## [7] "j =386 peak =286 savedpeak = 49566"
## [8] "j =386 peak =321 savedpeak = 49601"
## [9] "j =386 peak =351 savedpeak = 49631"
## [10] "j =386 peak =404 savedpeak = 49684"
## [11] "j =386 peak =419 savedpeak = 49699"
## [1] "j =387 peak =101 savedpeak = 49509"
## [2] "j =387 peak =140 savedpeak = 49548"
## [3] "j =387 peak =158 savedpeak = 49566"
## [4] "j =387 peak =193 savedpeak = 49601"
## [5] "j =387 peak =223 savedpeak = 49631"
## [6] "j =387 peak =276 savedpeak = 49684"
## [7] "j =387 peak =291 savedpeak = 49699"
## [8] "j =387 peak =325 savedpeak = 49733"
## [9] "j =387 peak =340 savedpeak = 49748"
## [10] "j =387 peak =392 savedpeak = 49800"
## [11] "j =387 peak =422 savedpeak = 49830"
## [1] "j =388 peak =95 savedpeak = 49631"
## [2] "j =388 peak =148 savedpeak = 49684"
## [3] "j =388 peak =163 savedpeak = 49699"
## [4] "j =388 peak =197 savedpeak = 49733"
## [5] "j =388 peak =212 savedpeak = 49748"
## [6] "j =388 peak =264 savedpeak = 49800"
## [7] "j =388 peak =294 savedpeak = 49830"
## [8] "j =388 peak =317 savedpeak = 49853"
## [9] "j =388 peak =352 savedpeak = 49888"
## [10] "j =388 peak =383 savedpeak = 49919"
## [11] "j =388 peak =407 savedpeak = 49943"
## [12] "j =388 peak =427 savedpeak = 49963"
## [1] "j =389 peak =84 savedpeak = 49748"
## [2] "j =389 peak =136 savedpeak = 49800"
## [3] "j =389 peak =166 savedpeak = 49830"
## [4] "j =389 peak =189 savedpeak = 49853"
## [5] "j =389 peak =224 savedpeak = 49888"
## [6] "j =389 peak =255 savedpeak = 49919"
## [7] "j =389 peak =279 savedpeak = 49943"
## [8] "j =389 peak =299 savedpeak = 49963"

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## [9] "j =389 peak =313 savedpeak = 49977"
## [10] "j =389 peak =329 savedpeak = 49993"
## [11] "j =389 peak =347 savedpeak = 50011"
## [12] "j =389 peak =388 savedpeak = 50052"
## [13] "j =389 peak =434 savedpeak = 50098"
## Spectrum exam_DNMR_Me2NCOMe has 0 peaks.
## [1] "j =7 peak =91 savedpeak = 859" "j =7 peak =117 savedpeak = 885"
## [3] "j =7 peak =158 savedpeak = 926" "j =7 peak =192 savedpeak = 960"
## [5] "j =7 peak =224 savedpeak = 992" "j =7 peak =250 savedpeak = 1018"
## [7] "j =7 peak =261 savedpeak = 1029" "j =7 peak =297 savedpeak = 1065"
## [9] "j =7 peak =347 savedpeak = 1115" "j =7 peak =365 savedpeak = 1133"
## [11] "j =7 peak =388 savedpeak = 1156" "j =7 peak =417 savedpeak = 1185"
## [1] "j =8 peak =96 savedpeak = 992" "j =8 peak =122 savedpeak = 1018"
## [3] "j =8 peak =133 savedpeak = 1029" "j =8 peak =169 savedpeak = 1065"
## [5] "j =8 peak =219 savedpeak = 1115" "j =8 peak =237 savedpeak = 1133"
## [7] "j =8 peak =260 savedpeak = 1156" "j =8 peak =289 savedpeak = 1185"
## [9] "j =8 peak =310 savedpeak = 1206" "j =8 peak =334 savedpeak = 1230"
## [11] "j =8 peak =380 savedpeak = 1276"
## [1] "j =9 peak =91 savedpeak = 1115" "j =9 peak =109 savedpeak = 1133"
## [3] "j =9 peak =132 savedpeak = 1156" "j =9 peak =161 savedpeak = 1185"
## [5] "j =9 peak =182 savedpeak = 1206" "j =9 peak =206 savedpeak = 1230"
## [7] "j =9 peak =252 savedpeak = 1276" "j =9 peak =312 savedpeak = 1336"
## [9] "j =9 peak =353 savedpeak = 1377" "j =9 peak =392 savedpeak = 1416"
## [1] "j =10 peak =78 savedpeak = 1230" "j =10 peak =124 savedpeak = 1276"
## [3] "j =10 peak =184 savedpeak = 1336" "j =10 peak =225 savedpeak = 1377"
## [5] "j =10 peak =264 savedpeak = 1416" "j =10 peak =348 savedpeak = 1500"
## [7] "j =10 peak =379 savedpeak = 1531" "j =10 peak =407 savedpeak = 1559"
## [9] "j =10 peak =434 savedpeak = 1586"
## [1] "j =11 peak =97 savedpeak = 1377" "j =11 peak =136 savedpeak = 1416"
## [3] "j =11 peak =220 savedpeak = 1500" "j =11 peak =256 savedpeak = 1536"
## [5] "j =11 peak =279 savedpeak = 1559" "j =11 peak =306 savedpeak = 1586"
## [7] "j =11 peak =320 savedpeak = 1600" "j =11 peak =345 savedpeak = 1625"
## [9] "j =11 peak =383 savedpeak = 1663" "j =11 peak =425 savedpeak = 1705"
## [1] "j =12 peak =92 savedpeak = 1500" "j =12 peak =128 savedpeak = 1536"
## [3] "j =12 peak =151 savedpeak = 1559" "j =12 peak =178 savedpeak = 1586"
## [5] "j =12 peak =192 savedpeak = 1600" "j =12 peak =217 savedpeak = 1625"
## [7] "j =12 peak =255 savedpeak = 1663" "j =12 peak =297 savedpeak = 1705"
## [9] "j =12 peak =318 savedpeak = 1726" "j =12 peak =338 savedpeak = 1746"
## [11] "j =12 peak =365 savedpeak = 1773" "j =12 peak =400 savedpeak = 1808"
## [13] "j =12 peak =433 savedpeak = 1841"
## [1] "j =19 peak =77 savedpeak = 2381" "j =19 peak =207 savedpeak = 2511"
## [3] "j =19 peak =243 savedpeak = 2547" "j =19 peak =351 savedpeak = 2655"
## [5] "j =19 peak =409 savedpeak = 2713"
## [1] "j =22 peak =350 savedpeak = 3038"
## [1] "j =23 peak =222 savedpeak = 3038" "j =23 peak =323 savedpeak = 3139"
## [3] "j =23 peak =388 savedpeak = 3204"
## [1] "j =24 peak =94 savedpeak = 3038" "j =24 peak =195 savedpeak = 3139"
## [3] "j =24 peak =260 savedpeak = 3204"
## [1] "j =25 peak =132 savedpeak = 3204" "j =25 peak =421 savedpeak = 3493"
## [1] "j =26 peak =91 savedpeak = 3291" "j =26 peak =293 savedpeak = 3493"
## [3] "j =26 peak =387 savedpeak = 3587"
## [1] "j =32 peak =153 savedpeak = 4121" "j =32 peak =348 savedpeak = 4316"
## [3] "j =32 peak =392 savedpeak = 4360"
## [1] "j =43 peak =139 savedpeak = 5515" "j =43 peak =175 savedpeak = 5551"

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## [3] "j =43 peak =219 savedpeak = 5595" "j =43 peak =284 savedpeak = 5660"
## [5] "j =43 peak =371 savedpeak = 5747"
## [1] "j =44 peak =156 savedpeak = 5660" "j =44 peak =243 savedpeak = 5747"
## [3] "j =44 peak =322 savedpeak = 5826" "j =44 peak =351 savedpeak = 5855"
## [1] "j =46 peak =376 savedpeak = 6136" "j =46 peak =406 savedpeak = 6166"
## [1] "j =47 peak =248 savedpeak = 6136" "j =47 peak =278 savedpeak = 6166"
## [1] "j =48 peak =120 savedpeak = 6136" "j =48 peak =150 savedpeak = 6166"
## [3] "j =48 peak =376 savedpeak = 6392"
## [1] "j =49 peak =248 savedpeak = 6392"
## [1] "j =50 peak =120 savedpeak = 6392"
## [1] "j =51 peak =229 savedpeak = 6629" "j =51 peak =314 savedpeak = 6714"
## [1] "j =52 peak =315 savedpeak = 6843"
## [1] "j =53 peak =187 savedpeak = 6843" "j =53 peak =365 savedpeak = 7021"
## [3] "j =53 peak =379 savedpeak = 7035" "j =53 peak =420 savedpeak = 7076"
## [5] "j =53 peak =431 savedpeak = 7087"
## [1] "j =54 peak =237 savedpeak = 7021" "j =54 peak =251 savedpeak = 7035"
## [3] "j =54 peak =292 savedpeak = 7076" "j =54 peak =303 savedpeak = 7087"
## [5] "j =54 peak =399 savedpeak = 7183" "j =54 peak =426 savedpeak = 7210"
## [1] "j =55 peak =109 savedpeak = 7021" "j =55 peak =123 savedpeak = 7035"
## [3] "j =55 peak =164 savedpeak = 7076" "j =55 peak =175 savedpeak = 7087"
## [5] "j =55 peak =271 savedpeak = 7183" "j =55 peak =298 savedpeak = 7210"
## [7] "j =55 peak =325 savedpeak = 7237"
## [1] "j =56 peak =143 savedpeak = 7183" "j =56 peak =170 savedpeak = 7210"
## [3] "j =56 peak =197 savedpeak = 7237"
## [1] "j =57 peak =100 savedpeak = 7268" "j =57 peak =133 savedpeak = 7301"
## [3] "j =57 peak =327 savedpeak = 7495"
## [1] "j =58 peak =117 savedpeak = 7413" "j =58 peak =138 savedpeak = 7434"
## [3] "j =58 peak =199 savedpeak = 7495" "j =58 peak =216 savedpeak = 7512"
## [5] "j =58 peak =301 savedpeak = 7597" "j =58 peak =331 savedpeak = 7627"
## [7] "j =58 peak =361 savedpeak = 7657" "j =58 peak =373 savedpeak = 7669"
## [9] "j =58 peak =413 savedpeak = 7709"
## [1] "j =59 peak =88 savedpeak = 7512" "j =59 peak =173 savedpeak = 7597"
## [3] "j =59 peak =203 savedpeak = 7627" "j =59 peak =233 savedpeak = 7657"
## [5] "j =59 peak =245 savedpeak = 7669" "j =59 peak =285 savedpeak = 7709"
## [7] "j =59 peak =310 savedpeak = 7734" "j =59 peak =342 savedpeak = 7766"
## [1] "j =60 peak =105 savedpeak = 7657" "j =60 peak =117 savedpeak = 7669"
## [3] "j =60 peak =157 savedpeak = 7709" "j =60 peak =177 savedpeak = 7729"
## [5] "j =60 peak =214 savedpeak = 7766"
## [1] "j =64 peak =100 savedpeak = 8164" "j =64 peak =130 savedpeak = 8194"
## [3] "j =64 peak =412 savedpeak = 8476" "j =64 peak =426 savedpeak = 8490"
## [1] "j =65 peak =284 savedpeak = 8476" "j =65 peak =298 savedpeak = 8490"
## [1] "j =66 peak =156 savedpeak = 8476" "j =66 peak =170 savedpeak = 8490"
## [1] "j =67 peak =332 savedpeak = 8780"
## [1] "j =68 peak =316 savedpeak = 8892"
## [1] "j =69 peak =188 savedpeak = 8892" "j =69 peak =352 savedpeak = 9056"
## [1] "j =70 peak =224 savedpeak = 9056"
## [1] "j =71 peak =96 savedpeak = 9056"
## [1] "j =72 peak =140 savedpeak = 9228" "j =72 peak =183 savedpeak = 9271"
## [3] "j =72 peak =242 savedpeak = 9330" "j =72 peak =342 savedpeak = 9430"
## [5] "j =72 peak =378 savedpeak = 9466"
## [1] "j =73 peak =114 savedpeak = 9330" "j =73 peak =185 savedpeak = 9401"
## [3] "j =73 peak =214 savedpeak = 9430" "j =73 peak =250 savedpeak = 9466"
## [5] "j =73 peak =301 savedpeak = 9517" "j =73 peak =330 savedpeak = 9546"
## [7] "j =73 peak =395 savedpeak = 9611"

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## [1] "j =74 peak =202 savedpeak = 9546" "j =74 peak =267 savedpeak = 9611"
## [3] "j =74 peak =353 savedpeak = 9697"
## [1] "j =75 peak =139 savedpeak = 9611" "j =75 peak =347 savedpeak = 9819"
## [3] "j =75 peak =374 savedpeak = 9846" "j =75 peak =401 savedpeak = 9873"
## [5] "j =75 peak =429 savedpeak = 9901"
## [1] "j =76 peak =219 savedpeak = 9819" "j =76 peak =246 savedpeak = 9846"
## [3] "j =76 peak =273 savedpeak = 9873" "j =76 peak =301 savedpeak = 9901"
## [5] "j =76 peak =327 savedpeak = 9927"
## [1] "j =77 peak =91 savedpeak = 9819" "j =77 peak =118 savedpeak = 9846"
## [3] "j =77 peak =145 savedpeak = 9873" "j =77 peak =173 savedpeak = 9901"
## [5] "j =77 peak =199 savedpeak = 9927" "j =77 peak =301 savedpeak = 10029"
## [7] "j =77 peak =388 savedpeak = 10116"
## [1] "j =78 peak =173 savedpeak = 10029" "j =78 peak =260 savedpeak = 10116"
## [3] "j =78 peak =353 savedpeak = 10209"
## [1] "j =79 peak =132 savedpeak = 10116" "j =79 peak =198 savedpeak = 10182"
## [3] "j =79 peak =225 savedpeak = 10209" "j =79 peak =342 savedpeak = 10326"
## [5] "j =79 peak =363 savedpeak = 10347" "j =79 peak =385 savedpeak = 10369"
## [1] "j =81 peak =86 savedpeak = 10326" "j =81 peak =107 savedpeak = 10347"
## [3] "j =81 peak =128 savedpeak = 10368" "j =81 peak =216 savedpeak = 10456"
## [5] "j =81 peak =243 savedpeak = 10483" "j =81 peak =395 savedpeak = 10635"
## [1] "j =82 peak =88 savedpeak = 10456" "j =82 peak =115 savedpeak = 10483"
## [3] "j =82 peak =267 savedpeak = 10635" "j =82 peak =392 savedpeak = 10760"
## [5] "j =82 peak =410 savedpeak = 10778"
## [1] "j =83 peak =139 savedpeak = 10635" "j =83 peak =261 savedpeak = 10757"
## [3] "j =83 peak =282 savedpeak = 10778" "j =83 peak =403 savedpeak = 10899"
## [1] "j =84 peak =133 savedpeak = 10757" "j =84 peak =154 savedpeak = 10778"
## [3] "j =84 peak =275 savedpeak = 10899" "j =84 peak =360 savedpeak = 10984"
## [5] "j =84 peak =395 savedpeak = 11019" "j =84 peak =423 savedpeak = 11047"
## [1] "j =85 peak =147 savedpeak = 10899" "j =85 peak =231 savedpeak = 10983"
## [3] "j =85 peak =266 savedpeak = 11018" "j =85 peak =295 savedpeak = 11047"
## [5] "j =85 peak =367 savedpeak = 11119"
## [1] "j =87 peak =111 savedpeak = 11119" "j =87 peak =169 savedpeak = 11177"
## [3] "j =87 peak =228 savedpeak = 11236" "j =87 peak =321 savedpeak = 11329"
## [1] "j =88 peak =100 savedpeak = 11236" "j =88 peak =193 savedpeak = 11329"
## [3] "j =88 peak =352 savedpeak = 11488" "j =88 peak =399 savedpeak = 11535"
## [5] "j =88 peak =407 savedpeak = 11543"
## [1] "j =89 peak =271 savedpeak = 11535" "j =89 peak =279 savedpeak = 11543"
## [3] "j =89 peak =325 savedpeak = 11589" "j =89 peak =405 savedpeak = 11669"
## [1] "j =90 peak =96 savedpeak = 11488" "j =90 peak =143 savedpeak = 11535"
## [3] "j =90 peak =151 savedpeak = 11543" "j =90 peak =197 savedpeak = 11589"
## [5] "j =90 peak =277 savedpeak = 11669" "j =90 peak =328 savedpeak = 11720"
## [1] "j =91 peak =148 savedpeak = 11668" "j =91 peak =200 savedpeak = 11720"
## [3] "j =91 peak =373 savedpeak = 11893"
## [1] "j =94 peak =90 savedpeak = 11994" "j =94 peak =183 savedpeak = 12087"
## [3] "j =94 peak =233 savedpeak = 12137" "j =94 peak =278 savedpeak = 12182"
## [5] "j =94 peak =313 savedpeak = 12217" "j =94 peak =356 savedpeak = 12260"
## [7] "j =94 peak =393 savedpeak = 12297" "j =94 peak =422 savedpeak = 12326"
## [1] "j =95 peak =105 savedpeak = 12137" "j =95 peak =150 savedpeak = 12182"
## [3] "j =95 peak =185 savedpeak = 12217" "j =95 peak =228 savedpeak = 12260"
## [5] "j =95 peak =265 savedpeak = 12297" "j =95 peak =294 savedpeak = 12326"
## [7] "j =95 peak =358 savedpeak = 12390" "j =95 peak =395 savedpeak = 12427"
## [1] "j =96 peak =100 savedpeak = 12260" "j =96 peak =137 savedpeak = 12297"
## [3] "j =96 peak =166 savedpeak = 12326" "j =96 peak =230 savedpeak = 12390"
## [5] "j =96 peak =267 savedpeak = 12427" "j =96 peak =310 savedpeak = 12470"

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## [7] "j =96 peak =347 savedpeak = 12507" "j =96 peak =376 savedpeak = 12536"
## [1] "j =97 peak =102 savedpeak = 12390" "j =97 peak =140 savedpeak = 12428"
## [3] "j =97 peak =182 savedpeak = 12470" "j =97 peak =219 savedpeak = 12507"
## [5] "j =97 peak =248 savedpeak = 12536" "j =97 peak =320 savedpeak = 12608"
## [7] "j =97 peak =370 savedpeak = 12658"
## [1] "j =98 peak =91 savedpeak = 12507" "j =98 peak =120 savedpeak = 12536"
## [3] "j =98 peak =192 savedpeak = 12608" "j =98 peak =242 savedpeak = 12658"
## [5] "j =98 peak =357 savedpeak = 12773" "j =98 peak =415 savedpeak = 12831"
## [1] "j =100 peak =101 savedpeak = 12773"
## [2] "j =100 peak =161 savedpeak = 12833"
## [3] "j =100 peak =202 savedpeak = 12874"
## [4] "j =100 peak =268 savedpeak = 12940"
## [5] "j =100 peak =303 savedpeak = 12975"
## [6] "j =100 peak =332 savedpeak = 13004"
## [7] "j =100 peak =375 savedpeak = 13047"
## [8] "j =100 peak =398 savedpeak = 13070"
## [9] "j =100 peak =425 savedpeak = 13097"
## [1] "j =101 peak =140 savedpeak = 12940"
## [2] "j =101 peak =175 savedpeak = 12975"
## [3] "j =101 peak =205 savedpeak = 13005"
## [4] "j =101 peak =247 savedpeak = 13047"
## [5] "j =101 peak =270 savedpeak = 13070"
## [6] "j =101 peak =297 savedpeak = 13097"
## [7] "j =101 peak =390 savedpeak = 13190"
## [1] "j =102 peak =120 savedpeak = 13048"
## [2] "j =102 peak =142 savedpeak = 13070"
## [3] "j =102 peak =166 savedpeak = 13094"
## [4] "j =102 peak =262 savedpeak = 13190"
## [5] "j =102 peak =329 savedpeak = 13257"
## [1] "j =103 peak =128 savedpeak = 13184"
## [2] "j =103 peak =201 savedpeak = 13257"
## [3] "j =103 peak =310 savedpeak = 13366"
## [4] "j =103 peak =378 savedpeak = 13434"
## [1] "j =122 peak =146 savedpeak = 15634"
## [2] "j =122 peak =204 savedpeak = 15692"
## [3] "j =122 peak =262 savedpeak = 15750"
## [1] "j =123 peak =76 savedpeak = 15692"
## [2] "j =123 peak =134 savedpeak = 15750"
## [3] "j =123 peak =379 savedpeak = 15995"
## [1] "j =124 peak =251 savedpeak = 15995"
## [2] "j =124 peak =367 savedpeak = 16111"
## [3] "j =124 peak =425 savedpeak = 16169"
## [1] "j =125 peak =123 savedpeak = 15995"
## [2] "j =125 peak =239 savedpeak = 16111"
## [3] "j =125 peak =297 savedpeak = 16169"
## [1] "j =126 peak =111 savedpeak = 16111"
## [2] "j =126 peak =169 savedpeak = 16169"
## [3] "j =126 peak =356 savedpeak = 16356"
## [1] "j =131 peak =106 savedpeak = 16746"
## [2] "j =131 peak =215 savedpeak = 16855"
## [3] "j =131 peak =294 savedpeak = 16934"
## [1] "j =132 peak =87 savedpeak = 16855"
## [2] "j =132 peak =166 savedpeak = 16934"
## [1] "j =133 peak =435 savedpeak = 17331"

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## [1] "j =137 peak =140 savedpeak = 17548"
## [1] "j =139 peak =339 savedpeak = 18003"
## [1] "j =140 peak =211 savedpeak = 18003"
## [1] "j =141 peak =83 savedpeak = 18003"
## [1] "j =142 peak =92 savedpeak = 18140"
## [2] "j =142 peak =237 savedpeak = 18285"
## [3] "j =142 peak =280 savedpeak = 18328"
## [4] "j =142 peak =338 savedpeak = 18386"
## [1] "j =143 peak =109 savedpeak = 18285"
## [2] "j =143 peak =152 savedpeak = 18328"
## [3] "j =143 peak =210 savedpeak = 18386"
## [4] "j =143 peak =232 savedpeak = 18408"
## [5] "j =143 peak =304 savedpeak = 18480"
## [1] "j =144 peak =82 savedpeak = 18386"
## [2] "j =144 peak =104 savedpeak = 18408"
## [3] "j =144 peak =176 savedpeak = 18480"
## [1] "j =153 peak =86 savedpeak = 19542"
## [2] "j =153 peak =122 savedpeak = 19578"
## [3] "j =153 peak =187 savedpeak = 19643"
## [4] "j =153 peak =432 savedpeak = 19888"
## [1] "j =154 peak =210 savedpeak = 19794"
## [2] "j =154 peak =304 savedpeak = 19888"
## [3] "j =154 peak =369 savedpeak = 19953"
## [1] "j =155 peak =82 savedpeak = 19794"
## [2] "j =155 peak =176 savedpeak = 19888"
## [3] "j =155 peak =241 savedpeak = 19953"
## [4] "j =155 peak =342 savedpeak = 20054"
## [5] "j =155 peak =429 savedpeak = 20141"
## [1] "j =156 peak =113 savedpeak = 19953"
## [2] "j =156 peak =214 savedpeak = 20054"
## [3] "j =156 peak =301 savedpeak = 20141"
## [4] "j =156 peak =417 savedpeak = 20257"
## [1] "j =157 peak =86 savedpeak = 20054"
## [2] "j =157 peak =173 savedpeak = 20141"
## [3] "j =157 peak =289 savedpeak = 20257"
## [4] "j =157 peak =346 savedpeak = 20314"
## [1] "j =158 peak =161 savedpeak = 20257"
## [2] "j =158 peak =218 savedpeak = 20314"
## [3] "j =158 peak =399 savedpeak = 20495"
## [1] "j =163 peak =142 savedpeak = 20878"
## [2] "j =163 peak =185 savedpeak = 20921"
## [3] "j =163 peak =207 savedpeak = 20943"
## [4] "j =163 peak =337 savedpeak = 21073"
## [5] "j =163 peak =366 savedpeak = 21102"
## [6] "j =163 peak =402 savedpeak = 21138"
## [1] "j =164 peak =238 savedpeak = 21102"
## [2] "j =164 peak =274 savedpeak = 21138"
## [3] "j =164 peak =317 savedpeak = 21181"
## [4] "j =164 peak =404 savedpeak = 21268"
## [1] "j =165 peak =81 savedpeak = 21073"
## [2] "j =165 peak =110 savedpeak = 21102"
## [3] "j =165 peak =146 savedpeak = 21138"
## [4] "j =165 peak =189 savedpeak = 21181"
## [5] "j =165 peak =276 savedpeak = 21268"

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## [6] "j =165 peak =355 savedpeak = 21347"
## [7] "j =165 peak =413 savedpeak = 21405"
## [1] "j =171 peak =136 savedpeak = 21896"
## [2] "j =171 peak =288 savedpeak = 22048"
## [3] "j =171 peak =367 savedpeak = 22127"
## [4] "j =171 peak =432 savedpeak = 22192"
## [1] "j =172 peak =160 savedpeak = 22048"
## [2] "j =172 peak =239 savedpeak = 22127"
## [3] "j =172 peak =304 savedpeak = 22192"
## [4] "j =172 peak =362 savedpeak = 22250"
## [5] "j =172 peak =397 savedpeak = 22285"
## [1] "j =173 peak =111 savedpeak = 22127"
## [2] "j =173 peak =176 savedpeak = 22192"
## [3] "j =173 peak =240 savedpeak = 22256"
## [4] "j =173 peak =269 savedpeak = 22285"
## [5] "j =173 peak =321 savedpeak = 22337"
## [6] "j =173 peak =356 savedpeak = 22372"
## [1] "j =177 peak =309 savedpeak = 22837"
## [2] "j =177 peak =432 savedpeak = 22960"
## [1] "j =178 peak =181 savedpeak = 22837"
## [2] "j =178 peak =304 savedpeak = 22960"
## [3] "j =178 peak =386 savedpeak = 23042"
## [1] "j =179 peak =176 savedpeak = 22960"
## [2] "j =179 peak =258 savedpeak = 23042"
## [1] "j =180 peak =130 savedpeak = 23042"
## [2] "j =180 peak =368 savedpeak = 23280"
## [3] "j =180 peak =422 savedpeak = 23334"
## [1] "j =181 peak =240 savedpeak = 23280"
## [2] "j =181 peak =294 savedpeak = 23334"
## [1] "j =182 peak =112 savedpeak = 23280"
## [2] "j =182 peak =166 savedpeak = 23334"
## [3] "j =182 peak =325 savedpeak = 23493"
## [4] "j =182 peak =360 savedpeak = 23528"
## [1] "j =183 peak =197 savedpeak = 23493"
## [2] "j =183 peak =232 savedpeak = 23528"
## [3] "j =183 peak =373 savedpeak = 23669"
## [4] "j =183 peak =435 savedpeak = 23731"
## [1] "j =184 peak =104 savedpeak = 23528"
## [2] "j =184 peak =245 savedpeak = 23669"
## [3] "j =184 peak =307 savedpeak = 23731"
## [1] "j =185 peak =117 savedpeak = 23669"
## [2] "j =185 peak =179 savedpeak = 23731"
## [3] "j =185 peak =219 savedpeak = 23771"
## [1] "j =186 peak =91 savedpeak = 23771"
## [1] "j =187 peak =406 savedpeak = 24214"
## [1] "j =188 peak =278 savedpeak = 24214"
## [2] "j =188 peak =311 savedpeak = 24247"
## [3] "j =188 peak =342 savedpeak = 24278"
## [1] "j =189 peak =150 savedpeak = 24214"
## [2] "j =189 peak =183 savedpeak = 24247"
## [3] "j =189 peak =214 savedpeak = 24278"
## [1] "j =190 peak =86 savedpeak = 24278"
## [1] "j =192 peak =332 savedpeak = 24780"
## [1] "j =193 peak =204 savedpeak = 24780"

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## [1] "j =194 peak =76 savedpeak = 24780"
## [1] "j =195 peak =82 savedpeak = 24914"
## [2] "j =195 peak =248 savedpeak = 25080"
## [3] "j =195 peak =380 savedpeak = 25212"
## [4] "j =195 peak =425 savedpeak = 25257"
## [1] "j =196 peak =366 savedpeak = 25326"
## [2] "j =196 peak =419 savedpeak = 25379"
## [1] "j =197 peak =238 savedpeak = 25326"
## [2] "j =197 peak =291 savedpeak = 25379"
## [3] "j =197 peak =346 savedpeak = 25434"
## [4] "j =197 peak =403 savedpeak = 25491"
## [1] "j =198 peak =110 savedpeak = 25326"
## [2] "j =198 peak =163 savedpeak = 25379"
## [3] "j =198 peak =218 savedpeak = 25434"
## [4] "j =198 peak =275 savedpeak = 25491"
## [5] "j =198 peak =431 savedpeak = 25647"
## [1] "j =199 peak =90 savedpeak = 25434"
## [2] "j =199 peak =147 savedpeak = 25491"
## [3] "j =199 peak =303 savedpeak = 25647"
## [1] "j =200 peak =175 savedpeak = 25647"
## [2] "j =200 peak =368 savedpeak = 25840"
## [3] "j =200 peak =398 savedpeak = 25870"
## [1] "j =201 peak =240 savedpeak = 25840"
## [2] "j =201 peak =270 savedpeak = 25870"
## [1] "j =202 peak =112 savedpeak = 25840"
## [2] "j =202 peak =142 savedpeak = 25870"
## [1] "j =203 peak =145 savedpeak = 26001"
## [2] "j =203 peak =235 savedpeak = 26091"
## [3] "j =203 peak =371 savedpeak = 26227"
## [4] "j =203 peak =412 savedpeak = 26268"
## [1] "j =204 peak =107 savedpeak = 26091"
## [2] "j =204 peak =243 savedpeak = 26227"
## [3] "j =204 peak =284 savedpeak = 26268"
## [1] "j =205 peak =115 savedpeak = 26227"
## [2] "j =205 peak =156 savedpeak = 26268"
## [1] "j =206 peak =80 savedpeak = 26320"
## [2] "j =206 peak =112 savedpeak = 26352"
## [3] "j =206 peak =155 savedpeak = 26395"
## [4] "j =206 peak =190 savedpeak = 26430"
## [5] "j =206 peak =207 savedpeak = 26447"
## [6] "j =206 peak =233 savedpeak = 26473"
## [7] "j =206 peak =254 savedpeak = 26494"
## [8] "j =206 peak =291 savedpeak = 26531"
## [9] "j =206 peak =306 savedpeak = 26546"
## [10] "j =206 peak =325 savedpeak = 26565"
## [11] "j =206 peak =337 savedpeak = 26577"
## [12] "j =206 peak =359 savedpeak = 26599"
## [13] "j =206 peak =396 savedpeak = 26636"
## [14] "j =206 peak =421 savedpeak = 26661"
## [1] "j =211 peak =128 savedpeak = 27008"
## [2] "j =211 peak =416 savedpeak = 27296"
## [1] "j =212 peak =288 savedpeak = 27296"
## [1] "j =213 peak =160 savedpeak = 27296"
## [1] "j =216 peak =84 savedpeak = 27604"

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## [2] "j =216 peak =96 savedpeak = 27616"
## [3] "j =216 peak =113 savedpeak = 27633"
## [4] "j =216 peak =153 savedpeak = 27673"
## [5] "j =216 peak =179 savedpeak = 27699"
## [6] "j =216 peak =238 savedpeak = 27758"
## [7] "j =216 peak =257 savedpeak = 27777"
## [8] "j =216 peak =270 savedpeak = 27790"
## [9] "j =216 peak =331 savedpeak = 27851"
## [10] "j =216 peak =347 savedpeak = 27867"
## [11] "j =216 peak =373 savedpeak = 27893"
## [12] "j =216 peak =412 savedpeak = 27932"
## [1] "j =217 peak =203 savedpeak = 27851"
## [2] "j =217 peak =244 savedpeak = 27892"
## [3] "j =217 peak =342 savedpeak = 27990"
## [4] "j =217 peak =362 savedpeak = 28010"
## [5] "j =217 peak =427 savedpeak = 28075"
## [1] "j =218 peak =116 savedpeak = 27892"
## [2] "j =218 peak =214 savedpeak = 27990"
## [3] "j =218 peak =234 savedpeak = 28010"
## [4] "j =218 peak =299 savedpeak = 28075"
## [5] "j =218 peak =366 savedpeak = 28142"
## [1] "j =219 peak =86 savedpeak = 27990"
## [2] "j =219 peak =106 savedpeak = 28010"
## [3] "j =219 peak =171 savedpeak = 28075"
## [4] "j =219 peak =238 savedpeak = 28142"
## [5] "j =219 peak =313 savedpeak = 28217"
## [6] "j =219 peak =377 savedpeak = 28281"
## [7] "j =219 peak =425 savedpeak = 28329"
## [1] "j =220 peak =110 savedpeak = 28142"
## [2] "j =220 peak =185 savedpeak = 28217"
## [3] "j =220 peak =249 savedpeak = 28281"
## [4] "j =220 peak =297 savedpeak = 28329"
## [1] "j =221 peak =121 savedpeak = 28281"
## [2] "j =221 peak =169 savedpeak = 28329"
## [3] "j =221 peak =358 savedpeak = 28518"
## [1] "j =237 peak =403 savedpeak = 30611"
## [1] "j =238 peak =275 savedpeak = 30611"
## [2] "j =238 peak =430 savedpeak = 30766"
## [1] "j =239 peak =147 savedpeak = 30611"
## [2] "j =239 peak =302 savedpeak = 30766"
## [1] "j =240 peak =405 savedpeak = 30997"
## [1] "j =241 peak =277 savedpeak = 30997"
## [2] "j =241 peak =313 savedpeak = 31033"
## [3] "j =241 peak =348 savedpeak = 31068"
## [1] "j =242 peak =149 savedpeak = 30997"
## [2] "j =242 peak =185 savedpeak = 31033"
## [3] "j =242 peak =220 savedpeak = 31068"
## [1] "j =243 peak =92 savedpeak = 31068"
## [1] "j =244 peak =92 savedpeak = 31196"
## [2] "j =244 peak =118 savedpeak = 31222"
## [3] "j =244 peak =143 savedpeak = 31247"
## [4] "j =244 peak =154 savedpeak = 31258"
## [5] "j =244 peak =196 savedpeak = 31300"
## [6] "j =244 peak =253 savedpeak = 31357"

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## [7] "j =244 peak =272 savedpeak = 31376"
## [8] "j =244 peak =298 savedpeak = 31402"
## [9] "j =244 peak =317 savedpeak = 31421"
## [10] "j =244 peak =359 savedpeak = 31463"
## [11] "j =244 peak =373 savedpeak = 31477"
## [12] "j =244 peak =403 savedpeak = 31507"
## [13] "j =244 peak =436 savedpeak = 31540"
## [1] "j =245 peak =125 savedpeak = 31357"
## [2] "j =245 peak =144 savedpeak = 31376"
## [3] "j =245 peak =170 savedpeak = 31402"
## [4] "j =245 peak =189 savedpeak = 31421"
## [5] "j =245 peak =231 savedpeak = 31463"
## [6] "j =245 peak =245 savedpeak = 31477"
## [7] "j =245 peak =275 savedpeak = 31507"
## [8] "j =245 peak =308 savedpeak = 31540"
## [9] "j =245 peak =327 savedpeak = 31559"
## [10] "j =245 peak =342 savedpeak = 31574"
## [11] "j =245 peak =367 savedpeak = 31599"
## [12] "j =245 peak =405 savedpeak = 31637"
## [13] "j =245 peak =424 savedpeak = 31656"
## [1] "j =246 peak =103 savedpeak = 31463"
## [2] "j =246 peak =117 savedpeak = 31477"
## [3] "j =246 peak =147 savedpeak = 31507"
## [4] "j =246 peak =180 savedpeak = 31540"
## [5] "j =246 peak =199 savedpeak = 31559"
## [6] "j =246 peak =214 savedpeak = 31574"
## [7] "j =246 peak =239 savedpeak = 31599"
## [8] "j =246 peak =277 savedpeak = 31637"
## [9] "j =246 peak =296 savedpeak = 31656"
## [10] "j =246 peak =335 savedpeak = 31695"
## [11] "j =246 peak =366 savedpeak = 31726"
## [12] "j =246 peak =388 savedpeak = 31748"
## [13] "j =246 peak =401 savedpeak = 31761"
## [14] "j =246 peak =436 savedpeak = 31796"
## [1] "j =247 peak =86 savedpeak = 31574"
## [2] "j =247 peak =111 savedpeak = 31599"
## [3] "j =247 peak =149 savedpeak = 31637"
## [4] "j =247 peak =168 savedpeak = 31656"
## [5] "j =247 peak =207 savedpeak = 31695"
## [6] "j =247 peak =238 savedpeak = 31726"
## [7] "j =247 peak =260 savedpeak = 31748"
## [8] "j =247 peak =273 savedpeak = 31761"
## [9] "j =247 peak =308 savedpeak = 31796"
## [10] "j =247 peak =315 savedpeak = 31803"
## [11] "j =247 peak =348 savedpeak = 31836"
## [12] "j =247 peak =361 savedpeak = 31849"
## [13] "j =247 peak =386 savedpeak = 31874"
## [14] "j =247 peak =417 savedpeak = 31905"
## [1] "j =248 peak =79 savedpeak = 31695"
## [2] "j =248 peak =110 savedpeak = 31726"
## [3] "j =248 peak =132 savedpeak = 31748"
## [4] "j =248 peak =145 savedpeak = 31761"
## [5] "j =248 peak =180 savedpeak = 31796"
## [6] "j =248 peak =187 savedpeak = 31803"

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## [7] "j =248 peak =220 savedpeak = 31836"
## [8] "j =248 peak =233 savedpeak = 31849"
## [9] "j =248 peak =258 savedpeak = 31874"
## [10] "j =248 peak =289 savedpeak = 31905"
## [11] "j =248 peak =342 savedpeak = 31958"
## [12] "j =248 peak =367 savedpeak = 31983"
## [13] "j =248 peak =394 savedpeak = 32010"
## [14] "j =248 peak =404 savedpeak = 32020"
## [15] "j =248 peak =427 savedpeak = 32043"
## [1] "j =249 peak =92 savedpeak = 31836"
## [2] "j =249 peak =105 savedpeak = 31849"
## [3] "j =249 peak =130 savedpeak = 31874"
## [4] "j =249 peak =161 savedpeak = 31905"
## [5] "j =249 peak =214 savedpeak = 31958"
## [6] "j =249 peak =239 savedpeak = 31983"
## [7] "j =249 peak =266 savedpeak = 32010"
## [8] "j =249 peak =276 savedpeak = 32020"
## [9] "j =249 peak =299 savedpeak = 32043"
## [10] "j =249 peak =329 savedpeak = 32073"
## [11] "j =249 peak =385 savedpeak = 32129"
## [12] "j =249 peak =404 savedpeak = 32148"
## [13] "j =249 peak =432 savedpeak = 32176"
## [1] "j =252 peak =103 savedpeak = 32231"
## [2] "j =252 peak =155 savedpeak = 32283"
## [3] "j =252 peak =167 savedpeak = 32295"
## [4] "j =252 peak =188 savedpeak = 32316"
## [5] "j =252 peak =216 savedpeak = 32344"
## [6] "j =252 peak =262 savedpeak = 32390"
## [7] "j =252 peak =275 savedpeak = 32403"
## [8] "j =252 peak =291 savedpeak = 32419"
## [9] "j =252 peak =317 savedpeak = 32445"
## [10] "j =252 peak =341 savedpeak = 32469"
## [11] "j =252 peak =365 savedpeak = 32493"
## [12] "j =252 peak =424 savedpeak = 32552"
## [1] "j =254 peak =85 savedpeak = 32469"
## [2] "j =254 peak =109 savedpeak = 32493"
## [3] "j =254 peak =168 savedpeak = 32552"
## [4] "j =254 peak =186 savedpeak = 32570"
## [5] "j =254 peak =248 savedpeak = 32632"
## [6] "j =254 peak =272 savedpeak = 32656"
## [7] "j =254 peak =288 savedpeak = 32672"
## [8] "j =254 peak =321 savedpeak = 32705"
## [9] "j =254 peak =344 savedpeak = 32728"
## [10] "j =254 peak =368 savedpeak = 32752"
## [11] "j =254 peak =385 savedpeak = 32769"
## [12] "j =254 peak =403 savedpeak = 32787"
## [13] "j =254 peak =418 savedpeak = 32802"
## [1] "j =255 peak =120 savedpeak = 32632"
## [2] "j =255 peak =144 savedpeak = 32656"
## [3] "j =255 peak =160 savedpeak = 32672"
## [4] "j =255 peak =193 savedpeak = 32705"
## [5] "j =255 peak =216 savedpeak = 32728"
## [6] "j =255 peak =240 savedpeak = 32752"
## [7] "j =255 peak =257 savedpeak = 32769"

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## [8] "j =255 peak =275 savedpeak = 32787"
## [9] "j =255 peak =290 savedpeak = 32802"
## [10] "j =255 peak =329 savedpeak = 32841"
## [11] "j =255 peak =375 savedpeak = 32887"
## [12] "j =255 peak =393 savedpeak = 32905"
## [13] "j =255 peak =433 savedpeak = 32945"
## [1] "j =259 peak =83 savedpeak = 33107"
## [2] "j =259 peak =114 savedpeak = 33138"
## [3] "j =259 peak =139 savedpeak = 33163"
## [4] "j =259 peak =189 savedpeak = 33213"
## [5] "j =259 peak =211 savedpeak = 33235"
## [6] "j =259 peak =229 savedpeak = 33253"
## [7] "j =259 peak =250 savedpeak = 33274"
## [8] "j =259 peak =283 savedpeak = 33307"
## [9] "j =259 peak =310 savedpeak = 33334"
## [10] "j =259 peak =339 savedpeak = 33363"
## [11] "j =259 peak =376 savedpeak = 33400"
## [12] "j =259 peak =403 savedpeak = 33427"
## [13] "j =259 peak =418 savedpeak = 33442"
## [1] "j =260 peak =83 savedpeak = 33235"
## [2] "j =260 peak =101 savedpeak = 33253"
## [3] "j =260 peak =122 savedpeak = 33274"
## [4] "j =260 peak =155 savedpeak = 33307"
## [5] "j =260 peak =182 savedpeak = 33334"
## [6] "j =260 peak =211 savedpeak = 33363"
## [7] "j =260 peak =248 savedpeak = 33400"
## [8] "j =260 peak =275 savedpeak = 33427"
## [9] "j =260 peak =290 savedpeak = 33442"
## [10] "j =260 peak =324 savedpeak = 33476"
## [11] "j =260 peak =340 savedpeak = 33492"
## [12] "j =260 peak =390 savedpeak = 33542"
## [13] "j =260 peak =431 savedpeak = 33583"
## [1] "j =269 peak =103 savedpeak = 34407"
## [2] "j =269 peak =117 savedpeak = 34421"
## [3] "j =269 peak =175 savedpeak = 34479"
## [4] "j =269 peak =206 savedpeak = 34510"
## [5] "j =269 peak =225 savedpeak = 34529"
## [6] "j =269 peak =264 savedpeak = 34568"
## [7] "j =269 peak =291 savedpeak = 34595"
## [8] "j =269 peak =305 savedpeak = 34609"
## [9] "j =269 peak =334 savedpeak = 34638"
## [10] "j =269 peak =390 savedpeak = 34694"
## [11] "j =269 peak =406 savedpeak = 34710"
## [12] "j =269 peak =415 savedpeak = 34719"
## [13] "j =269 peak =431 savedpeak = 34735"
## [1] "j =270 peak =78 savedpeak = 34510"
## [2] "j =270 peak =97 savedpeak = 34529"
## [3] "j =270 peak =136 savedpeak = 34568"
## [4] "j =270 peak =163 savedpeak = 34595"
## [5] "j =270 peak =177 savedpeak = 34609"
## [6] "j =270 peak =206 savedpeak = 34638"
## [7] "j =270 peak =262 savedpeak = 34694"
## [8] "j =270 peak =278 savedpeak = 34710"
## [9] "j =270 peak =287 savedpeak = 34719"

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## [10] "j =270 peak =303 savedpeak = 34735"
## [11] "j =270 peak =361 savedpeak = 34793"
## [12] "j =270 peak =379 savedpeak = 34811"
## [13] "j =270 peak =408 savedpeak = 34840"
## [14] "j =270 peak =425 savedpeak = 34857"
## [1] "j =271 peak =78 savedpeak = 34638"
## [2] "j =271 peak =150 savedpeak = 34710"
## [3] "j =271 peak =159 savedpeak = 34719"
## [4] "j =271 peak =175 savedpeak = 34735"
## [5] "j =271 peak =233 savedpeak = 34793"
## [6] "j =271 peak =251 savedpeak = 34811"
## [7] "j =271 peak =280 savedpeak = 34840"
## [8] "j =271 peak =297 savedpeak = 34857"
## [9] "j =271 peak =311 savedpeak = 34871"
## [10] "j =271 peak =327 savedpeak = 34887"
## [11] "j =271 peak =343 savedpeak = 34903"
## [12] "j =271 peak =384 savedpeak = 34944"
## [13] "j =271 peak =430 savedpeak = 34990"
## [1] "j =273 peak =87 savedpeak = 34903"
## [2] "j =273 peak =128 savedpeak = 34944"
## [3] "j =273 peak =174 savedpeak = 34990"
## [4] "j =273 peak =195 savedpeak = 35011"
## [5] "j =273 peak =218 savedpeak = 35034"
## [6] "j =273 peak =263 savedpeak = 35079"
## [7] "j =273 peak =297 savedpeak = 35113"
## [8] "j =273 peak =317 savedpeak = 35133"
## [9] "j =273 peak =351 savedpeak = 35167"
## [10] "j =273 peak =378 savedpeak = 35194"
## [11] "j =273 peak =435 savedpeak = 35251"
## [1] "j =274 peak =90 savedpeak = 35034"
## [2] "j =274 peak =135 savedpeak = 35079"
## [3] "j =274 peak =169 savedpeak = 35113"
## [4] "j =274 peak =189 savedpeak = 35133"
## [5] "j =274 peak =223 savedpeak = 35167"
## [6] "j =274 peak =250 savedpeak = 35194"
## [7] "j =274 peak =307 savedpeak = 35251"
## [8] "j =274 peak =326 savedpeak = 35270"
## [9] "j =274 peak =365 savedpeak = 35309"
## [10] "j =274 peak =392 savedpeak = 35336"
## [11] "j =274 peak =412 savedpeak = 35356"
## [1] "j =278 peak =85 savedpeak = 35541"
## [2] "j =278 peak =97 savedpeak = 35553"
## [3] "j =278 peak =145 savedpeak = 35601"
## [4] "j =278 peak =182 savedpeak = 35638"
## [5] "j =278 peak =208 savedpeak = 35664"
## [6] "j =278 peak =245 savedpeak = 35701"
## [7] "j =278 peak =336 savedpeak = 35792"
## [8] "j =278 peak =361 savedpeak = 35817"
## [9] "j =278 peak =391 savedpeak = 35847"
## [10] "j =278 peak =411 savedpeak = 35867"
## [1] "j =279 peak =80 savedpeak = 35664"
## [2] "j =279 peak =117 savedpeak = 35701"
## [3] "j =279 peak =208 savedpeak = 35792"
## [4] "j =279 peak =233 savedpeak = 35817"

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## [5] "j =279 peak =263 savedpeak = 35847"
## [6] "j =279 peak =283 savedpeak = 35867"
## [7] "j =279 peak =310 savedpeak = 35894"
## [8] "j =279 peak =375 savedpeak = 35959"
## [9] "j =279 peak =408 savedpeak = 35992"
## [10] "j =279 peak =426 savedpeak = 36010"
## [1] "j =280 peak =80 savedpeak = 35792"
## [2] "j =280 peak =105 savedpeak = 35817"
## [3] "j =280 peak =135 savedpeak = 35847"
## [4] "j =280 peak =155 savedpeak = 35867"
## [5] "j =280 peak =182 savedpeak = 35894"
## [6] "j =280 peak =247 savedpeak = 35959"
## [7] "j =280 peak =280 savedpeak = 35992"
## [8] "j =280 peak =298 savedpeak = 36010"
## [9] "j =280 peak =319 savedpeak = 36031"
## [10] "j =280 peak =339 savedpeak = 36051"
## [11] "j =280 peak =351 savedpeak = 36063"
## [12] "j =280 peak =390 savedpeak = 36102"
## [13] "j =280 peak =422 savedpeak = 36134"
## [1] "j =281 peak =76 savedpeak = 35916"
## [2] "j =281 peak =119 savedpeak = 35959"
## [3] "j =281 peak =152 savedpeak = 35992"
## [4] "j =281 peak =170 savedpeak = 36010"
## [5] "j =281 peak =191 savedpeak = 36031"
## [6] "j =281 peak =211 savedpeak = 36051"
## [7] "j =281 peak =223 savedpeak = 36063"
## [8] "j =281 peak =262 savedpeak = 36102"
## [9] "j =281 peak =294 savedpeak = 36134"
## [10] "j =281 peak =333 savedpeak = 36173"
## [11] "j =281 peak =374 savedpeak = 36214"
## [12] "j =281 peak =396 savedpeak = 36236"
## [1] "j =282 peak =83 savedpeak = 36051"
## [2] "j =282 peak =95 savedpeak = 36063"
## [3] "j =282 peak =134 savedpeak = 36102"
## [4] "j =282 peak =166 savedpeak = 36134"
## [5] "j =282 peak =205 savedpeak = 36173"
## [6] "j =282 peak =246 savedpeak = 36214"
## [7] "j =282 peak =268 savedpeak = 36236"
## [8] "j =282 peak =315 savedpeak = 36283"
## [9] "j =282 peak =348 savedpeak = 36316"
## [10] "j =282 peak =364 savedpeak = 36332"
## [11] "j =282 peak =390 savedpeak = 36358"
## [12] "j =282 peak =406 savedpeak = 36374"
## [13] "j =282 peak =425 savedpeak = 36393"
## [1] "j =283 peak =77 savedpeak = 36173"
## [2] "j =283 peak =118 savedpeak = 36214"
## [3] "j =283 peak =140 savedpeak = 36236"
## [4] "j =283 peak =187 savedpeak = 36283"
## [5] "j =283 peak =220 savedpeak = 36316"
## [6] "j =283 peak =236 savedpeak = 36332"
## [7] "j =283 peak =262 savedpeak = 36358"
## [8] "j =283 peak =278 savedpeak = 36374"
## [9] "j =283 peak =297 savedpeak = 36393"
## [10] "j =283 peak =325 savedpeak = 36421"

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## [11] "j =283 peak =348 savedpeak = 36444"
## [12] "j =283 peak =360 savedpeak = 36456"
## [13] "j =283 peak =382 savedpeak = 36478"
## [14] "j =283 peak =402 savedpeak = 36498"
## [15] "j =283 peak =415 savedpeak = 36511"
## [1] "j =289 peak =134 savedpeak = 36998"
## [2] "j =289 peak =275 savedpeak = 37139"
## [3] "j =289 peak =357 savedpeak = 37221"
## [1] "j =290 peak =147 savedpeak = 37139"
## [2] "j =290 peak =229 savedpeak = 37221"
## [3] "j =290 peak =357 savedpeak = 37349"
## [4] "j =290 peak =435 savedpeak = 37427"
## [1] "j =291 peak =101 savedpeak = 37221"
## [2] "j =291 peak =229 savedpeak = 37349"
## [3] "j =291 peak =307 savedpeak = 37427"
## [1] "j =292 peak =101 savedpeak = 37349"
## [2] "j =292 peak =179 savedpeak = 37427"
## [3] "j =292 peak =412 savedpeak = 37660"
## [1] "j =293 peak =107 savedpeak = 37483"
## [2] "j =293 peak =128 savedpeak = 37504"
## [3] "j =293 peak =168 savedpeak = 37544"
## [4] "j =293 peak =194 savedpeak = 37570"
## [5] "j =293 peak =240 savedpeak = 37616"
## [6] "j =293 peak =277 savedpeak = 37653"
## [7] "j =293 peak =307 savedpeak = 37683"
## [8] "j =293 peak =323 savedpeak = 37699"
## [9] "j =293 peak =369 savedpeak = 37745"
## [10] "j =293 peak =409 savedpeak = 37785"
## [11] "j =293 peak =423 savedpeak = 37799"
## [1] "j =294 peak =112 savedpeak = 37616"
## [2] "j =294 peak =155 savedpeak = 37659"
## [3] "j =294 peak =179 savedpeak = 37683"
## [4] "j =294 peak =195 savedpeak = 37699"
## [5] "j =294 peak =241 savedpeak = 37745"
## [6] "j =294 peak =281 savedpeak = 37785"
## [7] "j =294 peak =295 savedpeak = 37799"
## [8] "j =294 peak =320 savedpeak = 37824"
## [9] "j =294 peak =344 savedpeak = 37848"
## [10] "j =294 peak =377 savedpeak = 37881"
## [11] "j =294 peak =406 savedpeak = 37910"
## [12] "j =294 peak =429 savedpeak = 37933"
## [1] "j =295 peak =113 savedpeak = 37745"
## [2] "j =295 peak =153 savedpeak = 37785"
## [3] "j =295 peak =167 savedpeak = 37799"
## [4] "j =295 peak =192 savedpeak = 37824"
## [5] "j =295 peak =216 savedpeak = 37848"
## [6] "j =295 peak =249 savedpeak = 37881"
## [7] "j =295 peak =278 savedpeak = 37910"
## [8] "j =295 peak =301 savedpeak = 37933"
## [9] "j =295 peak =316 savedpeak = 37948"
## [10] "j =295 peak =341 savedpeak = 37973"
## [11] "j =295 peak =361 savedpeak = 37993"
## [12] "j =295 peak =374 savedpeak = 38006"
## [13] "j =295 peak =393 savedpeak = 38025"

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## [14] "j =295 peak =404 savedpeak = 38036"
## [15] "j =295 peak =422 savedpeak = 38054"
## [1] "j =296 peak =88 savedpeak = 37848"
## [2] "j =296 peak =121 savedpeak = 37881"
## [3] "j =296 peak =150 savedpeak = 37910"
## [4] "j =296 peak =173 savedpeak = 37933"
## [5] "j =296 peak =188 savedpeak = 37948"
## [6] "j =296 peak =213 savedpeak = 37973"
## [7] "j =296 peak =233 savedpeak = 37993"
## [8] "j =296 peak =246 savedpeak = 38006"
## [9] "j =296 peak =265 savedpeak = 38025"
## [10] "j =296 peak =276 savedpeak = 38036"
## [11] "j =296 peak =294 savedpeak = 38054"
## [12] "j =296 peak =329 savedpeak = 38089"
## [13] "j =296 peak =344 savedpeak = 38104"
## [14] "j =296 peak =385 savedpeak = 38145"
## [15] "j =296 peak =424 savedpeak = 38184"
## [1] "j =303 peak =98 savedpeak = 38754"
## [2] "j =303 peak =125 savedpeak = 38781"
## [3] "j =303 peak =164 savedpeak = 38820"
## [4] "j =303 peak =187 savedpeak = 38843"
## [5] "j =303 peak =228 savedpeak = 38884"
## [6] "j =303 peak =276 savedpeak = 38932"
## [7] "j =303 peak =297 savedpeak = 38953"
## [8] "j =303 peak =314 savedpeak = 38970"
## [9] "j =303 peak =359 savedpeak = 39015"
## [10] "j =303 peak =380 savedpeak = 39036"
## [11] "j =303 peak =393 savedpeak = 39049"
## [1] "j =304 peak =100 savedpeak = 38884"
## [2] "j =304 peak =148 savedpeak = 38932"
## [3] "j =304 peak =169 savedpeak = 38953"
## [4] "j =304 peak =186 savedpeak = 38970"
## [5] "j =304 peak =231 savedpeak = 39015"
## [6] "j =304 peak =252 savedpeak = 39036"
## [7] "j =304 peak =265 savedpeak = 39049"
## [8] "j =304 peak =310 savedpeak = 39094"
## [9] "j =304 peak =332 savedpeak = 39116"
## [10] "j =304 peak =347 savedpeak = 39131"
## [11] "j =304 peak =369 savedpeak = 39153"
## [12] "j =304 peak =394 savedpeak = 39178"
## [13] "j =304 peak =419 savedpeak = 39203"
## [1] "j =305 peak =103 savedpeak = 39015"
## [2] "j =305 peak =124 savedpeak = 39036"
## [3] "j =305 peak =137 savedpeak = 39049"
## [4] "j =305 peak =182 savedpeak = 39094"
## [5] "j =305 peak =204 savedpeak = 39116"
## [6] "j =305 peak =219 savedpeak = 39131"
## [7] "j =305 peak =241 savedpeak = 39153"
## [8] "j =305 peak =266 savedpeak = 39178"
## [9] "j =305 peak =291 savedpeak = 39203"
## [10] "j =305 peak =315 savedpeak = 39227"
## [11] "j =305 peak =333 savedpeak = 39245"
## [12] "j =305 peak =380 savedpeak = 39292"
## [13] "j =305 peak =395 savedpeak = 39307"

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## [14] "j =305 peak =431 savedpeak = 39343"
## [1] "j =306 peak =76 savedpeak = 39116"
## [2] "j =306 peak =91 savedpeak = 39131"
## [3] "j =306 peak =113 savedpeak = 39153"
## [4] "j =306 peak =138 savedpeak = 39178"
## [5] "j =306 peak =163 savedpeak = 39203"
## [6] "j =306 peak =187 savedpeak = 39227"
## [7] "j =306 peak =205 savedpeak = 39245"
## [8] "j =306 peak =252 savedpeak = 39292"
## [9] "j =306 peak =267 savedpeak = 39307"
## [10] "j =306 peak =303 savedpeak = 39343"
## [11] "j =306 peak =328 savedpeak = 39368"
## [12] "j =306 peak =364 savedpeak = 39404"
## [13] "j =306 peak =380 savedpeak = 39420"
## [14] "j =306 peak =404 savedpeak = 39444"
## [15] "j =306 peak =427 savedpeak = 39467"
## [1] "j =307 peak =77 savedpeak = 39245"
## [2] "j =307 peak =124 savedpeak = 39292"
## [3] "j =307 peak =139 savedpeak = 39307"
## [4] "j =307 peak =175 savedpeak = 39343"
## [5] "j =307 peak =200 savedpeak = 39368"
## [6] "j =307 peak =236 savedpeak = 39404"
## [7] "j =307 peak =252 savedpeak = 39420"
## [8] "j =307 peak =276 savedpeak = 39444"
## [9] "j =307 peak =299 savedpeak = 39467"
## [10] "j =307 peak =328 savedpeak = 39496"
## [11] "j =307 peak =347 savedpeak = 39515"
## [12] "j =307 peak =377 savedpeak = 39545"
## [13] "j =307 peak =397 savedpeak = 39565"
## [14] "j =307 peak =424 savedpeak = 39592"
## [1] "j =308 peak =107 savedpeak = 39403"
## [2] "j =308 peak =145 savedpeak = 39441"
## [3] "j =308 peak =198 savedpeak = 39494"
## [4] "j =308 peak =224 savedpeak = 39520"
## [5] "j =308 peak =269 savedpeak = 39565"
## [6] "j =308 peak =354 savedpeak = 39650"
## [7] "j =308 peak =406 savedpeak = 39702"
## [1] "j =309 peak =141 savedpeak = 39565"
## [2] "j =309 peak =226 savedpeak = 39650"
## [3] "j =309 peak =278 savedpeak = 39702"
## [4] "j =309 peak =375 savedpeak = 39799"
## [5] "j =309 peak =399 savedpeak = 39823"
## [1] "j =310 peak =98 savedpeak = 39650"
## [2] "j =310 peak =150 savedpeak = 39702"
## [3] "j =310 peak =247 savedpeak = 39799"
## [4] "j =310 peak =271 savedpeak = 39823"
## [5] "j =310 peak =312 savedpeak = 39864"
## [6] "j =310 peak =341 savedpeak = 39893"
## [7] "j =310 peak =403 savedpeak = 39955"
## [8] "j =310 peak =427 savedpeak = 39979"
## [1] "j =311 peak =119 savedpeak = 39799"
## [2] "j =311 peak =143 savedpeak = 39823"
## [3] "j =311 peak =184 savedpeak = 39864"
## [4] "j =311 peak =213 savedpeak = 39893"

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## [5] "j =311 peak =275 savedpeak = 39955"
## [6] "j =311 peak =299 savedpeak = 39979"
## [7] "j =311 peak =336 savedpeak = 40016"
## [8] "j =311 peak =373 savedpeak = 40053"
## [9] "j =311 peak =424 savedpeak = 40104"
## [1] "j =312 peak =80 savedpeak = 39888"
## [2] "j =312 peak =117 savedpeak = 39925"
## [3] "j =312 peak =140 savedpeak = 39948"
## [4] "j =312 peak =171 savedpeak = 39979"
## [5] "j =312 peak =210 savedpeak = 40018"
## [6] "j =312 peak =244 savedpeak = 40052"
## [7] "j =312 peak =260 savedpeak = 40068"
## [8] "j =312 peak =274 savedpeak = 40082"
## [9] "j =312 peak =297 savedpeak = 40105"
## [10] "j =312 peak =335 savedpeak = 40143"
## [11] "j =312 peak =370 savedpeak = 40178"
## [12] "j =312 peak =382 savedpeak = 40190"
## [13] "j =312 peak =410 savedpeak = 40218"
## [14] "j =312 peak =424 savedpeak = 40232"
## [15] "j =312 peak =433 savedpeak = 40241"
## [1] "j =313 peak =82 savedpeak = 40018"
## [2] "j =313 peak =116 savedpeak = 40052"
## [3] "j =313 peak =132 savedpeak = 40068"
## [4] "j =313 peak =146 savedpeak = 40082"
## [5] "j =313 peak =169 savedpeak = 40105"
## [6] "j =313 peak =207 savedpeak = 40143"
## [7] "j =313 peak =242 savedpeak = 40178"
## [8] "j =313 peak =254 savedpeak = 40190"
## [9] "j =313 peak =282 savedpeak = 40218"
## [10] "j =313 peak =296 savedpeak = 40232"
## [11] "j =313 peak =305 savedpeak = 40241"
## [12] "j =313 peak =330 savedpeak = 40266"
## [13] "j =313 peak =348 savedpeak = 40284"
## [14] "j =313 peak =365 savedpeak = 40301"
## [15] "j =313 peak =397 savedpeak = 40333"
## [16] "j =313 peak =410 savedpeak = 40346"
## [1] "j =315 peak =92 savedpeak = 40284"
## [2] "j =315 peak =109 savedpeak = 40301"
## [3] "j =315 peak =141 savedpeak = 40333"
## [4] "j =315 peak =154 savedpeak = 40346"
## [5] "j =315 peak =186 savedpeak = 40378"
## [6] "j =315 peak =204 savedpeak = 40396"
## [7] "j =315 peak =228 savedpeak = 40420"
## [8] "j =315 peak =270 savedpeak = 40462"
## [9] "j =315 peak =310 savedpeak = 40502"
## [10] "j =315 peak =331 savedpeak = 40523"
## [11] "j =315 peak =347 savedpeak = 40539"
## [12] "j =315 peak =375 savedpeak = 40567"
## [13] "j =315 peak =413 savedpeak = 40605"
## [14] "j =315 peak =421 savedpeak = 40613"
## [1] "j =316 peak =76 savedpeak = 40396"
## [2] "j =316 peak =100 savedpeak = 40420"
## [3] "j =316 peak =142 savedpeak = 40462"
## [4] "j =316 peak =182 savedpeak = 40502"

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## [5] "j =316 peak =203 savedpeak = 40523"
## [6] "j =316 peak =219 savedpeak = 40539"
## [7] "j =316 peak =247 savedpeak = 40567"
## [8] "j =316 peak =262 savedpeak = 40582"
## [9] "j =316 peak =285 savedpeak = 40605"
## [10] "j =316 peak =293 savedpeak = 40613"
## [11] "j =316 peak =322 savedpeak = 40642"
## [12] "j =316 peak =355 savedpeak = 40675"
## [13] "j =316 peak =417 savedpeak = 40737"
## [1] "j =317 peak =91 savedpeak = 40539"
## [2] "j =317 peak =119 savedpeak = 40567"
## [3] "j =317 peak =134 savedpeak = 40582"
## [4] "j =317 peak =157 savedpeak = 40605"
## [5] "j =317 peak =165 savedpeak = 40613"
## [6] "j =317 peak =194 savedpeak = 40642"
## [7] "j =317 peak =227 savedpeak = 40675"
## [8] "j =317 peak =289 savedpeak = 40737"
## [9] "j =317 peak =314 savedpeak = 40762"
## [10] "j =317 peak =357 savedpeak = 40805"
## [11] "j =317 peak =379 savedpeak = 40827"
## [12] "j =317 peak =402 savedpeak = 40850"
## [13] "j =317 peak =419 savedpeak = 40867"
## [14] "j =317 peak =437 savedpeak = 40885"
## [1] "j =318 peak =99 savedpeak = 40675"
## [2] "j =318 peak =161 savedpeak = 40737"
## [3] "j =318 peak =186 savedpeak = 40762"
## [4] "j =318 peak =229 savedpeak = 40805"
## [5] "j =318 peak =251 savedpeak = 40827"
## [6] "j =318 peak =274 savedpeak = 40850"
## [7] "j =318 peak =291 savedpeak = 40867"
## [8] "j =318 peak =309 savedpeak = 40885"
## [9] "j =318 peak =350 savedpeak = 40926"
## [10] "j =318 peak =387 savedpeak = 40963"
## [11] "j =318 peak =409 savedpeak = 40985"
## [12] "j =318 peak =436 savedpeak = 41012"
## [1] "j =319 peak =101 savedpeak = 40805"
## [2] "j =319 peak =123 savedpeak = 40827"
## [3] "j =319 peak =146 savedpeak = 40850"
## [4] "j =319 peak =163 savedpeak = 40867"
## [5] "j =319 peak =181 savedpeak = 40885"
## [6] "j =319 peak =222 savedpeak = 40926"
## [7] "j =319 peak =259 savedpeak = 40963"
## [8] "j =319 peak =281 savedpeak = 40985"
## [9] "j =319 peak =308 savedpeak = 41012"
## [10] "j =319 peak =336 savedpeak = 41040"
## [11] "j =319 peak =370 savedpeak = 41074"
## [12] "j =319 peak =405 savedpeak = 41109"
## [13] "j =319 peak =415 savedpeak = 41119"
## [1] "j =320 peak =94 savedpeak = 40926"
## [2] "j =320 peak =131 savedpeak = 40963"
## [3] "j =320 peak =153 savedpeak = 40985"
## [4] "j =320 peak =180 savedpeak = 41012"
## [5] "j =320 peak =208 savedpeak = 41040"
## [6] "j =320 peak =242 savedpeak = 41074"

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## [7] "j =320 peak =277 savedpeak = 41109"
## [8] "j =320 peak =287 savedpeak = 41119"
## [9] "j =320 peak =355 savedpeak = 41187"
## [10] "j =320 peak =382 savedpeak = 41214"
## [11] "j =320 peak =412 savedpeak = 41244"
## [1] "j =322 peak =99 savedpeak = 41187"
## [2] "j =322 peak =126 savedpeak = 41214"
## [3] "j =322 peak =156 savedpeak = 41244"
## [4] "j =322 peak =198 savedpeak = 41286"
## [5] "j =322 peak =215 savedpeak = 41303"
## [6] "j =322 peak =283 savedpeak = 41371"
## [7] "j =322 peak =316 savedpeak = 41404"
## [8] "j =322 peak =340 savedpeak = 41428"
## [9] "j =322 peak =368 savedpeak = 41456"
## [10] "j =322 peak =409 savedpeak = 41497"
## [1] "j =325 peak =84 savedpeak = 41556"
## [2] "j =325 peak =109 savedpeak = 41581"
## [3] "j =325 peak =141 savedpeak = 41613"
## [4] "j =325 peak =158 savedpeak = 41630"
## [5] "j =325 peak =182 savedpeak = 41654"
## [6] "j =325 peak =223 savedpeak = 41695"
## [7] "j =325 peak =241 savedpeak = 41713"
## [8] "j =325 peak =300 savedpeak = 41772"
## [9] "j =325 peak =323 savedpeak = 41795"
## [10] "j =325 peak =344 savedpeak = 41816"
## [11] "j =325 peak =351 savedpeak = 41823"
## [12] "j =325 peak =378 savedpeak = 41850"
## [13] "j =325 peak =416 savedpeak = 41888"
## [1] "j =326 peak =95 savedpeak = 41695"
## [2] "j =326 peak =113 savedpeak = 41713"
## [3] "j =326 peak =172 savedpeak = 41772"
## [4] "j =326 peak =195 savedpeak = 41795"
## [5] "j =326 peak =216 savedpeak = 41816"
## [6] "j =326 peak =223 savedpeak = 41823"
## [7] "j =326 peak =250 savedpeak = 41850"
## [8] "j =326 peak =288 savedpeak = 41888"
## [9] "j =326 peak =315 savedpeak = 41915"
## [10] "j =326 peak =345 savedpeak = 41945"
## [11] "j =326 peak =390 savedpeak = 41990"
## [12] "j =326 peak =414 savedpeak = 42014"
## [13] "j =326 peak =429 savedpeak = 42029"
## [1] "j =327 peak =88 savedpeak = 41816"
## [2] "j =327 peak =95 savedpeak = 41823"
## [3] "j =327 peak =122 savedpeak = 41850"
## [4] "j =327 peak =160 savedpeak = 41888"
## [5] "j =327 peak =187 savedpeak = 41915"
## [6] "j =327 peak =217 savedpeak = 41945"
## [7] "j =327 peak =262 savedpeak = 41990"
## [8] "j =327 peak =286 savedpeak = 42014"
## [9] "j =327 peak =301 savedpeak = 42029"
## [10] "j =327 peak =324 savedpeak = 42052"
## [11] "j =327 peak =360 savedpeak = 42088"
## [12] "j =327 peak =380 savedpeak = 42108"
## [13] "j =327 peak =395 savedpeak = 42123"

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## [1] "j =328 peak =89 savedpeak = 41945"
## [2] "j =328 peak =134 savedpeak = 41990"
## [3] "j =328 peak =158 savedpeak = 42014"
## [4] "j =328 peak =173 savedpeak = 42029"
## [5] "j =328 peak =196 savedpeak = 42052"
## [6] "j =328 peak =232 savedpeak = 42088"
## [7] "j =328 peak =252 savedpeak = 42108"
## [8] "j =328 peak =267 savedpeak = 42123"
## [9] "j =328 peak =311 savedpeak = 42167"
## [10] "j =328 peak =325 savedpeak = 42181"
## [11] "j =328 peak =350 savedpeak = 42206"
## [12] "j =328 peak =410 savedpeak = 42266"
## [1] "j =330 peak =94 savedpeak = 42206"
## [2] "j =330 peak =154 savedpeak = 42266"
## [3] "j =330 peak =183 savedpeak = 42295"
## [4] "j =330 peak =218 savedpeak = 42330"
## [5] "j =330 peak =279 savedpeak = 42391"
## [6] "j =330 peak =317 savedpeak = 42429"
## [7] "j =330 peak =334 savedpeak = 42446"
## [8] "j =330 peak =352 savedpeak = 42464"
## [9] "j =330 peak =368 savedpeak = 42480"
## [10] "j =330 peak =422 savedpeak = 42534"
## [1] "j =331 peak =90 savedpeak = 42330"
## [2] "j =331 peak =151 savedpeak = 42391"
## [3] "j =331 peak =189 savedpeak = 42429"
## [4] "j =331 peak =206 savedpeak = 42446"
## [5] "j =331 peak =224 savedpeak = 42464"
## [6] "j =331 peak =240 savedpeak = 42480"
## [7] "j =331 peak =294 savedpeak = 42534"
## [8] "j =331 peak =348 savedpeak = 42588"
## [9] "j =331 peak =365 savedpeak = 42605"
## [10] "j =331 peak =382 savedpeak = 42622"
## [11] "j =331 peak =408 savedpeak = 42648"
## [12] "j =331 peak =417 savedpeak = 42657"
## [13] "j =331 peak =437 savedpeak = 42677"
## [1] "j =332 peak =78 savedpeak = 42446"
## [2] "j =332 peak =96 savedpeak = 42464"
## [3] "j =332 peak =112 savedpeak = 42480"
## [4] "j =332 peak =166 savedpeak = 42534"
## [5] "j =332 peak =220 savedpeak = 42588"
## [6] "j =332 peak =237 savedpeak = 42605"
## [7] "j =332 peak =254 savedpeak = 42622"
## [8] "j =332 peak =280 savedpeak = 42648"
## [9] "j =332 peak =289 savedpeak = 42657"
## [10] "j =332 peak =309 savedpeak = 42677"
## [11] "j =332 peak =334 savedpeak = 42702"
## [12] "j =332 peak =350 savedpeak = 42718"
## [13] "j =332 peak =362 savedpeak = 42730"
## [14] "j =332 peak =381 savedpeak = 42749"
## [15] "j =332 peak =402 savedpeak = 42770"
## [16] "j =332 peak =428 savedpeak = 42796"
## [1] "j =333 peak =92 savedpeak = 42588"
## [2] "j =333 peak =109 savedpeak = 42605"
## [3] "j =333 peak =126 savedpeak = 42622"

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## [4] "j =333 peak =152 savedpeak = 42648"
## [5] "j =333 peak =161 savedpeak = 42657"
## [6] "j =333 peak =181 savedpeak = 42677"
## [7] "j =333 peak =206 savedpeak = 42702"
## [8] "j =333 peak =222 savedpeak = 42718"
## [9] "j =333 peak =234 savedpeak = 42730"
## [10] "j =333 peak =253 savedpeak = 42749"
## [11] "j =333 peak =274 savedpeak = 42770"
## [12] "j =333 peak =300 savedpeak = 42796"
## [13] "j =333 peak =319 savedpeak = 42815"
## [14] "j =333 peak =339 savedpeak = 42835"
## [15] "j =333 peak =354 savedpeak = 42850"
## [16] "j =333 peak =380 savedpeak = 42876"
## [17] "j =333 peak =395 savedpeak = 42891"
## [18] "j =333 peak =413 savedpeak = 42909"
## [19] "j =333 peak =433 savedpeak = 42929"
## [1] "j =334 peak =78 savedpeak = 42702"
## [2] "j =334 peak =94 savedpeak = 42718"
## [3] "j =334 peak =106 savedpeak = 42730"
## [4] "j =334 peak =125 savedpeak = 42749"
## [5] "j =334 peak =146 savedpeak = 42770"
## [6] "j =334 peak =172 savedpeak = 42796"
## [7] "j =334 peak =191 savedpeak = 42815"
## [8] "j =334 peak =211 savedpeak = 42835"
## [9] "j =334 peak =226 savedpeak = 42850"
## [10] "j =334 peak =252 savedpeak = 42876"
## [11] "j =334 peak =267 savedpeak = 42891"
## [12] "j =334 peak =285 savedpeak = 42909"
## [13] "j =334 peak =305 savedpeak = 42929"
## [14] "j =334 peak =354 savedpeak = 42978"
## [15] "j =334 peak =378 savedpeak = 43002"
## [16] "j =334 peak =422 savedpeak = 43046"
## [1] "j =335 peak =83 savedpeak = 42835"
## [2] "j =335 peak =98 savedpeak = 42850"
## [3] "j =335 peak =124 savedpeak = 42876"
## [4] "j =335 peak =139 savedpeak = 42891"
## [5] "j =335 peak =157 savedpeak = 42909"
## [6] "j =335 peak =177 savedpeak = 42929"
## [7] "j =335 peak =226 savedpeak = 42978"
## [8] "j =335 peak =250 savedpeak = 43002"
## [9] "j =335 peak =294 savedpeak = 43046"
## [10] "j =335 peak =318 savedpeak = 43070"
## [11] "j =335 peak =346 savedpeak = 43098"
## [12] "j =335 peak =383 savedpeak = 43135"
## [13] "j =335 peak =421 savedpeak = 43173"
## [1] "j =337 peak =90 savedpeak = 43098"
## [2] "j =337 peak =127 savedpeak = 43135"
## [3] "j =337 peak =165 savedpeak = 43173"
## [4] "j =337 peak =198 savedpeak = 43206"
## [5] "j =337 peak =228 savedpeak = 43236"
## [6] "j =337 peak =258 savedpeak = 43266"
## [7] "j =337 peak =278 savedpeak = 43286"
## [8] "j =337 peak =316 savedpeak = 43324"
## [9] "j =337 peak =333 savedpeak = 43341"

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## [10] "j =337 peak =367 savedpeak = 43375"
## [11] "j =337 peak =390 savedpeak = 43398"
## [12] "j =337 peak =411 savedpeak = 43419"
## [13] "j =337 peak =433 savedpeak = 43441"
## [1] "j =341 peak =117 savedpeak = 43637"
## [2] "j =341 peak =172 savedpeak = 43692"
## [3] "j =341 peak =209 savedpeak = 43729"
## [4] "j =341 peak =256 savedpeak = 43776"
## [5] "j =341 peak =312 savedpeak = 43832"
## [6] "j =341 peak =377 savedpeak = 43897"
## [7] "j =341 peak =401 savedpeak = 43921"
## [1] "j =342 peak =81 savedpeak = 43729"
## [2] "j =342 peak =128 savedpeak = 43776"
## [3] "j =342 peak =184 savedpeak = 43832"
## [4] "j =342 peak =249 savedpeak = 43897"
## [5] "j =342 peak =273 savedpeak = 43921"
## [6] "j =342 peak =328 savedpeak = 43976"
## [7] "j =342 peak =404 savedpeak = 44052"
## [1] "j =343 peak =121 savedpeak = 43897"
## [2] "j =343 peak =145 savedpeak = 43921"
## [3] "j =343 peak =200 savedpeak = 43976"
## [4] "j =343 peak =276 savedpeak = 44052"
## [5] "j =343 peak =338 savedpeak = 44114"
## [6] "j =343 peak =398 savedpeak = 44174"
## [1] "j =344 peak =148 savedpeak = 44052"
## [2] "j =344 peak =210 savedpeak = 44114"
## [3] "j =344 peak =270 savedpeak = 44174"
## [4] "j =344 peak =319 savedpeak = 44223"
## [5] "j =344 peak =343 savedpeak = 44247"
## [6] "j =344 peak =361 savedpeak = 44265"
## [7] "j =344 peak =380 savedpeak = 44284"
## [1] "j =345 peak =88 savedpeak = 44120"
## [2] "j =345 peak =109 savedpeak = 44141"
## [3] "j =345 peak =142 savedpeak = 44174"
## [4] "j =345 peak =159 savedpeak = 44191"
## [5] "j =345 peak =172 savedpeak = 44204"
## [6] "j =345 peak =189 savedpeak = 44221"
## [7] "j =345 peak =215 savedpeak = 44247"
## [8] "j =345 peak =232 savedpeak = 44264"
## [9] "j =345 peak =252 savedpeak = 44284"
## [10] "j =345 peak =312 savedpeak = 44344"
## [11] "j =345 peak =359 savedpeak = 44391"
## [12] "j =345 peak =389 savedpeak = 44421"
## [13] "j =345 peak =418 savedpeak = 44450"
## [1] "j =346 peak =87 savedpeak = 44247"
## [2] "j =346 peak =104 savedpeak = 44264"
## [3] "j =346 peak =124 savedpeak = 44284"
## [4] "j =346 peak =184 savedpeak = 44344"
## [5] "j =346 peak =231 savedpeak = 44391"
## [6] "j =346 peak =261 savedpeak = 44421"
## [7] "j =346 peak =290 savedpeak = 44450"
## [8] "j =346 peak =317 savedpeak = 44477"
## [9] "j =346 peak =342 savedpeak = 44502"
## [10] "j =346 peak =355 savedpeak = 44515"

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## [11] "j =346 peak =373 savedpeak = 44533"
## [12] "j =346 peak =397 savedpeak = 44557"
## [13] "j =346 peak =416 savedpeak = 44576"
## [14] "j =346 peak =432 savedpeak = 44592"
## [1] "j =347 peak =103 savedpeak = 44391"
## [2] "j =347 peak =133 savedpeak = 44421"
## [3] "j =347 peak =162 savedpeak = 44450"
## [4] "j =347 peak =189 savedpeak = 44477"
## [5] "j =347 peak =214 savedpeak = 44502"
## [6] "j =347 peak =227 savedpeak = 44515"
## [7] "j =347 peak =245 savedpeak = 44533"
## [8] "j =347 peak =269 savedpeak = 44557"
## [9] "j =347 peak =288 savedpeak = 44576"
## [10] "j =347 peak =304 savedpeak = 44592"
## [11] "j =347 peak =329 savedpeak = 44617"
## [12] "j =347 peak =348 savedpeak = 44636"
## [13] "j =347 peak =368 savedpeak = 44656"
## [14] "j =347 peak =401 savedpeak = 44689"
## [15] "j =347 peak =429 savedpeak = 44717"
## [1] "j =348 peak =86 savedpeak = 44502"
## [2] "j =348 peak =99 savedpeak = 44515"
## [3] "j =348 peak =117 savedpeak = 44533"
## [4] "j =348 peak =141 savedpeak = 44557"
## [5] "j =348 peak =160 savedpeak = 44576"
## [6] "j =348 peak =176 savedpeak = 44592"
## [7] "j =348 peak =201 savedpeak = 44617"
## [8] "j =348 peak =220 savedpeak = 44636"
## [9] "j =348 peak =240 savedpeak = 44656"
## [10] "j =348 peak =273 savedpeak = 44689"
## [11] "j =348 peak =301 savedpeak = 44717"
## [12] "j =348 peak =318 savedpeak = 44734"
## [13] "j =348 peak =338 savedpeak = 44754"
## [14] "j =348 peak =386 savedpeak = 44802"
## [15] "j =348 peak =417 savedpeak = 44833"
## [1] "j =349 peak =92 savedpeak = 44636"
## [2] "j =349 peak =112 savedpeak = 44656"
## [3] "j =349 peak =145 savedpeak = 44689"
## [4] "j =349 peak =173 savedpeak = 44717"
## [5] "j =349 peak =190 savedpeak = 44734"
## [6] "j =349 peak =210 savedpeak = 44754"
## [7] "j =349 peak =258 savedpeak = 44802"
## [8] "j =349 peak =289 savedpeak = 44833"
## [9] "j =349 peak =337 savedpeak = 44881"
## [10] "j =349 peak =350 savedpeak = 44894"
## [11] "j =349 peak =370 savedpeak = 44914"
## [12] "j =349 peak =396 savedpeak = 44940"
## [13] "j =349 peak =408 savedpeak = 44952"
## [1] "j =356 peak =122 savedpeak = 45562"
## [2] "j =356 peak =142 savedpeak = 45582"
## [3] "j =356 peak =173 savedpeak = 45613"
## [4] "j =356 peak =211 savedpeak = 45651"
## [5] "j =356 peak =223 savedpeak = 45663"
## [6] "j =356 peak =261 savedpeak = 45701"
## [7] "j =356 peak =295 savedpeak = 45735"

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## [8] "j =356 peak =330 savedpeak = 45770"
## [9] "j =356 peak =353 savedpeak = 45793"
## [10] "j =356 peak =382 savedpeak = 45822"
## [11] "j =356 peak =398 savedpeak = 45838"
## [1] "j =363 peak =83 savedpeak = 46419"
## [2] "j =363 peak =135 savedpeak = 46471"
## [3] "j =363 peak =162 savedpeak = 46498"
## [4] "j =363 peak =193 savedpeak = 46529"
## [5] "j =363 peak =206 savedpeak = 46542"
## [6] "j =363 peak =228 savedpeak = 46564"
## [7] "j =363 peak =242 savedpeak = 46578"
## [8] "j =363 peak =262 savedpeak = 46598"
## [9] "j =363 peak =296 savedpeak = 46632"
## [10] "j =363 peak =338 savedpeak = 46674"
## [11] "j =363 peak =365 savedpeak = 46701"
## [12] "j =363 peak =407 savedpeak = 46743"
## [13] "j =363 peak =430 savedpeak = 46766"
## [1] "j =369 peak =86 savedpeak = 47190"
## [2] "j =369 peak =153 savedpeak = 47257"
## [3] "j =369 peak =195 savedpeak = 47299"
## [4] "j =369 peak =239 savedpeak = 47343"
## [5] "j =369 peak =256 savedpeak = 47360"
## [6] "j =369 peak =292 savedpeak = 47396"
## [7] "j =369 peak =337 savedpeak = 47441"
## [8] "j =369 peak =364 savedpeak = 47468"
## [9] "j =369 peak =389 savedpeak = 47493"
## [10] "j =369 peak =430 savedpeak = 47534"
## [1] "j =370 peak =111 savedpeak = 47343"
## [2] "j =370 peak =128 savedpeak = 47360"
## [3] "j =370 peak =164 savedpeak = 47396"
## [4] "j =370 peak =209 savedpeak = 47441"
## [5] "j =370 peak =236 savedpeak = 47468"
## [6] "j =370 peak =261 savedpeak = 47493"
## [7] "j =370 peak =302 savedpeak = 47534"
## [8] "j =370 peak =322 savedpeak = 47554"
## [9] "j =370 peak =346 savedpeak = 47578"
## [10] "j =370 peak =358 savedpeak = 47590"
## [11] "j =370 peak =380 savedpeak = 47612"
## [12] "j =370 peak =404 savedpeak = 47636"
## [13] "j =370 peak =424 savedpeak = 47656"
## [1] "j =371 peak =81 savedpeak = 47441"
## [2] "j =371 peak =108 savedpeak = 47468"
## [3] "j =371 peak =133 savedpeak = 47493"
## [4] "j =371 peak =174 savedpeak = 47534"
## [5] "j =371 peak =194 savedpeak = 47554"
## [6] "j =371 peak =218 savedpeak = 47578"
## [7] "j =371 peak =230 savedpeak = 47590"
## [8] "j =371 peak =252 savedpeak = 47612"
## [9] "j =371 peak =276 savedpeak = 47636"
## [10] "j =371 peak =296 savedpeak = 47656"
## [11] "j =371 peak =310 savedpeak = 47670"
## [12] "j =371 peak =344 savedpeak = 47704"
## [13] "j =371 peak =368 savedpeak = 47728"
## [14] "j =371 peak =402 savedpeak = 47762"

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## [15] "j =371 peak =428 savedpeak = 47788"
## [1] "j =377 peak =112 savedpeak = 48240"
## [2] "j =377 peak =139 savedpeak = 48267"
## [3] "j =377 peak =164 savedpeak = 48292"
## [4] "j =377 peak =198 savedpeak = 48326"
## [5] "j =377 peak =225 savedpeak = 48353"
## [6] "j =377 peak =258 savedpeak = 48386"
## [7] "j =377 peak =302 savedpeak = 48430"
## [8] "j =377 peak =328 savedpeak = 48456"
## [9] "j =377 peak =362 savedpeak = 48490"
## [10] "j =377 peak =388 savedpeak = 48516"
## [11] "j =377 peak =406 savedpeak = 48534"
## [12] "j =377 peak =433 savedpeak = 48561"
## [1] "j =379 peak =106 savedpeak = 48490"
## [2] "j =379 peak =132 savedpeak = 48516"
## [3] "j =379 peak =150 savedpeak = 48534"
## [4] "j =379 peak =177 savedpeak = 48561"
## [5] "j =379 peak =195 savedpeak = 48579"
## [6] "j =379 peak =236 savedpeak = 48620"
## [7] "j =379 peak =259 savedpeak = 48643"
## [8] "j =379 peak =302 savedpeak = 48686"
## [9] "j =379 peak =315 savedpeak = 48699"
## [10] "j =379 peak =353 savedpeak = 48737"
## [11] "j =379 peak =394 savedpeak = 48778"
## [12] "j =379 peak =403 savedpeak = 48787"
## [13] "j =379 peak =435 savedpeak = 48819"
## [1] "j =380 peak =108 savedpeak = 48620"
## [2] "j =380 peak =117 savedpeak = 48629"
## [3] "j =380 peak =131 savedpeak = 48643"
## [4] "j =380 peak =174 savedpeak = 48686"
## [5] "j =380 peak =187 savedpeak = 48699"
## [6] "j =380 peak =225 savedpeak = 48737"
## [7] "j =380 peak =266 savedpeak = 48778"
## [8] "j =380 peak =275 savedpeak = 48787"
## [9] "j =380 peak =307 savedpeak = 48819"
## [10] "j =380 peak =345 savedpeak = 48857"
## [11] "j =380 peak =360 savedpeak = 48872"
## [12] "j =380 peak =383 savedpeak = 48895"
## [13] "j =380 peak =428 savedpeak = 48940"
## [1] "j =381 peak =97 savedpeak = 48737"
## [2] "j =381 peak =138 savedpeak = 48778"
## [3] "j =381 peak =147 savedpeak = 48787"
## [4] "j =381 peak =179 savedpeak = 48819"
## [5] "j =381 peak =217 savedpeak = 48857"
## [6] "j =381 peak =232 savedpeak = 48872"
## [7] "j =381 peak =255 savedpeak = 48895"
## [8] "j =381 peak =300 savedpeak = 48940"
## [9] "j =381 peak =314 savedpeak = 48954"
## [10] "j =381 peak =354 savedpeak = 48994"
## [11] "j =381 peak =380 savedpeak = 49020"
## [12] "j =381 peak =410 savedpeak = 49050"
## [1] "j =382 peak =89 savedpeak = 48857"
## [2] "j =382 peak =104 savedpeak = 48872"
## [3] "j =382 peak =127 savedpeak = 48895"

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## [4] "j =382 peak =172 savedpeak = 48940"
## [5] "j =382 peak =186 savedpeak = 48954"
## [6] "j =382 peak =226 savedpeak = 48994"
## [7] "j =382 peak =252 savedpeak = 49020"
## [8] "j =382 peak =282 savedpeak = 49050"
## [9] "j =382 peak =356 savedpeak = 49124"
## [10] "j =382 peak =382 savedpeak = 49150"
## [11] "j =382 peak =403 savedpeak = 49171"
## [12] "j =382 peak =419 savedpeak = 49187"
## [1] "j =390 peak =80 savedpeak = 49872"
## [2] "j =390 peak =114 savedpeak = 49906"
## [3] "j =390 peak =148 savedpeak = 49940"
## [4] "j =390 peak =164 savedpeak = 49956"
## [5] "j =390 peak =178 savedpeak = 49970"
## [6] "j =390 peak =213 savedpeak = 50005"
## [7] "j =390 peak =244 savedpeak = 50036"
## [8] "j =390 peak =275 savedpeak = 50067"
## [9] "j =390 peak =310 savedpeak = 50102"
## [10] "j =390 peak =330 savedpeak = 50122"
## [11] "j =390 peak =347 savedpeak = 50139"
## [12] "j =390 peak =368 savedpeak = 50160"
## [13] "j =390 peak =396 savedpeak = 50188"
## [14] "j =390 peak =413 savedpeak = 50205"
## [1] "j =391 peak =85 savedpeak = 50005"
## [2] "j =391 peak =116 savedpeak = 50036"
## [3] "j =391 peak =147 savedpeak = 50067"
## [4] "j =391 peak =182 savedpeak = 50102"
## [5] "j =391 peak =202 savedpeak = 50122"
## [6] "j =391 peak =219 savedpeak = 50139"
## [7] "j =391 peak =240 savedpeak = 50160"
## [8] "j =391 peak =268 savedpeak = 50188"
## [9] "j =391 peak =285 savedpeak = 50205"
## [10] "j =391 peak =319 savedpeak = 50239"
## [11] "j =391 peak =355 savedpeak = 50275"
## [12] "j =391 peak =373 savedpeak = 50293"
## [13] "j =391 peak =395 savedpeak = 50315"
## [1] "j =392 peak =91 savedpeak = 50139"
## [2] "j =392 peak =112 savedpeak = 50160"
## [3] "j =392 peak =140 savedpeak = 50188"
## [4] "j =392 peak =157 savedpeak = 50205"
## [5] "j =392 peak =191 savedpeak = 50239"
## [6] "j =392 peak =227 savedpeak = 50275"
## [7] "j =392 peak =245 savedpeak = 50293"
## [8] "j =392 peak =267 savedpeak = 50315"
## [9] "j =392 peak =314 savedpeak = 50362"
## [10] "j =392 peak =327 savedpeak = 50375"
## [11] "j =392 peak =358 savedpeak = 50406"
## [12] "j =392 peak =378 savedpeak = 50426"
## [13] "j =392 peak =404 savedpeak = 50452"
## [14] "j =392 peak =413 savedpeak = 50461"
## [15] "j =392 peak =433 savedpeak = 50481"
## Spectrum exam1d_13C has 43 peaks.
## [1] "j =7 peak =91 savedpeak = 859" "j =7 peak =117 savedpeak = 885"
## [3] "j =7 peak =158 savedpeak = 926" "j =7 peak =192 savedpeak = 960"

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## [5] "j =7 peak =224 savedpeak = 992" "j =7 peak =250 savedpeak = 1018"
## [7] "j =7 peak =261 savedpeak = 1029" "j =7 peak =297 savedpeak = 1065"
## [9] "j =7 peak =347 savedpeak = 1115" "j =7 peak =365 savedpeak = 1133"
## [11] "j =7 peak =388 savedpeak = 1156" "j =7 peak =417 savedpeak = 1185"
## [1] "j =8 peak =96 savedpeak = 992" "j =8 peak =122 savedpeak = 1018"
## [3] "j =8 peak =133 savedpeak = 1029" "j =8 peak =169 savedpeak = 1065"
## [5] "j =8 peak =219 savedpeak = 1115" "j =8 peak =237 savedpeak = 1133"
## [7] "j =8 peak =260 savedpeak = 1156" "j =8 peak =289 savedpeak = 1185"
## [9] "j =8 peak =310 savedpeak = 1206" "j =8 peak =334 savedpeak = 1230"
## [11] "j =8 peak =380 savedpeak = 1276"
## [1] "j =9 peak =91 savedpeak = 1115" "j =9 peak =109 savedpeak = 1133"
## [3] "j =9 peak =132 savedpeak = 1156" "j =9 peak =161 savedpeak = 1185"
## [5] "j =9 peak =182 savedpeak = 1206" "j =9 peak =206 savedpeak = 1230"
## [7] "j =9 peak =252 savedpeak = 1276" "j =9 peak =312 savedpeak = 1336"
## [9] "j =9 peak =353 savedpeak = 1377" "j =9 peak =392 savedpeak = 1416"
## [1] "j =10 peak =78 savedpeak = 1230" "j =10 peak =124 savedpeak = 1276"
## [3] "j =10 peak =184 savedpeak = 1336" "j =10 peak =225 savedpeak = 1377"
## [5] "j =10 peak =264 savedpeak = 1416" "j =10 peak =348 savedpeak = 1500"
## [7] "j =10 peak =379 savedpeak = 1531" "j =10 peak =407 savedpeak = 1559"
## [9] "j =10 peak =434 savedpeak = 1586"
## [1] "j =11 peak =97 savedpeak = 1377" "j =11 peak =136 savedpeak = 1416"
## [3] "j =11 peak =220 savedpeak = 1500" "j =11 peak =256 savedpeak = 1536"
## [5] "j =11 peak =279 savedpeak = 1559" "j =11 peak =306 savedpeak = 1586"
## [7] "j =11 peak =320 savedpeak = 1600" "j =11 peak =345 savedpeak = 1625"
## [9] "j =11 peak =383 savedpeak = 1663" "j =11 peak =425 savedpeak = 1705"
## [1] "j =12 peak =92 savedpeak = 1500" "j =12 peak =128 savedpeak = 1536"
## [3] "j =12 peak =151 savedpeak = 1559" "j =12 peak =178 savedpeak = 1586"
## [5] "j =12 peak =192 savedpeak = 1600" "j =12 peak =217 savedpeak = 1625"
## [7] "j =12 peak =255 savedpeak = 1663" "j =12 peak =297 savedpeak = 1705"
## [9] "j =12 peak =318 savedpeak = 1726" "j =12 peak =338 savedpeak = 1746"
## [11] "j =12 peak =365 savedpeak = 1773" "j =12 peak =400 savedpeak = 1808"
## [13] "j =12 peak =433 savedpeak = 1841"
## [1] "j =19 peak =77 savedpeak = 2381" "j =19 peak =207 savedpeak = 2511"
## [3] "j =19 peak =243 savedpeak = 2547" "j =19 peak =351 savedpeak = 2655"
## [5] "j =19 peak =409 savedpeak = 2713"
## [1] "j =22 peak =350 savedpeak = 3038"
## [1] "j =23 peak =222 savedpeak = 3038" "j =23 peak =323 savedpeak = 3139"
## [3] "j =23 peak =388 savedpeak = 3204"
## [1] "j =24 peak =94 savedpeak = 3038" "j =24 peak =195 savedpeak = 3139"
## [3] "j =24 peak =260 savedpeak = 3204"
## [1] "j =25 peak =132 savedpeak = 3204" "j =25 peak =219 savedpeak = 3291"
## [3] "j =25 peak =421 savedpeak = 3493"
## [1] "j =26 peak =91 savedpeak = 3291" "j =26 peak =293 savedpeak = 3493"
## [3] "j =26 peak =387 savedpeak = 3587"
## [1] "j =32 peak =153 savedpeak = 4121" "j =32 peak =348 savedpeak = 4316"
## [3] "j =32 peak =392 savedpeak = 4360"
## [1] "j =38 peak =86 savedpeak = 4822" "j =38 peak =180 savedpeak = 4916"
## [3] "j =38 peak =216 savedpeak = 4952" "j =38 peak =238 savedpeak = 4974"
## [5] "j =38 peak =274 savedpeak = 5010" "j =38 peak =317 savedpeak = 5053"
## [7] "j =38 peak =389 savedpeak = 5125"
## [1] "j =41 peak =291 savedpeak = 5411" "j =41 peak =307 savedpeak = 5427"
## [3] "j =41 peak =373 savedpeak = 5493" "j =41 peak =420 savedpeak = 5540"
## [1] "j =42 peak =126 savedpeak = 5374" "j =42 peak =163 savedpeak = 5411"
## [3] "j =42 peak =179 savedpeak = 5427" "j =42 peak =201 savedpeak = 5449"

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## [5] "j =42 peak =245 savedpeak = 5493" "j =42 peak =292 savedpeak = 5540"
## [7] "j =42 peak =312 savedpeak = 5560" "j =42 peak =325 savedpeak = 5573"
## [9] "j =42 peak =348 savedpeak = 5596" "j =42 peak =392 savedpeak = 5640"
## [11] "j =42 peak =408 savedpeak = 5656"
## [1] "j =43 peak =117 savedpeak = 5493" "j =43 peak =164 savedpeak = 5540"
## [3] "j =43 peak =184 savedpeak = 5560" "j =43 peak =197 savedpeak = 5573"
## [5] "j =43 peak =220 savedpeak = 5596" "j =43 peak =264 savedpeak = 5640"
## [7] "j =43 peak =280 savedpeak = 5656" "j =43 peak =333 savedpeak = 5709"
## [9] "j =43 peak =348 savedpeak = 5724" "j =43 peak =394 savedpeak = 5770"
## [1] "j =44 peak =92 savedpeak = 5596" "j =44 peak =136 savedpeak = 5640"
## [3] "j =44 peak =152 savedpeak = 5656" "j =44 peak =205 savedpeak = 5709"
## [5] "j =44 peak =220 savedpeak = 5724" "j =44 peak =266 savedpeak = 5770"
## [7] "j =44 peak =323 savedpeak = 5827" "j =44 peak =355 savedpeak = 5859"
## [9] "j =44 peak =375 savedpeak = 5879" "j =44 peak =410 savedpeak = 5914"
## [1] "j =45 peak =138 savedpeak = 5770" "j =45 peak =247 savedpeak = 5879"
## [3] "j =45 peak =415 savedpeak = 6047"
## [1] "j =46 peak =406 savedpeak = 6166"
## [1] "j =47 peak =278 savedpeak = 6166"
## [1] "j =48 peak =150 savedpeak = 6166"
## [1] "j =50 peak =128 savedpeak = 6400" "j =50 peak =141 savedpeak = 6413"
## [3] "j =50 peak =173 savedpeak = 6445" "j =50 peak =289 savedpeak = 6561"
## [5] "j =50 peak =426 savedpeak = 6698"
## [1] "j =52 peak =315 savedpeak = 6843"
## [1] "j =53 peak =187 savedpeak = 6843" "j =53 peak =365 savedpeak = 7021"
## [3] "j =53 peak =383 savedpeak = 7039" "j =53 peak =419 savedpeak = 7075"
## [5] "j =53 peak =431 savedpeak = 7087"
## [1] "j =54 peak =237 savedpeak = 7021" "j =54 peak =255 savedpeak = 7039"
## [3] "j =54 peak =291 savedpeak = 7075" "j =54 peak =303 savedpeak = 7087"
## [5] "j =54 peak =400 savedpeak = 7184"
## [1] "j =55 peak =109 savedpeak = 7021" "j =55 peak =127 savedpeak = 7039"
## [3] "j =55 peak =163 savedpeak = 7075" "j =55 peak =175 savedpeak = 7087"
## [5] "j =55 peak =272 savedpeak = 7184" "j =55 peak =337 savedpeak = 7249"
## [7] "j =55 peak =356 savedpeak = 7268" "j =55 peak =390 savedpeak = 7302"
## [9] "j =55 peak =415 savedpeak = 7327"
## [1] "j =56 peak =112 savedpeak = 7152" "j =56 peak =144 savedpeak = 7184"
## [3] "j =56 peak =209 savedpeak = 7249" "j =56 peak =228 savedpeak = 7268"
## [5] "j =56 peak =262 savedpeak = 7302" "j =56 peak =287 savedpeak = 7327"
## [7] "j =56 peak =373 savedpeak = 7413" "j =56 peak =394 savedpeak = 7434"
## [1] "j =57 peak =81 savedpeak = 7249" "j =57 peak =100 savedpeak = 7268"
## [3] "j =57 peak =134 savedpeak = 7302" "j =57 peak =159 savedpeak = 7327"
## [5] "j =57 peak =193 savedpeak = 7361" "j =57 peak =245 savedpeak = 7413"
## [7] "j =57 peak =266 savedpeak = 7434" "j =57 peak =327 savedpeak = 7495"
## [9] "j =57 peak =344 savedpeak = 7512" "j =57 peak =429 savedpeak = 7597"
## [1] "j =58 peak =90 savedpeak = 7386" "j =58 peak =117 savedpeak = 7413"
## [3] "j =58 peak =138 savedpeak = 7434" "j =58 peak =199 savedpeak = 7495"
## [5] "j =58 peak =216 savedpeak = 7512" "j =58 peak =301 savedpeak = 7597"
## [7] "j =58 peak =331 savedpeak = 7627" "j =58 peak =361 savedpeak = 7657"
## [9] "j =58 peak =373 savedpeak = 7669" "j =58 peak =413 savedpeak = 7709"
## [1] "j =59 peak =88 savedpeak = 7512" "j =59 peak =173 savedpeak = 7597"
## [3] "j =59 peak =203 savedpeak = 7627" "j =59 peak =233 savedpeak = 7657"
## [5] "j =59 peak =245 savedpeak = 7669" "j =59 peak =285 savedpeak = 7709"
## [7] "j =59 peak =310 savedpeak = 7734" "j =59 peak =338 savedpeak = 7762"
## [9] "j =59 peak =360 savedpeak = 7784" "j =59 peak =420 savedpeak = 7844"
## [1] "j =60 peak =105 savedpeak = 7657" "j =60 peak =117 savedpeak = 7669"

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## [3] "j =60 peak =157 savedpeak = 7709" "j =60 peak =182 savedpeak = 7734"
## [5] "j =60 peak =210 savedpeak = 7762" "j =60 peak =292 savedpeak = 7844"
## [1] "j =65 peak =114 savedpeak = 8306" "j =65 peak =285 savedpeak = 8477"
## [3] "j =65 peak =299 savedpeak = 8491"
## [1] "j =66 peak =120 savedpeak = 8440" "j =66 peak =157 savedpeak = 8477"
## [3] "j =66 peak =171 savedpeak = 8491"
## [1] "j =69 peak =91 savedpeak = 8795" "j =69 peak =105 savedpeak = 8809"
## [3] "j =69 peak =187 savedpeak = 8891" "j =69 peak =406 savedpeak = 9110"
## [1] "j =70 peak =112 savedpeak = 8944" "j =70 peak =255 savedpeak = 9087"
## [3] "j =70 peak =278 savedpeak = 9110" "j =70 peak =346 savedpeak = 9178"
## [1] "j =71 peak =127 savedpeak = 9087" "j =71 peak =150 savedpeak = 9110"
## [3] "j =71 peak =218 savedpeak = 9178" "j =71 peak =238 savedpeak = 9198"
## [5] "j =71 peak =284 savedpeak = 9244" "j =71 peak =302 savedpeak = 9262"
## [7] "j =71 peak =365 savedpeak = 9325"
## [1] "j =72 peak =90 savedpeak = 9178" "j =72 peak =110 savedpeak = 9198"
## [3] "j =72 peak =156 savedpeak = 9244" "j =72 peak =174 savedpeak = 9262"
## [5] "j =72 peak =237 savedpeak = 9325" "j =72 peak =311 savedpeak = 9399"
## [7] "j =72 peak =335 savedpeak = 9423" "j =72 peak =384 savedpeak = 9472"
## [9] "j =72 peak =406 savedpeak = 9494" "j =72 peak =419 savedpeak = 9507"
## [1] "j =73 peak =109 savedpeak = 9325" "j =73 peak =183 savedpeak = 9399"
## [3] "j =73 peak =207 savedpeak = 9423" "j =73 peak =256 savedpeak = 9472"
## [5] "j =73 peak =291 savedpeak = 9507" "j =73 peak =320 savedpeak = 9536"
## [7] "j =73 peak =342 savedpeak = 9558" "j =73 peak =380 savedpeak = 9596"
## [1] "j =74 peak =79 savedpeak = 9423" "j =74 peak =128 savedpeak = 9472"
## [3] "j =74 peak =163 savedpeak = 9507" "j =74 peak =192 savedpeak = 9536"
## [5] "j =74 peak =214 savedpeak = 9558" "j =74 peak =252 savedpeak = 9596"
## [7] "j =74 peak =319 savedpeak = 9663" "j =74 peak =341 savedpeak = 9685"
## [9] "j =74 peak =364 savedpeak = 9708"
## [1] "j =75 peak =86 savedpeak = 9558" "j =75 peak =124 savedpeak = 9596"
## [3] "j =75 peak =191 savedpeak = 9663" "j =75 peak =213 savedpeak = 9685"
## [5] "j =75 peak =236 savedpeak = 9708" "j =75 peak =335 savedpeak = 9807"
## [7] "j =75 peak =349 savedpeak = 9821" "j =75 peak =367 savedpeak = 9839"
## [1] "j =76 peak =85 savedpeak = 9685" "j =76 peak =108 savedpeak = 9708"
## [3] "j =76 peak =207 savedpeak = 9807" "j =76 peak =221 savedpeak = 9821"
## [5] "j =76 peak =239 savedpeak = 9839" "j =76 peak =315 savedpeak = 9915"
## [7] "j =76 peak =333 savedpeak = 9933" "j =76 peak =393 savedpeak = 9993"
## [9] "j =76 peak =427 savedpeak = 10027"
## [1] "j =77 peak =79 savedpeak = 9807"
## [2] "j =77 peak =93 savedpeak = 9821"
## [3] "j =77 peak =111 savedpeak = 9839"
## [4] "j =77 peak =187 savedpeak = 9915"
## [5] "j =77 peak =205 savedpeak = 9933"
## [6] "j =77 peak =238 savedpeak = 9966"
## [7] "j =77 peak =265 savedpeak = 9993"
## [8] "j =77 peak =299 savedpeak = 10027"
## [9] "j =77 peak =331 savedpeak = 10059"
## [10] "j =77 peak =382 savedpeak = 10110"
## [11] "j =77 peak =406 savedpeak = 10134"
## [1] "j =78 peak =77 savedpeak = 9933"
## [2] "j =78 peak =110 savedpeak = 9966"
## [3] "j =78 peak =137 savedpeak = 9993"
## [4] "j =78 peak =171 savedpeak = 10027"
## [5] "j =78 peak =203 savedpeak = 10059"
## [6] "j =78 peak =254 savedpeak = 10110"

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## [7] "j =78 peak =278 savedpeak = 10134"
## [8] "j =78 peak =323 savedpeak = 10179"
## [9] "j =78 peak =344 savedpeak = 10200"
## [10] "j =78 peak =366 savedpeak = 10222"
## [11] "j =78 peak =390 savedpeak = 10246"
## [12] "j =78 peak =425 savedpeak = 10281"
## [1] "j =79 peak =126 savedpeak = 10110" "j =79 peak =150 savedpeak = 10134"
## [3] "j =79 peak =195 savedpeak = 10179" "j =79 peak =216 savedpeak = 10200"
## [5] "j =79 peak =238 savedpeak = 10222" "j =79 peak =262 savedpeak = 10246"
## [7] "j =79 peak =315 savedpeak = 10299" "j =79 peak =386 savedpeak = 10370"
## [9] "j =79 peak =437 savedpeak = 10421"
## [1] "j =80 peak =88 savedpeak = 10200"
## [2] "j =80 peak =110 savedpeak = 10222"
## [3] "j =80 peak =139 savedpeak = 10251"
## [4] "j =80 peak =258 savedpeak = 10370"
## [5] "j =80 peak =309 savedpeak = 10421"
## [6] "j =80 peak =358 savedpeak = 10470"
## [7] "j =80 peak =370 savedpeak = 10482"
## [8] "j =80 peak =382 savedpeak = 10494"
## [9] "j =80 peak =421 savedpeak = 10533"
## [10] "j =80 peak =432 savedpeak = 10544"
## [1] "j =82 peak =390 savedpeak = 10758" "j =82 peak =410 savedpeak = 10778"
## [1] "j =83 peak =261 savedpeak = 10757" "j =83 peak =282 savedpeak = 10778"
## [1] "j =84 peak =133 savedpeak = 10757" "j =84 peak =157 savedpeak = 10781"
## [1] "j =86 peak =103 savedpeak = 10983" "j =86 peak =161 savedpeak = 11041"
## [3] "j =86 peak =280 savedpeak = 11160" "j =86 peak =361 savedpeak = 11241"
## [5] "j =86 peak =376 savedpeak = 11256" "j =86 peak =388 savedpeak = 11268"
## [1] "j =87 peak =152 savedpeak = 11160" "j =87 peak =233 savedpeak = 11241"
## [3] "j =87 peak =248 savedpeak = 11256" "j =87 peak =260 savedpeak = 11268"
## [5] "j =87 peak =307 savedpeak = 11315" "j =87 peak =381 savedpeak = 11389"
## [1] "j =88 peak =399 savedpeak = 11535"
## [1] "j =89 peak =271 savedpeak = 11535"
## [1] "j =90 peak =143 savedpeak = 11535"
## [1] "j =92 peak =93 savedpeak = 11741" "j =92 peak =120 savedpeak = 11768"
## [3] "j =92 peak =156 savedpeak = 11804" "j =92 peak =169 savedpeak = 11817"
## [5] "j =92 peak =194 savedpeak = 11842" "j =92 peak =258 savedpeak = 11906"
## [7] "j =92 peak =278 savedpeak = 11926" "j =92 peak =330 savedpeak = 11978"
## [1] "j =94 peak =95 savedpeak = 11999"
## [2] "j =94 peak =111 savedpeak = 12015"
## [3] "j =94 peak =148 savedpeak = 12052"
## [4] "j =94 peak =228 savedpeak = 12132"
## [5] "j =94 peak =250 savedpeak = 12154"
## [6] "j =94 peak =289 savedpeak = 12193"
## [7] "j =94 peak =321 savedpeak = 12225"
## [8] "j =94 peak =345 savedpeak = 12249"
## [9] "j =94 peak =369 savedpeak = 12273"
## [10] "j =94 peak =395 savedpeak = 12299"
## [11] "j =94 peak =413 savedpeak = 12317"
## [12] "j =94 peak =430 savedpeak = 12334"
## [1] "j =101 peak =135 savedpeak = 12935"
## [2] "j =101 peak =170 savedpeak = 12970"
## [3] "j =101 peak =220 savedpeak = 13020"
## [4] "j =101 peak =236 savedpeak = 13036"
## [5] "j =101 peak =254 savedpeak = 13054"

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## [6] "j =101 peak =276 savedpeak = 13076"
## [7] "j =101 peak =302 savedpeak = 13102"
## [8] "j =101 peak =366 savedpeak = 13166"
## [9] "j =101 peak =422 savedpeak = 13222"
## [1] "j =103 peak =378 savedpeak = 13434"
## [1] "j =104 peak =250 savedpeak = 13434"
## [1] "j =105 peak =122 savedpeak = 13434"
## [1] "j =106 peak =80 savedpeak = 13520"
## [2] "j =106 peak =99 savedpeak = 13539"
## [3] "j =106 peak =122 savedpeak = 13562"
## [4] "j =106 peak =152 savedpeak = 13592"
## [5] "j =106 peak =200 savedpeak = 13640"
## [6] "j =106 peak =301 savedpeak = 13741"
## [7] "j =106 peak =325 savedpeak = 13765"
## [8] "j =106 peak =374 savedpeak = 13814"
## [9] "j =106 peak =417 savedpeak = 13857"
## [1] "j =110 peak =91 savedpeak = 14043"
## [2] "j =110 peak =107 savedpeak = 14059"
## [3] "j =110 peak =135 savedpeak = 14087"
## [4] "j =110 peak =187 savedpeak = 14139"
## [5] "j =110 peak =212 savedpeak = 14164"
## [6] "j =110 peak =238 savedpeak = 14190"
## [7] "j =110 peak =280 savedpeak = 14232"
## [8] "j =110 peak =302 savedpeak = 14254"
## [9] "j =110 peak =327 savedpeak = 14279"
## [10] "j =110 peak =408 savedpeak = 14360"
## [1] "j =111 peak =84 savedpeak = 14164"
## [2] "j =111 peak =110 savedpeak = 14190"
## [3] "j =111 peak =152 savedpeak = 14232"
## [4] "j =111 peak =174 savedpeak = 14254"
## [5] "j =111 peak =199 savedpeak = 14279"
## [6] "j =111 peak =251 savedpeak = 14331"
## [7] "j =111 peak =280 savedpeak = 14360"
## [8] "j =111 peak =343 savedpeak = 14423"
## [9] "j =111 peak =367 savedpeak = 14447"
## [10] "j =111 peak =380 savedpeak = 14460"
## [11] "j =111 peak =397 savedpeak = 14477"
## [12] "j =111 peak =421 savedpeak = 14501"
## [1] "j =120 peak =111 savedpeak = 15343"
## [2] "j =120 peak =146 savedpeak = 15378"
## [3] "j =120 peak =169 savedpeak = 15401"
## [4] "j =120 peak =193 savedpeak = 15425"
## [5] "j =120 peak =209 savedpeak = 15441"
## [6] "j =120 peak =257 savedpeak = 15489"
## [7] "j =120 peak =281 savedpeak = 15513"
## [8] "j =120 peak =288 savedpeak = 15520"
## [9] "j =120 peak =314 savedpeak = 15546"
## [10] "j =120 peak =353 savedpeak = 15585"
## [11] "j =120 peak =377 savedpeak = 15609"
## [12] "j =120 peak =395 savedpeak = 15627"
## [13] "j =120 peak =417 savedpeak = 15649"
## [1] "j =121 peak =81 savedpeak = 15441"
## [2] "j =121 peak =129 savedpeak = 15489"
## [3] "j =121 peak =153 savedpeak = 15513"

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## [4] "j =121 peak =160 savedpeak = 15520"
## [5] "j =121 peak =186 savedpeak = 15546"
## [6] "j =121 peak =225 savedpeak = 15585"
## [7] "j =121 peak =249 savedpeak = 15609"
## [8] "j =121 peak =267 savedpeak = 15627"
## [9] "j =121 peak =289 savedpeak = 15649"
## [10] "j =121 peak =344 savedpeak = 15704"
## [11] "j =121 peak =406 savedpeak = 15766"
## [12] "j =121 peak =422 savedpeak = 15782"
## [1] "j =122 peak =97 savedpeak = 15585"
## [2] "j =122 peak =121 savedpeak = 15609"
## [3] "j =122 peak =139 savedpeak = 15627"
## [4] "j =122 peak =161 savedpeak = 15649"
## [5] "j =122 peak =216 savedpeak = 15704"
## [6] "j =122 peak =278 savedpeak = 15766"
## [7] "j =122 peak =349 savedpeak = 15837"
## [8] "j =122 peak =365 savedpeak = 15853"
## [9] "j =122 peak =389 savedpeak = 15877"
## [10] "j =122 peak =428 savedpeak = 15916"
## [1] "j =123 peak =88 savedpeak = 15704"
## [2] "j =123 peak =150 savedpeak = 15766"
## [3] "j =123 peak =221 savedpeak = 15837"
## [4] "j =123 peak =237 savedpeak = 15853"
## [5] "j =123 peak =261 savedpeak = 15877"
## [6] "j =123 peak =300 savedpeak = 15916"
## [7] "j =123 peak =314 savedpeak = 15930"
## [8] "j =123 peak =337 savedpeak = 15953"
## [9] "j =123 peak =352 savedpeak = 15968"
## [10] "j =123 peak =384 savedpeak = 16000"
## [11] "j =123 peak =428 savedpeak = 16044"
## [1] "j =124 peak =93 savedpeak = 15837"
## [2] "j =124 peak =109 savedpeak = 15853"
## [3] "j =124 peak =133 savedpeak = 15877"
## [4] "j =124 peak =172 savedpeak = 15916"
## [5] "j =124 peak =186 savedpeak = 15930"
## [6] "j =124 peak =209 savedpeak = 15953"
## [7] "j =124 peak =224 savedpeak = 15968"
## [8] "j =124 peak =256 savedpeak = 16000"
## [9] "j =124 peak =300 savedpeak = 16044"
## [10] "j =124 peak =320 savedpeak = 16064"
## [11] "j =124 peak =341 savedpeak = 16085"
## [12] "j =124 peak =369 savedpeak = 16113"
## [13] "j =124 peak =395 savedpeak = 16139"
## [14] "j =124 peak =424 savedpeak = 16168"
## [1] "j =125 peak =81 savedpeak = 15953"
## [2] "j =125 peak =96 savedpeak = 15968"
## [3] "j =125 peak =128 savedpeak = 16000"
## [4] "j =125 peak =172 savedpeak = 16044"
## [5] "j =125 peak =192 savedpeak = 16064"
## [6] "j =125 peak =213 savedpeak = 16085"
## [7] "j =125 peak =241 savedpeak = 16113"
## [8] "j =125 peak =267 savedpeak = 16139"
## [9] "j =125 peak =296 savedpeak = 16168"
## [10] "j =125 peak =322 savedpeak = 16194"

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## [11] "j =125 peak =358 savedpeak = 16230"
## [12] "j =125 peak =372 savedpeak = 16244"
## [1] "j =126 peak =85 savedpeak = 16085"
## [2] "j =126 peak =113 savedpeak = 16113"
## [3] "j =126 peak =139 savedpeak = 16139"
## [4] "j =126 peak =168 savedpeak = 16168"
## [5] "j =126 peak =194 savedpeak = 16194"
## [6] "j =126 peak =230 savedpeak = 16230"
## [7] "j =126 peak =244 savedpeak = 16244"
## [8] "j =126 peak =267 savedpeak = 16267"
## [9] "j =126 peak =331 savedpeak = 16331"
## [10] "j =126 peak =396 savedpeak = 16396"
## [1] "j =127 peak =102 savedpeak = 16230"
## [2] "j =127 peak =116 savedpeak = 16244"
## [3] "j =127 peak =139 savedpeak = 16267"
## [4] "j =127 peak =151 savedpeak = 16279"
## [5] "j =127 peak =203 savedpeak = 16331"
## [6] "j =127 peak =268 savedpeak = 16396"
## [7] "j =127 peak =311 savedpeak = 16439"
## [8] "j =127 peak =339 savedpeak = 16467"
## [9] "j =127 peak =368 savedpeak = 16496"
## [10] "j =127 peak =406 savedpeak = 16534"
## [1] "j =129 peak =83 savedpeak = 16467"
## [2] "j =129 peak =112 savedpeak = 16496"
## [3] "j =129 peak =150 savedpeak = 16534"
## [4] "j =129 peak =214 savedpeak = 16598"
## [5] "j =129 peak =249 savedpeak = 16633"
## [6] "j =129 peak =274 savedpeak = 16658"
## [7] "j =129 peak =287 savedpeak = 16671"
## [8] "j =129 peak =340 savedpeak = 16724"
## [9] "j =129 peak =350 savedpeak = 16734"
## [10] "j =129 peak =377 savedpeak = 16761"
## [11] "j =129 peak =406 savedpeak = 16790"
## [1] "j =132 peak =383 savedpeak = 17151"
## [1] "j =133 peak =313 savedpeak = 17209"
## [2] "j =133 peak =399 savedpeak = 17295"
## [3] "j =133 peak =435 savedpeak = 17331"
## [1] "j =137 peak =140 savedpeak = 17548"
## [1] "j =139 peak =339 savedpeak = 18003"
## [1] "j =140 peak =211 savedpeak = 18003"
## [1] "j =141 peak =83 savedpeak = 18003"
## [1] "j =142 peak =92 savedpeak = 18140"
## [2] "j =142 peak =237 savedpeak = 18285"
## [3] "j =142 peak =280 savedpeak = 18328"
## [4] "j =142 peak =338 savedpeak = 18386"
## [1] "j =143 peak =109 savedpeak = 18285"
## [2] "j =143 peak =152 savedpeak = 18328"
## [3] "j =143 peak =210 savedpeak = 18386"
## [4] "j =143 peak =232 savedpeak = 18408"
## [5] "j =143 peak =304 savedpeak = 18480"
## [1] "j =144 peak =82 savedpeak = 18386"
## [2] "j =144 peak =104 savedpeak = 18408"
## [3] "j =144 peak =176 savedpeak = 18480"
## [1] "j =155 peak =82 savedpeak = 19794"

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## [2] "j =155 peak =176 savedpeak = 19888"
## [3] "j =155 peak =241 savedpeak = 19953"
## [4] "j =155 peak =342 savedpeak = 20054"
## [5] "j =155 peak =429 savedpeak = 20141"
## [1] "j =156 peak =113 savedpeak = 19953"
## [2] "j =156 peak =214 savedpeak = 20054"
## [3] "j =156 peak =301 savedpeak = 20141"
## [4] "j =156 peak =417 savedpeak = 20257"
## [1] "j =157 peak =86 savedpeak = 20054"
## [2] "j =157 peak =173 savedpeak = 20141"
## [3] "j =157 peak =289 savedpeak = 20257"
## [4] "j =157 peak =346 savedpeak = 20314"
## [1] "j =165 peak =81 savedpeak = 21073"
## [2] "j =165 peak =110 savedpeak = 21102"
## [3] "j =165 peak =146 savedpeak = 21138"
## [4] "j =165 peak =189 savedpeak = 21181"
## [5] "j =165 peak =276 savedpeak = 21268"
## [6] "j =165 peak =355 savedpeak = 21347"
## [7] "j =165 peak =413 savedpeak = 21405"
## [1] "j =171 peak =136 savedpeak = 21896"
## [2] "j =171 peak =288 savedpeak = 22048"
## [3] "j =171 peak =367 savedpeak = 22127"
## [4] "j =171 peak =432 savedpeak = 22192"
## [1] "j =172 peak =160 savedpeak = 22048"
## [2] "j =172 peak =239 savedpeak = 22127"
## [3] "j =172 peak =304 savedpeak = 22192"
## [4] "j =172 peak =362 savedpeak = 22250"
## [5] "j =172 peak =397 savedpeak = 22285"
## [1] "j =177 peak =309 savedpeak = 22837"
## [2] "j =177 peak =432 savedpeak = 22960"
## [1] "j =178 peak =181 savedpeak = 22837"
## [2] "j =178 peak =304 savedpeak = 22960"
## [3] "j =178 peak =386 savedpeak = 23042"
## [1] "j =179 peak =176 savedpeak = 22960"
## [2] "j =179 peak =258 savedpeak = 23042"
## [1] "j =180 peak =130 savedpeak = 23042"
## [2] "j =180 peak =368 savedpeak = 23280"
## [3] "j =180 peak =422 savedpeak = 23334"
## [1] "j =181 peak =240 savedpeak = 23280"
## [2] "j =181 peak =294 savedpeak = 23334"
## [1] "j =182 peak =112 savedpeak = 23280"
## [2] "j =182 peak =166 savedpeak = 23334"
## [3] "j =182 peak =325 savedpeak = 23493"
## [4] "j =182 peak =360 savedpeak = 23528"
## [1] "j =183 peak =197 savedpeak = 23493"
## [2] "j =183 peak =232 savedpeak = 23528"
## [3] "j =183 peak =373 savedpeak = 23669"
## [4] "j =183 peak =435 savedpeak = 23731"
## [1] "j =184 peak =104 savedpeak = 23528"
## [2] "j =184 peak =245 savedpeak = 23669"
## [3] "j =184 peak =307 savedpeak = 23731"
## [1] "j =185 peak =117 savedpeak = 23669"
## [2] "j =185 peak =179 savedpeak = 23731"
## [3] "j =185 peak =219 savedpeak = 23771"

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## [1] "j =186 peak =91 savedpeak = 23771"
## [1] "j =187 peak =406 savedpeak = 24214"
## [1] "j =188 peak =278 savedpeak = 24214"
## [2] "j =188 peak =311 savedpeak = 24247"
## [3] "j =188 peak =342 savedpeak = 24278"
## [1] "j =189 peak =150 savedpeak = 24214"
## [2] "j =189 peak =183 savedpeak = 24247"
## [3] "j =189 peak =214 savedpeak = 24278"
## [1] "j =190 peak =86 savedpeak = 24278"
## [1] "j =192 peak =332 savedpeak = 24780"
## [1] "j =193 peak =204 savedpeak = 24780"
## [1] "j =194 peak =76 savedpeak = 24780"
## [1] "j =195 peak =82 savedpeak = 24914"
## [2] "j =195 peak =248 savedpeak = 25080"
## [3] "j =195 peak =380 savedpeak = 25212"
## [4] "j =195 peak =425 savedpeak = 25257"
## [1] "j =196 peak =366 savedpeak = 25326"
## [2] "j =196 peak =419 savedpeak = 25379"
## [1] "j =197 peak =238 savedpeak = 25326"
## [2] "j =197 peak =291 savedpeak = 25379"
## [3] "j =197 peak =346 savedpeak = 25434"
## [4] "j =197 peak =403 savedpeak = 25491"
## [1] "j =198 peak =110 savedpeak = 25326"
## [2] "j =198 peak =163 savedpeak = 25379"
## [3] "j =198 peak =218 savedpeak = 25434"
## [4] "j =198 peak =275 savedpeak = 25491"
## [5] "j =198 peak =431 savedpeak = 25647"
## [1] "j =199 peak =90 savedpeak = 25434"
## [2] "j =199 peak =147 savedpeak = 25491"
## [3] "j =199 peak =303 savedpeak = 25647"
## [1] "j =200 peak =175 savedpeak = 25647"
## [2] "j =200 peak =368 savedpeak = 25840"
## [3] "j =200 peak =398 savedpeak = 25870"
## [1] "j =201 peak =240 savedpeak = 25840"
## [2] "j =201 peak =270 savedpeak = 25870"
## [1] "j =202 peak =112 savedpeak = 25840"
## [2] "j =202 peak =142 savedpeak = 25870"
## [1] "j =203 peak =145 savedpeak = 26001"
## [2] "j =203 peak =235 savedpeak = 26091"
## [3] "j =203 peak =371 savedpeak = 26227"
## [4] "j =203 peak =412 savedpeak = 26268"
## [1] "j =204 peak =107 savedpeak = 26091"
## [2] "j =204 peak =243 savedpeak = 26227"
## [3] "j =204 peak =284 savedpeak = 26268"
## [1] "j =205 peak =115 savedpeak = 26227"
## [2] "j =205 peak =156 savedpeak = 26268"
## [1] "j =206 peak =80 savedpeak = 26320"
## [2] "j =206 peak =112 savedpeak = 26352"
## [3] "j =206 peak =155 savedpeak = 26395"
## [4] "j =206 peak =190 savedpeak = 26430"
## [5] "j =206 peak =207 savedpeak = 26447"
## [6] "j =206 peak =233 savedpeak = 26473"
## [7] "j =206 peak =254 savedpeak = 26494"
## [8] "j =206 peak =291 savedpeak = 26531"

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## [9] "j =206 peak =306 savedpeak = 26546"
## [10] "j =206 peak =325 savedpeak = 26565"
## [11] "j =206 peak =337 savedpeak = 26577"
## [12] "j =206 peak =359 savedpeak = 26599"
## [13] "j =206 peak =396 savedpeak = 26636"
## [14] "j =206 peak =421 savedpeak = 26661"
## [1] "j =211 peak =128 savedpeak = 27008"
## [2] "j =211 peak =416 savedpeak = 27296"
## [1] "j =212 peak =288 savedpeak = 27296"
## [1] "j =213 peak =160 savedpeak = 27296"
## [1] "j =216 peak =84 savedpeak = 27604"
## [2] "j =216 peak =96 savedpeak = 27616"
## [3] "j =216 peak =113 savedpeak = 27633"
## [4] "j =216 peak =153 savedpeak = 27673"
## [5] "j =216 peak =179 savedpeak = 27699"
## [6] "j =216 peak =238 savedpeak = 27758"
## [7] "j =216 peak =257 savedpeak = 27777"
## [8] "j =216 peak =270 savedpeak = 27790"
## [9] "j =216 peak =331 savedpeak = 27851"
## [10] "j =216 peak =347 savedpeak = 27867"
## [11] "j =216 peak =373 savedpeak = 27893"
## [12] "j =216 peak =412 savedpeak = 27932"
## [1] "j =217 peak =203 savedpeak = 27851"
## [2] "j =217 peak =244 savedpeak = 27892"
## [3] "j =217 peak =342 savedpeak = 27990"
## [4] "j =217 peak =362 savedpeak = 28010"
## [5] "j =217 peak =427 savedpeak = 28075"
## [1] "j =218 peak =116 savedpeak = 27892"
## [2] "j =218 peak =214 savedpeak = 27990"
## [3] "j =218 peak =234 savedpeak = 28010"
## [4] "j =218 peak =299 savedpeak = 28075"
## [5] "j =218 peak =366 savedpeak = 28142"
## [1] "j =219 peak =86 savedpeak = 27990"
## [2] "j =219 peak =106 savedpeak = 28010"
## [3] "j =219 peak =171 savedpeak = 28075"
## [4] "j =219 peak =238 savedpeak = 28142"
## [5] "j =219 peak =313 savedpeak = 28217"
## [6] "j =219 peak =377 savedpeak = 28281"
## [7] "j =219 peak =425 savedpeak = 28329"
## [1] "j =220 peak =110 savedpeak = 28142"
## [2] "j =220 peak =185 savedpeak = 28217"
## [3] "j =220 peak =249 savedpeak = 28281"
## [4] "j =220 peak =297 savedpeak = 28329"
## [1] "j =221 peak =121 savedpeak = 28281"
## [2] "j =221 peak =169 savedpeak = 28329"
## [3] "j =221 peak =358 savedpeak = 28518"
## [1] "j =237 peak =403 savedpeak = 30611"
## [1] "j =238 peak =275 savedpeak = 30611"
## [2] "j =238 peak =430 savedpeak = 30766"
## [1] "j =239 peak =147 savedpeak = 30611"
## [2] "j =239 peak =302 savedpeak = 30766"
## [1] "j =240 peak =405 savedpeak = 30997"
## [1] "j =241 peak =277 savedpeak = 30997"
## [2] "j =241 peak =313 savedpeak = 31033"

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## [3] "j =241 peak =348 savedpeak = 31068"
## [1] "j =242 peak =149 savedpeak = 30997"
## [2] "j =242 peak =185 savedpeak = 31033"
## [3] "j =242 peak =220 savedpeak = 31068"
## [1] "j =243 peak =92 savedpeak = 31068"
## [1] "j =244 peak =92 savedpeak = 31196"
## [2] "j =244 peak =118 savedpeak = 31222"
## [3] "j =244 peak =143 savedpeak = 31247"
## [4] "j =244 peak =154 savedpeak = 31258"
## [5] "j =244 peak =196 savedpeak = 31300"
## [6] "j =244 peak =253 savedpeak = 31357"
## [7] "j =244 peak =272 savedpeak = 31376"
## [8] "j =244 peak =298 savedpeak = 31402"
## [9] "j =244 peak =317 savedpeak = 31421"
## [10] "j =244 peak =359 savedpeak = 31463"
## [11] "j =244 peak =373 savedpeak = 31477"
## [12] "j =244 peak =403 savedpeak = 31507"
## [13] "j =244 peak =436 savedpeak = 31540"
## [1] "j =245 peak =125 savedpeak = 31357"
## [2] "j =245 peak =144 savedpeak = 31376"
## [3] "j =245 peak =170 savedpeak = 31402"
## [4] "j =245 peak =189 savedpeak = 31421"
## [5] "j =245 peak =231 savedpeak = 31463"
## [6] "j =245 peak =245 savedpeak = 31477"
## [7] "j =245 peak =275 savedpeak = 31507"
## [8] "j =245 peak =308 savedpeak = 31540"
## [9] "j =245 peak =327 savedpeak = 31559"
## [10] "j =245 peak =342 savedpeak = 31574"
## [11] "j =245 peak =367 savedpeak = 31599"
## [12] "j =245 peak =405 savedpeak = 31637"
## [13] "j =245 peak =424 savedpeak = 31656"
## [1] "j =246 peak =103 savedpeak = 31463"
## [2] "j =246 peak =117 savedpeak = 31477"
## [3] "j =246 peak =147 savedpeak = 31507"
## [4] "j =246 peak =180 savedpeak = 31540"
## [5] "j =246 peak =199 savedpeak = 31559"
## [6] "j =246 peak =214 savedpeak = 31574"
## [7] "j =246 peak =239 savedpeak = 31599"
## [8] "j =246 peak =277 savedpeak = 31637"
## [9] "j =246 peak =296 savedpeak = 31656"
## [10] "j =246 peak =335 savedpeak = 31695"
## [11] "j =246 peak =366 savedpeak = 31726"
## [12] "j =246 peak =388 savedpeak = 31748"
## [13] "j =246 peak =401 savedpeak = 31761"
## [14] "j =246 peak =436 savedpeak = 31796"
## [1] "j =247 peak =86 savedpeak = 31574"
## [2] "j =247 peak =111 savedpeak = 31599"
## [3] "j =247 peak =149 savedpeak = 31637"
## [4] "j =247 peak =168 savedpeak = 31656"
## [5] "j =247 peak =207 savedpeak = 31695"
## [6] "j =247 peak =238 savedpeak = 31726"
## [7] "j =247 peak =260 savedpeak = 31748"
## [8] "j =247 peak =273 savedpeak = 31761"
## [9] "j =247 peak =308 savedpeak = 31796"

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## [10] "j =247 peak =315 savedpeak = 31803"
## [11] "j =247 peak =348 savedpeak = 31836"
## [12] "j =247 peak =361 savedpeak = 31849"
## [13] "j =247 peak =386 savedpeak = 31874"
## [14] "j =247 peak =417 savedpeak = 31905"
## [1] "j =248 peak =79 savedpeak = 31695"
## [2] "j =248 peak =110 savedpeak = 31726"
## [3] "j =248 peak =132 savedpeak = 31748"
## [4] "j =248 peak =145 savedpeak = 31761"
## [5] "j =248 peak =180 savedpeak = 31796"
## [6] "j =248 peak =187 savedpeak = 31803"
## [7] "j =248 peak =220 savedpeak = 31836"
## [8] "j =248 peak =233 savedpeak = 31849"
## [9] "j =248 peak =258 savedpeak = 31874"
## [10] "j =248 peak =289 savedpeak = 31905"
## [11] "j =248 peak =342 savedpeak = 31958"
## [12] "j =248 peak =367 savedpeak = 31983"
## [13] "j =248 peak =394 savedpeak = 32010"
## [14] "j =248 peak =404 savedpeak = 32020"
## [15] "j =248 peak =427 savedpeak = 32043"
## [1] "j =249 peak =92 savedpeak = 31836"
## [2] "j =249 peak =105 savedpeak = 31849"
## [3] "j =249 peak =130 savedpeak = 31874"
## [4] "j =249 peak =161 savedpeak = 31905"
## [5] "j =249 peak =214 savedpeak = 31958"
## [6] "j =249 peak =239 savedpeak = 31983"
## [7] "j =249 peak =266 savedpeak = 32010"
## [8] "j =249 peak =276 savedpeak = 32020"
## [9] "j =249 peak =299 savedpeak = 32043"
## [10] "j =249 peak =329 savedpeak = 32073"
## [11] "j =249 peak =385 savedpeak = 32129"
## [12] "j =249 peak =404 savedpeak = 32148"
## [13] "j =249 peak =432 savedpeak = 32176"
## [1] "j =252 peak =103 savedpeak = 32231"
## [2] "j =252 peak =155 savedpeak = 32283"
## [3] "j =252 peak =167 savedpeak = 32295"
## [4] "j =252 peak =188 savedpeak = 32316"
## [5] "j =252 peak =216 savedpeak = 32344"
## [6] "j =252 peak =262 savedpeak = 32390"
## [7] "j =252 peak =275 savedpeak = 32403"
## [8] "j =252 peak =291 savedpeak = 32419"
## [9] "j =252 peak =317 savedpeak = 32445"
## [10] "j =252 peak =341 savedpeak = 32469"
## [11] "j =252 peak =365 savedpeak = 32493"
## [12] "j =252 peak =424 savedpeak = 32552"
## [1] "j =254 peak =85 savedpeak = 32469"
## [2] "j =254 peak =109 savedpeak = 32493"
## [3] "j =254 peak =168 savedpeak = 32552"
## [4] "j =254 peak =186 savedpeak = 32570"
## [5] "j =254 peak =248 savedpeak = 32632"
## [6] "j =254 peak =272 savedpeak = 32656"
## [7] "j =254 peak =288 savedpeak = 32672"
## [8] "j =254 peak =321 savedpeak = 32705"
## [9] "j =254 peak =344 savedpeak = 32728"

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## [10] "j =254 peak =368 savedpeak = 32752"
## [11] "j =254 peak =385 savedpeak = 32769"
## [12] "j =254 peak =403 savedpeak = 32787"
## [13] "j =254 peak =418 savedpeak = 32802"
## [1] "j =255 peak =120 savedpeak = 32632"
## [2] "j =255 peak =144 savedpeak = 32656"
## [3] "j =255 peak =160 savedpeak = 32672"
## [4] "j =255 peak =193 savedpeak = 32705"
## [5] "j =255 peak =216 savedpeak = 32728"
## [6] "j =255 peak =240 savedpeak = 32752"
## [7] "j =255 peak =257 savedpeak = 32769"
## [8] "j =255 peak =275 savedpeak = 32787"
## [9] "j =255 peak =290 savedpeak = 32802"
## [10] "j =255 peak =329 savedpeak = 32841"
## [11] "j =255 peak =375 savedpeak = 32887"
## [12] "j =255 peak =393 savedpeak = 32905"
## [13] "j =255 peak =433 savedpeak = 32945"
## [1] "j =259 peak =83 savedpeak = 33107"
## [2] "j =259 peak =114 savedpeak = 33138"
## [3] "j =259 peak =139 savedpeak = 33163"
## [4] "j =259 peak =189 savedpeak = 33213"
## [5] "j =259 peak =211 savedpeak = 33235"
## [6] "j =259 peak =229 savedpeak = 33253"
## [7] "j =259 peak =250 savedpeak = 33274"
## [8] "j =259 peak =283 savedpeak = 33307"
## [9] "j =259 peak =310 savedpeak = 33334"
## [10] "j =259 peak =339 savedpeak = 33363"
## [11] "j =259 peak =376 savedpeak = 33400"
## [12] "j =259 peak =403 savedpeak = 33427"
## [13] "j =259 peak =418 savedpeak = 33442"
## [1] "j =260 peak =83 savedpeak = 33235"
## [2] "j =260 peak =101 savedpeak = 33253"
## [3] "j =260 peak =122 savedpeak = 33274"
## [4] "j =260 peak =155 savedpeak = 33307"
## [5] "j =260 peak =182 savedpeak = 33334"
## [6] "j =260 peak =211 savedpeak = 33363"
## [7] "j =260 peak =248 savedpeak = 33400"
## [8] "j =260 peak =275 savedpeak = 33427"
## [9] "j =260 peak =290 savedpeak = 33442"
## [10] "j =260 peak =324 savedpeak = 33476"
## [11] "j =260 peak =340 savedpeak = 33492"
## [12] "j =260 peak =390 savedpeak = 33542"
## [13] "j =260 peak =431 savedpeak = 33583"
## [1] "j =269 peak =103 savedpeak = 34407"
## [2] "j =269 peak =117 savedpeak = 34421"
## [3] "j =269 peak =175 savedpeak = 34479"
## [4] "j =269 peak =206 savedpeak = 34510"
## [5] "j =269 peak =225 savedpeak = 34529"
## [6] "j =269 peak =264 savedpeak = 34568"
## [7] "j =269 peak =291 savedpeak = 34595"
## [8] "j =269 peak =305 savedpeak = 34609"
## [9] "j =269 peak =334 savedpeak = 34638"
## [10] "j =269 peak =390 savedpeak = 34694"
## [11] "j =269 peak =406 savedpeak = 34710"

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## [12] "j =269 peak =415 savedpeak = 34719"
## [13] "j =269 peak =431 savedpeak = 34735"
## [1] "j =270 peak =78 savedpeak = 34510"
## [2] "j =270 peak =97 savedpeak = 34529"
## [3] "j =270 peak =136 savedpeak = 34568"
## [4] "j =270 peak =163 savedpeak = 34595"
## [5] "j =270 peak =177 savedpeak = 34609"
## [6] "j =270 peak =206 savedpeak = 34638"
## [7] "j =270 peak =262 savedpeak = 34694"
## [8] "j =270 peak =278 savedpeak = 34710"
## [9] "j =270 peak =287 savedpeak = 34719"
## [10] "j =270 peak =303 savedpeak = 34735"
## [11] "j =270 peak =361 savedpeak = 34793"
## [12] "j =270 peak =379 savedpeak = 34811"
## [13] "j =270 peak =408 savedpeak = 34840"
## [14] "j =270 peak =425 savedpeak = 34857"
## [1] "j =271 peak =78 savedpeak = 34638"
## [2] "j =271 peak =150 savedpeak = 34710"
## [3] "j =271 peak =159 savedpeak = 34719"
## [4] "j =271 peak =175 savedpeak = 34735"
## [5] "j =271 peak =233 savedpeak = 34793"
## [6] "j =271 peak =251 savedpeak = 34811"
## [7] "j =271 peak =280 savedpeak = 34840"
## [8] "j =271 peak =297 savedpeak = 34857"
## [9] "j =271 peak =311 savedpeak = 34871"
## [10] "j =271 peak =327 savedpeak = 34887"
## [11] "j =271 peak =343 savedpeak = 34903"
## [12] "j =271 peak =384 savedpeak = 34944"
## [13] "j =271 peak =430 savedpeak = 34990"
## [1] "j =273 peak =87 savedpeak = 34903"
## [2] "j =273 peak =128 savedpeak = 34944"
## [3] "j =273 peak =174 savedpeak = 34990"
## [4] "j =273 peak =195 savedpeak = 35011"
## [5] "j =273 peak =218 savedpeak = 35034"
## [6] "j =273 peak =263 savedpeak = 35079"
## [7] "j =273 peak =297 savedpeak = 35113"
## [8] "j =273 peak =317 savedpeak = 35133"
## [9] "j =273 peak =351 savedpeak = 35167"
## [10] "j =273 peak =378 savedpeak = 35194"
## [11] "j =273 peak =435 savedpeak = 35251"
## [1] "j =274 peak =90 savedpeak = 35034"
## [2] "j =274 peak =135 savedpeak = 35079"
## [3] "j =274 peak =169 savedpeak = 35113"
## [4] "j =274 peak =189 savedpeak = 35133"
## [5] "j =274 peak =223 savedpeak = 35167"
## [6] "j =274 peak =250 savedpeak = 35194"
## [7] "j =274 peak =307 savedpeak = 35251"
## [8] "j =274 peak =326 savedpeak = 35270"
## [9] "j =274 peak =365 savedpeak = 35309"
## [10] "j =274 peak =392 savedpeak = 35336"
## [11] "j =274 peak =412 savedpeak = 35356"
## [1] "j =278 peak =85 savedpeak = 35541"
## [2] "j =278 peak =97 savedpeak = 35553"
## [3] "j =278 peak =145 savedpeak = 35601"

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## [4] "j =278 peak =182 savedpeak = 35638"
## [5] "j =278 peak =208 savedpeak = 35664"
## [6] "j =278 peak =245 savedpeak = 35701"
## [7] "j =278 peak =336 savedpeak = 35792"
## [8] "j =278 peak =361 savedpeak = 35817"
## [9] "j =278 peak =391 savedpeak = 35847"
## [10] "j =278 peak =411 savedpeak = 35867"
## [1] "j =279 peak =80 savedpeak = 35664"
## [2] "j =279 peak =117 savedpeak = 35701"
## [3] "j =279 peak =208 savedpeak = 35792"
## [4] "j =279 peak =233 savedpeak = 35817"
## [5] "j =279 peak =263 savedpeak = 35847"
## [6] "j =279 peak =283 savedpeak = 35867"
## [7] "j =279 peak =310 savedpeak = 35894"
## [8] "j =279 peak =375 savedpeak = 35959"
## [9] "j =279 peak =408 savedpeak = 35992"
## [10] "j =279 peak =426 savedpeak = 36010"
## [1] "j =280 peak =80 savedpeak = 35792"
## [2] "j =280 peak =105 savedpeak = 35817"
## [3] "j =280 peak =135 savedpeak = 35847"
## [4] "j =280 peak =155 savedpeak = 35867"
## [5] "j =280 peak =182 savedpeak = 35894"
## [6] "j =280 peak =247 savedpeak = 35959"
## [7] "j =280 peak =280 savedpeak = 35992"
## [8] "j =280 peak =298 savedpeak = 36010"
## [9] "j =280 peak =319 savedpeak = 36031"
## [10] "j =280 peak =339 savedpeak = 36051"
## [11] "j =280 peak =351 savedpeak = 36063"
## [12] "j =280 peak =390 savedpeak = 36102"
## [13] "j =280 peak =422 savedpeak = 36134"
## [1] "j =281 peak =76 savedpeak = 35916"
## [2] "j =281 peak =119 savedpeak = 35959"
## [3] "j =281 peak =152 savedpeak = 35992"
## [4] "j =281 peak =170 savedpeak = 36010"
## [5] "j =281 peak =191 savedpeak = 36031"
## [6] "j =281 peak =211 savedpeak = 36051"
## [7] "j =281 peak =223 savedpeak = 36063"
## [8] "j =281 peak =262 savedpeak = 36102"
## [9] "j =281 peak =294 savedpeak = 36134"
## [10] "j =281 peak =333 savedpeak = 36173"
## [11] "j =281 peak =374 savedpeak = 36214"
## [12] "j =281 peak =396 savedpeak = 36236"
## [1] "j =282 peak =83 savedpeak = 36051"
## [2] "j =282 peak =95 savedpeak = 36063"
## [3] "j =282 peak =134 savedpeak = 36102"
## [4] "j =282 peak =166 savedpeak = 36134"
## [5] "j =282 peak =205 savedpeak = 36173"
## [6] "j =282 peak =246 savedpeak = 36214"
## [7] "j =282 peak =268 savedpeak = 36236"
## [8] "j =282 peak =315 savedpeak = 36283"
## [9] "j =282 peak =348 savedpeak = 36316"
## [10] "j =282 peak =364 savedpeak = 36332"
## [11] "j =282 peak =390 savedpeak = 36358"
## [12] "j =282 peak =406 savedpeak = 36374"

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## [13] "j =282 peak =425 savedpeak = 36393"
## [1] "j =283 peak =77 savedpeak = 36173"
## [2] "j =283 peak =118 savedpeak = 36214"
## [3] "j =283 peak =140 savedpeak = 36236"
## [4] "j =283 peak =187 savedpeak = 36283"
## [5] "j =283 peak =220 savedpeak = 36316"
## [6] "j =283 peak =236 savedpeak = 36332"
## [7] "j =283 peak =262 savedpeak = 36358"
## [8] "j =283 peak =278 savedpeak = 36374"
## [9] "j =283 peak =297 savedpeak = 36393"
## [10] "j =283 peak =325 savedpeak = 36421"
## [11] "j =283 peak =348 savedpeak = 36444"
## [12] "j =283 peak =360 savedpeak = 36456"
## [13] "j =283 peak =382 savedpeak = 36478"
## [14] "j =283 peak =402 savedpeak = 36498"
## [15] "j =283 peak =415 savedpeak = 36511"
## [1] "j =289 peak =134 savedpeak = 36998"
## [2] "j =289 peak =275 savedpeak = 37139"
## [3] "j =289 peak =357 savedpeak = 37221"
## [1] "j =290 peak =147 savedpeak = 37139"
## [2] "j =290 peak =229 savedpeak = 37221"
## [3] "j =290 peak =357 savedpeak = 37349"
## [4] "j =290 peak =435 savedpeak = 37427"
## [1] "j =291 peak =101 savedpeak = 37221"
## [2] "j =291 peak =229 savedpeak = 37349"
## [3] "j =291 peak =307 savedpeak = 37427"
## [1] "j =292 peak =101 savedpeak = 37349"
## [2] "j =292 peak =179 savedpeak = 37427"
## [3] "j =292 peak =412 savedpeak = 37660"
## [1] "j =293 peak =107 savedpeak = 37483"
## [2] "j =293 peak =128 savedpeak = 37504"
## [3] "j =293 peak =168 savedpeak = 37544"
## [4] "j =293 peak =194 savedpeak = 37570"
## [5] "j =293 peak =240 savedpeak = 37616"
## [6] "j =293 peak =277 savedpeak = 37653"
## [7] "j =293 peak =307 savedpeak = 37683"
## [8] "j =293 peak =323 savedpeak = 37699"
## [9] "j =293 peak =369 savedpeak = 37745"
## [10] "j =293 peak =409 savedpeak = 37785"
## [11] "j =293 peak =423 savedpeak = 37799"
## [1] "j =294 peak =112 savedpeak = 37616"
## [2] "j =294 peak =155 savedpeak = 37659"
## [3] "j =294 peak =179 savedpeak = 37683"
## [4] "j =294 peak =195 savedpeak = 37699"
## [5] "j =294 peak =241 savedpeak = 37745"
## [6] "j =294 peak =281 savedpeak = 37785"
## [7] "j =294 peak =295 savedpeak = 37799"
## [8] "j =294 peak =320 savedpeak = 37824"
## [9] "j =294 peak =344 savedpeak = 37848"
## [10] "j =294 peak =377 savedpeak = 37881"
## [11] "j =294 peak =406 savedpeak = 37910"
## [12] "j =294 peak =429 savedpeak = 37933"
## [1] "j =295 peak =113 savedpeak = 37745"
## [2] "j =295 peak =153 savedpeak = 37785"

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## [3] "j =295 peak =167 savedpeak = 37799"
## [4] "j =295 peak =192 savedpeak = 37824"
## [5] "j =295 peak =216 savedpeak = 37848"
## [6] "j =295 peak =249 savedpeak = 37881"
## [7] "j =295 peak =278 savedpeak = 37910"
## [8] "j =295 peak =301 savedpeak = 37933"
## [9] "j =295 peak =316 savedpeak = 37948"
## [10] "j =295 peak =341 savedpeak = 37973"
## [11] "j =295 peak =361 savedpeak = 37993"
## [12] "j =295 peak =374 savedpeak = 38006"
## [13] "j =295 peak =393 savedpeak = 38025"
## [14] "j =295 peak =404 savedpeak = 38036"
## [15] "j =295 peak =422 savedpeak = 38054"
## [1] "j =296 peak =88 savedpeak = 37848"
## [2] "j =296 peak =121 savedpeak = 37881"
## [3] "j =296 peak =150 savedpeak = 37910"
## [4] "j =296 peak =173 savedpeak = 37933"
## [5] "j =296 peak =188 savedpeak = 37948"
## [6] "j =296 peak =213 savedpeak = 37973"
## [7] "j =296 peak =233 savedpeak = 37993"
## [8] "j =296 peak =246 savedpeak = 38006"
## [9] "j =296 peak =265 savedpeak = 38025"
## [10] "j =296 peak =276 savedpeak = 38036"
## [11] "j =296 peak =294 savedpeak = 38054"
## [12] "j =296 peak =329 savedpeak = 38089"
## [13] "j =296 peak =344 savedpeak = 38104"
## [14] "j =296 peak =385 savedpeak = 38145"
## [15] "j =296 peak =424 savedpeak = 38184"
## [1] "j =303 peak =98 savedpeak = 38754"
## [2] "j =303 peak =125 savedpeak = 38781"
## [3] "j =303 peak =164 savedpeak = 38820"
## [4] "j =303 peak =187 savedpeak = 38843"
## [5] "j =303 peak =228 savedpeak = 38884"
## [6] "j =303 peak =276 savedpeak = 38932"
## [7] "j =303 peak =297 savedpeak = 38953"
## [8] "j =303 peak =314 savedpeak = 38970"
## [9] "j =303 peak =359 savedpeak = 39015"
## [10] "j =303 peak =380 savedpeak = 39036"
## [11] "j =303 peak =393 savedpeak = 39049"
## [1] "j =304 peak =100 savedpeak = 38884"
## [2] "j =304 peak =148 savedpeak = 38932"
## [3] "j =304 peak =169 savedpeak = 38953"
## [4] "j =304 peak =186 savedpeak = 38970"
## [5] "j =304 peak =231 savedpeak = 39015"
## [6] "j =304 peak =252 savedpeak = 39036"
## [7] "j =304 peak =265 savedpeak = 39049"
## [8] "j =304 peak =310 savedpeak = 39094"
## [9] "j =304 peak =332 savedpeak = 39116"
## [10] "j =304 peak =347 savedpeak = 39131"
## [11] "j =304 peak =369 savedpeak = 39153"
## [12] "j =304 peak =394 savedpeak = 39178"
## [13] "j =304 peak =419 savedpeak = 39203"
## [1] "j =305 peak =103 savedpeak = 39015"
## [2] "j =305 peak =124 savedpeak = 39036"

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## [3] "j =305 peak =137 savedpeak = 39049"
## [4] "j =305 peak =182 savedpeak = 39094"
## [5] "j =305 peak =204 savedpeak = 39116"
## [6] "j =305 peak =219 savedpeak = 39131"
## [7] "j =305 peak =241 savedpeak = 39153"
## [8] "j =305 peak =266 savedpeak = 39178"
## [9] "j =305 peak =291 savedpeak = 39203"
## [10] "j =305 peak =315 savedpeak = 39227"
## [11] "j =305 peak =333 savedpeak = 39245"
## [12] "j =305 peak =380 savedpeak = 39292"
## [13] "j =305 peak =395 savedpeak = 39307"
## [14] "j =305 peak =431 savedpeak = 39343"
## [1] "j =306 peak =76 savedpeak = 39116"
## [2] "j =306 peak =91 savedpeak = 39131"
## [3] "j =306 peak =113 savedpeak = 39153"
## [4] "j =306 peak =138 savedpeak = 39178"
## [5] "j =306 peak =163 savedpeak = 39203"
## [6] "j =306 peak =187 savedpeak = 39227"
## [7] "j =306 peak =205 savedpeak = 39245"
## [8] "j =306 peak =252 savedpeak = 39292"
## [9] "j =306 peak =267 savedpeak = 39307"
## [10] "j =306 peak =303 savedpeak = 39343"
## [11] "j =306 peak =328 savedpeak = 39368"
## [12] "j =306 peak =364 savedpeak = 39404"
## [13] "j =306 peak =380 savedpeak = 39420"
## [14] "j =306 peak =404 savedpeak = 39444"
## [15] "j =306 peak =427 savedpeak = 39467"
## [1] "j =307 peak =77 savedpeak = 39245"
## [2] "j =307 peak =124 savedpeak = 39292"
## [3] "j =307 peak =139 savedpeak = 39307"
## [4] "j =307 peak =175 savedpeak = 39343"
## [5] "j =307 peak =200 savedpeak = 39368"
## [6] "j =307 peak =236 savedpeak = 39404"
## [7] "j =307 peak =252 savedpeak = 39420"
## [8] "j =307 peak =276 savedpeak = 39444"
## [9] "j =307 peak =299 savedpeak = 39467"
## [10] "j =307 peak =328 savedpeak = 39496"
## [11] "j =307 peak =347 savedpeak = 39515"
## [12] "j =307 peak =377 savedpeak = 39545"
## [13] "j =307 peak =397 savedpeak = 39565"
## [14] "j =307 peak =424 savedpeak = 39592"
## [1] "j =308 peak =107 savedpeak = 39403"
## [2] "j =308 peak =145 savedpeak = 39441"
## [3] "j =308 peak =198 savedpeak = 39494"
## [4] "j =308 peak =224 savedpeak = 39520"
## [5] "j =308 peak =269 savedpeak = 39565"
## [6] "j =308 peak =354 savedpeak = 39650"
## [7] "j =308 peak =406 savedpeak = 39702"
## [1] "j =309 peak =141 savedpeak = 39565"
## [2] "j =309 peak =226 savedpeak = 39650"
## [3] "j =309 peak =278 savedpeak = 39702"
## [4] "j =309 peak =375 savedpeak = 39799"
## [5] "j =309 peak =399 savedpeak = 39823"
## [1] "j =310 peak =98 savedpeak = 39650"

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## [2] "j =310 peak =150 savedpeak = 39702"
## [3] "j =310 peak =247 savedpeak = 39799"
## [4] "j =310 peak =271 savedpeak = 39823"
## [5] "j =310 peak =312 savedpeak = 39864"
## [6] "j =310 peak =341 savedpeak = 39893"
## [7] "j =310 peak =403 savedpeak = 39955"
## [8] "j =310 peak =427 savedpeak = 39979"
## [1] "j =311 peak =119 savedpeak = 39799"
## [2] "j =311 peak =143 savedpeak = 39823"
## [3] "j =311 peak =184 savedpeak = 39864"
## [4] "j =311 peak =213 savedpeak = 39893"
## [5] "j =311 peak =275 savedpeak = 39955"
## [6] "j =311 peak =299 savedpeak = 39979"
## [7] "j =311 peak =336 savedpeak = 40016"
## [8] "j =311 peak =373 savedpeak = 40053"
## [9] "j =311 peak =424 savedpeak = 40104"
## [1] "j =312 peak =80 savedpeak = 39888"
## [2] "j =312 peak =117 savedpeak = 39925"
## [3] "j =312 peak =140 savedpeak = 39948"
## [4] "j =312 peak =171 savedpeak = 39979"
## [5] "j =312 peak =210 savedpeak = 40018"
## [6] "j =312 peak =244 savedpeak = 40052"
## [7] "j =312 peak =260 savedpeak = 40068"
## [8] "j =312 peak =274 savedpeak = 40082"
## [9] "j =312 peak =297 savedpeak = 40105"
## [10] "j =312 peak =335 savedpeak = 40143"
## [11] "j =312 peak =370 savedpeak = 40178"
## [12] "j =312 peak =382 savedpeak = 40190"
## [13] "j =312 peak =410 savedpeak = 40218"
## [14] "j =312 peak =424 savedpeak = 40232"
## [15] "j =312 peak =433 savedpeak = 40241"
## [1] "j =313 peak =82 savedpeak = 40018"
## [2] "j =313 peak =116 savedpeak = 40052"
## [3] "j =313 peak =132 savedpeak = 40068"
## [4] "j =313 peak =146 savedpeak = 40082"
## [5] "j =313 peak =169 savedpeak = 40105"
## [6] "j =313 peak =207 savedpeak = 40143"
## [7] "j =313 peak =242 savedpeak = 40178"
## [8] "j =313 peak =254 savedpeak = 40190"
## [9] "j =313 peak =282 savedpeak = 40218"
## [10] "j =313 peak =296 savedpeak = 40232"
## [11] "j =313 peak =305 savedpeak = 40241"
## [12] "j =313 peak =330 savedpeak = 40266"
## [13] "j =313 peak =348 savedpeak = 40284"
## [14] "j =313 peak =365 savedpeak = 40301"
## [15] "j =313 peak =397 savedpeak = 40333"
## [16] "j =313 peak =410 savedpeak = 40346"
## [1] "j =315 peak =92 savedpeak = 40284"
## [2] "j =315 peak =109 savedpeak = 40301"
## [3] "j =315 peak =141 savedpeak = 40333"
## [4] "j =315 peak =154 savedpeak = 40346"
## [5] "j =315 peak =186 savedpeak = 40378"
## [6] "j =315 peak =204 savedpeak = 40396"
## [7] "j =315 peak =228 savedpeak = 40420"

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## [8] "j =315 peak =270 savedpeak = 40462"
## [9] "j =315 peak =310 savedpeak = 40502"
## [10] "j =315 peak =331 savedpeak = 40523"
## [11] "j =315 peak =347 savedpeak = 40539"
## [12] "j =315 peak =375 savedpeak = 40567"
## [13] "j =315 peak =413 savedpeak = 40605"
## [14] "j =315 peak =421 savedpeak = 40613"
## [1] "j =316 peak =76 savedpeak = 40396"
## [2] "j =316 peak =100 savedpeak = 40420"
## [3] "j =316 peak =142 savedpeak = 40462"
## [4] "j =316 peak =182 savedpeak = 40502"
## [5] "j =316 peak =203 savedpeak = 40523"
## [6] "j =316 peak =219 savedpeak = 40539"
## [7] "j =316 peak =247 savedpeak = 40567"
## [8] "j =316 peak =262 savedpeak = 40582"
## [9] "j =316 peak =285 savedpeak = 40605"
## [10] "j =316 peak =293 savedpeak = 40613"
## [11] "j =316 peak =322 savedpeak = 40642"
## [12] "j =316 peak =355 savedpeak = 40675"
## [13] "j =316 peak =417 savedpeak = 40737"
## [1] "j =317 peak =91 savedpeak = 40539"
## [2] "j =317 peak =119 savedpeak = 40567"
## [3] "j =317 peak =134 savedpeak = 40582"
## [4] "j =317 peak =157 savedpeak = 40605"
## [5] "j =317 peak =165 savedpeak = 40613"
## [6] "j =317 peak =194 savedpeak = 40642"
## [7] "j =317 peak =227 savedpeak = 40675"
## [8] "j =317 peak =289 savedpeak = 40737"
## [9] "j =317 peak =314 savedpeak = 40762"
## [10] "j =317 peak =357 savedpeak = 40805"
## [11] "j =317 peak =379 savedpeak = 40827"
## [12] "j =317 peak =402 savedpeak = 40850"
## [13] "j =317 peak =419 savedpeak = 40867"
## [14] "j =317 peak =437 savedpeak = 40885"
## [1] "j =318 peak =99 savedpeak = 40675"
## [2] "j =318 peak =161 savedpeak = 40737"
## [3] "j =318 peak =186 savedpeak = 40762"
## [4] "j =318 peak =229 savedpeak = 40805"
## [5] "j =318 peak =251 savedpeak = 40827"
## [6] "j =318 peak =274 savedpeak = 40850"
## [7] "j =318 peak =291 savedpeak = 40867"
## [8] "j =318 peak =309 savedpeak = 40885"
## [9] "j =318 peak =350 savedpeak = 40926"
## [10] "j =318 peak =387 savedpeak = 40963"
## [11] "j =318 peak =409 savedpeak = 40985"
## [12] "j =318 peak =436 savedpeak = 41012"
## [1] "j =319 peak =101 savedpeak = 40805"
## [2] "j =319 peak =123 savedpeak = 40827"
## [3] "j =319 peak =146 savedpeak = 40850"
## [4] "j =319 peak =163 savedpeak = 40867"
## [5] "j =319 peak =181 savedpeak = 40885"
## [6] "j =319 peak =222 savedpeak = 40926"
## [7] "j =319 peak =259 savedpeak = 40963"
## [8] "j =319 peak =281 savedpeak = 40985"

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## [9] "j =319 peak =308 savedpeak = 41012"
## [10] "j =319 peak =336 savedpeak = 41040"
## [11] "j =319 peak =370 savedpeak = 41074"
## [12] "j =319 peak =405 savedpeak = 41109"
## [13] "j =319 peak =415 savedpeak = 41119"
## [1] "j =320 peak =94 savedpeak = 40926"
## [2] "j =320 peak =131 savedpeak = 40963"
## [3] "j =320 peak =153 savedpeak = 40985"
## [4] "j =320 peak =180 savedpeak = 41012"
## [5] "j =320 peak =208 savedpeak = 41040"
## [6] "j =320 peak =242 savedpeak = 41074"
## [7] "j =320 peak =277 savedpeak = 41109"
## [8] "j =320 peak =287 savedpeak = 41119"
## [9] "j =320 peak =355 savedpeak = 41187"
## [10] "j =320 peak =382 savedpeak = 41214"
## [11] "j =320 peak =412 savedpeak = 41244"
## [1] "j =322 peak =99 savedpeak = 41187"
## [2] "j =322 peak =126 savedpeak = 41214"
## [3] "j =322 peak =156 savedpeak = 41244"
## [4] "j =322 peak =198 savedpeak = 41286"
## [5] "j =322 peak =215 savedpeak = 41303"
## [6] "j =322 peak =283 savedpeak = 41371"
## [7] "j =322 peak =316 savedpeak = 41404"
## [8] "j =322 peak =340 savedpeak = 41428"
## [9] "j =322 peak =368 savedpeak = 41456"
## [10] "j =322 peak =409 savedpeak = 41497"
## [1] "j =325 peak =84 savedpeak = 41556"
## [2] "j =325 peak =109 savedpeak = 41581"
## [3] "j =325 peak =141 savedpeak = 41613"
## [4] "j =325 peak =158 savedpeak = 41630"
## [5] "j =325 peak =182 savedpeak = 41654"
## [6] "j =325 peak =223 savedpeak = 41695"
## [7] "j =325 peak =241 savedpeak = 41713"
## [8] "j =325 peak =300 savedpeak = 41772"
## [9] "j =325 peak =323 savedpeak = 41795"
## [10] "j =325 peak =344 savedpeak = 41816"
## [11] "j =325 peak =351 savedpeak = 41823"
## [12] "j =325 peak =378 savedpeak = 41850"
## [13] "j =325 peak =416 savedpeak = 41888"
## [1] "j =326 peak =95 savedpeak = 41695"
## [2] "j =326 peak =113 savedpeak = 41713"
## [3] "j =326 peak =172 savedpeak = 41772"
## [4] "j =326 peak =195 savedpeak = 41795"
## [5] "j =326 peak =216 savedpeak = 41816"
## [6] "j =326 peak =223 savedpeak = 41823"
## [7] "j =326 peak =250 savedpeak = 41850"
## [8] "j =326 peak =288 savedpeak = 41888"
## [9] "j =326 peak =315 savedpeak = 41915"
## [10] "j =326 peak =345 savedpeak = 41945"
## [11] "j =326 peak =390 savedpeak = 41990"
## [12] "j =326 peak =414 savedpeak = 42014"
## [13] "j =326 peak =429 savedpeak = 42029"
## [1] "j =327 peak =88 savedpeak = 41816"
## [2] "j =327 peak =95 savedpeak = 41823"

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## [3] "j =327 peak =122 savedpeak = 41850"
## [4] "j =327 peak =160 savedpeak = 41888"
## [5] "j =327 peak =187 savedpeak = 41915"
## [6] "j =327 peak =217 savedpeak = 41945"
## [7] "j =327 peak =262 savedpeak = 41990"
## [8] "j =327 peak =286 savedpeak = 42014"
## [9] "j =327 peak =301 savedpeak = 42029"
## [10] "j =327 peak =324 savedpeak = 42052"
## [11] "j =327 peak =360 savedpeak = 42088"
## [12] "j =327 peak =380 savedpeak = 42108"
## [13] "j =327 peak =395 savedpeak = 42123"
## [1] "j =328 peak =89 savedpeak = 41945"
## [2] "j =328 peak =134 savedpeak = 41990"
## [3] "j =328 peak =158 savedpeak = 42014"
## [4] "j =328 peak =173 savedpeak = 42029"
## [5] "j =328 peak =196 savedpeak = 42052"
## [6] "j =328 peak =232 savedpeak = 42088"
## [7] "j =328 peak =252 savedpeak = 42108"
## [8] "j =328 peak =267 savedpeak = 42123"
## [9] "j =328 peak =311 savedpeak = 42167"
## [10] "j =328 peak =325 savedpeak = 42181"
## [11] "j =328 peak =350 savedpeak = 42206"
## [12] "j =328 peak =410 savedpeak = 42266"
## [1] "j =330 peak =94 savedpeak = 42206"
## [2] "j =330 peak =154 savedpeak = 42266"
## [3] "j =330 peak =183 savedpeak = 42295"
## [4] "j =330 peak =218 savedpeak = 42330"
## [5] "j =330 peak =279 savedpeak = 42391"
## [6] "j =330 peak =317 savedpeak = 42429"
## [7] "j =330 peak =334 savedpeak = 42446"
## [8] "j =330 peak =352 savedpeak = 42464"
## [9] "j =330 peak =368 savedpeak = 42480"
## [10] "j =330 peak =422 savedpeak = 42534"
## [1] "j =331 peak =90 savedpeak = 42330"
## [2] "j =331 peak =151 savedpeak = 42391"
## [3] "j =331 peak =189 savedpeak = 42429"
## [4] "j =331 peak =206 savedpeak = 42446"
## [5] "j =331 peak =224 savedpeak = 42464"
## [6] "j =331 peak =240 savedpeak = 42480"
## [7] "j =331 peak =294 savedpeak = 42534"
## [8] "j =331 peak =348 savedpeak = 42588"
## [9] "j =331 peak =365 savedpeak = 42605"
## [10] "j =331 peak =382 savedpeak = 42622"
## [11] "j =331 peak =408 savedpeak = 42648"
## [12] "j =331 peak =417 savedpeak = 42657"
## [13] "j =331 peak =437 savedpeak = 42677"
## [1] "j =332 peak =78 savedpeak = 42446"
## [2] "j =332 peak =96 savedpeak = 42464"
## [3] "j =332 peak =112 savedpeak = 42480"
## [4] "j =332 peak =166 savedpeak = 42534"
## [5] "j =332 peak =220 savedpeak = 42588"
## [6] "j =332 peak =237 savedpeak = 42605"
## [7] "j =332 peak =254 savedpeak = 42622"
## [8] "j =332 peak =280 savedpeak = 42648"

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## [9] "j =332 peak =289 savedpeak = 42657"
## [10] "j =332 peak =309 savedpeak = 42677"
## [11] "j =332 peak =334 savedpeak = 42702"
## [12] "j =332 peak =350 savedpeak = 42718"
## [13] "j =332 peak =362 savedpeak = 42730"
## [14] "j =332 peak =381 savedpeak = 42749"
## [15] "j =332 peak =402 savedpeak = 42770"
## [16] "j =332 peak =428 savedpeak = 42796"
## [1] "j =333 peak =92 savedpeak = 42588"
## [2] "j =333 peak =109 savedpeak = 42605"
## [3] "j =333 peak =126 savedpeak = 42622"
## [4] "j =333 peak =152 savedpeak = 42648"
## [5] "j =333 peak =161 savedpeak = 42657"
## [6] "j =333 peak =181 savedpeak = 42677"
## [7] "j =333 peak =206 savedpeak = 42702"
## [8] "j =333 peak =222 savedpeak = 42718"
## [9] "j =333 peak =234 savedpeak = 42730"
## [10] "j =333 peak =253 savedpeak = 42749"
## [11] "j =333 peak =274 savedpeak = 42770"
## [12] "j =333 peak =300 savedpeak = 42796"
## [13] "j =333 peak =319 savedpeak = 42815"
## [14] "j =333 peak =339 savedpeak = 42835"
## [15] "j =333 peak =354 savedpeak = 42850"
## [16] "j =333 peak =380 savedpeak = 42876"
## [17] "j =333 peak =395 savedpeak = 42891"
## [18] "j =333 peak =413 savedpeak = 42909"
## [19] "j =333 peak =433 savedpeak = 42929"
## [1] "j =334 peak =78 savedpeak = 42702"
## [2] "j =334 peak =94 savedpeak = 42718"
## [3] "j =334 peak =106 savedpeak = 42730"
## [4] "j =334 peak =125 savedpeak = 42749"
## [5] "j =334 peak =146 savedpeak = 42770"
## [6] "j =334 peak =172 savedpeak = 42796"
## [7] "j =334 peak =191 savedpeak = 42815"
## [8] "j =334 peak =211 savedpeak = 42835"
## [9] "j =334 peak =226 savedpeak = 42850"
## [10] "j =334 peak =252 savedpeak = 42876"
## [11] "j =334 peak =267 savedpeak = 42891"
## [12] "j =334 peak =285 savedpeak = 42909"
## [13] "j =334 peak =305 savedpeak = 42929"
## [14] "j =334 peak =354 savedpeak = 42978"
## [15] "j =334 peak =378 savedpeak = 43002"
## [16] "j =334 peak =422 savedpeak = 43046"
## [1] "j =335 peak =83 savedpeak = 42835"
## [2] "j =335 peak =98 savedpeak = 42850"
## [3] "j =335 peak =124 savedpeak = 42876"
## [4] "j =335 peak =139 savedpeak = 42891"
## [5] "j =335 peak =157 savedpeak = 42909"
## [6] "j =335 peak =177 savedpeak = 42929"
## [7] "j =335 peak =226 savedpeak = 42978"
## [8] "j =335 peak =250 savedpeak = 43002"
## [9] "j =335 peak =294 savedpeak = 43046"
## [10] "j =335 peak =318 savedpeak = 43070"
## [11] "j =335 peak =346 savedpeak = 43098"

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## [12] "j =335 peak =383 savedpeak = 43135"
## [13] "j =335 peak =421 savedpeak = 43173"
## [1] "j =337 peak =90 savedpeak = 43098"
## [2] "j =337 peak =127 savedpeak = 43135"
## [3] "j =337 peak =165 savedpeak = 43173"
## [4] "j =337 peak =198 savedpeak = 43206"
## [5] "j =337 peak =228 savedpeak = 43236"
## [6] "j =337 peak =258 savedpeak = 43266"
## [7] "j =337 peak =278 savedpeak = 43286"
## [8] "j =337 peak =316 savedpeak = 43324"
## [9] "j =337 peak =333 savedpeak = 43341"
## [10] "j =337 peak =367 savedpeak = 43375"
## [11] "j =337 peak =390 savedpeak = 43398"
## [12] "j =337 peak =411 savedpeak = 43419"
## [13] "j =337 peak =433 savedpeak = 43441"
## [1] "j =341 peak =117 savedpeak = 43637"
## [2] "j =341 peak =172 savedpeak = 43692"
## [3] "j =341 peak =209 savedpeak = 43729"
## [4] "j =341 peak =256 savedpeak = 43776"
## [5] "j =341 peak =312 savedpeak = 43832"
## [6] "j =341 peak =377 savedpeak = 43897"
## [7] "j =341 peak =401 savedpeak = 43921"
## [1] "j =342 peak =81 savedpeak = 43729"
## [2] "j =342 peak =128 savedpeak = 43776"
## [3] "j =342 peak =184 savedpeak = 43832"
## [4] "j =342 peak =249 savedpeak = 43897"
## [5] "j =342 peak =273 savedpeak = 43921"
## [6] "j =342 peak =328 savedpeak = 43976"
## [7] "j =342 peak =404 savedpeak = 44052"
## [1] "j =343 peak =121 savedpeak = 43897"
## [2] "j =343 peak =145 savedpeak = 43921"
## [3] "j =343 peak =200 savedpeak = 43976"
## [4] "j =343 peak =276 savedpeak = 44052"
## [5] "j =343 peak =338 savedpeak = 44114"
## [6] "j =343 peak =398 savedpeak = 44174"
## [1] "j =344 peak =148 savedpeak = 44052"
## [2] "j =344 peak =210 savedpeak = 44114"
## [3] "j =344 peak =270 savedpeak = 44174"
## [4] "j =344 peak =319 savedpeak = 44223"
## [5] "j =344 peak =343 savedpeak = 44247"
## [6] "j =344 peak =361 savedpeak = 44265"
## [7] "j =344 peak =380 savedpeak = 44284"
## [1] "j =345 peak =88 savedpeak = 44120"
## [2] "j =345 peak =109 savedpeak = 44141"
## [3] "j =345 peak =142 savedpeak = 44174"
## [4] "j =345 peak =159 savedpeak = 44191"
## [5] "j =345 peak =172 savedpeak = 44204"
## [6] "j =345 peak =189 savedpeak = 44221"
## [7] "j =345 peak =215 savedpeak = 44247"
## [8] "j =345 peak =232 savedpeak = 44264"
## [9] "j =345 peak =252 savedpeak = 44284"
## [10] "j =345 peak =312 savedpeak = 44344"
## [11] "j =345 peak =359 savedpeak = 44391"
## [12] "j =345 peak =389 savedpeak = 44421"

```

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## [13] "j =345 peak =418 savedpeak = 44450"
## [1] "j =346 peak =87 savedpeak = 44247"
## [2] "j =346 peak =104 savedpeak = 44264"
## [3] "j =346 peak =124 savedpeak = 44284"
## [4] "j =346 peak =184 savedpeak = 44344"
## [5] "j =346 peak =231 savedpeak = 44391"
## [6] "j =346 peak =261 savedpeak = 44421"
## [7] "j =346 peak =290 savedpeak = 44450"
## [8] "j =346 peak =317 savedpeak = 44477"
## [9] "j =346 peak =342 savedpeak = 44502"
## [10] "j =346 peak =355 savedpeak = 44515"
## [11] "j =346 peak =373 savedpeak = 44533"
## [12] "j =346 peak =397 savedpeak = 44557"
## [13] "j =346 peak =416 savedpeak = 44576"
## [14] "j =346 peak =432 savedpeak = 44592"
## [1] "j =347 peak =103 savedpeak = 44391"
## [2] "j =347 peak =133 savedpeak = 44421"
## [3] "j =347 peak =162 savedpeak = 44450"
## [4] "j =347 peak =189 savedpeak = 44477"
## [5] "j =347 peak =214 savedpeak = 44502"
## [6] "j =347 peak =227 savedpeak = 44515"
## [7] "j =347 peak =245 savedpeak = 44533"
## [8] "j =347 peak =269 savedpeak = 44557"
## [9] "j =347 peak =288 savedpeak = 44576"
## [10] "j =347 peak =304 savedpeak = 44592"
## [11] "j =347 peak =329 savedpeak = 44617"
## [12] "j =347 peak =348 savedpeak = 44636"
## [13] "j =347 peak =368 savedpeak = 44656"
## [14] "j =347 peak =401 savedpeak = 44689"
## [15] "j =347 peak =429 savedpeak = 44717"
## [1] "j =348 peak =86 savedpeak = 44502"
## [2] "j =348 peak =99 savedpeak = 44515"
## [3] "j =348 peak =117 savedpeak = 44533"
## [4] "j =348 peak =141 savedpeak = 44557"
## [5] "j =348 peak =160 savedpeak = 44576"
## [6] "j =348 peak =176 savedpeak = 44592"
## [7] "j =348 peak =201 savedpeak = 44617"
## [8] "j =348 peak =220 savedpeak = 44636"
## [9] "j =348 peak =240 savedpeak = 44656"
## [10] "j =348 peak =273 savedpeak = 44689"
## [11] "j =348 peak =301 savedpeak = 44717"
## [12] "j =348 peak =318 savedpeak = 44734"
## [13] "j =348 peak =338 savedpeak = 44754"
## [14] "j =348 peak =386 savedpeak = 44802"
## [15] "j =348 peak =417 savedpeak = 44833"
## [1] "j =349 peak =92 savedpeak = 44636"
## [2] "j =349 peak =112 savedpeak = 44656"
## [3] "j =349 peak =145 savedpeak = 44689"
## [4] "j =349 peak =173 savedpeak = 44717"
## [5] "j =349 peak =190 savedpeak = 44734"
## [6] "j =349 peak =210 savedpeak = 44754"
## [7] "j =349 peak =258 savedpeak = 44802"
## [8] "j =349 peak =289 savedpeak = 44833"
## [9] "j =349 peak =337 savedpeak = 44881"

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## [10] "j =349 peak =350 savedpeak = 44894"
## [11] "j =349 peak =370 savedpeak = 44914"
## [12] "j =349 peak =396 savedpeak = 44940"
## [13] "j =349 peak =408 savedpeak = 44952"
## [1] "j =356 peak =122 savedpeak = 45562"
## [2] "j =356 peak =142 savedpeak = 45582"
## [3] "j =356 peak =173 savedpeak = 45613"
## [4] "j =356 peak =211 savedpeak = 45651"
## [5] "j =356 peak =223 savedpeak = 45663"
## [6] "j =356 peak =261 savedpeak = 45701"
## [7] "j =356 peak =295 savedpeak = 45735"
## [8] "j =356 peak =330 savedpeak = 45770"
## [9] "j =356 peak =353 savedpeak = 45793"
## [10] "j =356 peak =382 savedpeak = 45822"
## [11] "j =356 peak =398 savedpeak = 45838"
## [1] "j =363 peak =83 savedpeak = 46419"
## [2] "j =363 peak =135 savedpeak = 46471"
## [3] "j =363 peak =162 savedpeak = 46498"
## [4] "j =363 peak =193 savedpeak = 46529"
## [5] "j =363 peak =206 savedpeak = 46542"
## [6] "j =363 peak =228 savedpeak = 46564"
## [7] "j =363 peak =242 savedpeak = 46578"
## [8] "j =363 peak =262 savedpeak = 46598"
## [9] "j =363 peak =296 savedpeak = 46632"
## [10] "j =363 peak =338 savedpeak = 46674"
## [11] "j =363 peak =365 savedpeak = 46701"
## [12] "j =363 peak =407 savedpeak = 46743"
## [13] "j =363 peak =430 savedpeak = 46766"
## [1] "j =369 peak =86 savedpeak = 47190"
## [2] "j =369 peak =153 savedpeak = 47257"
## [3] "j =369 peak =195 savedpeak = 47299"
## [4] "j =369 peak =239 savedpeak = 47343"
## [5] "j =369 peak =256 savedpeak = 47360"
## [6] "j =369 peak =292 savedpeak = 47396"
## [7] "j =369 peak =337 savedpeak = 47441"
## [8] "j =369 peak =364 savedpeak = 47468"
## [9] "j =369 peak =389 savedpeak = 47493"
## [10] "j =369 peak =430 savedpeak = 47534"
## [1] "j =370 peak =111 savedpeak = 47343"
## [2] "j =370 peak =128 savedpeak = 47360"
## [3] "j =370 peak =164 savedpeak = 47396"
## [4] "j =370 peak =209 savedpeak = 47441"
## [5] "j =370 peak =236 savedpeak = 47468"
## [6] "j =370 peak =261 savedpeak = 47493"
## [7] "j =370 peak =302 savedpeak = 47534"
## [8] "j =370 peak =322 savedpeak = 47554"
## [9] "j =370 peak =346 savedpeak = 47578"
## [10] "j =370 peak =358 savedpeak = 47590"
## [11] "j =370 peak =380 savedpeak = 47612"
## [12] "j =370 peak =404 savedpeak = 47636"
## [13] "j =370 peak =424 savedpeak = 47656"
## [1] "j =371 peak =81 savedpeak = 47441"
## [2] "j =371 peak =108 savedpeak = 47468"
## [3] "j =371 peak =133 savedpeak = 47493"

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## [4] "j =371 peak =174 savedpeak = 47534"
## [5] "j =371 peak =194 savedpeak = 47554"
## [6] "j =371 peak =218 savedpeak = 47578"
## [7] "j =371 peak =230 savedpeak = 47590"
## [8] "j =371 peak =252 savedpeak = 47612"
## [9] "j =371 peak =276 savedpeak = 47636"
## [10] "j =371 peak =296 savedpeak = 47656"
## [11] "j =371 peak =310 savedpeak = 47670"
## [12] "j =371 peak =344 savedpeak = 47704"
## [13] "j =371 peak =368 savedpeak = 47728"
## [14] "j =371 peak =402 savedpeak = 47762"
## [15] "j =371 peak =428 savedpeak = 47788"
## [1] "j =377 peak =112 savedpeak = 48240"
## [2] "j =377 peak =139 savedpeak = 48267"
## [3] "j =377 peak =164 savedpeak = 48292"
## [4] "j =377 peak =198 savedpeak = 48326"
## [5] "j =377 peak =225 savedpeak = 48353"
## [6] "j =377 peak =258 savedpeak = 48386"
## [7] "j =377 peak =302 savedpeak = 48430"
## [8] "j =377 peak =328 savedpeak = 48456"
## [9] "j =377 peak =362 savedpeak = 48490"
## [10] "j =377 peak =388 savedpeak = 48516"
## [11] "j =377 peak =406 savedpeak = 48534"
## [12] "j =377 peak =433 savedpeak = 48561"
## [1] "j =379 peak =106 savedpeak = 48490"
## [2] "j =379 peak =132 savedpeak = 48516"
## [3] "j =379 peak =150 savedpeak = 48534"
## [4] "j =379 peak =177 savedpeak = 48561"
## [5] "j =379 peak =195 savedpeak = 48579"
## [6] "j =379 peak =236 savedpeak = 48620"
## [7] "j =379 peak =259 savedpeak = 48643"
## [8] "j =379 peak =302 savedpeak = 48686"
## [9] "j =379 peak =315 savedpeak = 48699"
## [10] "j =379 peak =353 savedpeak = 48737"
## [11] "j =379 peak =394 savedpeak = 48778"
## [12] "j =379 peak =403 savedpeak = 48787"
## [13] "j =379 peak =435 savedpeak = 48819"
## [1] "j =380 peak =108 savedpeak = 48620"
## [2] "j =380 peak =117 savedpeak = 48629"
## [3] "j =380 peak =131 savedpeak = 48643"
## [4] "j =380 peak =174 savedpeak = 48686"
## [5] "j =380 peak =187 savedpeak = 48699"
## [6] "j =380 peak =225 savedpeak = 48737"
## [7] "j =380 peak =266 savedpeak = 48778"
## [8] "j =380 peak =275 savedpeak = 48787"
## [9] "j =380 peak =307 savedpeak = 48819"
## [10] "j =380 peak =345 savedpeak = 48857"
## [11] "j =380 peak =360 savedpeak = 48872"
## [12] "j =380 peak =383 savedpeak = 48895"
## [13] "j =380 peak =428 savedpeak = 48940"
## [1] "j =381 peak =97 savedpeak = 48737"
## [2] "j =381 peak =138 savedpeak = 48778"
## [3] "j =381 peak =147 savedpeak = 48787"
## [4] "j =381 peak =179 savedpeak = 48819"

```

```

## [5] "j =381 peak =217 savedpeak = 48857"
## [6] "j =381 peak =232 savedpeak = 48872"
## [7] "j =381 peak =255 savedpeak = 48895"
## [8] "j =381 peak =300 savedpeak = 48940"
## [9] "j =381 peak =314 savedpeak = 48954"
## [10] "j =381 peak =354 savedpeak = 48994"
## [11] "j =381 peak =380 savedpeak = 49020"
## [12] "j =381 peak =410 savedpeak = 49050"
## [1] "j =382 peak =89 savedpeak = 48857"
## [2] "j =382 peak =104 savedpeak = 48872"
## [3] "j =382 peak =127 savedpeak = 48895"
## [4] "j =382 peak =172 savedpeak = 48940"
## [5] "j =382 peak =186 savedpeak = 48954"
## [6] "j =382 peak =226 savedpeak = 48994"
## [7] "j =382 peak =252 savedpeak = 49020"
## [8] "j =382 peak =282 savedpeak = 49050"
## [9] "j =382 peak =356 savedpeak = 49124"
## [10] "j =382 peak =382 savedpeak = 49150"
## [11] "j =382 peak =403 savedpeak = 49171"
## [12] "j =382 peak =419 savedpeak = 49187"
## [1] "j =390 peak =80 savedpeak = 49872"
## [2] "j =390 peak =114 savedpeak = 49906"
## [3] "j =390 peak =148 savedpeak = 49940"
## [4] "j =390 peak =164 savedpeak = 49956"
## [5] "j =390 peak =178 savedpeak = 49970"
## [6] "j =390 peak =213 savedpeak = 50005"
## [7] "j =390 peak =244 savedpeak = 50036"
## [8] "j =390 peak =275 savedpeak = 50067"
## [9] "j =390 peak =310 savedpeak = 50102"
## [10] "j =390 peak =330 savedpeak = 50122"
## [11] "j =390 peak =347 savedpeak = 50139"
## [12] "j =390 peak =368 savedpeak = 50160"
## [13] "j =390 peak =396 savedpeak = 50188"
## [14] "j =390 peak =413 savedpeak = 50205"
## [1] "j =391 peak =85 savedpeak = 50005"
## [2] "j =391 peak =116 savedpeak = 50036"
## [3] "j =391 peak =147 savedpeak = 50067"
## [4] "j =391 peak =182 savedpeak = 50102"
## [5] "j =391 peak =202 savedpeak = 50122"
## [6] "j =391 peak =219 savedpeak = 50139"
## [7] "j =391 peak =240 savedpeak = 50160"
## [8] "j =391 peak =268 savedpeak = 50188"
## [9] "j =391 peak =285 savedpeak = 50205"
## [10] "j =391 peak =319 savedpeak = 50239"
## [11] "j =391 peak =355 savedpeak = 50275"
## [12] "j =391 peak =373 savedpeak = 50293"
## [13] "j =391 peak =395 savedpeak = 50315"
## [1] "j =392 peak =91 savedpeak = 50139"
## [2] "j =392 peak =112 savedpeak = 50160"
## [3] "j =392 peak =140 savedpeak = 50188"
## [4] "j =392 peak =157 savedpeak = 50205"
## [5] "j =392 peak =191 savedpeak = 50239"
## [6] "j =392 peak =227 savedpeak = 50275"
## [7] "j =392 peak =245 savedpeak = 50293"

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```
## [8] "j =392 peak =267 savedpeak = 50315"
## [9] "j =392 peak =314 savedpeak = 50362"
## [10] "j =392 peak =327 savedpeak = 50375"
## [11] "j =392 peak =358 savedpeak = 50406"
## [12] "j =392 peak =378 savedpeak = 50426"
## [13] "j =392 peak =404 savedpeak = 50452"
## [14] "j =392 peak =413 savedpeak = 50461"
## [15] "j =392 peak =433 savedpeak = 50481"
## Spectrum exam1d_1H has 38 peaks.
```

Plot the aligned peaks

```
plot_spectra_simple(aligned_peaks, main = "Aligned spectra peaks", xlab = "ppm", ylab = "intensity")
```

